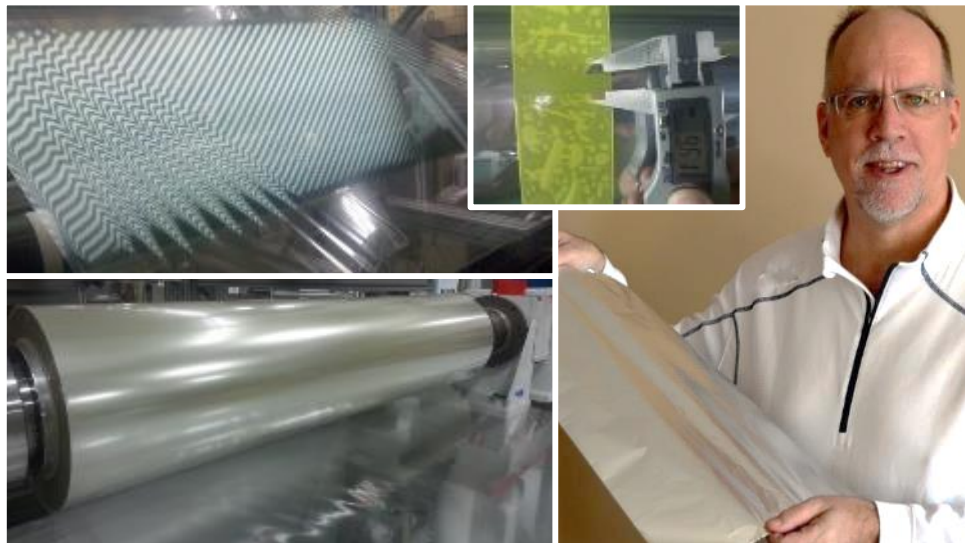




5.0 Lateral Control

Tracking, Wrinkling, Spreading, Guiding

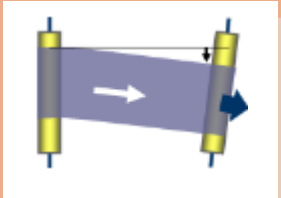
Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgumbe Road, Saint Paul, MN 55116

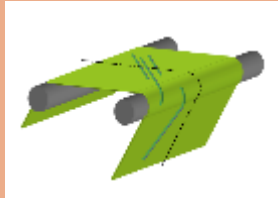
tjwalker@tjwa.com www.webhandling.com 651.686.5400

WH Fundamentals

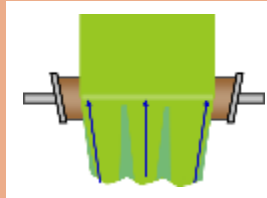
5: Lateral Motion: Tracking, Wrinkling, Spreading, Guiding



Tracking



Wrinkling



Spreading



Guiding

+ Lateral Motion

- ✓ What cause a web to shift laterally?
- ✓ What web guide is best for a process?
- ✓ What causes web wrinkles?
- ✓ What eliminates wrinkles?

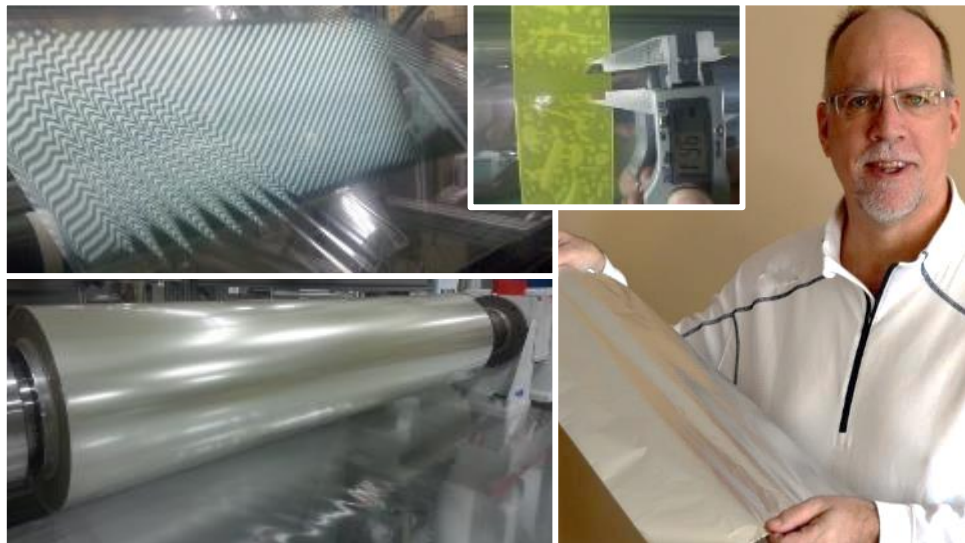
5.0 Lateral Control

5.1 Web Tracking Rules

5.2 Wrinkling

5.3 Spreading

5.4 Guiding



5.1 Web Tracking Rules

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgumbe Road, Saint Paul, MN 55116

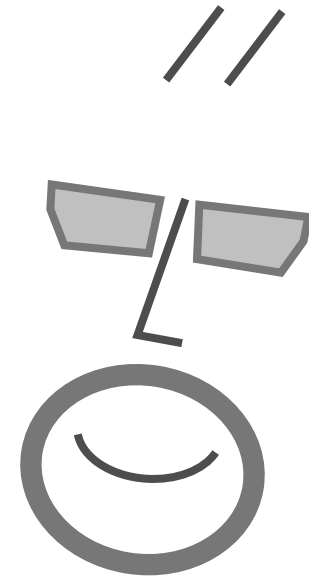
tjwalker@tjwa.com www.webhandling.com 651.686.5400

5.1 Learning Objectives

- ▶ How do the following cause a running web to shift laterally:
 - Roller level or square to centerline?
 - Roller diameter variations?
 - Web bagginess or skew
 - Thickness profile?
 - Uneven nip load?
- ▶ Why is traction (slip vs. stick) important to understanding how webs shift laterally?
- ▶ What is moment transfer?

Importance of Web Tracking

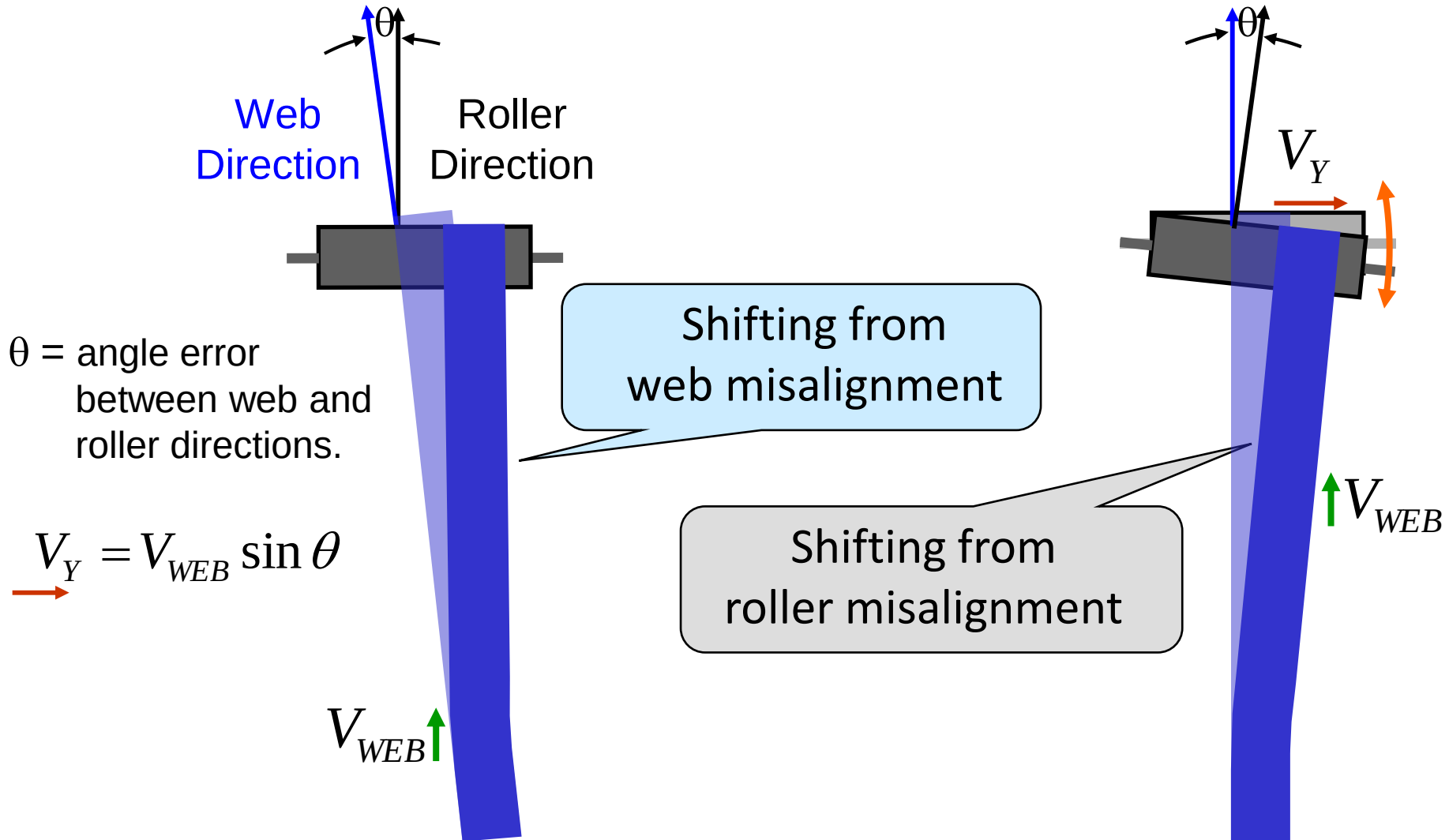
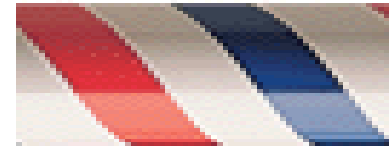
Understanding why webs shift left or right, when, and how much is the key to preventing wrinkles and proper use of spreading rollers and web guides.



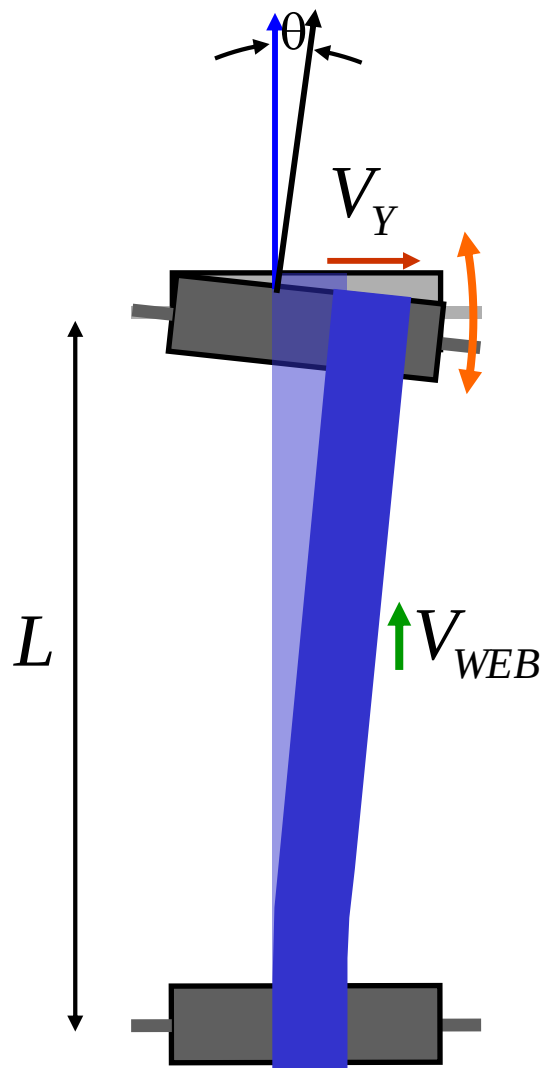
Tracking Philosophy

- ▶ Understand what makes webs shift left, right, or towards centerline.
- ▶ In traction (stick) web tracking is more reliable and predictable than non-traction (slip).
- ▶ If you want to avoid lateral tracking, use short spans ($L/W < 1$); if you want to shift the web laterally, use long spans ($L/W > 3$).
- ▶ Force to bend lateral goes up until the angle that creates a slack edge, and then does not increase much more.

Helical Wrapping



Helical Wrapping



$$V_Y = V_{WEB} \sin \theta$$

End pivoting rollers is common in belt guiding, but not fast enough for web guiding.

Example:

$$V_{web} = 6 \text{ m/min} = 0.1 \text{ m/s} = 100 \text{ mm/s}$$

$$\theta = 1 \text{ mm/m} = 0.001 \text{ radians} = 0.057 \text{ degrees}$$

$$V_Y = (100 \text{ mm/s})(0.001) = 0.1 \text{ mm/s}$$

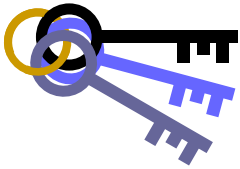
$$L = 1000 \text{ mm}$$

The web will shift initially at a speed of 0.1mm/s, but as the web to roller angle decreases, the web lateral speed will slow down.

The web steady-state position will be:

$$(2/3)L\theta = (0.667)(1000)(0.001) = 0.67 \text{ mm}$$

This will take 30s to make the full shift.



KEY CONCEPT



Parallel Entry Rule

The most important rule in web handling!

Webs will try to enter each downstream roller parallel to the roller's rotation.

The parallel entry rule is dependent on web-to-roller traction.

If parallel entry requires a web shift, the roller must have traction forces high enough to bend the web.

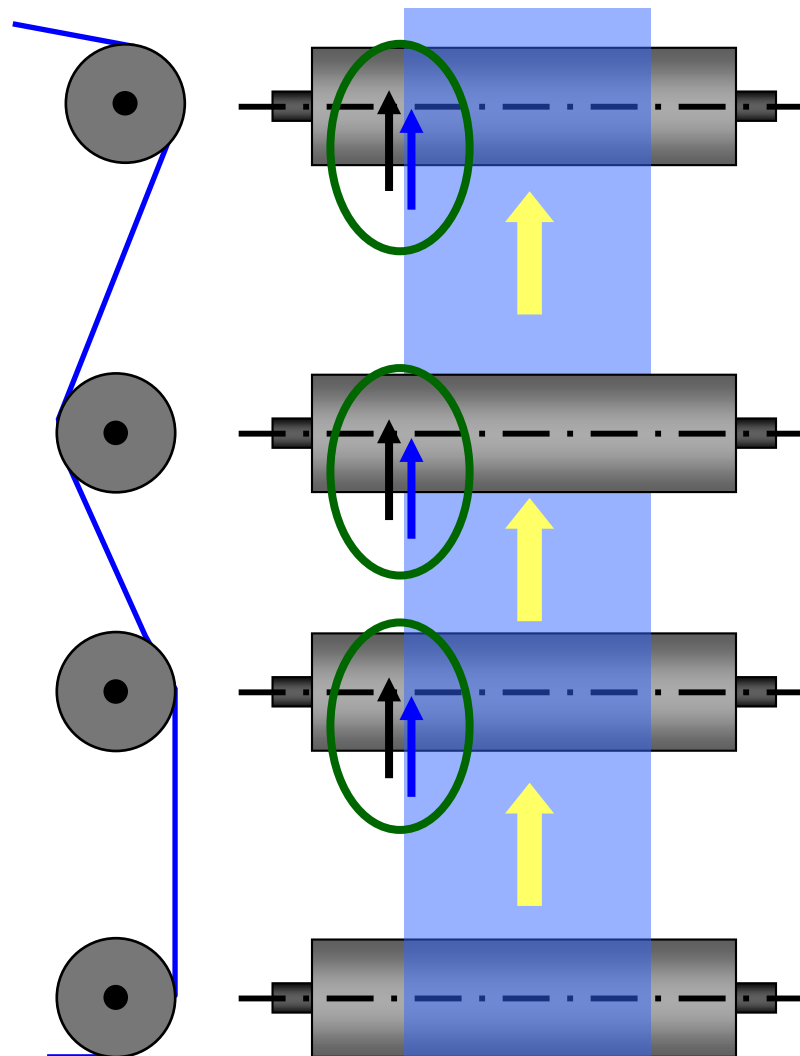
If the traction is too low or the roller is slipping, the web will not follow the parallel entry rule.

Parallel Entry Rule

Webs will try to enter each downstream roller parallel to the roller's rotation.

Roller Rotation Angle ↑

Web Entry Angle ↑

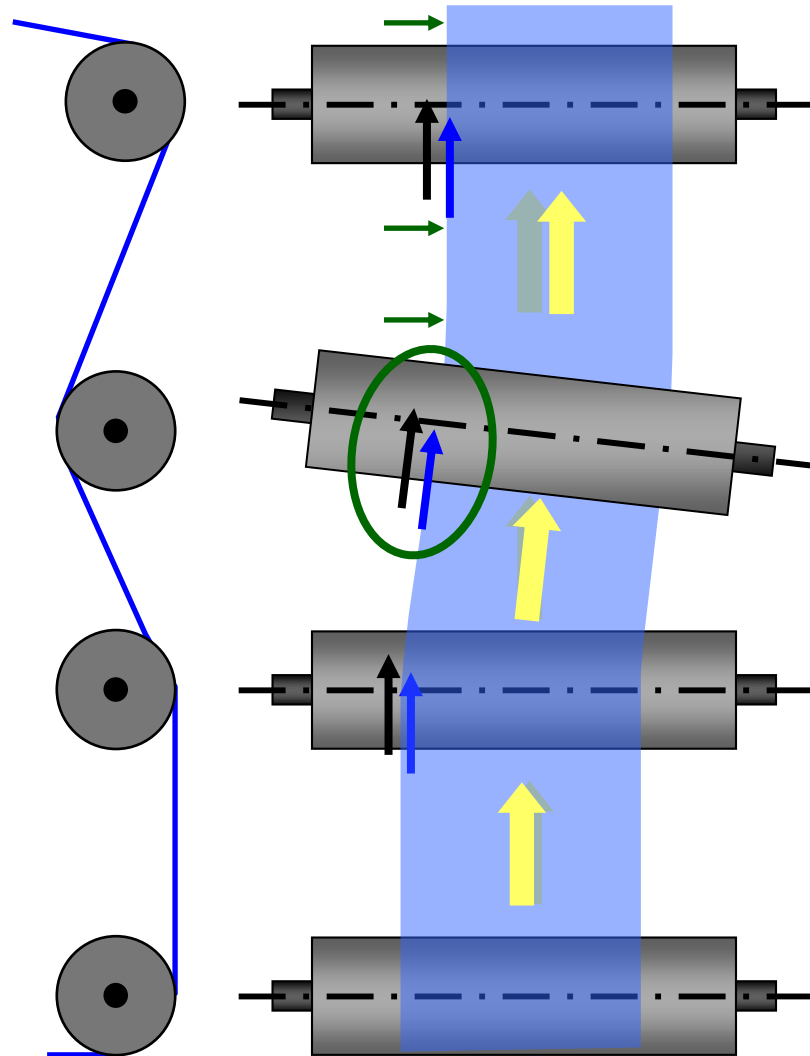


Parallel Entry Rule

Webs will try to enter each downstream roller parallel to the roller's rotation.

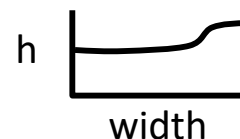
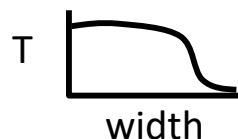
**Roller Rotation
Angle** ↑

**Web Entry
Angle** ↑



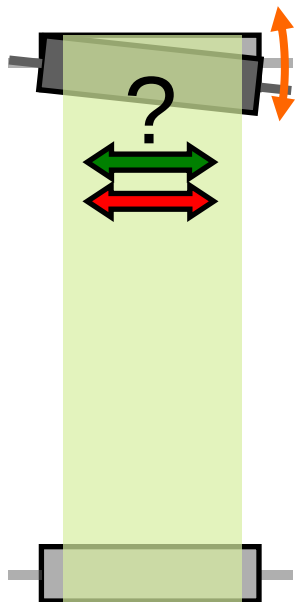
Summary - Five Tracking Rules

Webs will shift laterally from any source of tension asymmetry. Shift direction and magnitude is a function of traction, geometry, and web mechanical properties.

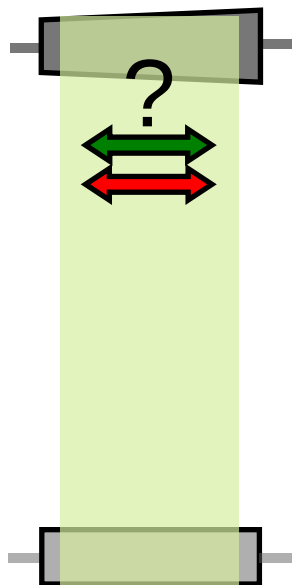


$$\theta_{\text{WEB}} \neq \theta_{\text{ROLLER}}$$

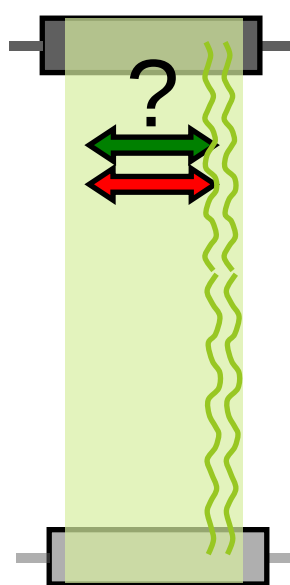
Misaligned
Roller



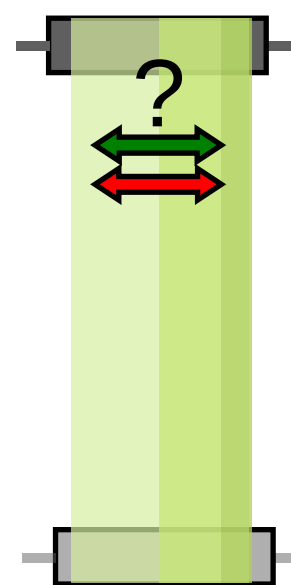
Diameter
Variations



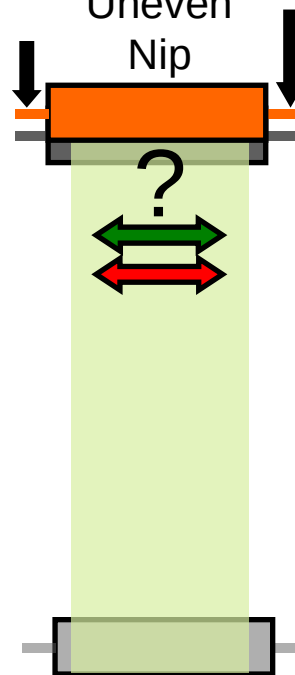
Baggy
Web

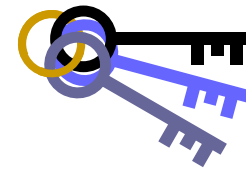


Thickness
Profile

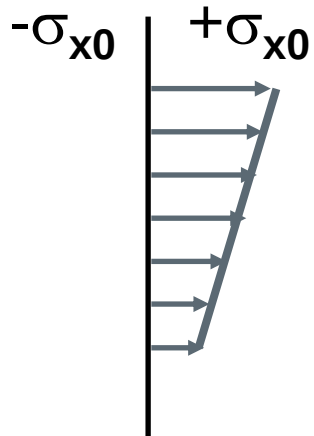


Uneven
Nip

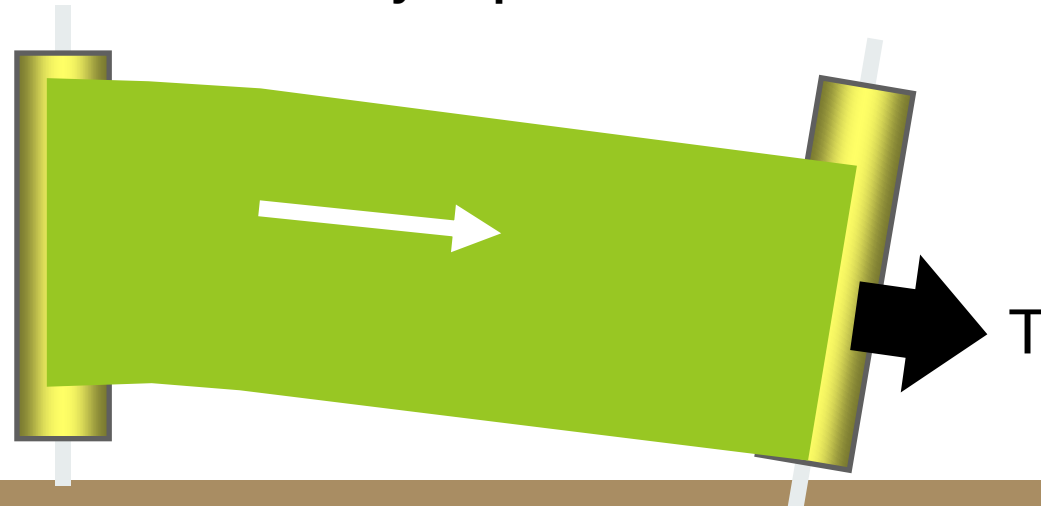
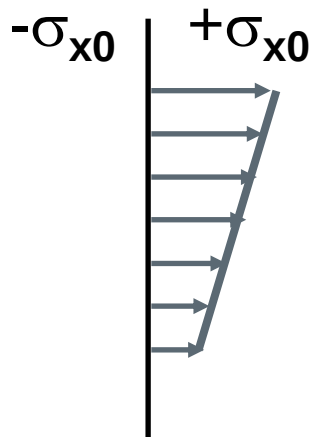




Web Bending Stress Analysis



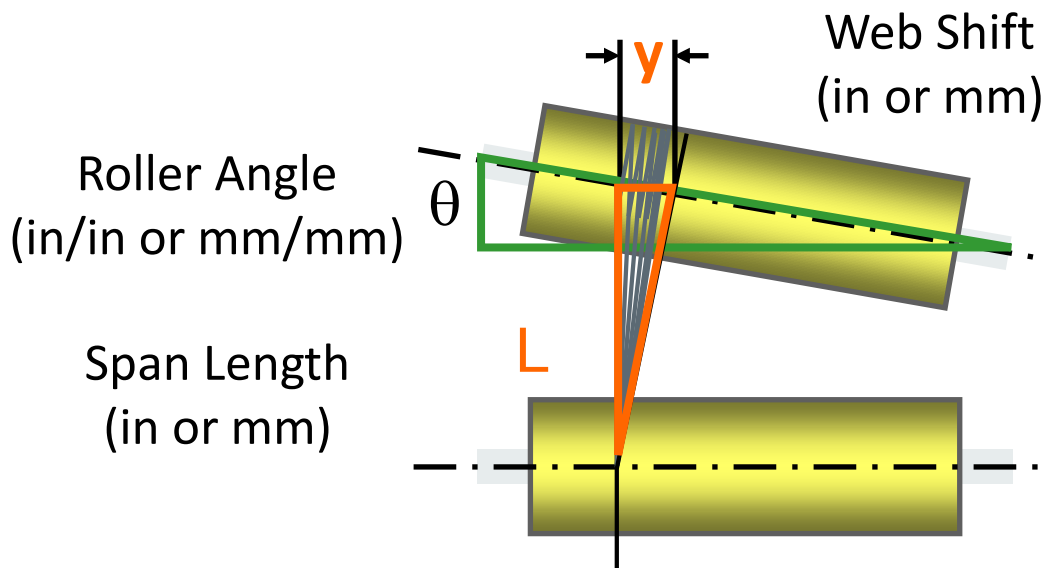
**Lateral bending requires lateral force.
Parallel entry requires web/roller traction.**



Tracking Result: Lateral Position

Two similar triangles estimate parallel entry tracking of a string.

Webs will shift as curves (not linear) and will shift 2/3 of strings.



String

$$\theta = \frac{y}{L}$$

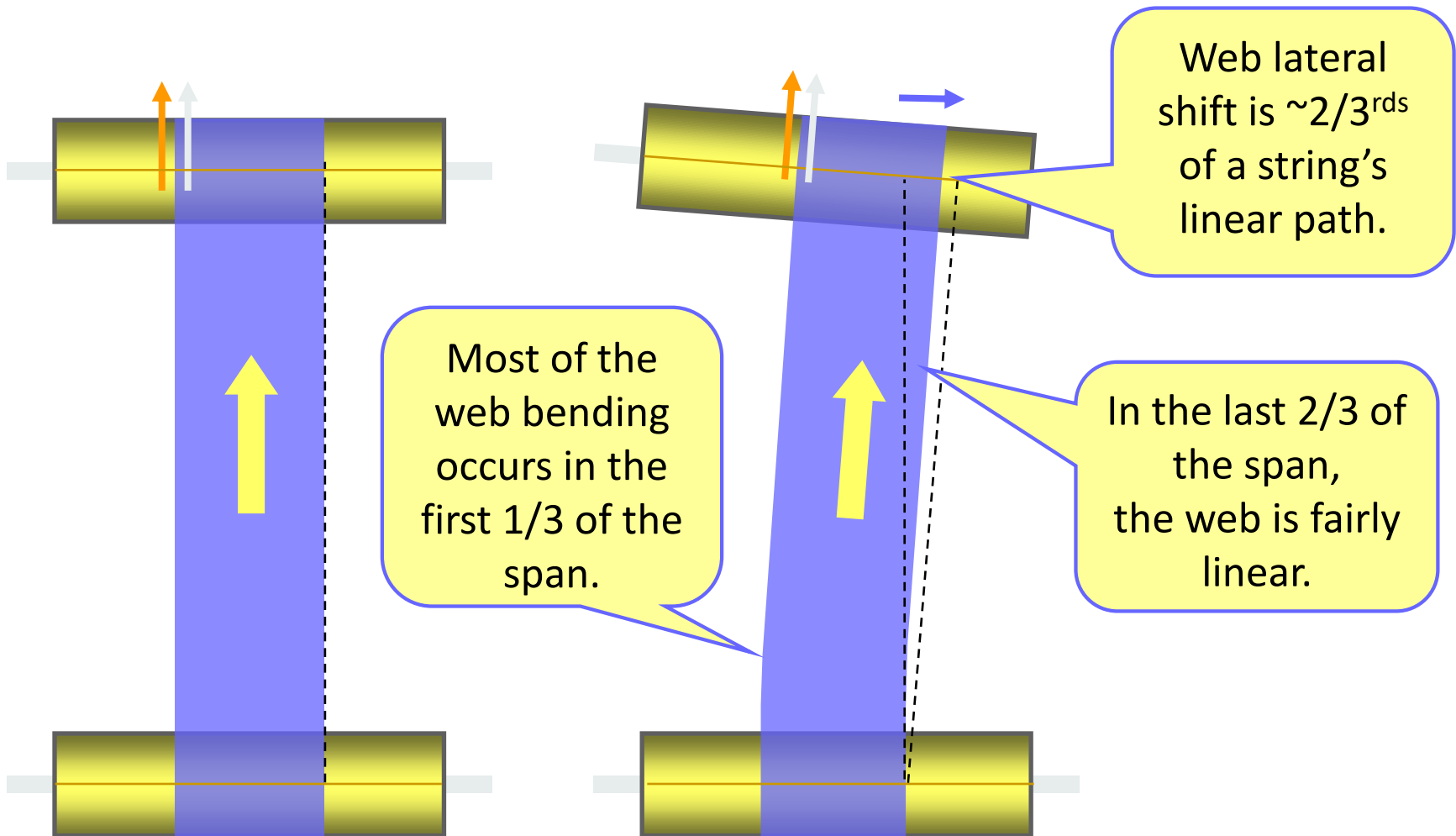
$$y = L \theta$$

Web

$$\frac{2}{3} \theta = \frac{y}{L}$$

$$y = \frac{2}{3} L \theta$$

Parallel Entry Tracking



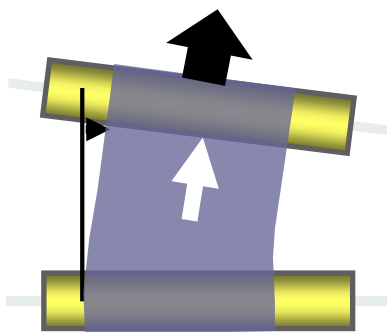
Tracking & Entry Span Length

As entry span length increases...

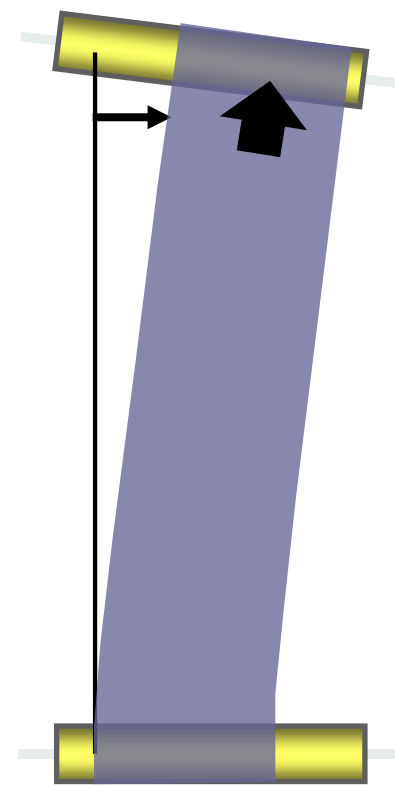
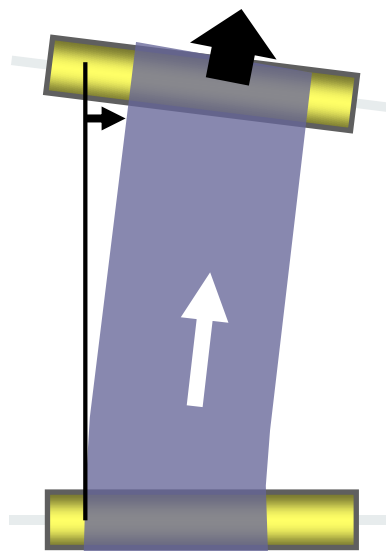
...the web is more flexible,

...shift more for the same roller angle,

...and required less force to shift the web.



Short Entry Span



Long Entry Span

Good or Poor Traction

What is...?



**Good
Traction
Tracking**



**Poor
Traction
Tracking**

Slip?

**Web bonds to
roller on
contact and
does not slip.**

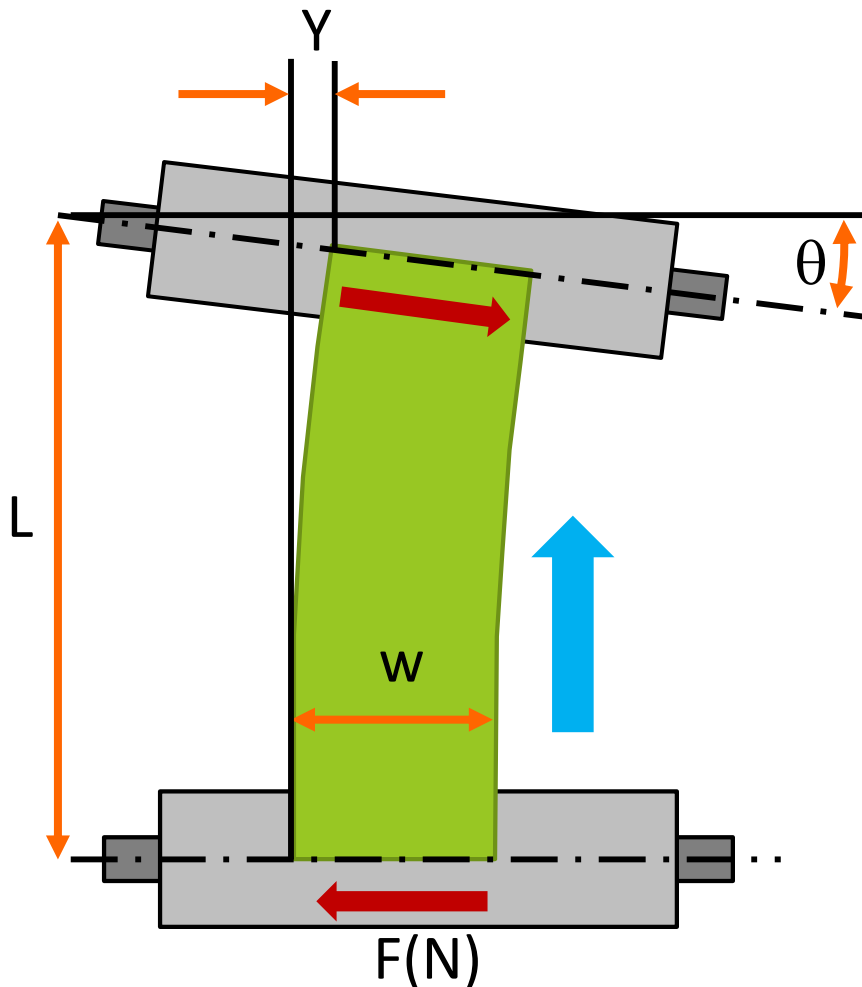
**Web forms
poor bond to
roller and may
slip.**

PER?

**Frictional force
is sufficient to
enforce PER.**

**Frictional force
is too low to
enforce PER.**

Lateral Tracking From Misalignment



- No slip
- Right-angle-rule

$$F = \frac{2EI\theta}{L^2} = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

$$Y(L) = \frac{2}{3} L \theta$$

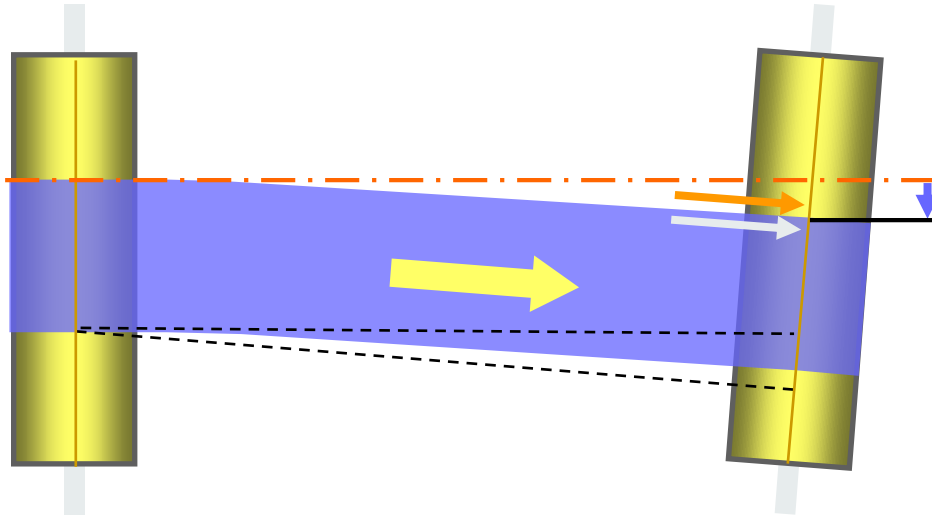
E = Young's modulus

t = web thickness

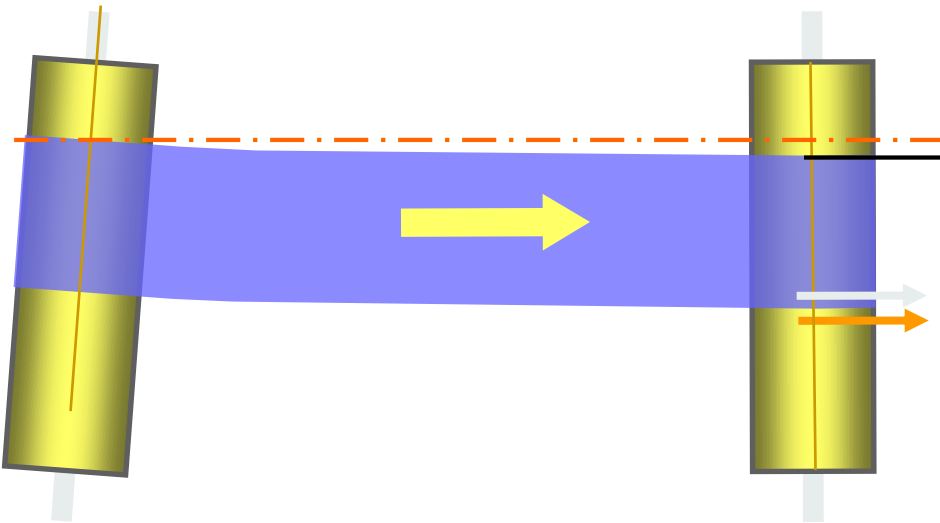
w = web width

$$I = \frac{1}{12} tw^3$$

Downstream Rollers Dominate

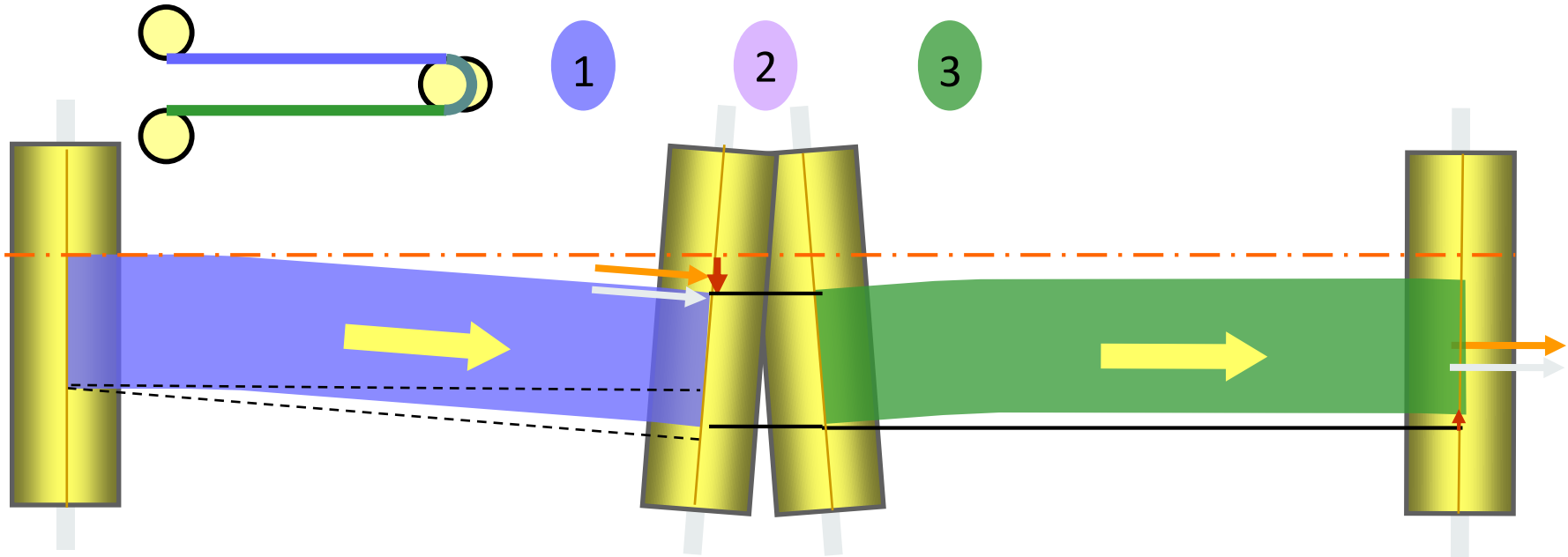


Downstream roller controls tracking direction.



Upstream roller has less effect on tracking direction.

One Misaligned Roller, Two Spans



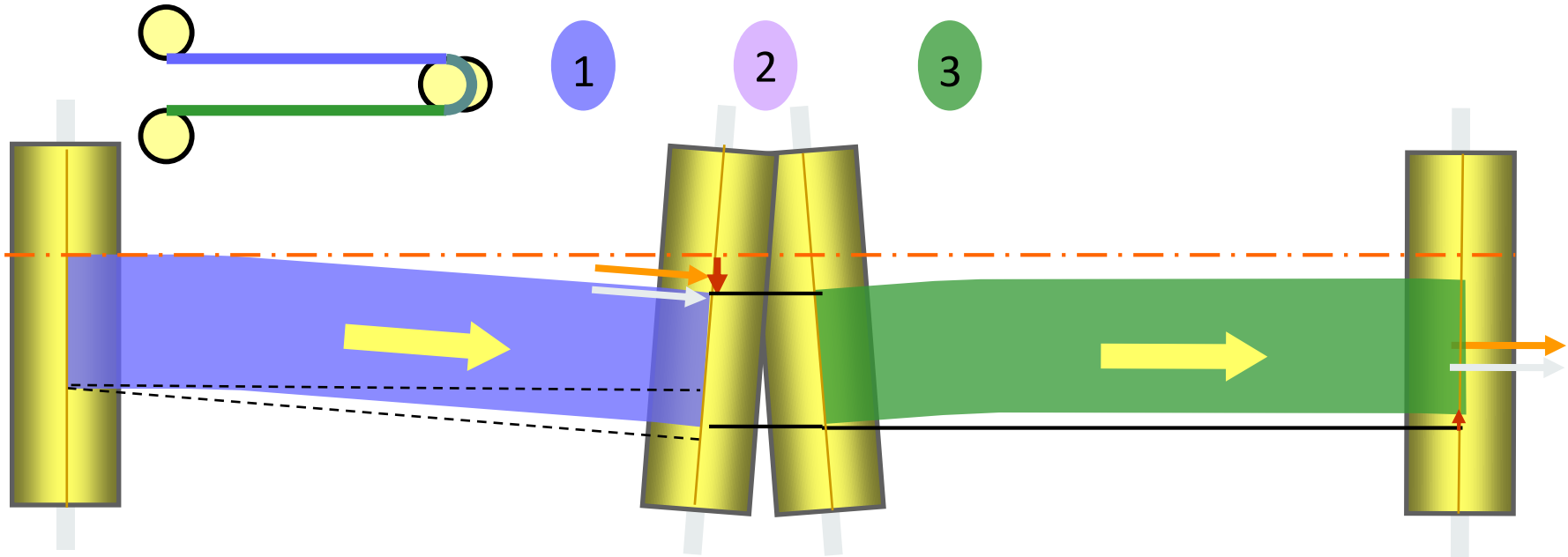
1 Tracking in entry span

3 Tracking in exit span

The parallel entry rule (PER)
(a.k.a. the normal entry rule)
(about 2/3 of straight line tracking)

This is also PER.
(For a string, this would be zero,
for a web, there is 'negative' steering.)

One Misaligned Roller, Two Spans



1 Tracking in entry span

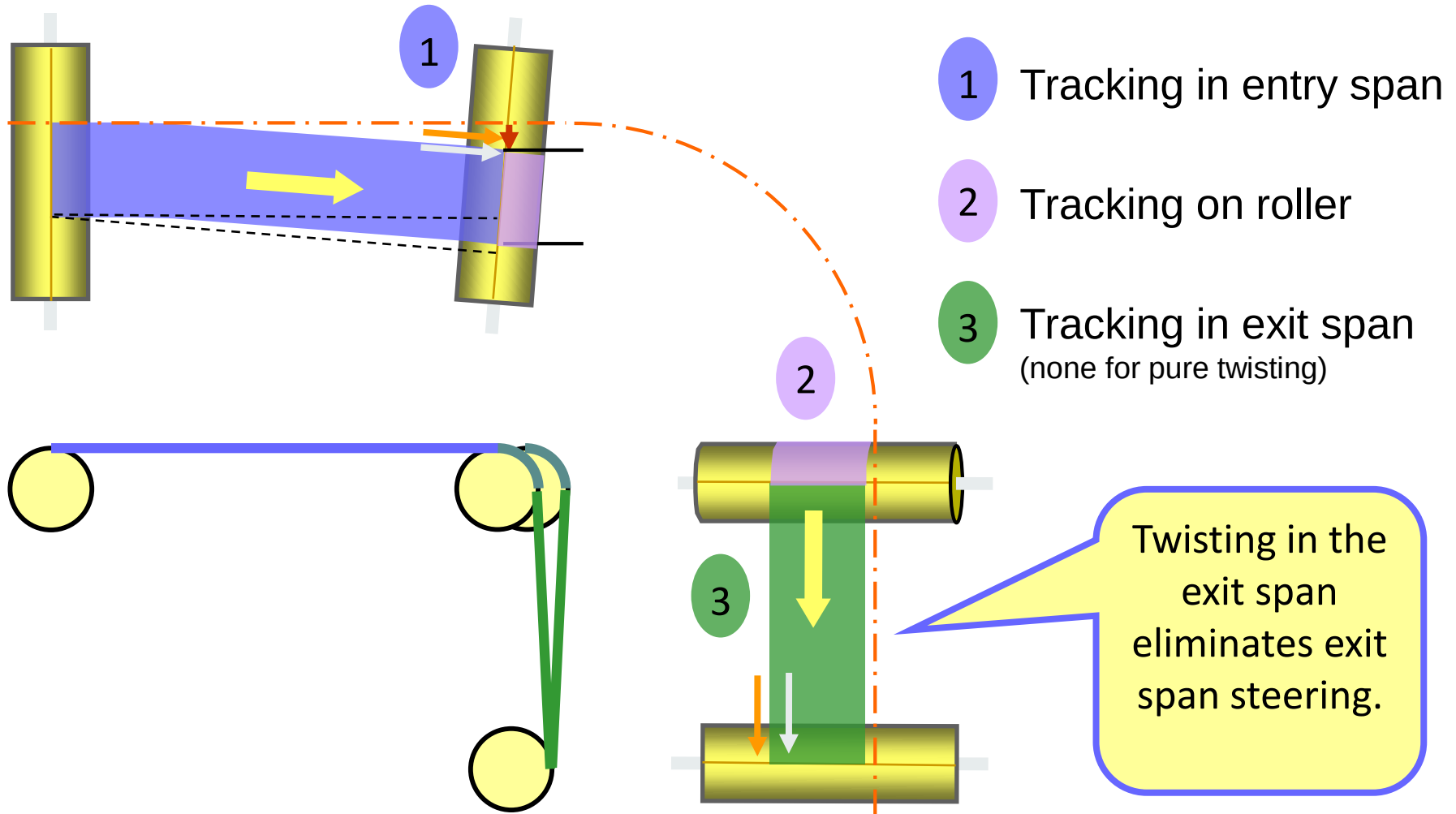
$$\delta_1 = \frac{2dL_1}{3W}$$

3 Tracking in exit span

$$\delta_2 = -\frac{dL_2}{3W}$$

$$\delta_{TOTAL} = \frac{d}{3W} (2L_1 - L_2)$$

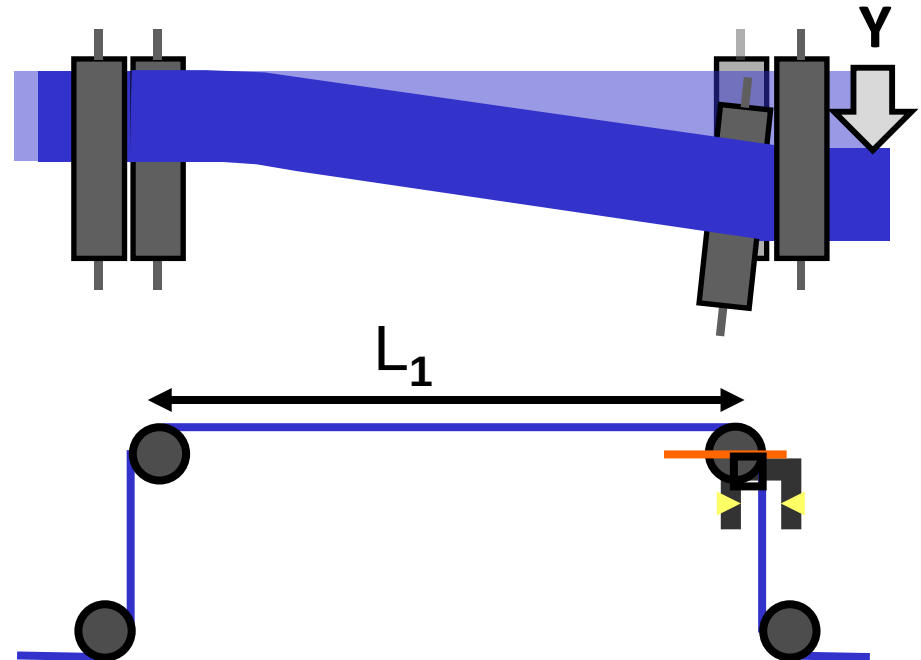
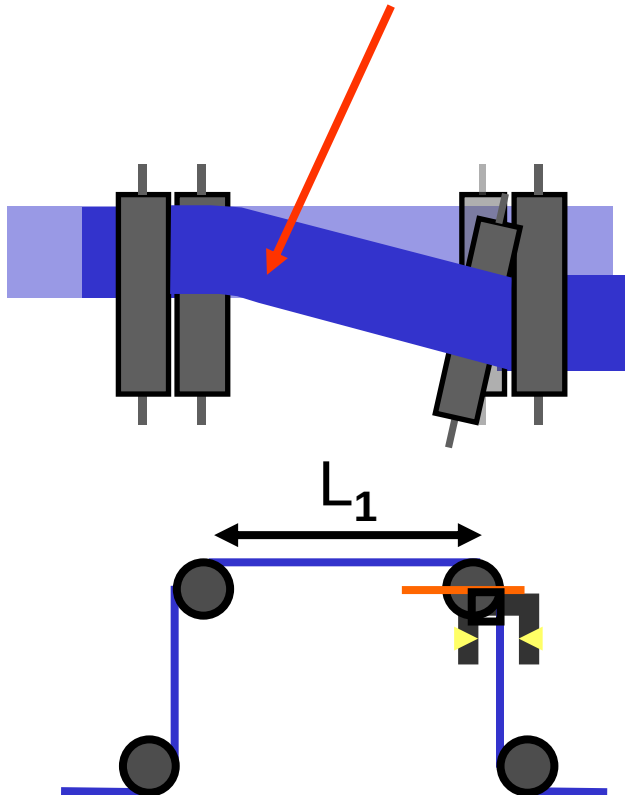
One Misaligned Roller, Two Spans



Avoiding Slack Edge

There will be a slack edge here if:

$$L_{1,\min} = \sqrt{\frac{3YtwE}{2T}} = \sqrt{\frac{3Yw}{2\varepsilon}}$$



Variable Definitions

$$L_{1,\min} = \sqrt{\frac{3YtwE}{2T}} = \sqrt{\frac{3Yw}{2\varepsilon}}$$

This equation can be rearranged to show the slack edge angle is a function of web strain and span length-to-width ratio.

$$\theta_{slack} = \varepsilon \left(\frac{L}{w} \right)$$

- Y = correction lateral shift (in.)
- δ = deflection or misalignment (in.)
- E = modulus (lb/in²)
- L = span length (in.)
- L₁ = entry span length (in.)
- L₂ = pre-entry span length (in.)
- t = thickness (in.)
- T = tension (lbf/in)
- w = width (in.)

Example Lateral Shift Calculations

We can calculate roller angle, lateral shift, and force required to create a slack edge.

$$\theta_{slack} = \varepsilon \left(\frac{L}{w} \right)$$

$$y = \frac{2}{3} L \theta$$

$$F = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

$$y_{slack} = \frac{2\varepsilon L^2}{3w} = \frac{2F_T}{3tE} \left(\frac{L}{w} \right)^2$$

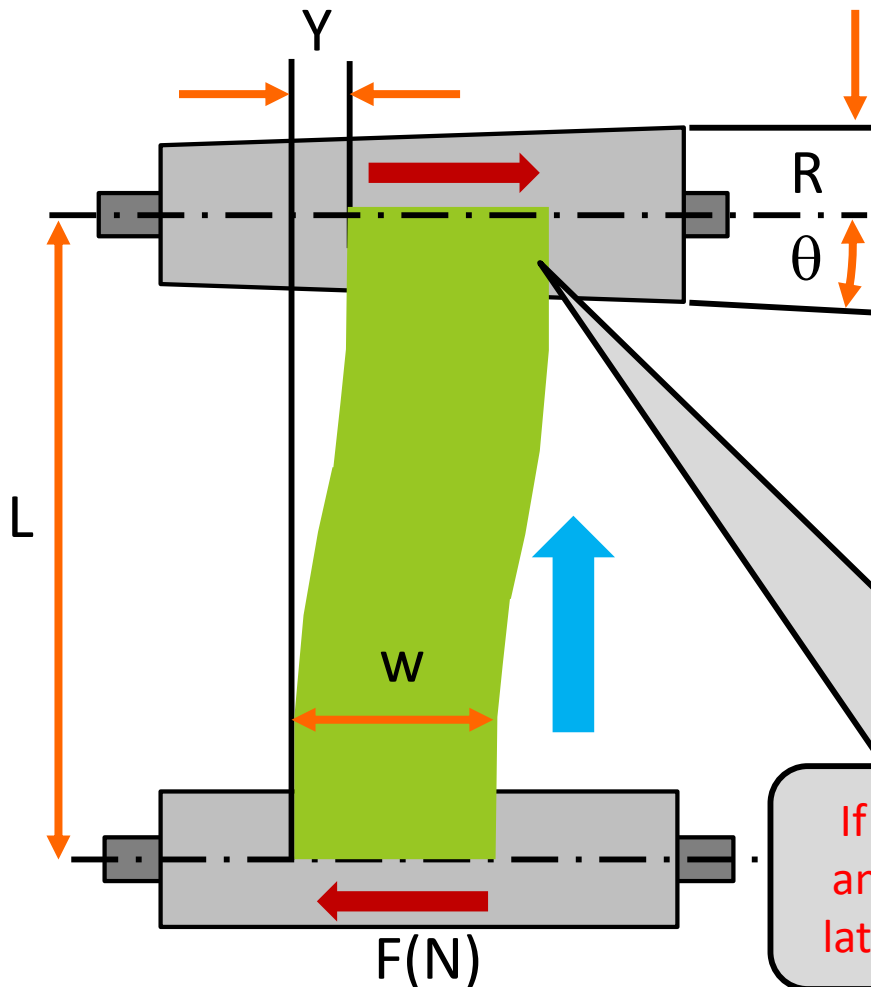
Y	= 0.5-in	= 12.5mm
E	= 500,000psi	= 3400MPa
L	= 40-in.	= 1000mm
t	= 0.002-in.	= 0.05mm
F _T	= 20 lbs	= 90N
w	= 10-in.	= 250mm.

Stress	= 1000 psi	= 0.7MPa
Strain	= 0.2%	= 0.2%
L/w	= 4	= 4
Y _{slack}	= 0.21-in	= 5mm
F _{slack}	= 1.9-lbs	= 8.5N

Slack Edge Angle

- The slack edge angle is defined as the point where web bending creates a change in tension of $\pm 100\%$ across the web's width, driving the low tension side of the web to zero tension and the high tension side of the web to 2X average tension.
- Bending the web at any angle beyond the slack edge angle creates slackness on the low tension side of the web.
- Bending the web to 2X the slack edge angle will put half the web width into slackness. Bending at 3X the slack edge angle will load all the tension on 1/3 of the web and put 2/3s of the web in slackness.
- A skew (a.k.a cambered) web will inherently carry the tension asymmetrically and will have a different angle whether the web bending is in the same or opposite slack edge as the web's natural curvature.

Lateral Tracking from Diameter Profile

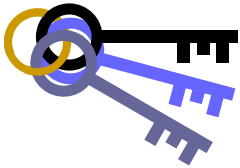


- No slip
- Right-angle-rule

$$F = \frac{2EI\theta_c}{RL} = EtY\left(\frac{w}{L}\right)^3$$

$$Y(L) = \frac{\theta_c L^2}{6R}$$

If friction from COF, tension, and wrap angle is less than F , lateral shift will be less than Y .



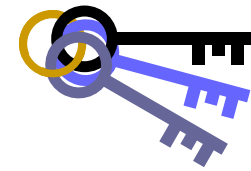
KEY CONCEPT



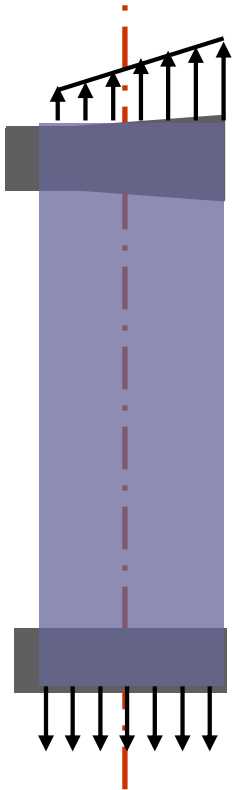
Diameter-Induced Tracking

A web will react to non-cylindrical roller's diameter variations. The combination of induced curvature and the parallel entry rule (PER) will cause a web to track laterally to the larger diameter of a downstream roller. However, if there is insufficient traction, the PER is ignored and the web will tend to track towards the smaller diameter position.

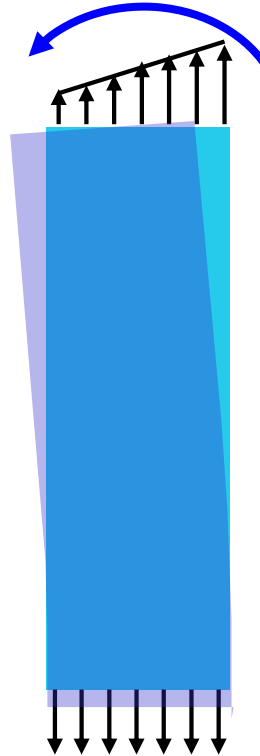
Use: Concave roller profiles will spread a web or prevent wrinkling. Convex (crowned) roller profiles will center a narrow web, but cause wrinkles in a wide web.



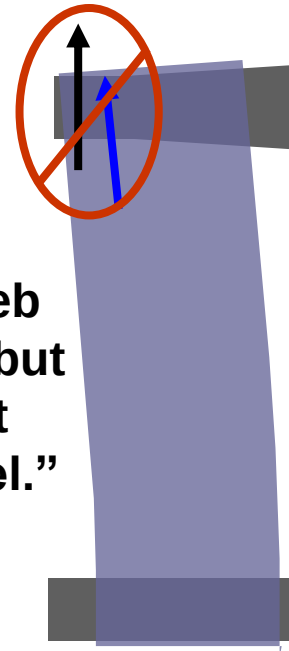
Tracking & Diameter Variations



Diameter variations strain the web, creating tension variations

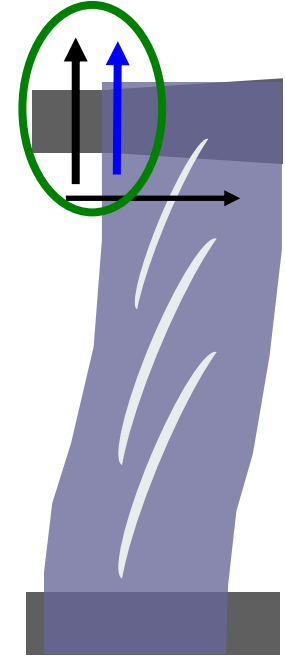


The uneven tension exerts a torque on the web.



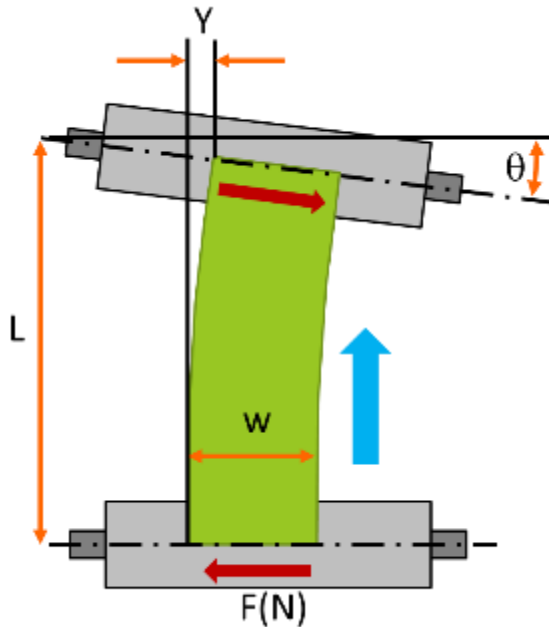
The web bends, but is not "parallel."

With low traction the web stays non-parallel.



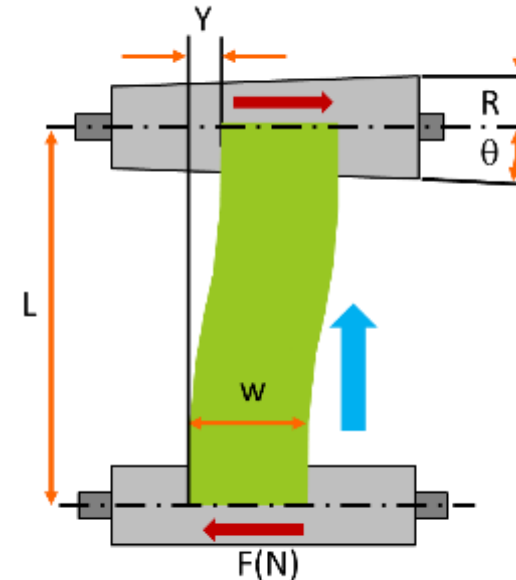
With high traction the web tracks to "parallel."

Lateral Tracking Comparison



$$F = \frac{2EI\theta}{L^2} = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

$$Y(L) = \frac{2}{3} L \theta$$



$$F = \frac{2EI\theta_c}{RL} = EtY \left(\frac{w}{L} \right)^3$$

$$Y(L) = \frac{\theta_c L^2}{6R}$$

Belt Sander Tracking

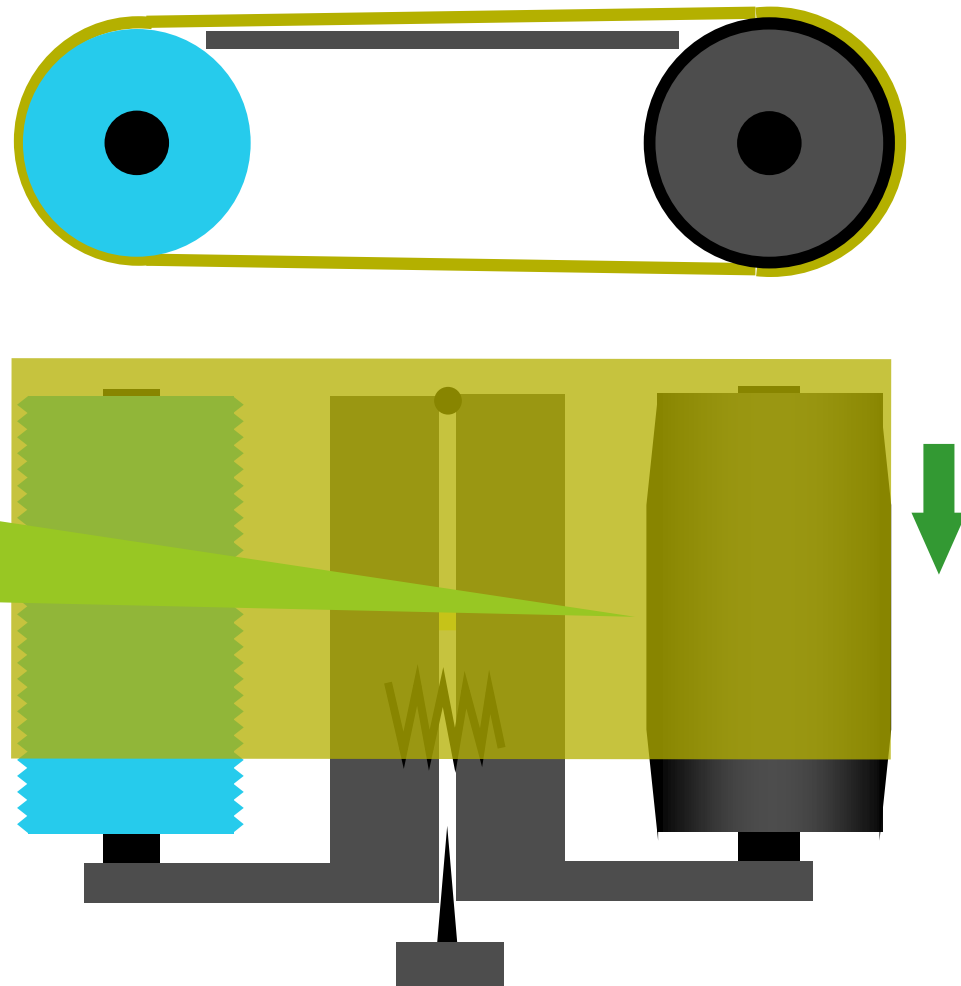
Grooved idler, with adjustable angle

Crowned, rubber driven roller



Tracking adjustment changes angle of grooved roller

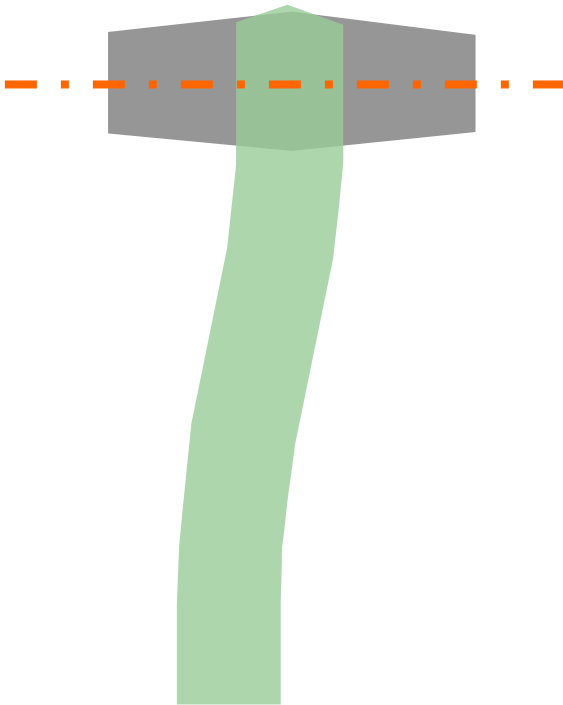
Belt Sander Tracking



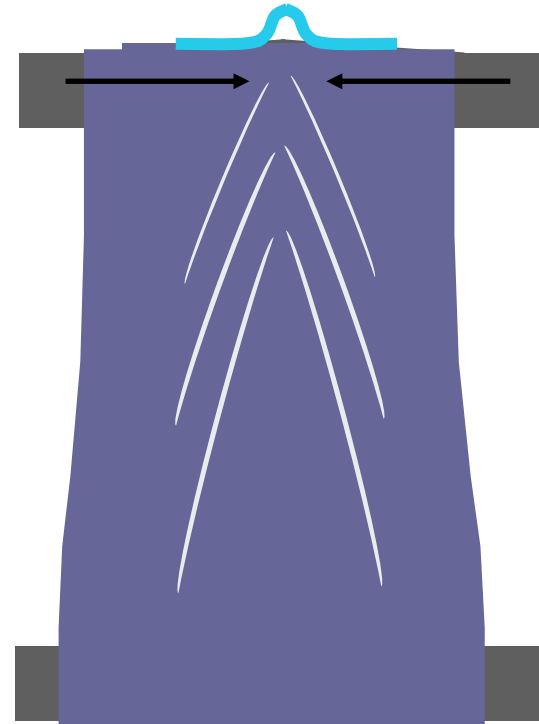
If the belt is off-center, the crowned roller and diameter-induced tracking will bring it back to center.

Crowned Rollers: Centering vs Wrinkling

A crowned roller is an effective passive centerline web guide for narrow, stiff webs.

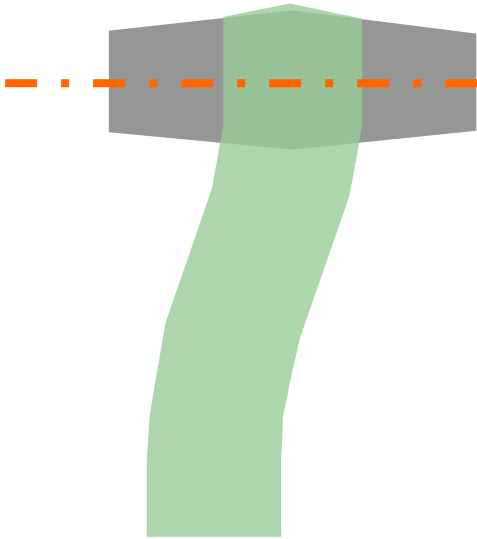


A crowned roller is an undesired wrinkle generator for wide, thin webs.

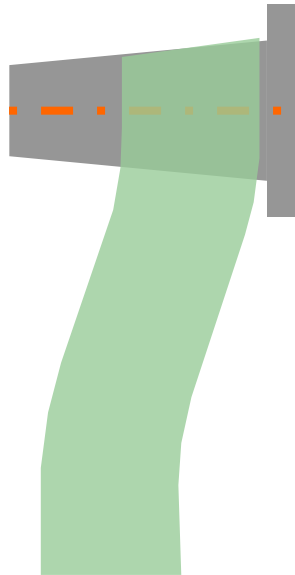


Passive Guiding

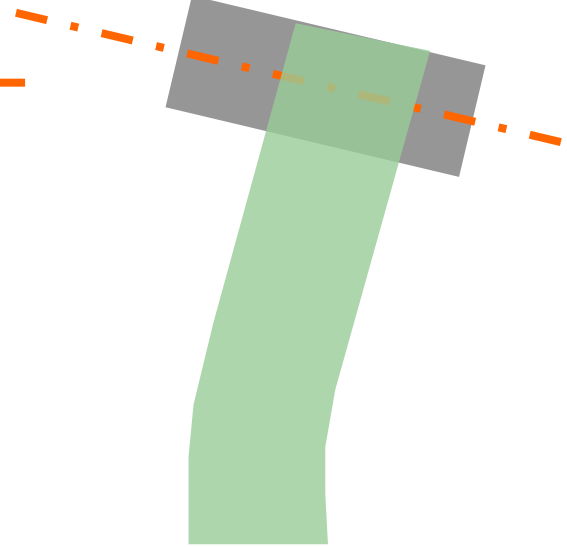
Crowned Rollers



Tapered Flanged Rollers

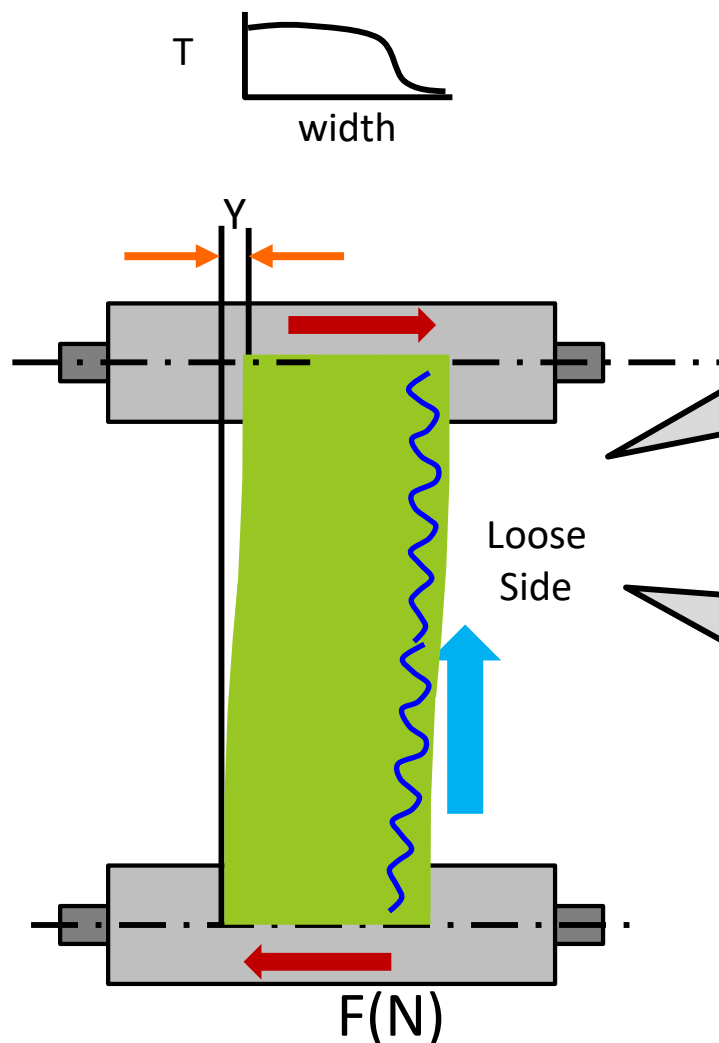


Skewed Rollers



- Best in spans with high L/W ratio.
- May create wrinkles in wide webs.

Lateral Tracking From **Web Skew** (One Side Baggy)



Lateral motion from web bagginess has been difficult to quantify or model.

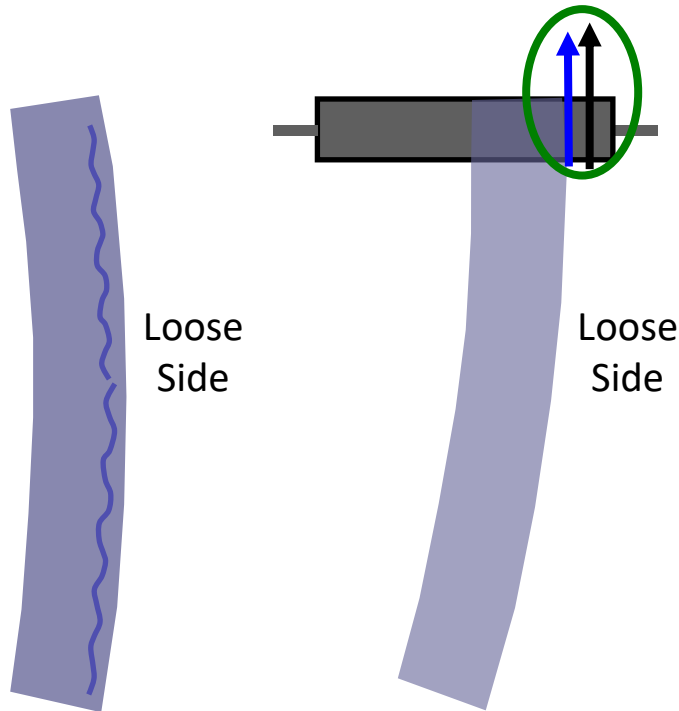
Some experiments have confirmed an asymmetrical baggy web shifts toward its baggy side. This is weak effect.

In wrinkling, baggy web is viewed as an additive effect to other causes.

- Bending with the bagginess is less wrinkle sensitive. Bending against the baggy will wrinkle easier.
- Baggy lanes will lower the shear needed to create troughing, the precursor to most wrinkles.

Baggy Web Tracking

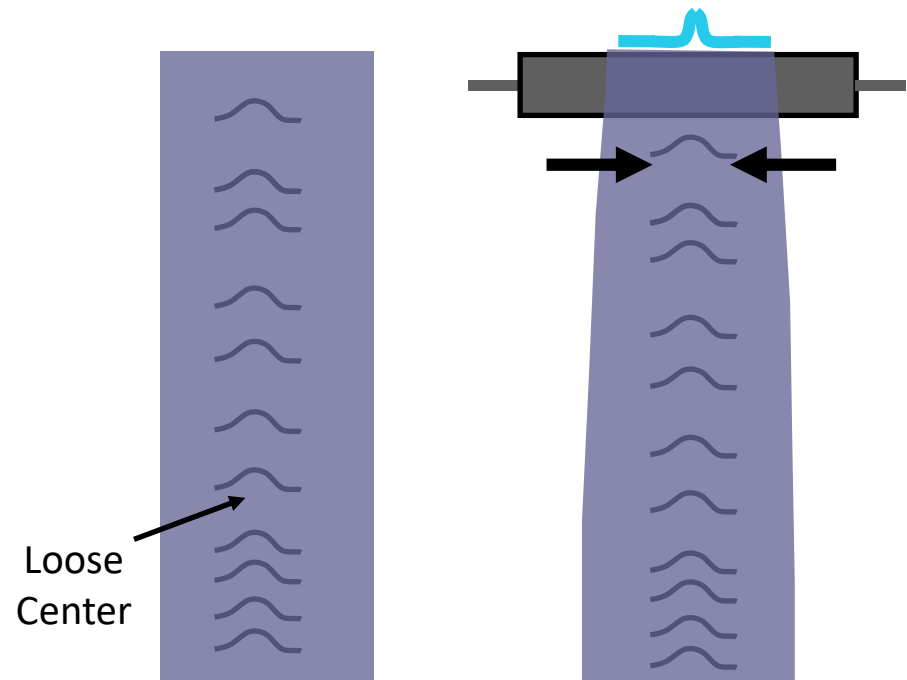
The curvature of a baggy edge web violates the parallel entry rule, tracking the web to the loose side.



Baggy Edge
Web

Baggy Edge
Web Tracking

A baggy center web will track from both edges to the center, creating a wrinkle.



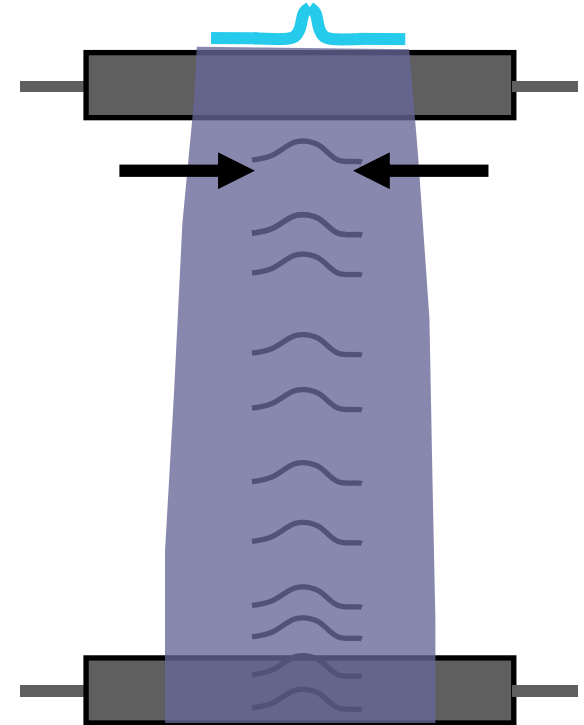
Baggy Center
Web

Baggy Center
Web Tracking

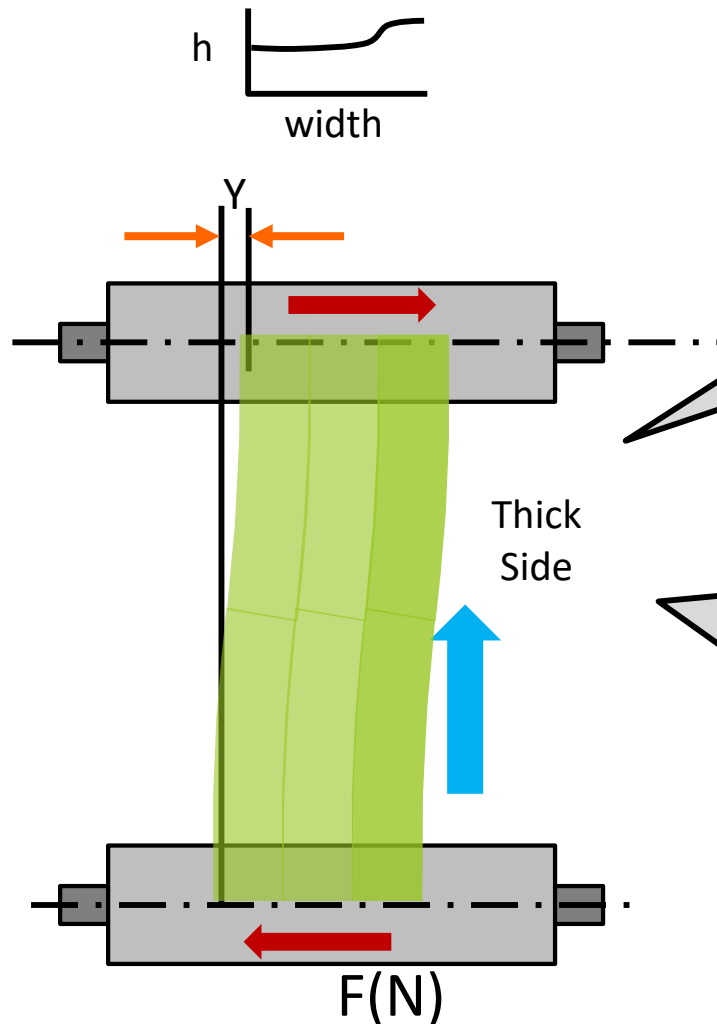
Baggy Center Wrinkling

Why does the web wrinkle in baggy lanes?

- Each half web tracks towards its long side (the center).
- Baggy lanes have low or no tension. Without tension stiffening, even the small compressive stress buckles the web.
- If the baggy lane had no tension, the tensioned portions on either side track as independent webs, not able to track outward, but free to track inward and wrinkle.



Lateral Tracking From Thickness Profile



Webs will move towards their thicker lanes, whether the added thickness is on the inside or outside of roller contact.

The thicker lanes are tensioned more from bending around a roller's radius, creating a moment that laterally bends the web away from the thick side. The bending web has an angle misaligned to the roller and tracks to the thick side to satisfy the parallel entry rule.

Lateral Motion From Thickness Profile

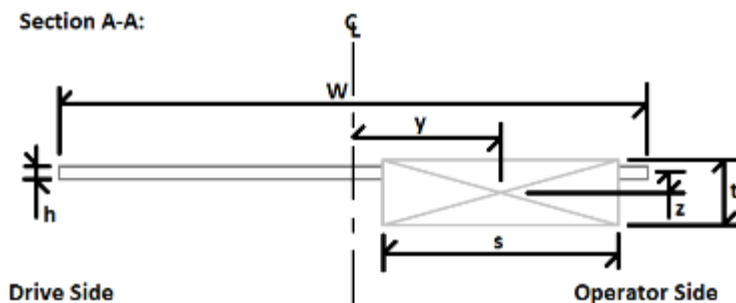
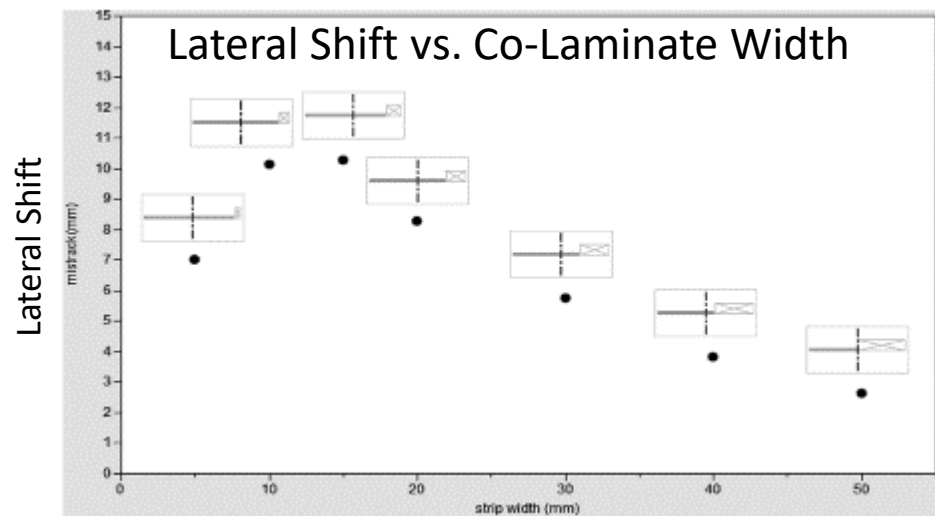
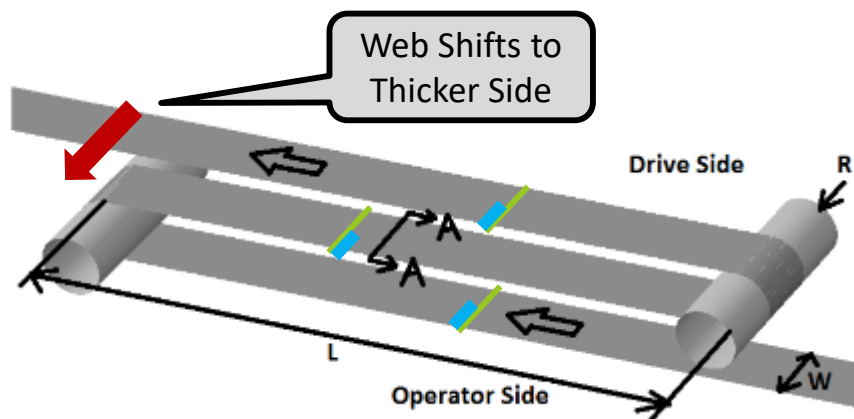


Figure 2 – Cross-section of Simulated Web with a Thick Section



Co-Laminate Width % of Overall Width

For most simulation runs the web shifted laterally toward the operator side of the web which contained the non-uniform thick section. If the thick section was more centered on the base web, generally there was a smaller lateral shift. In a small set of runs, the web shifted in the opposite direction (see Table 5).

LATERAL DYNAMICS SIMULATION OF WEBS HAVING CROSS-MACHINE DIRECTION VARIATION

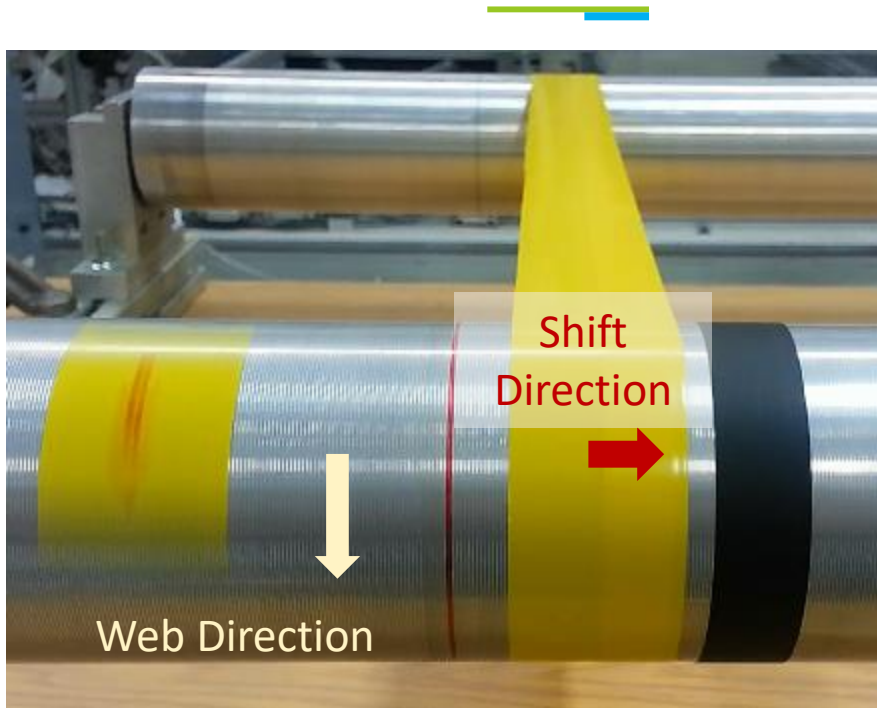
By

Stephen Lange, Michael Looney, and Jeremy Carrie
Procter & Gamble

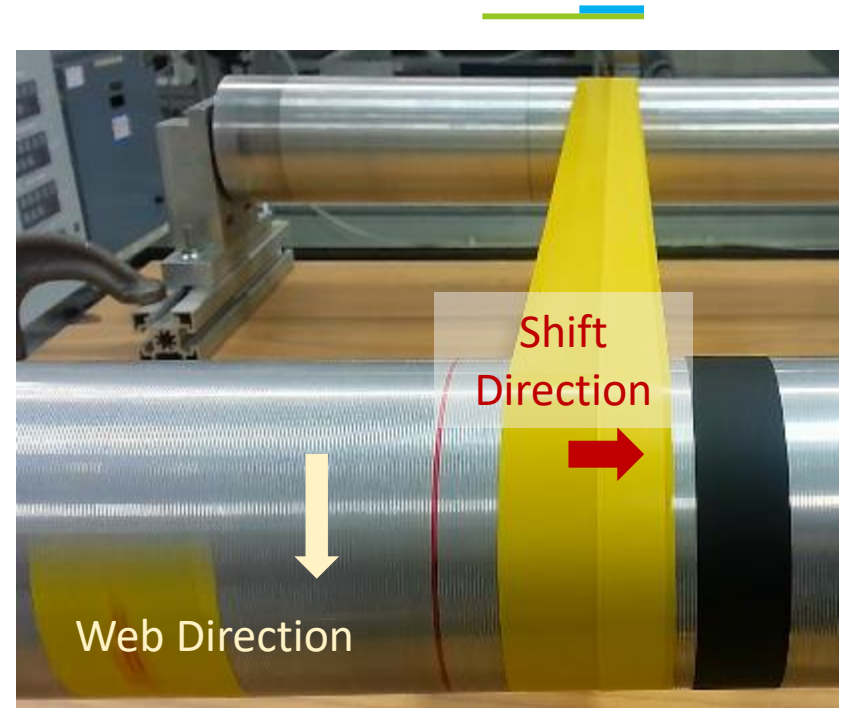
IWEB 2013

Belt Shifting From Thickness Profile

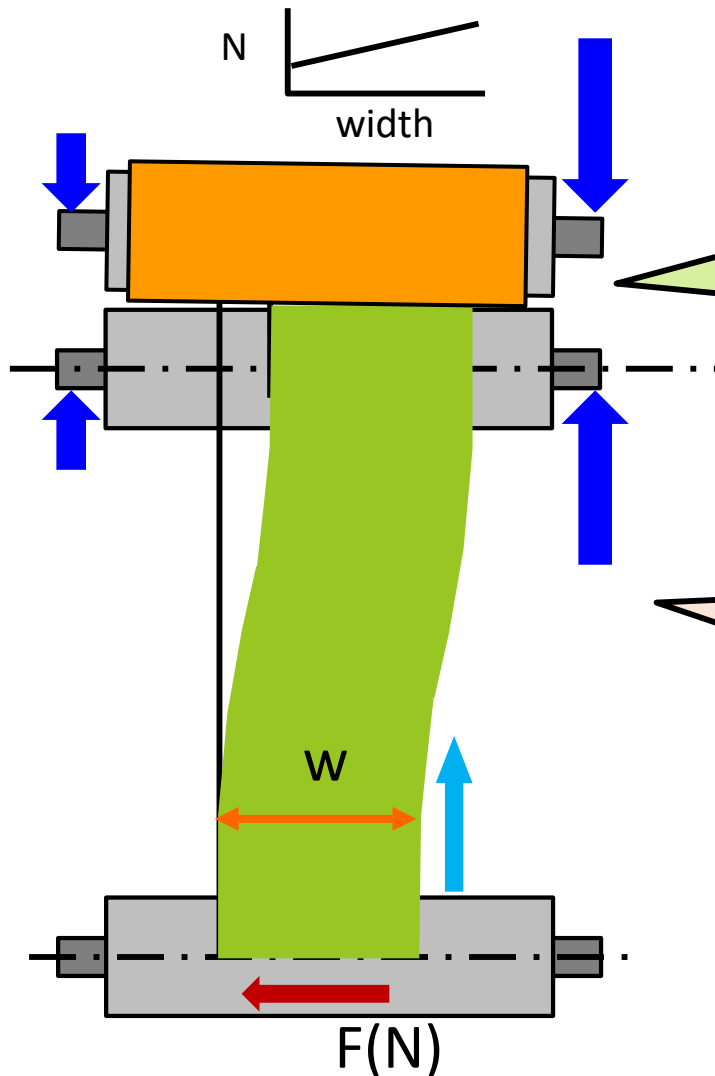
Belt shifting with thickness variation on inside of roller wrap.



Belt shifting with thickness variation on outside of roller wrap.



Lateral Tracking From Uneven Nip Load



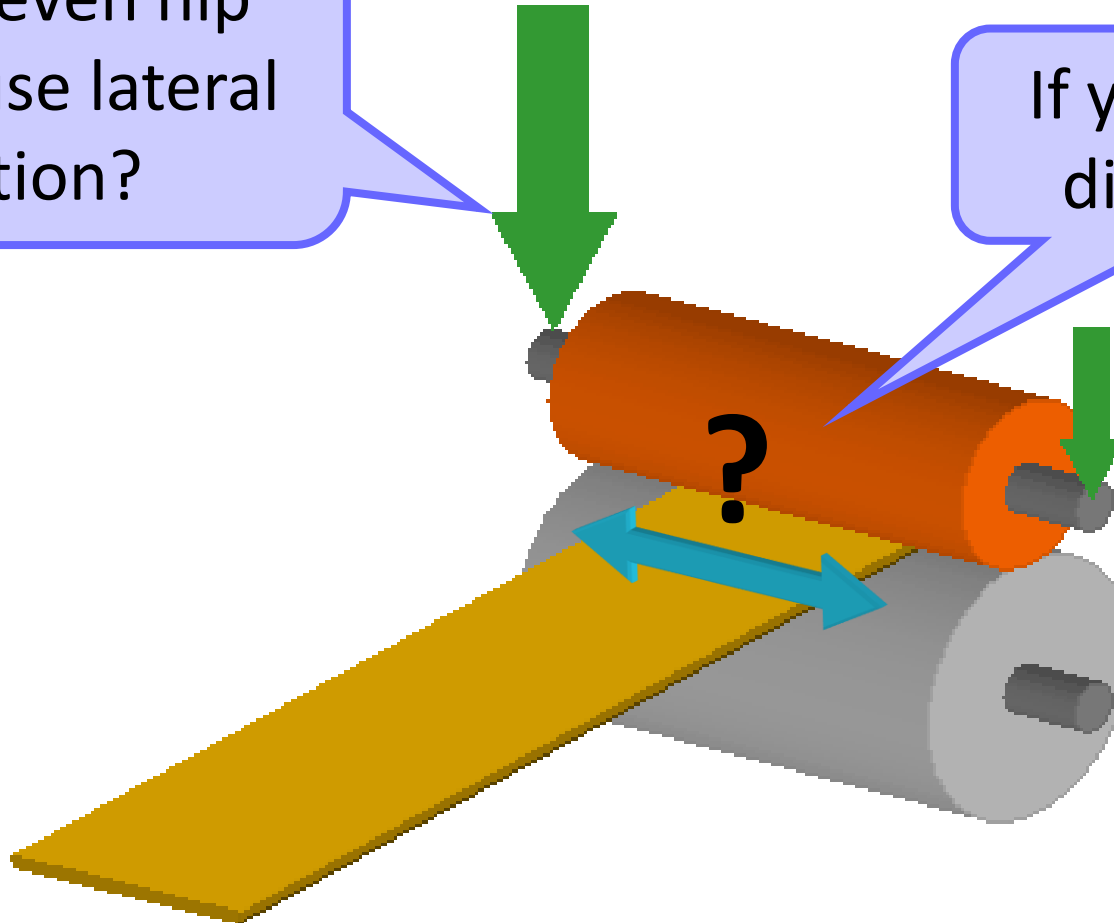
Webs will move towards higher nip load if they follow the rubber elongation speed differential and web entry path is perpendicular to nip misalignment.

Webs will move away from the higher nip load when slipping from calender or milling process or lubrication.

Nip Tracking: The Elusive 5th Rule?

Will uneven nip load cause lateral motion?

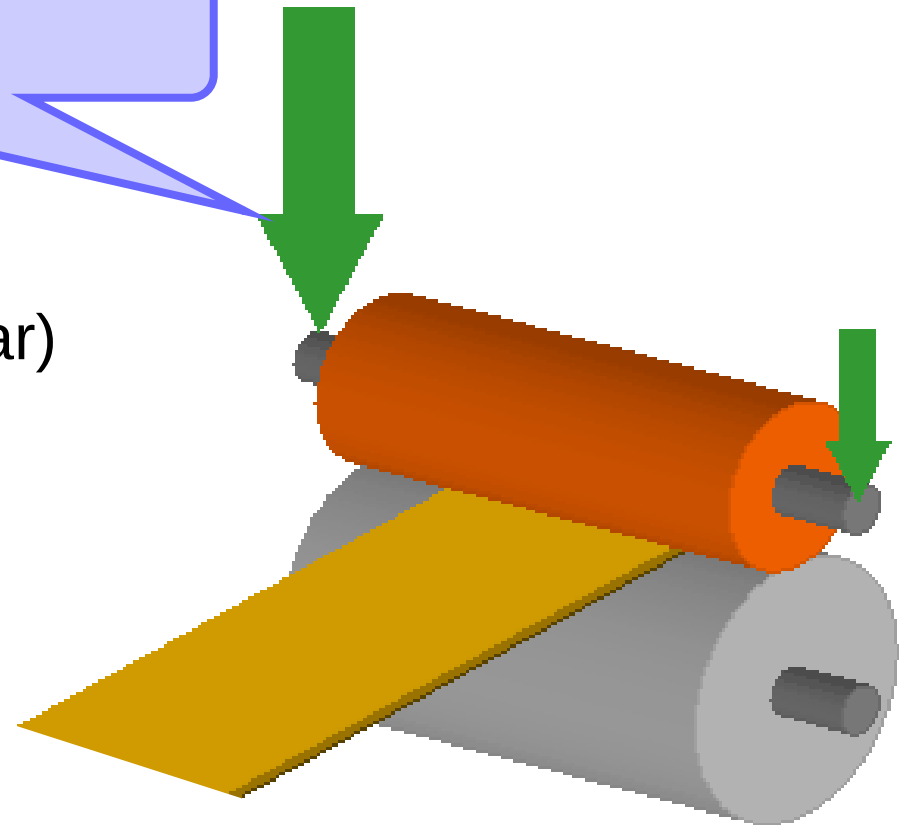
If yes, which direction?



Nip Variation Causes

What cause nip load to vary?

- Misalignment
- Diameter variation (from wear)
- Product thickness variations
- Uneven supply pressure
- Uneven friction
- Deflection
- Crowning
- Uneven tension, if nipping roller is wrapped

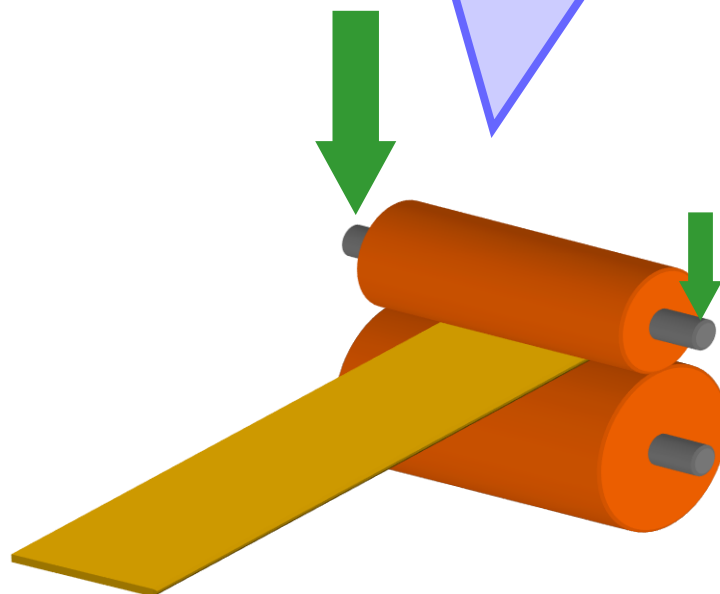
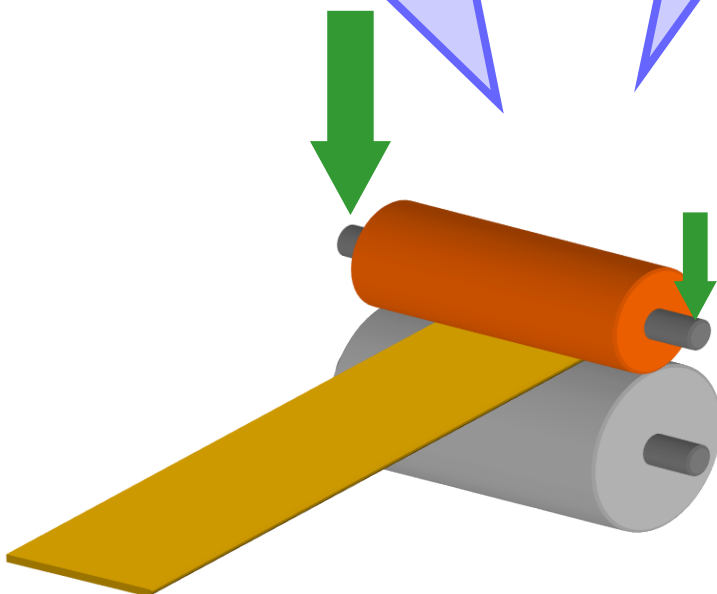


Lateral Motion at Nips

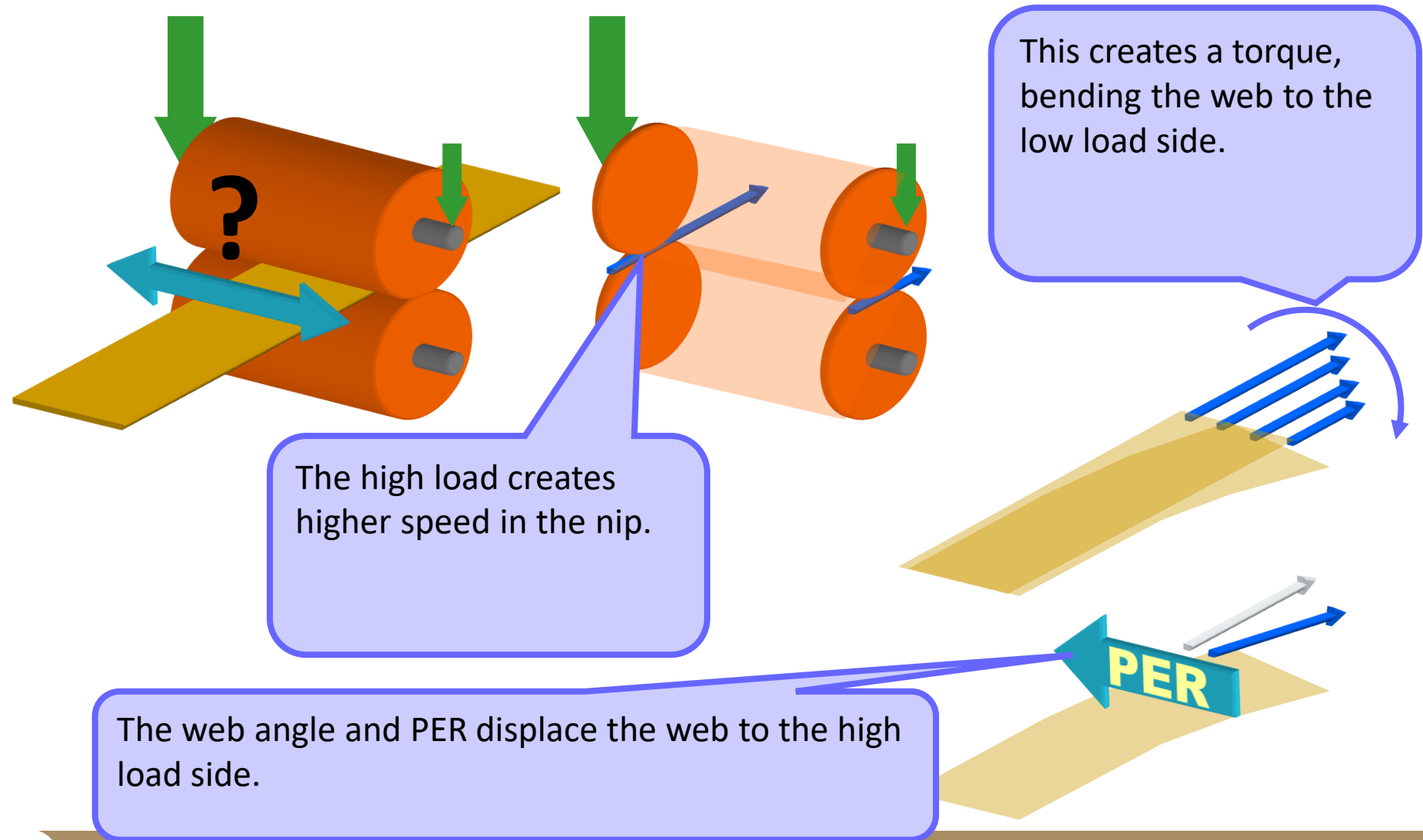
Will uneven nip load cause lateral motion?

If yes, which direction?

A rubber-rubber nip is easier to predict.

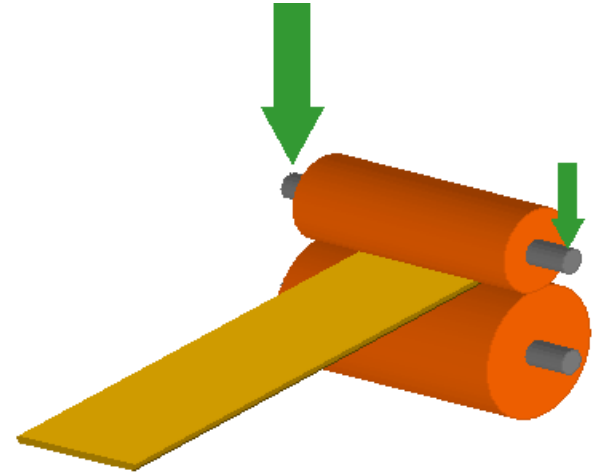
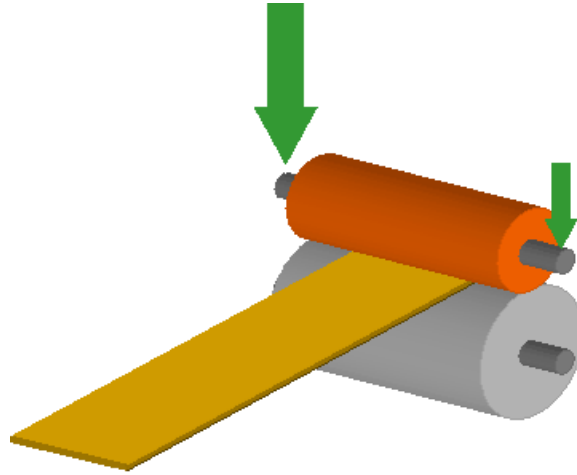


Lateral Motion at Nips



Nip-Induced Tracking

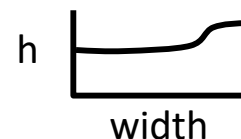
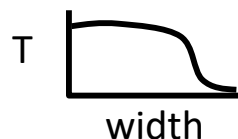
Why is Nip-Induced Tracking Difficult to Predict:



1. Nip-induced MD rubber strain
2. Nip-induced MD web strain
3. Will the web stick to steel or rubber roller surface?
4. Will pre-nip wrap friction isolate the entry span from nip forces?
5. Does roller deflection and entry angle steer the web?
6. Does the web stick or slide in the nip? (Will PER apply?)
7. Downstream tension slipping through the nip.

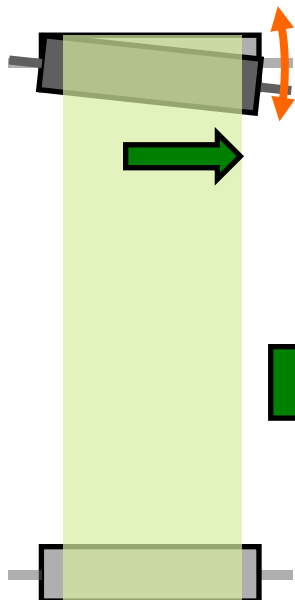
Summary - Five Tracking Rules

CD Shifting with Good Traction

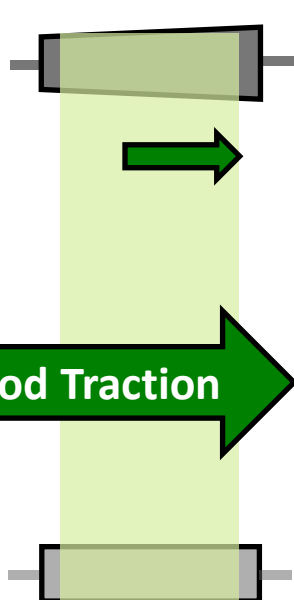


$$\theta_{\text{WEB}} \neq \theta_{\text{ROLLER}}$$

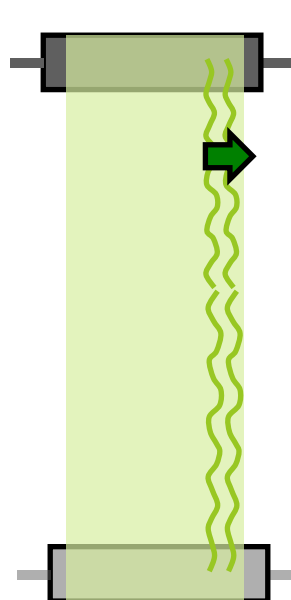
Misaligned Roller



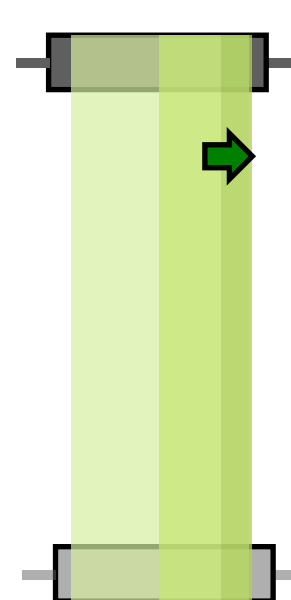
Diameter Variations



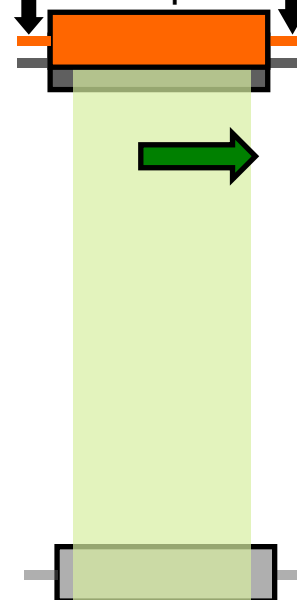
Baggy Web



Thickness Profile



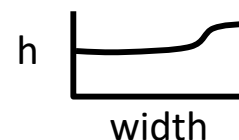
Uneven Nip



Good Traction

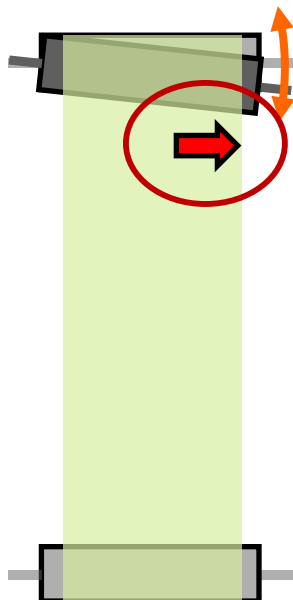
Summary - Five Tracking Rules

CD Shifting with Poor Traction

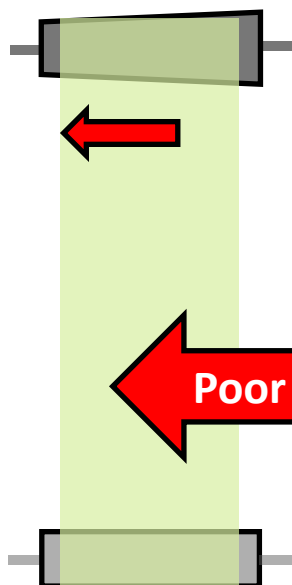


$$\theta_{\text{WEB}} \neq \theta_{\text{ROLLER}}$$

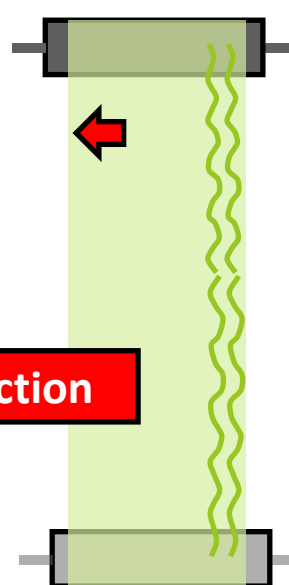
Misaligned Roller



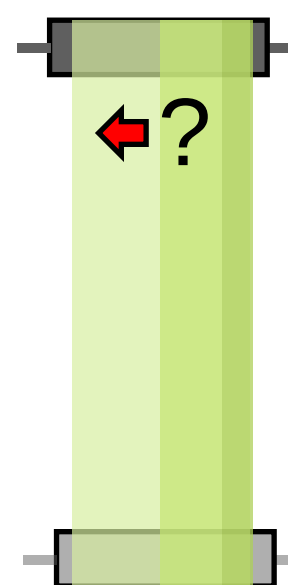
Diameter Variations



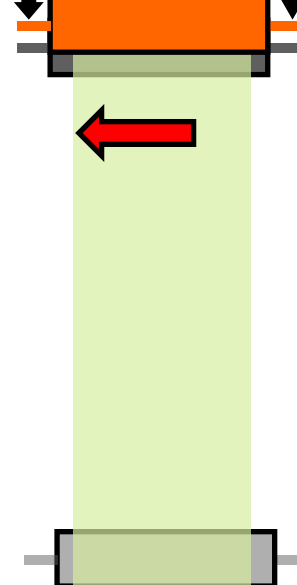
Baggy Web



Thickness Profile



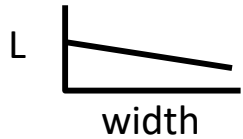
Uneven Nip



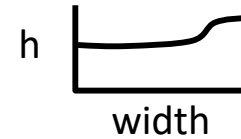
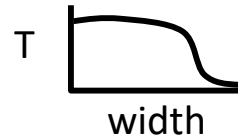
Poor Traction

Summary - Five Tracking Rules

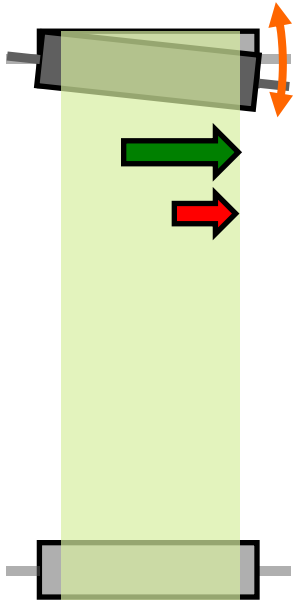
Top causes of web CD shifting.



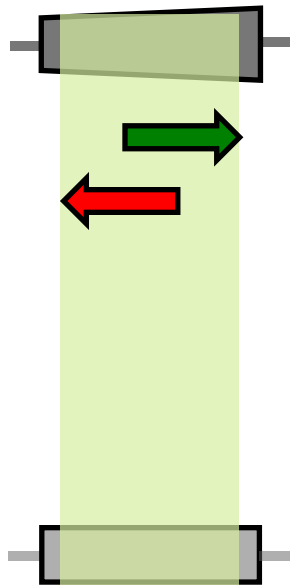
$$\theta_{\text{WEB}} \neq \theta_{\text{ROLLER}}$$



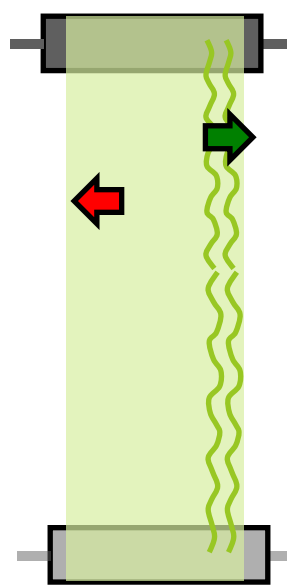
Misaligned
Roller



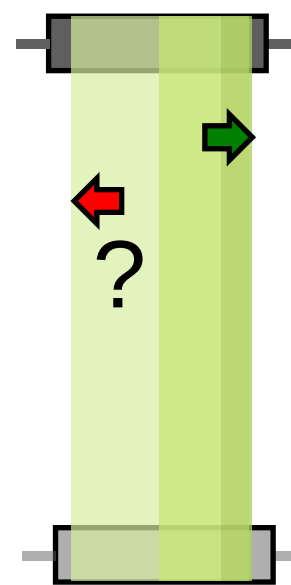
Diameter
Variations



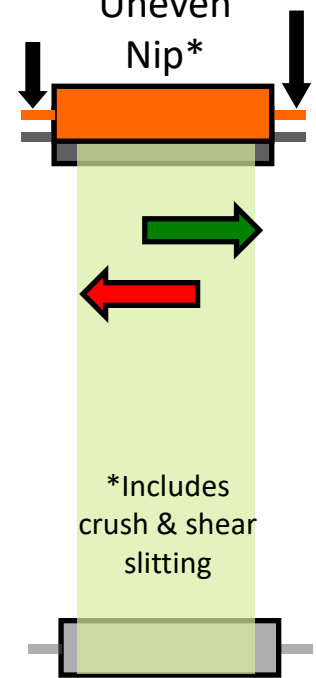
Baggy
Web



Thickness
Profile



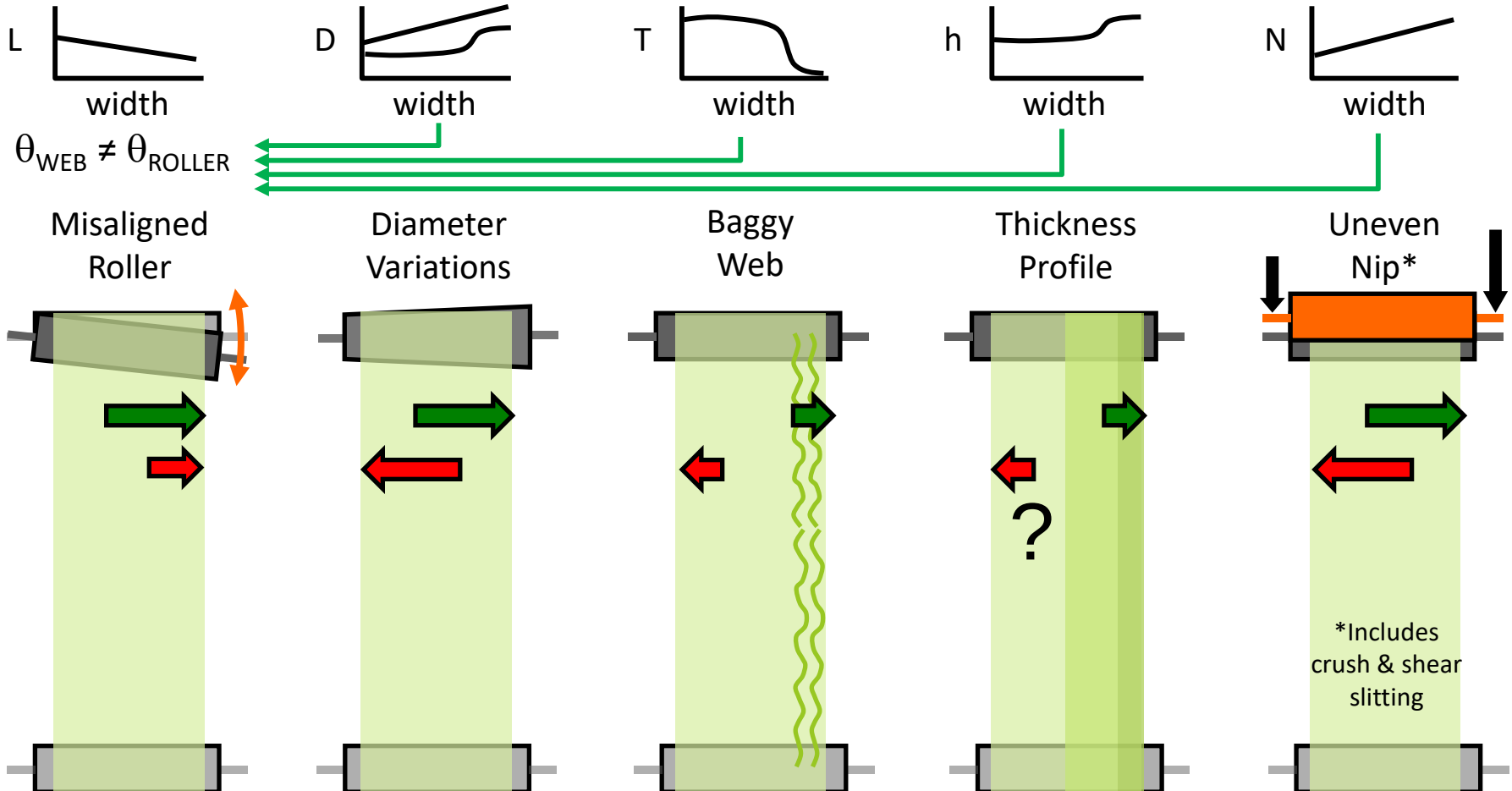
Uneven
Nip*



*Includes
crush & shear
slitting

Summary - Five Tracking Rules

All five rules are essentially Case 1: Anything that induces an angle difference between web and roller surface will cause CD shifting.



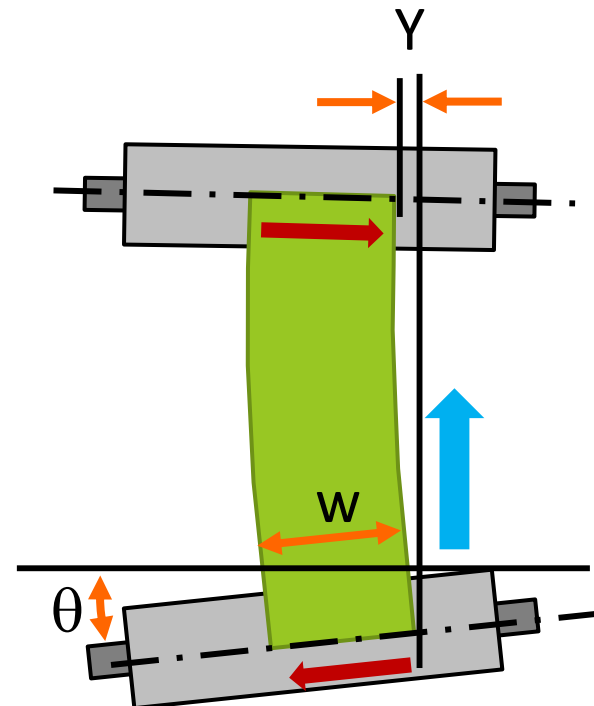
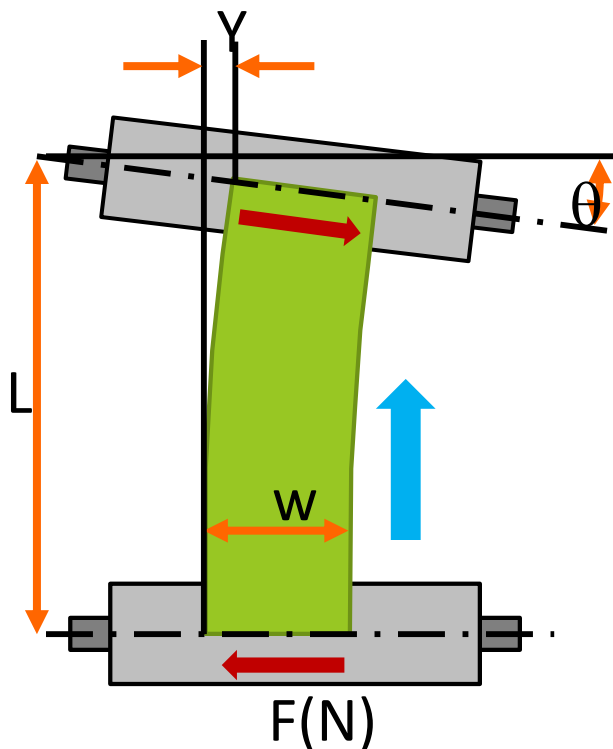
Other Lateral Motion Causes

- ▶ Can upstream variations shift and wrinkle a web?
- ▶ Can shifting and wrinkling occur in a span of a perfectly uniform web and two perfectly cylindrical, aligned rollers?

Upstream Rollers Tracking Causes

► Roller misalignment:

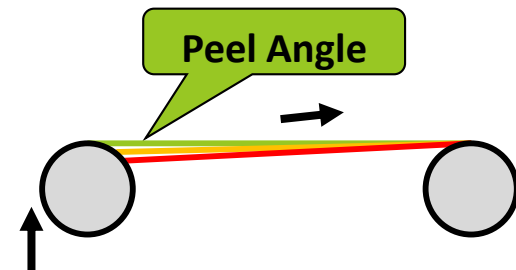
If two rollers are misaligned, the induced bending and stress variations are independent of how the misalignment is oriented relative to machine centerline.



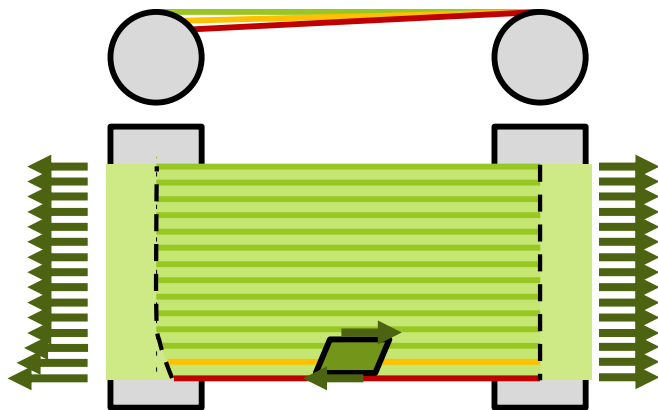
Upstream Rollers Wrinkle Causes

CD peel force variations:

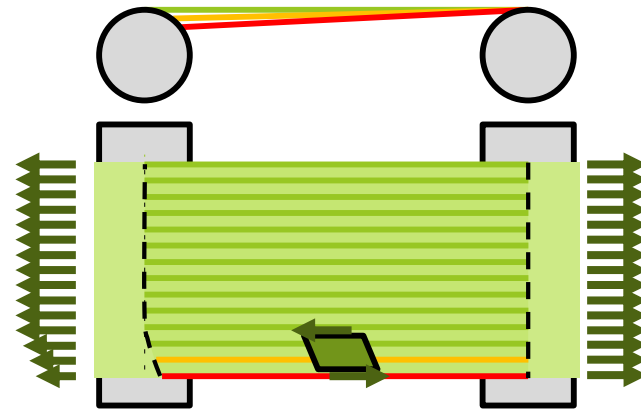
- ▶ Uneven peeling can create shear in the downstream web span.
- ▶ Peel angle can vary due to CD adhesion variations or web bagginess (CD tension variations).



$$\theta = \cos^{-1} \left(\frac{T_{ADH}}{T_{WH}} \right)$$



Peel angle changing from peel force profile.



Peel angle changing from baggy web tension profile.

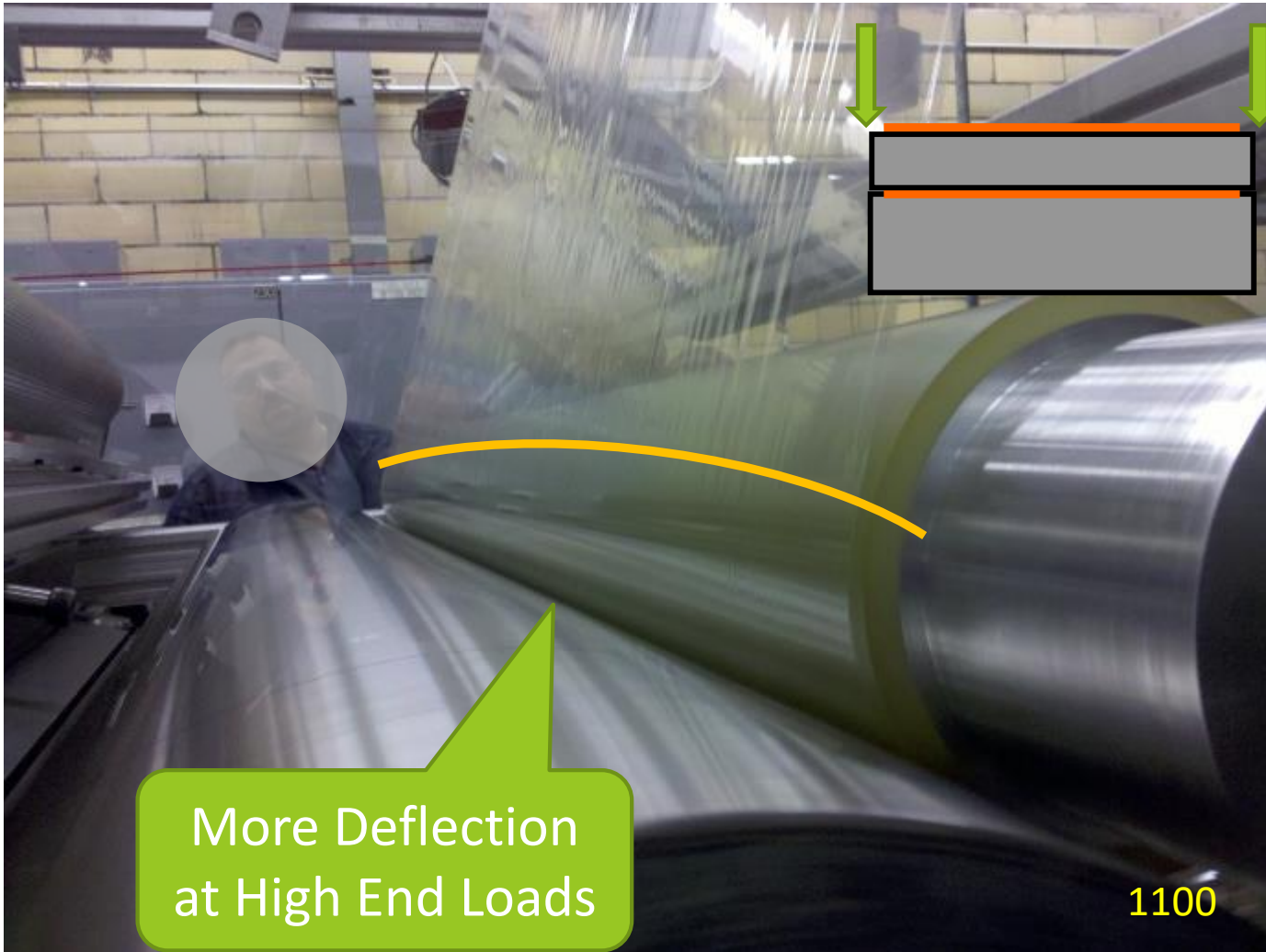
No Skew, Nip Med. & Even, High Tension

Nip Far
= 150 lbf

Nip Near
= 150 lbf

Skew
= 0

Tension
= 0.6 PLI



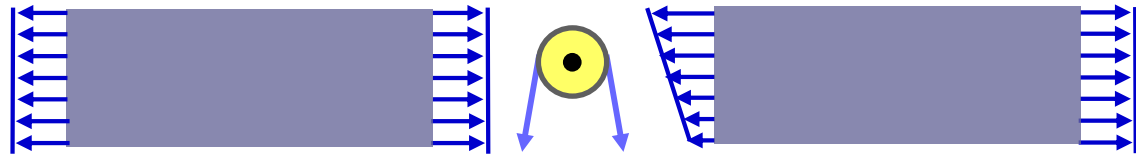
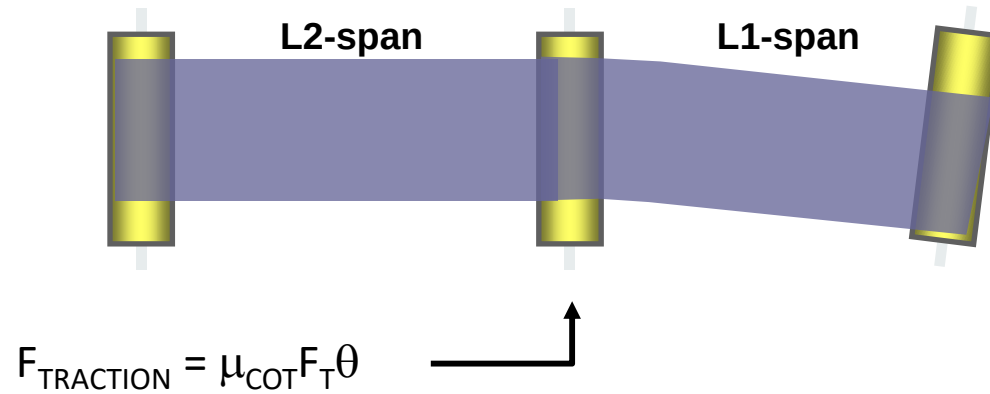
No Skew, Nip Med. & Even, High Tension



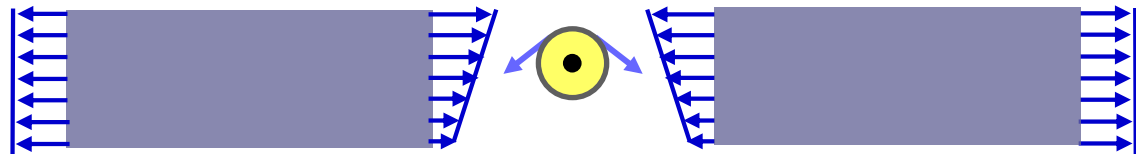
Moment Transfer

Moment transfer is more common in wide, stiff web, and short spans.

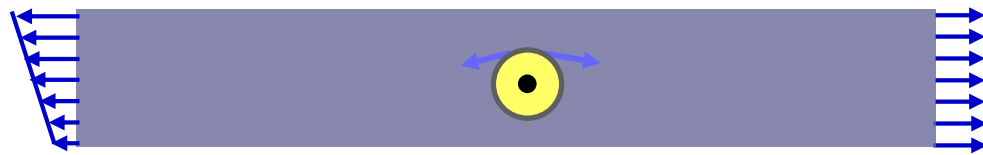
No moment transfer



Moment transfer + parallel entry

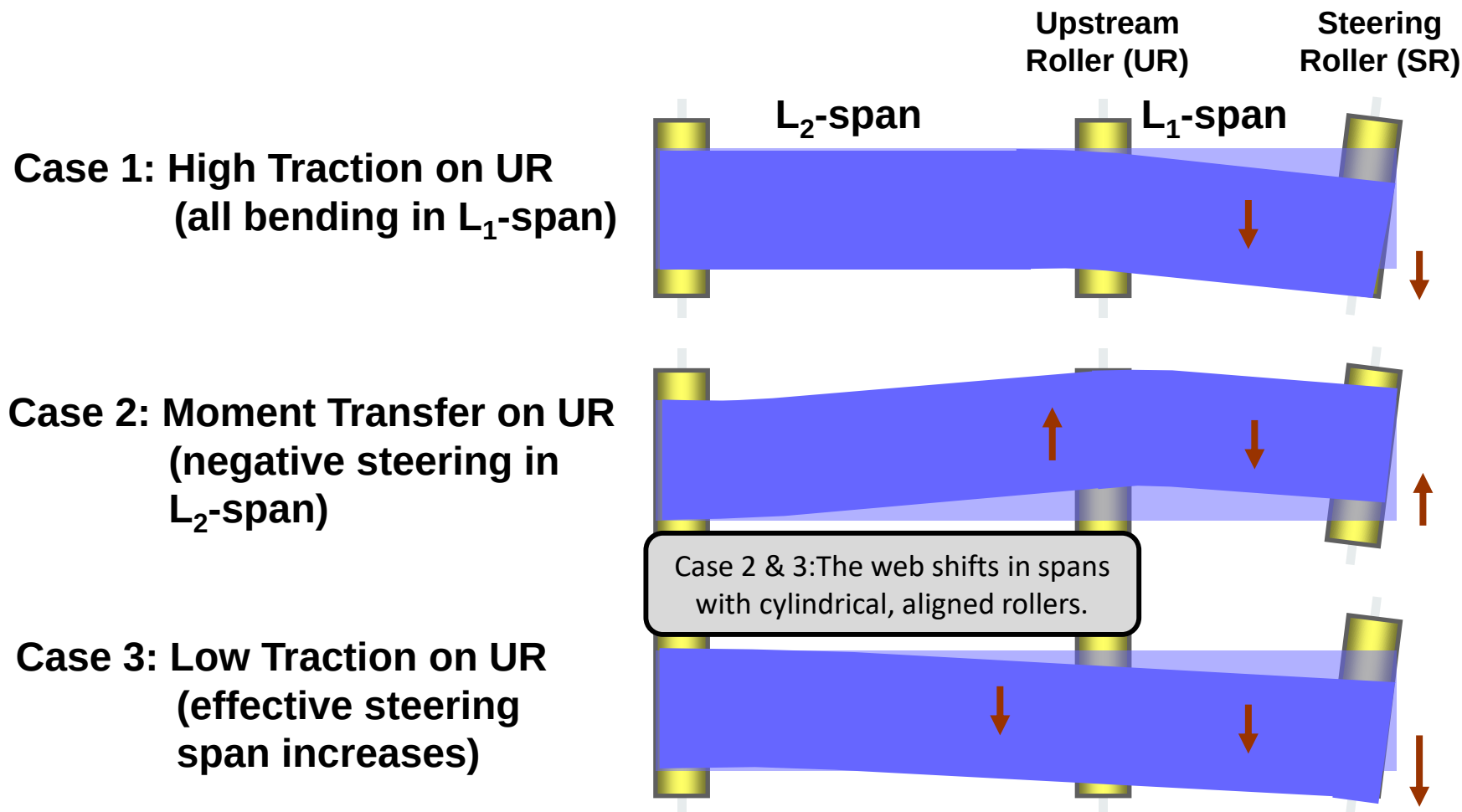


Moment transfer + No parallel entry



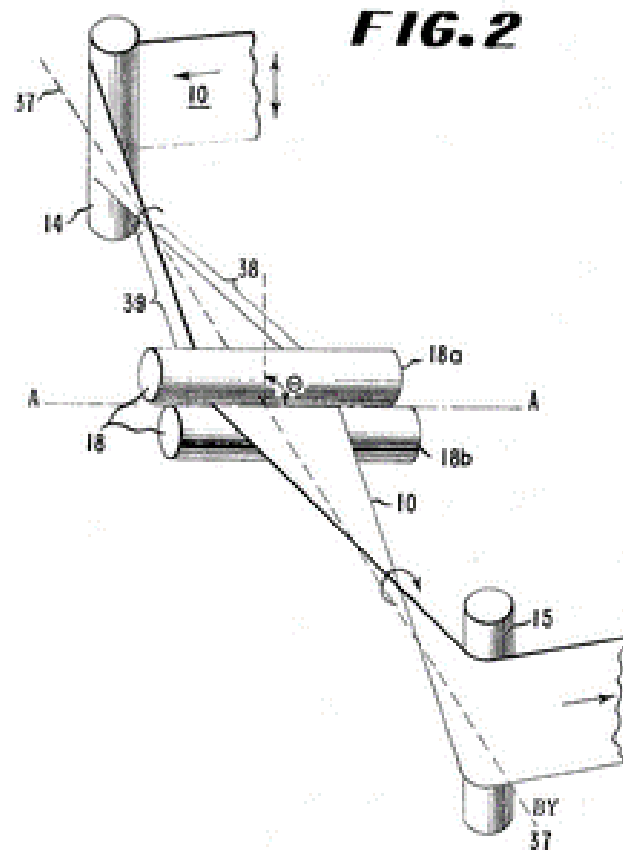
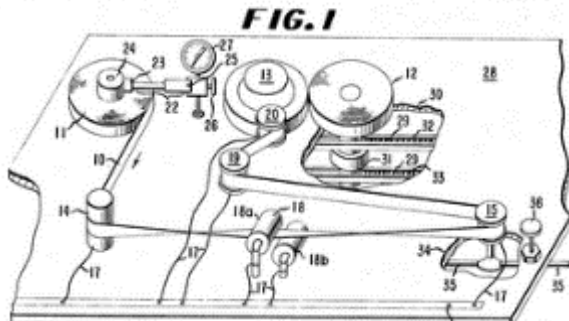
Unstable Web Path and Steering

Moment transfer and $L_2 > L_1$ can lead to negative steering.



Centering by Twisting

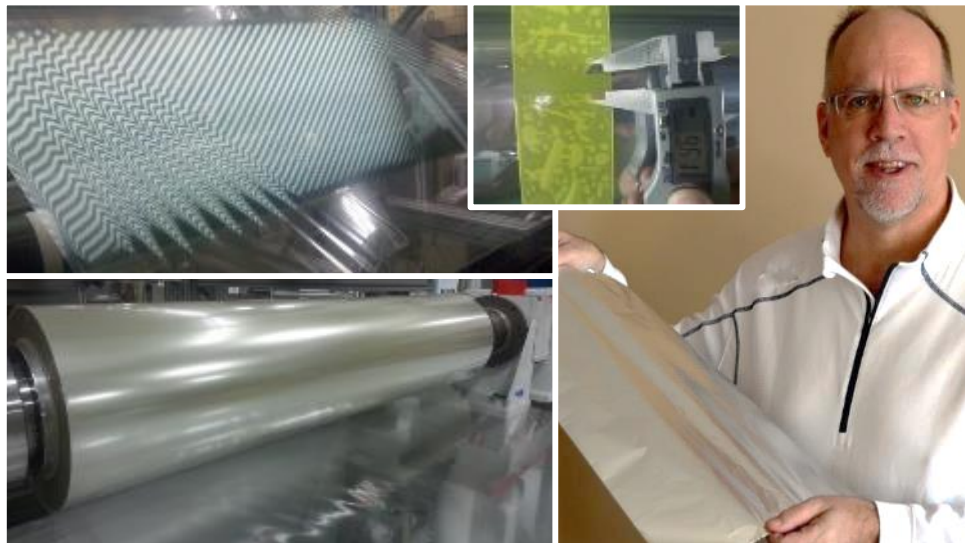
March 14, 1967
S. E. AMOS
WEB REWINDER
3,309,037
Filed Oct. 8, 1964
2 Sheets-Sheet 1



INVENTOR
STEPHEN EARL AMOS

Robert W. Black

ATTORNEY



5.2 Wrinkling

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

5.2 Learning Objectives

- ▶ What is a trough? A wrinkle? A crease?
- ▶ How do you find the origin of a wrinkling problem?
- ▶ What are possible immediate and longer-term remedies to eliminate a wrinkle?
- ▶ What is shape stiffening?
- ▶ What causes a shear wrinkle?, a converging wrinkle?, an expansion wrinkle?, an accumulation wrinkle?
- ▶ Why is traction important to wrinkling?
- ▶ How do span and wraps affect wrinkles?
- ▶ How can wrinkles be detected in a moving web?

Definitions

- ▶ Buckling – Any imperfection in the web shape that is non-planar in a span between rollers or non-cylindrical (or semi-cylindrical) in web-roller contact zone or in the layers of a winding or wound roll. Buckling occurs when compressive strains or stresses exceed a critical buckling point.
- ▶ Critical Buckling – When a web buckles is a function of many factors, including, but not limited to: elastic modulus, thickness, tension, Poisson's ratio, shape, and geometry (length, width).

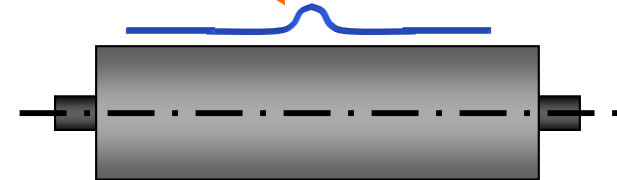
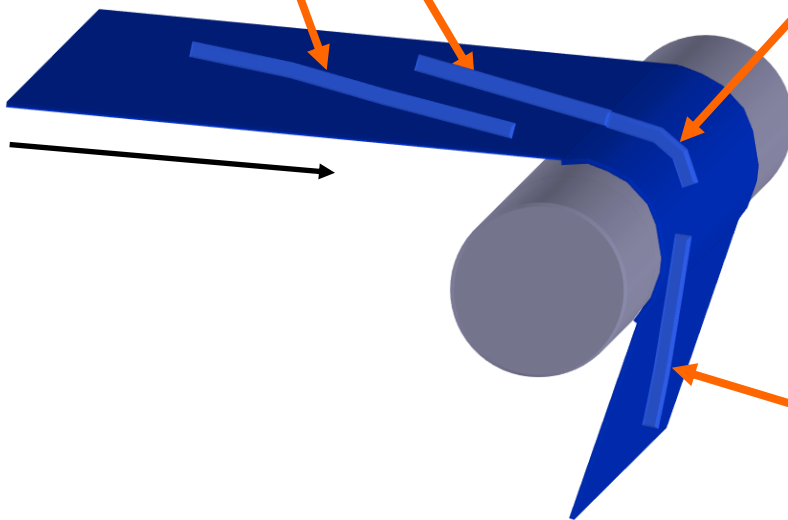
Definitions

- ▶ Troughing – The web buckles in the span between rollers. The buckling of a troughing web span is sometimes called a soft wrinkle, because it usually does not create a permanent defect in the web.
- ▶ Wrinkling – The web buckling while in contact with a roller.
- ▶ Creases or Hard Wrinkles – The permanent defect in a web usually forming during wrinkling (during the buckling of web-roller contact).

Definitions

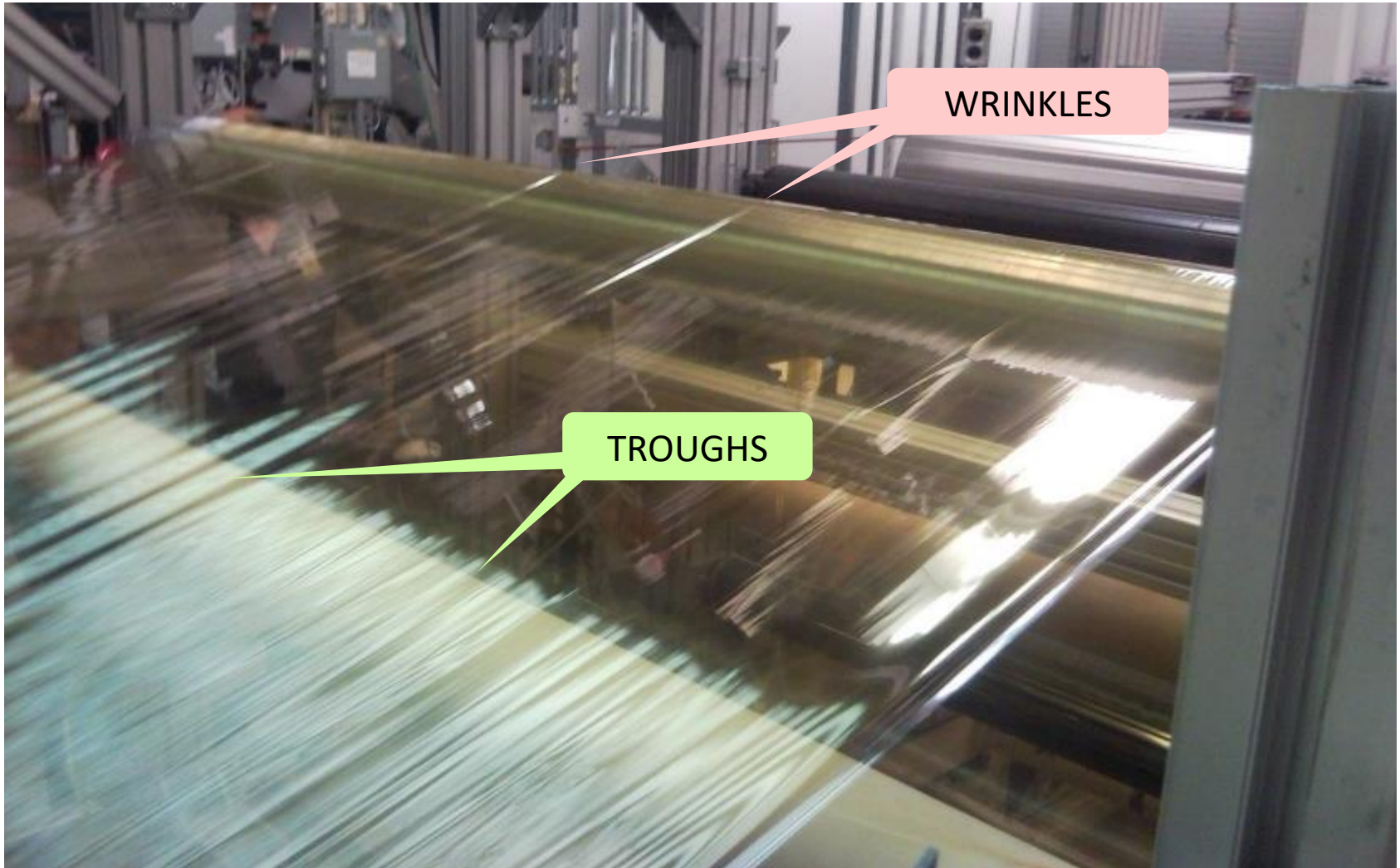
Troughs = Web buckled in span

Wrinkles = Web buckled on roller



Crease = Permanent deformation from wrinkling

Troughs and Wrinkles



Beginning Thoughts on Troughing/Wrinkling

- ▶ Troughing occurs under low compressive stresses, is quite common, and rarely leads to permanent damage of the web.
- ▶ Wrinkling require moderate compressive stresses delivered via web-roller traction, need not be commonplace, and typically leads to permanent creasing of the web or web breaks in brittle webs.
- ▶ Wrinkles occur with nearly invisible amounts of bending.

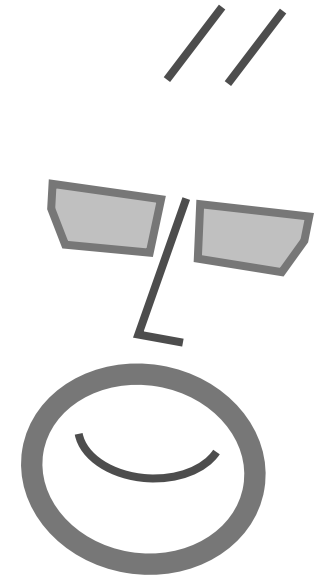
Find the Wrinkles Origin

1. Walk the line
2. Find the first wrinkle
3. Draw some pictures, take pictures and videos
4. Note where wrinkles occur
5. Check roller alignment and diameter variations where wrinkles form
6. Observe wrinkles at low speed

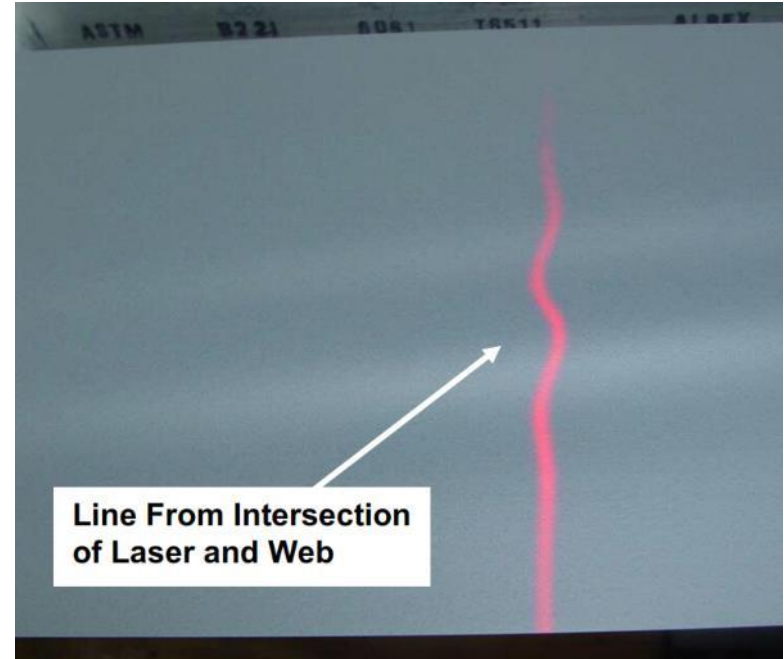
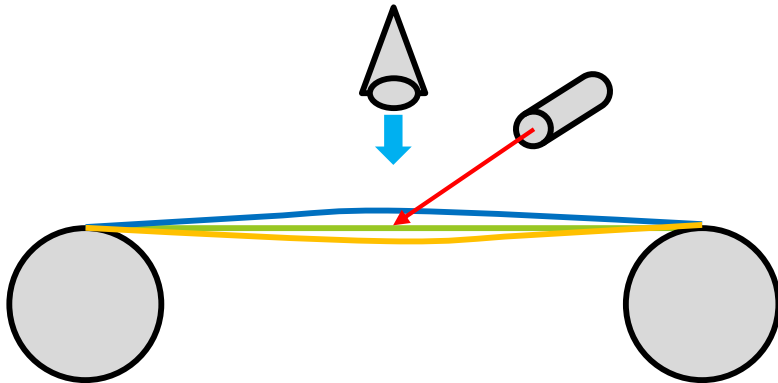
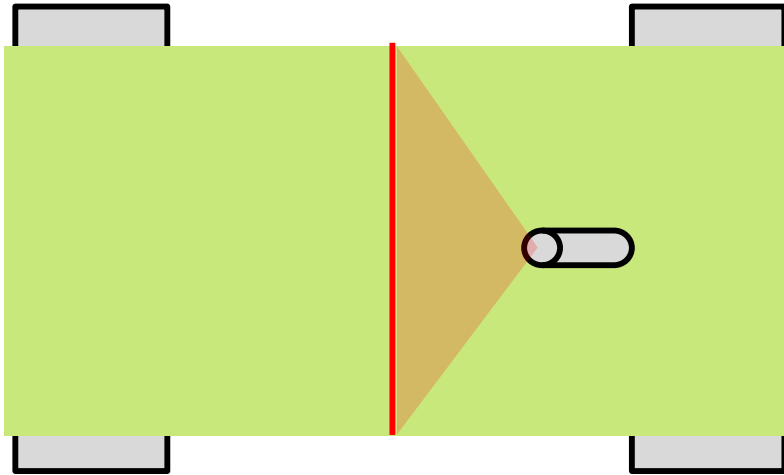
Detecting Wrinkles

How can troughs and wrinkles be detected in a moving web?

- Direct visual observations (low angle lighting helps).
- Direction physical observations (be careful).
- Web width measurement.
- Pen laser beam parallel and close to wrapped roller's surface.
- Physical samples from end of process.



Angle Linescan Laser Trough Detection



SINGLE SPAN WEB BUCKLING DUE TO

ROLLER IMPERFECTIONS IN WEB

PROCESS MACHINERY

Joseph Biesel
OK State U, Dec. 2006

Strobe Light and High Speed Webs

Handheld or full width high frequency strobe lights can help view wrinkling or buckling in winding rolls.

They are especially useful for viewing opaque or matte webs in high speed processes.



Lith-O-Light 5 Approx. \$1500

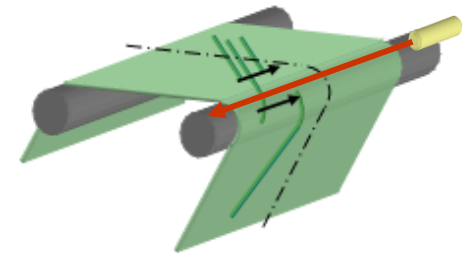
www.unilux.com

4B2: Measuring MD Wrinkles

- Most wrinkles are measured by viewing or counting the creases in the web. Note:
 - ✓ The angle and position any troughs in the span immediately upstream of where the wrinkles forms,
 - ✓ The angle and crossweb position of the wrinkles.
 - ✓ Whether the wrinkle shifts or walks laterally.
- View wrinkling in high speed processes by projecting a visible laser beam crossweb at a small spacing above the roller's surface. Any wrinkling that buckling into the laser's path will reflect the laser light for easy observation.



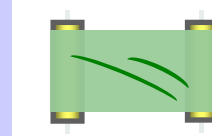
**MD Buckles
On Rollers**



4A: Buckled Web in Span

Definition:

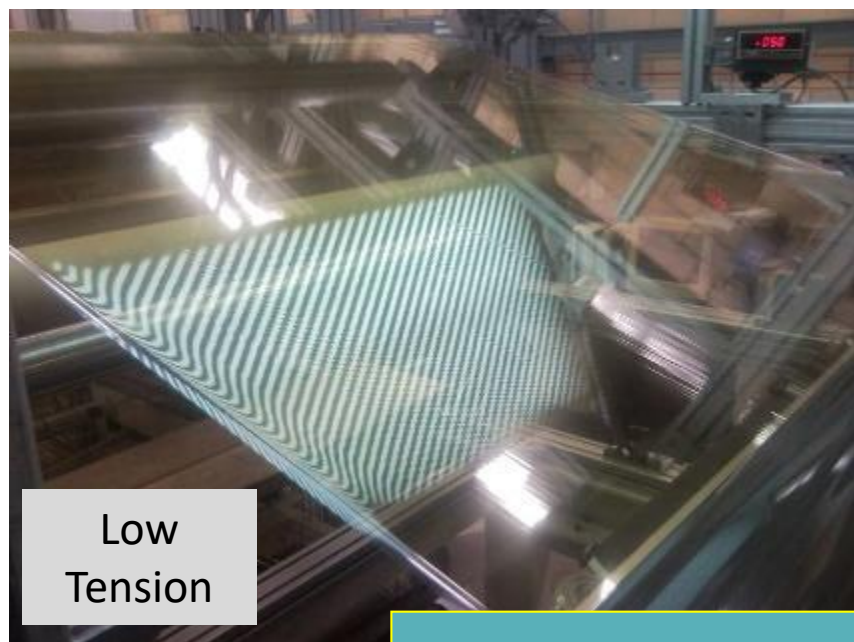
The web is out of plane in the span between rollers.



**Buckled Web in
Spans**

Measuring Troughing:

- Visual inspection viewing web with low angled lighting.
- Noting frequency and amplitude of troughs can help separate temporary from permanent troughs.
- A low angle line scan laser can highlight troughing severity.
- Zebra light reflection highlights buckling in reflective webs.



Pics 368, 369, 370

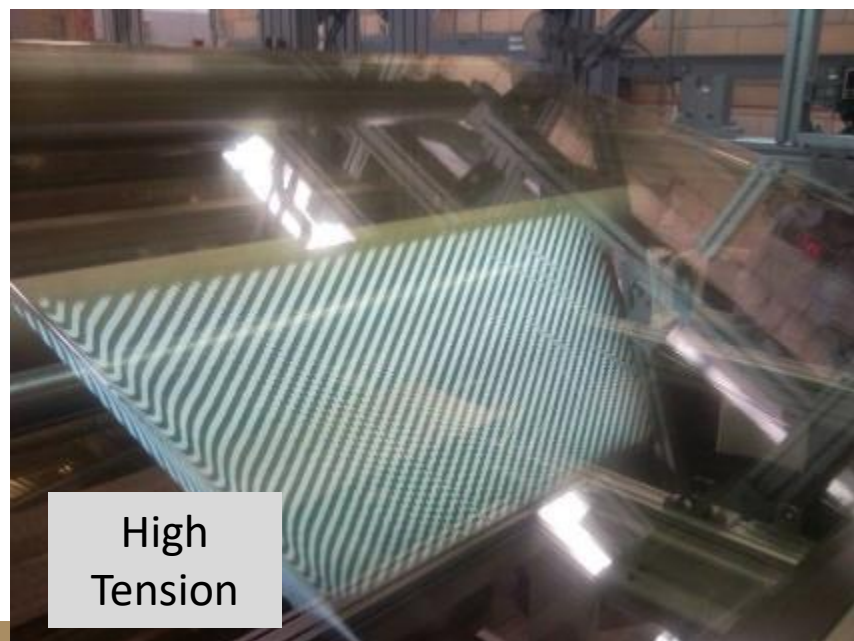
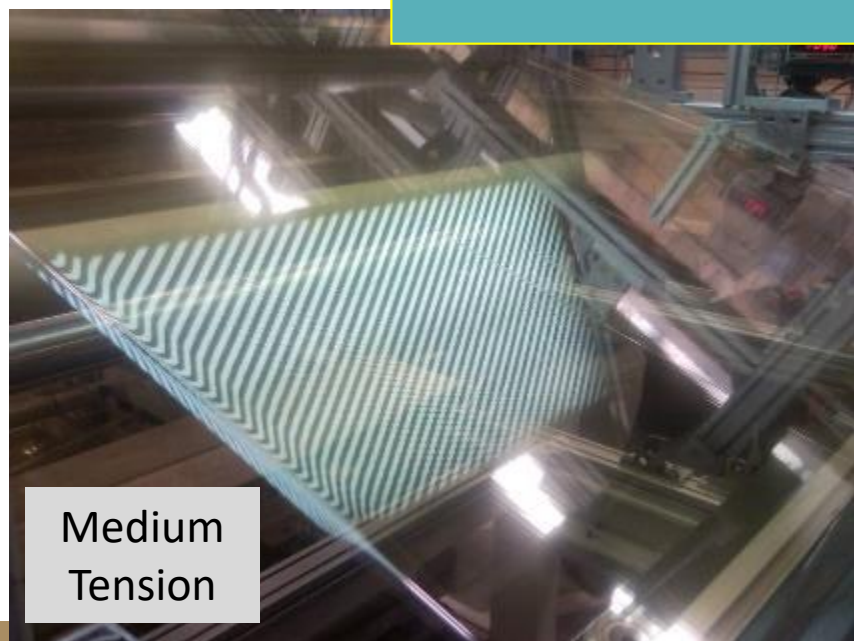
Roller: Straight AV

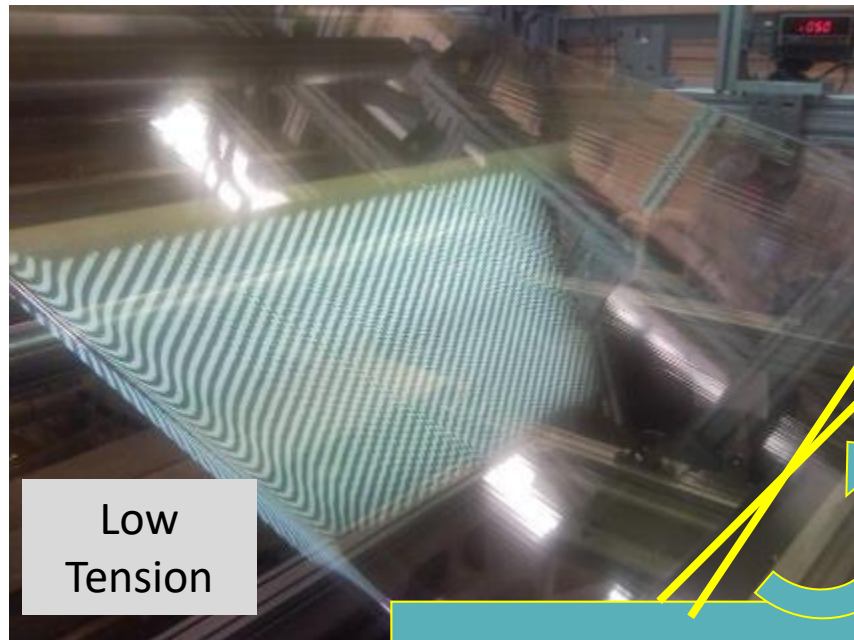
PET, 12 micron, 58-in

Tension = 25, 50, 80 lbs

Alignment Error = 0 mils

Entry Span = 50-in.





Low
Tension

Pics 371, 372, 373

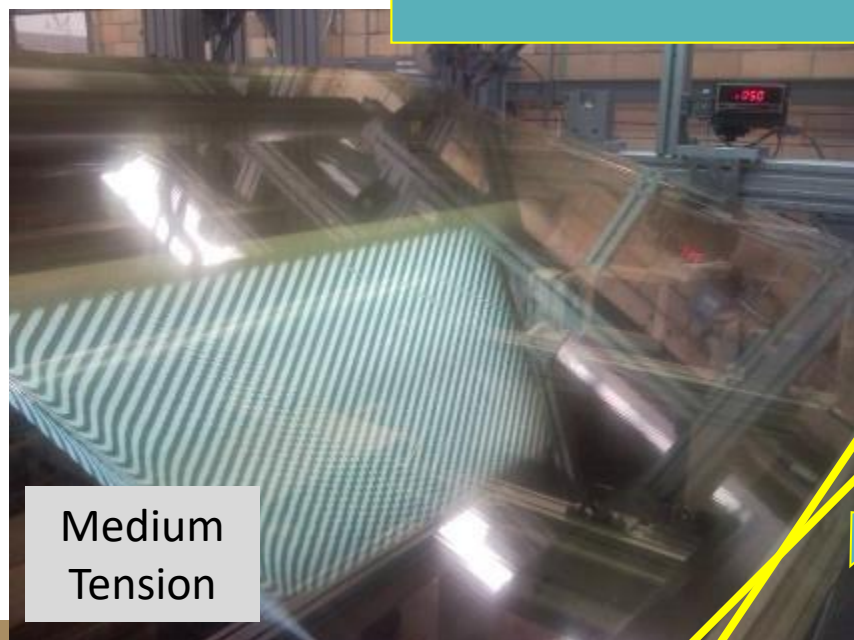
Roller: Straight AV

PET, 12 micron, 58-in

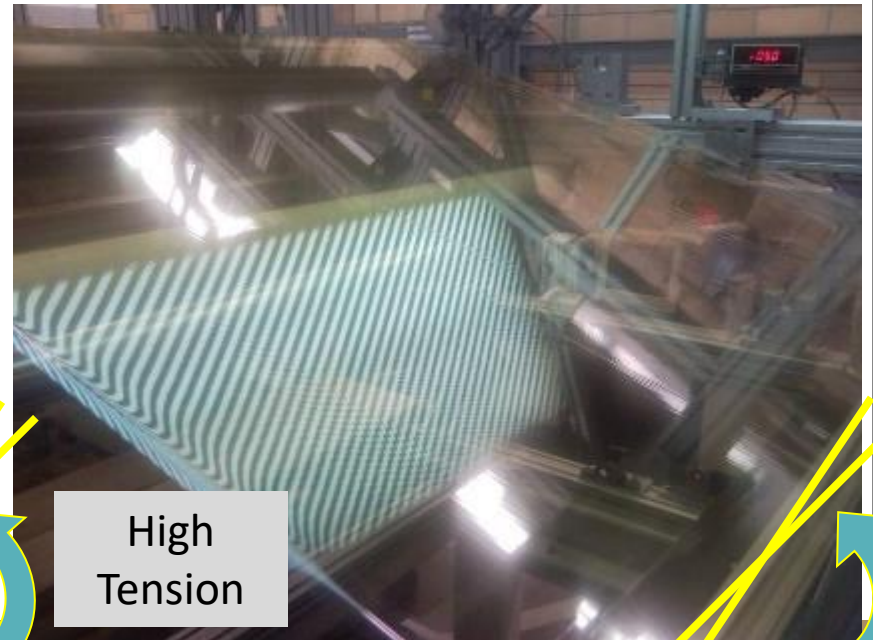
Tension = 25, 50, 80 lbs

Alignment Error = 5 mils

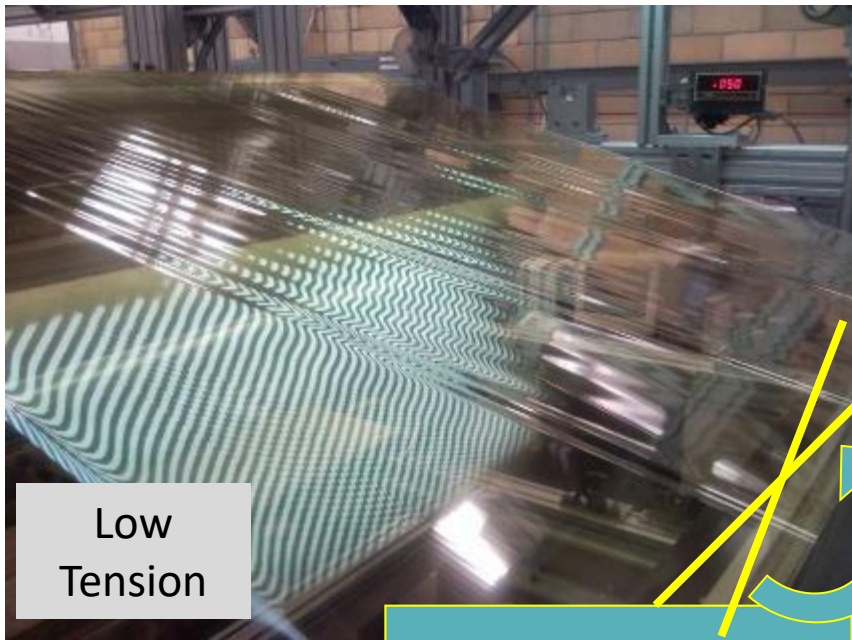
Entry Span = 50-in.



Medium
Tension



High
Tension



Pics 378, 379, 381

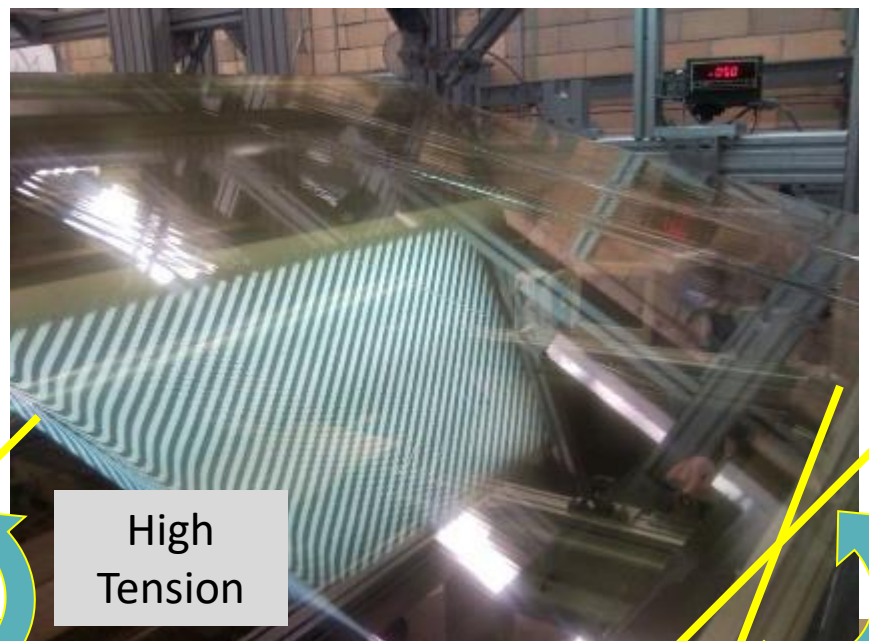
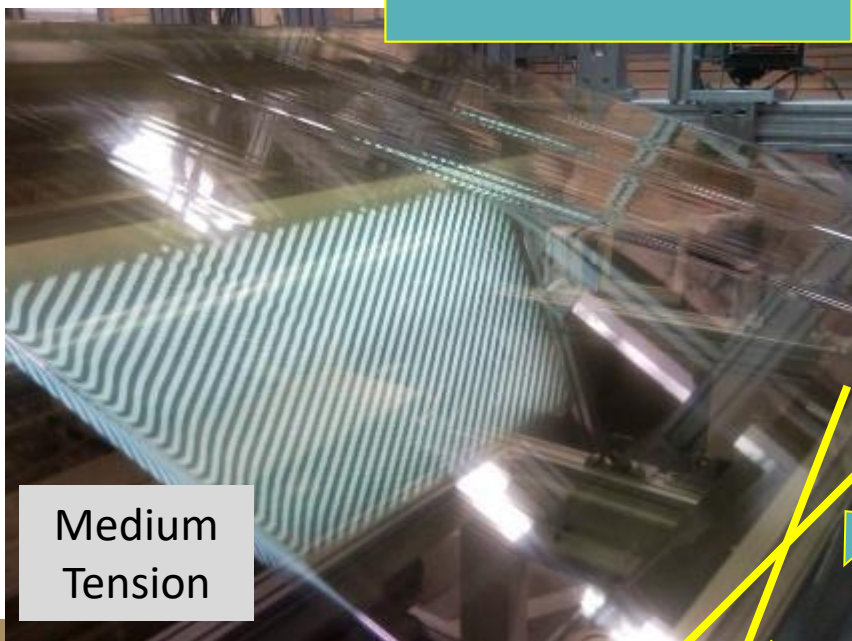
Roller: Straight AV

PET, 12 micron, 58-in

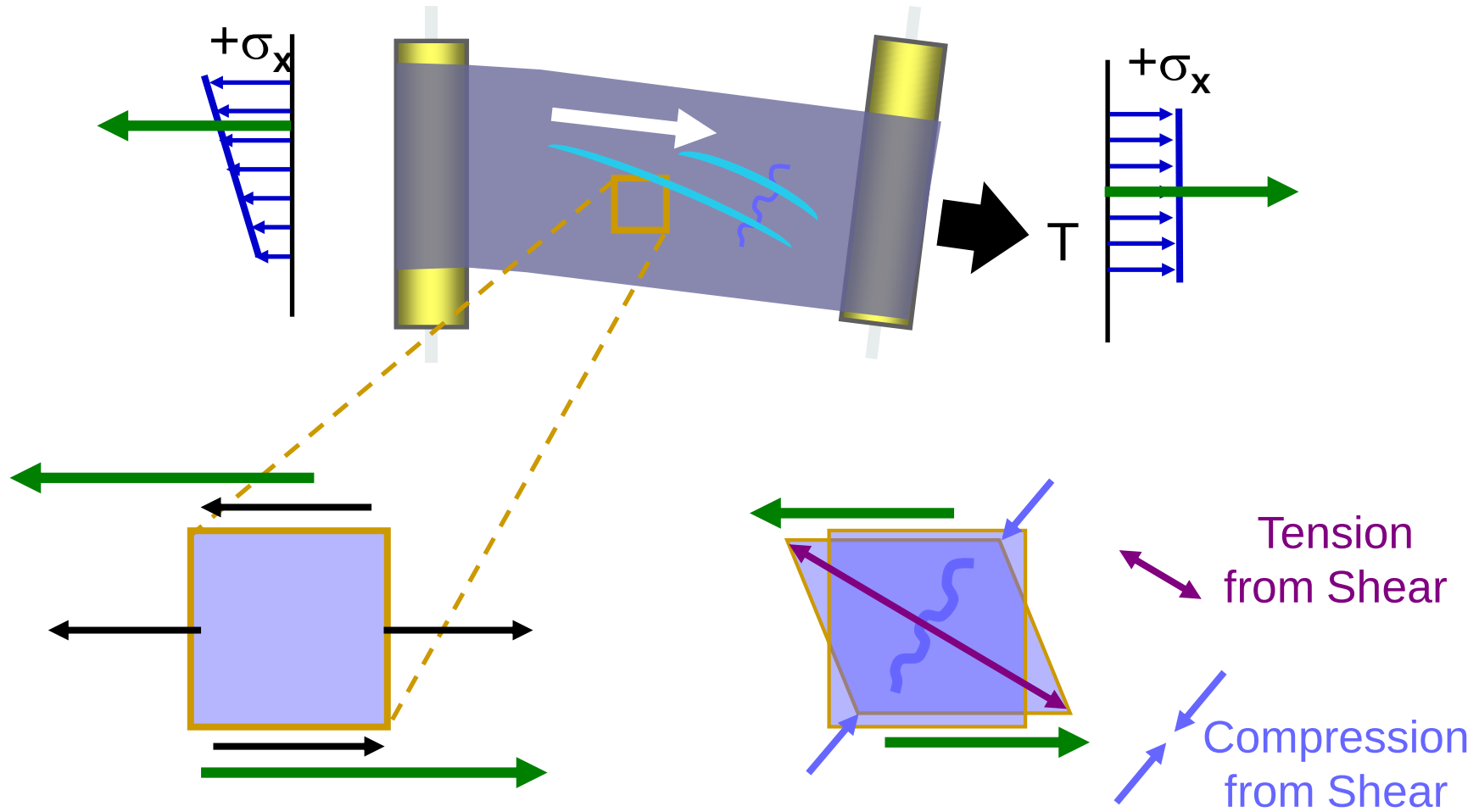
Tension = 25, 50, 80 lbs

Alignment Error = **50 mils**

Entry Span = 50-in.

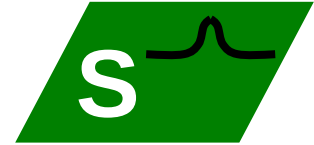


Deformation from Shear & Bending



Learn Troughing Sign Language

Troughs and wrinkles are much easier to see than web bending or roller misalignment.



This web is 'signing' to you. What is it saying?

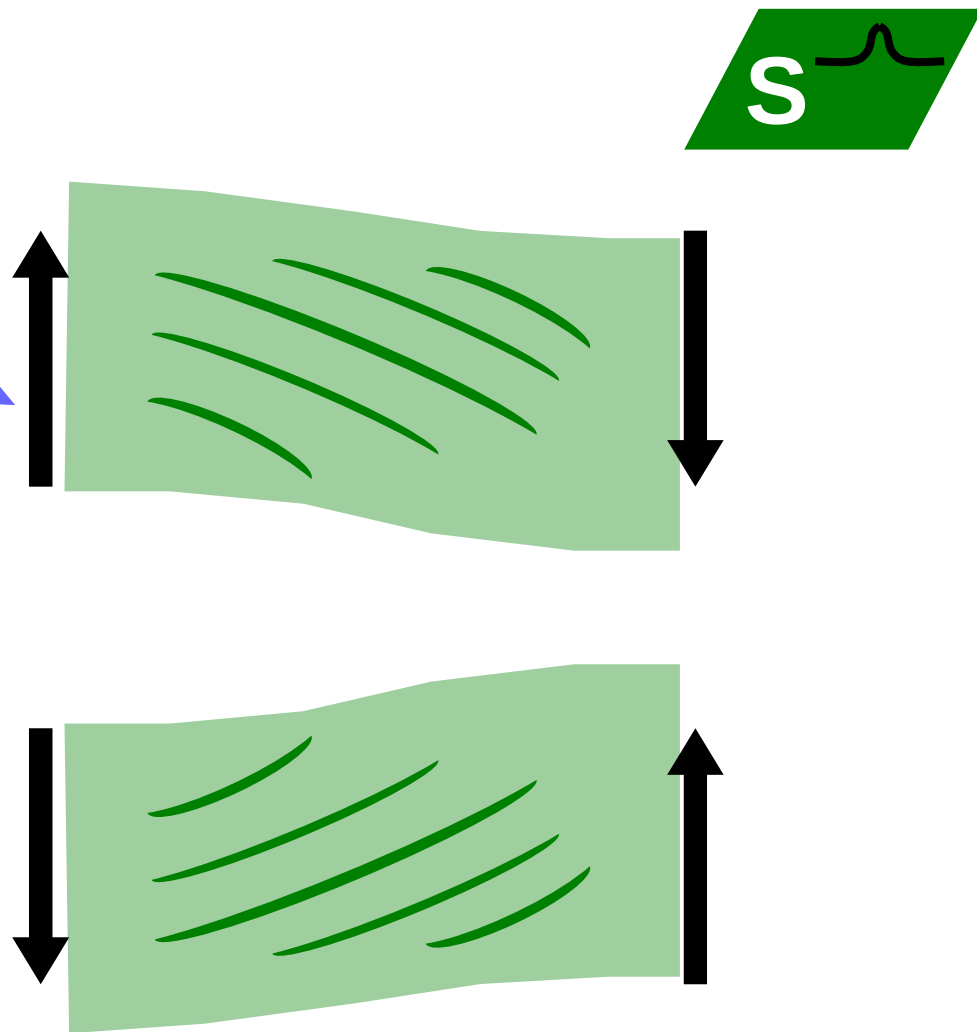


What is this web saying?

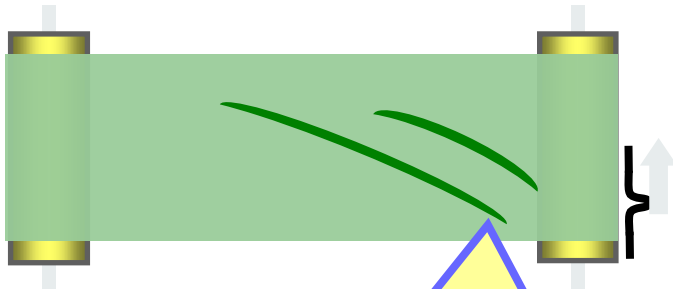


Learn Troughing Sign Language

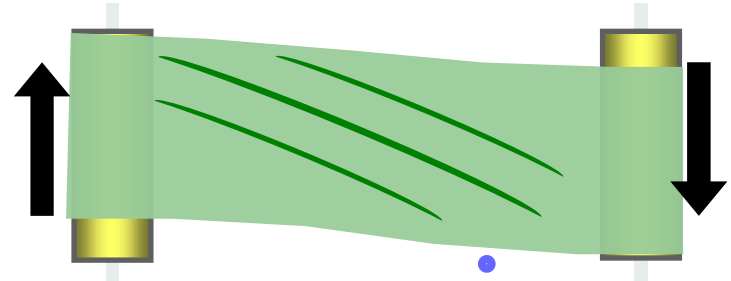
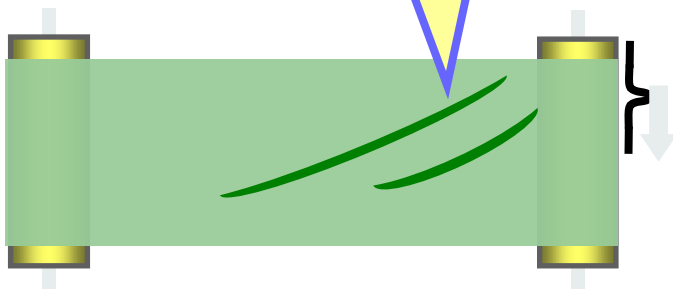
If you take a sheet of paper, napkin, or part of your shirt and shear it, you will see these same 'signs'



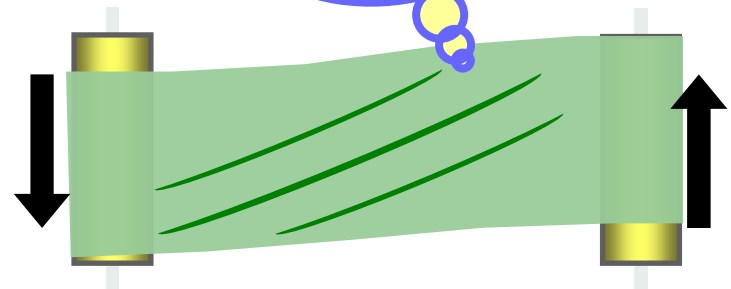
Learn Troughing Sign Language



When you see
angled troughs...



...think bending &...
...figure out why.



Buckling in Spans (Troughing)

Critical Stress to Buckle:

$$\sigma_{ycrit} = -1.73 \left(\frac{h}{L} \right) \sqrt{\sigma_x E_{iso}}$$

Increases directly with thickness, tension^{1/2}, modulus^{1/2}, and inversely with length

$$\sigma_{ycrit} = - \left(\frac{\pi h}{L} \right) \left(\frac{\sigma_x E_{iso}}{3(1-\nu^2)} \right)^{1/2} \xleftarrow{\text{Isotropic Webs}} \xrightarrow{\text{Orthotropic Webs}} \sigma_{ycrit} = - \left(\frac{\pi h}{L} \right) \left(\frac{\sigma_x E_y}{3(1-\nu_{xy}\nu_{yx})} \right)^{1/2}$$

Wavelength of MD wrinkles:

$$\lambda = \frac{1.95(Lh)^{1/2}}{\epsilon_x^{1/4}}$$

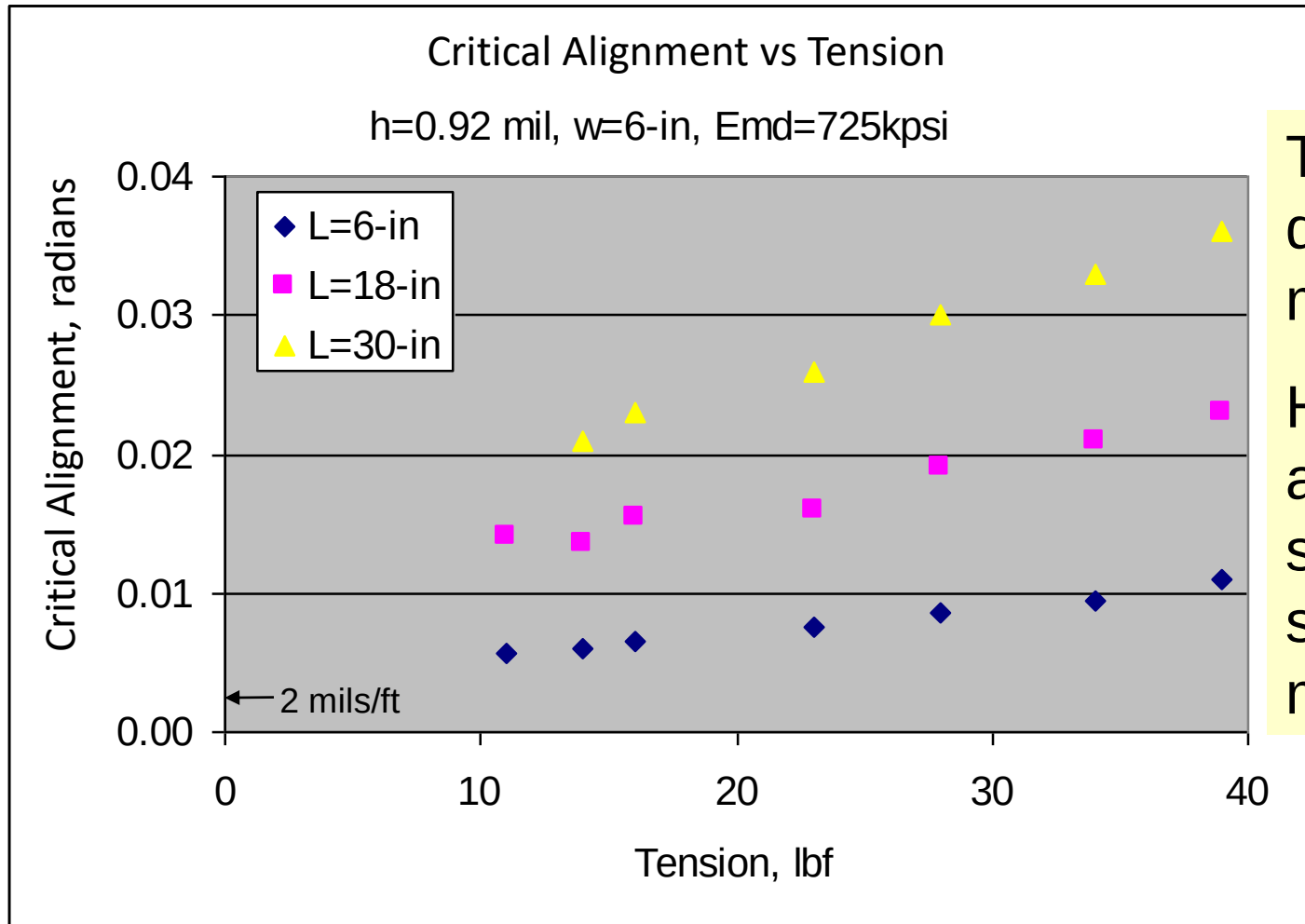
Increases directly with length^{1/2}, thickness^{1/2} and inversely strain^{1/4} (strain = tension / thickness x modulus)

Amplitude of MD wrinkles:

$$a = \frac{\lambda}{\pi} \sqrt{\epsilon}$$

Increases directly with wavelength and strain^{1/2}

Troughs and Roller Alignment



Troughs form due to roller misalignment.

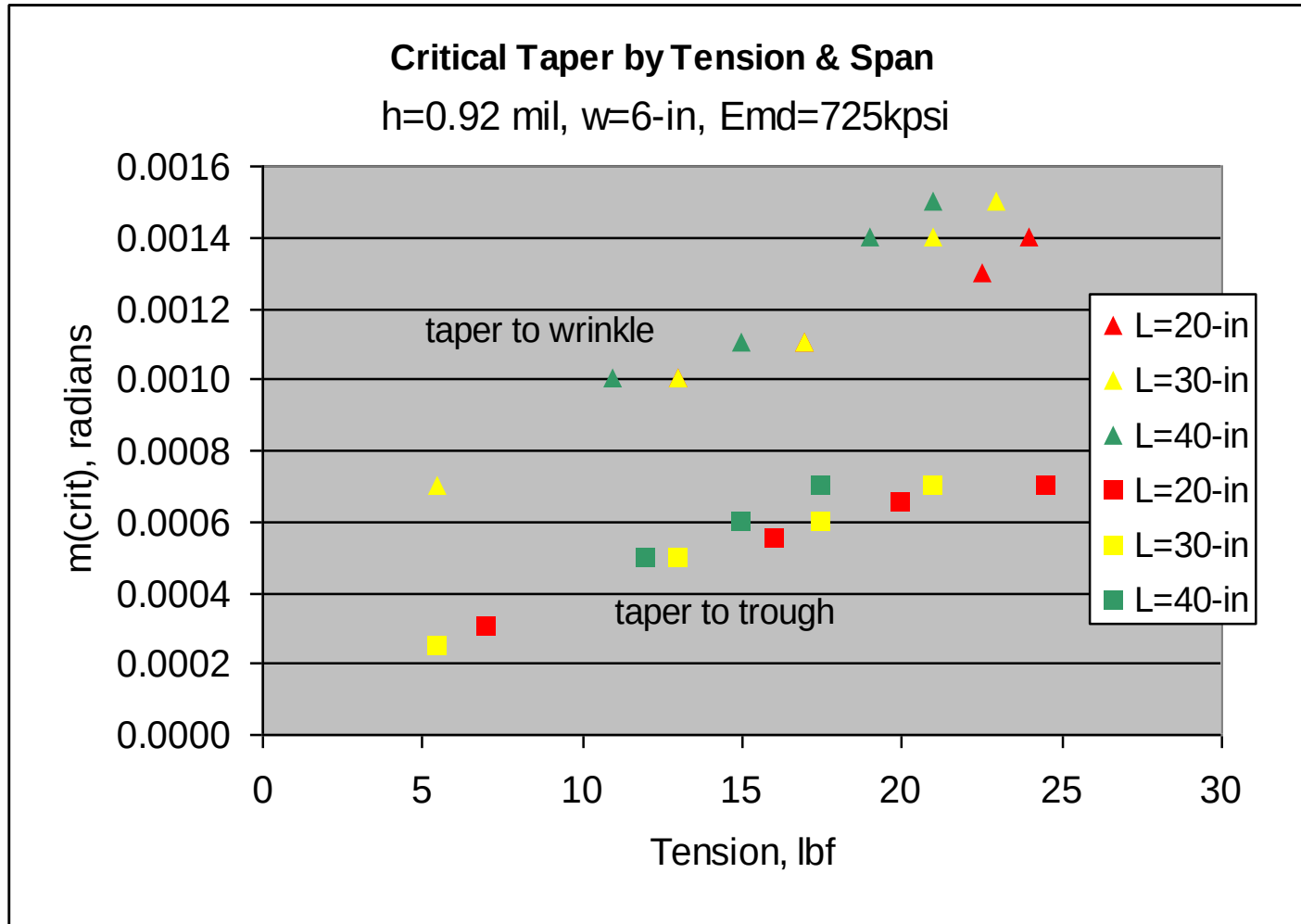
Higher tension and longer spans reduce sensitivity to misalignment

Predicting Web Wrinkles on Rollers, Beisel, Yurtcu, Good, Proceeding of the Tenth International Conf. on Web Handling, June 2009

TJWA Inc., Copyright 2015

www.webhandling.com

Troughs and Roller Taper



Troughs form due to roller taper.

Higher tension reduces sensitivity to roller taper, but span length has little effect.

Predicting Web Wrinkles on Rollers, Beisel, Yurtcu, Good, Proceeding of the Tenth International Conf.

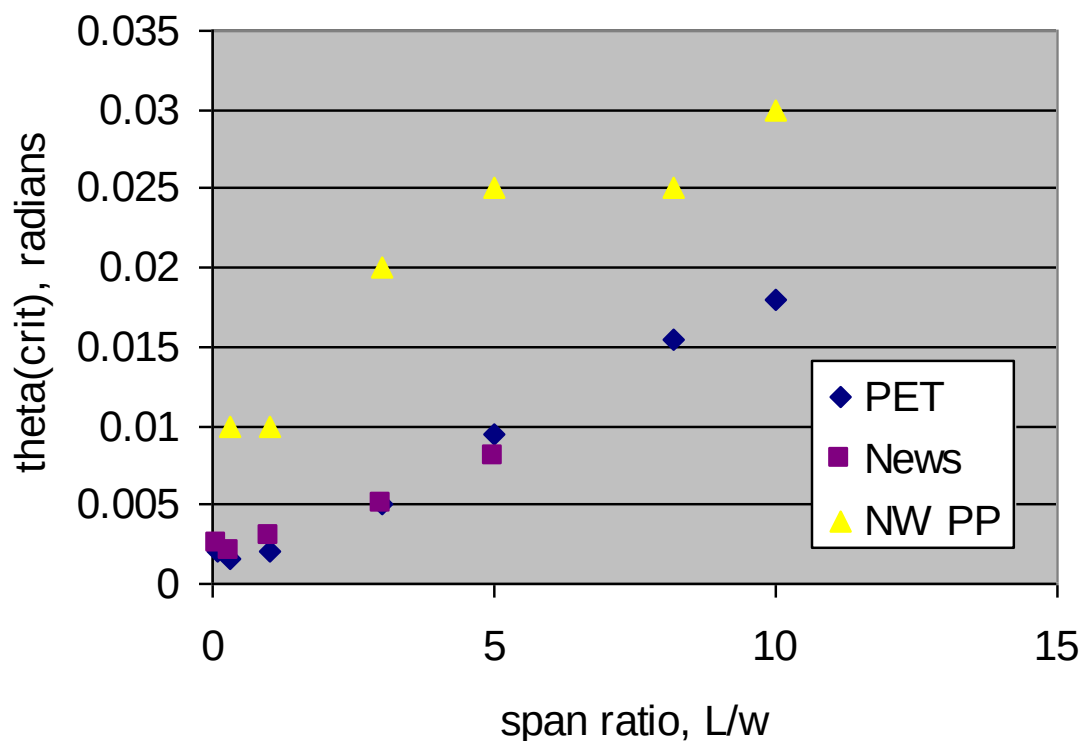
on Web Handling, June 2009

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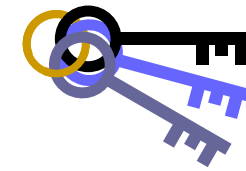
Troughing Misalignment vs. Span Length

Critical Tram for Troughing



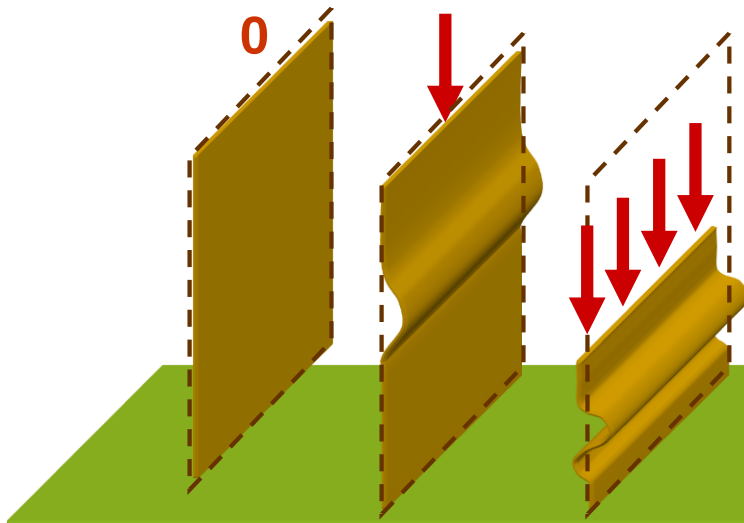
Sensitivity to
absolute
misalignment drops
with longer spans.

Predicting Web Wrinkles on Rollers, Beisel, Yurtcu, Good, Proceeding of the Tenth International Conf. on Web Handling, June 2009

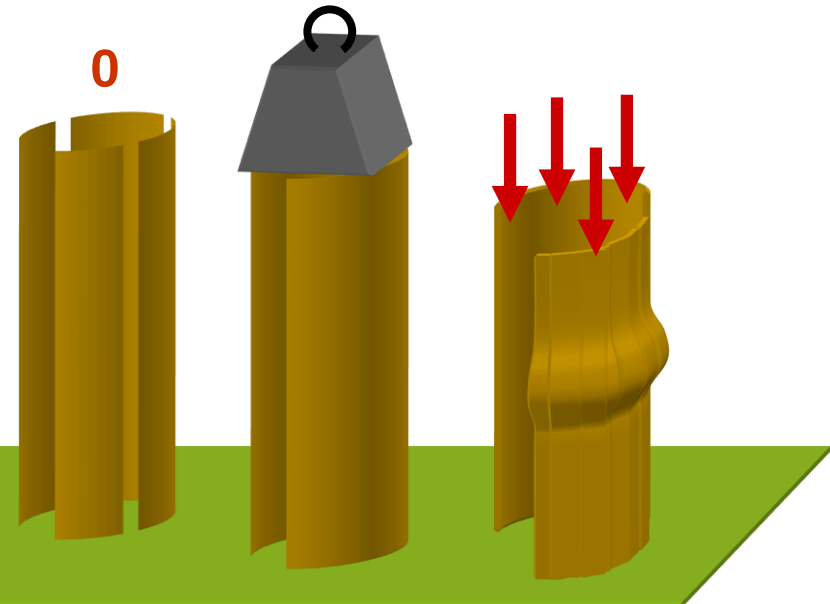


Shape Stiffness

A web in a planar shape quickly buckles under compression.

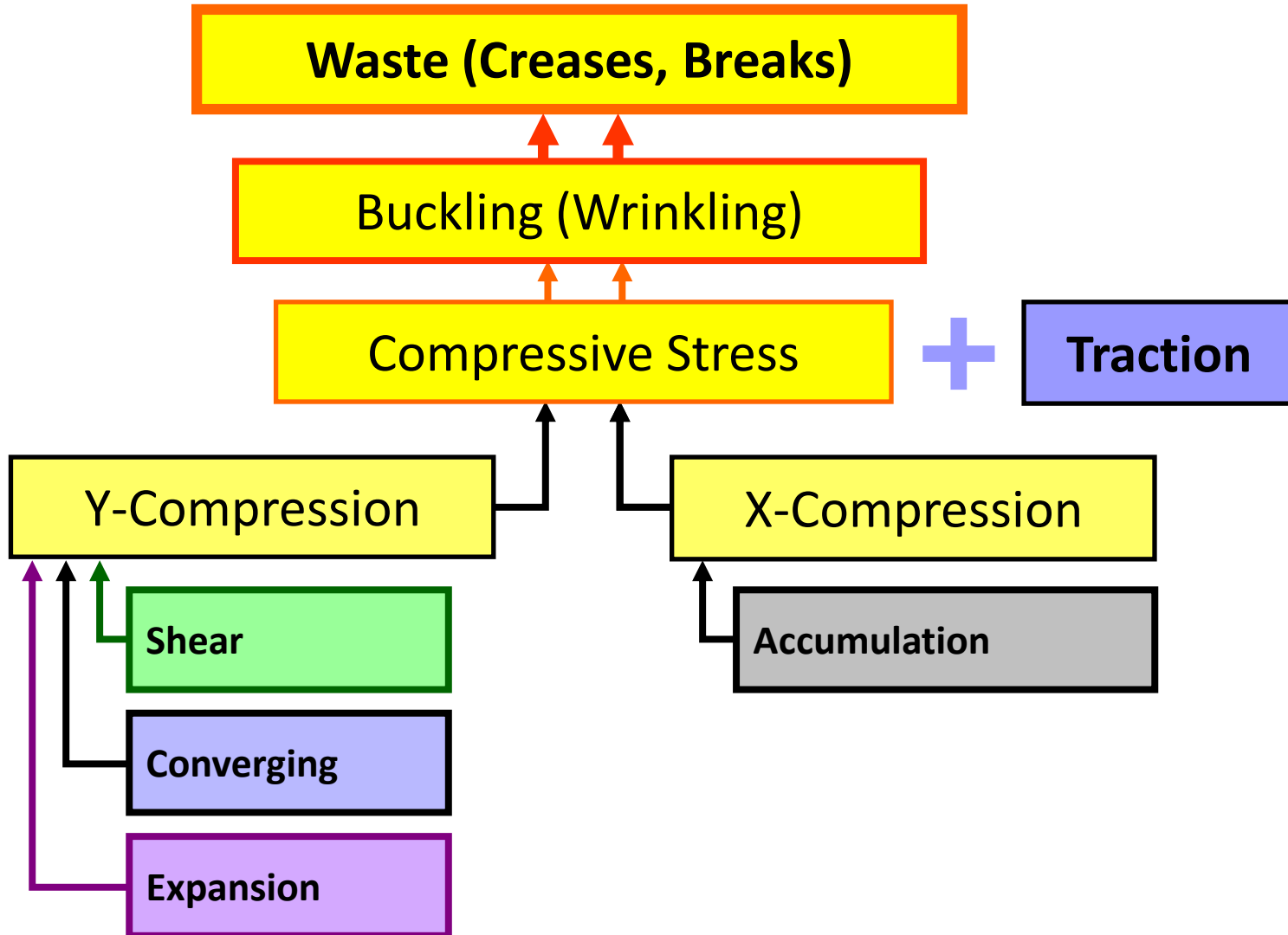


A web in a cylindrical shape supports higher compressive loads.

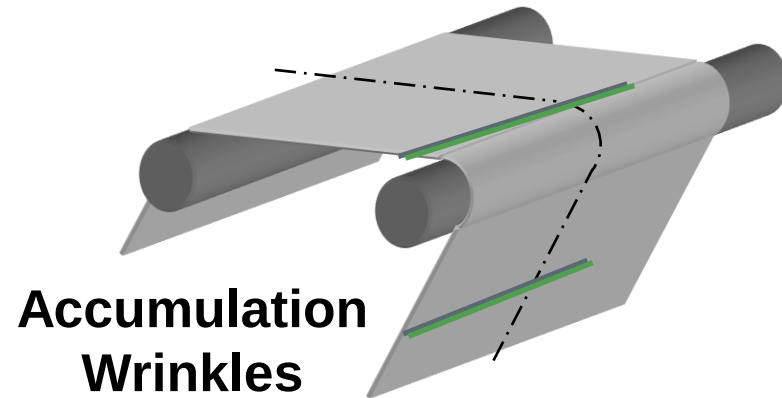
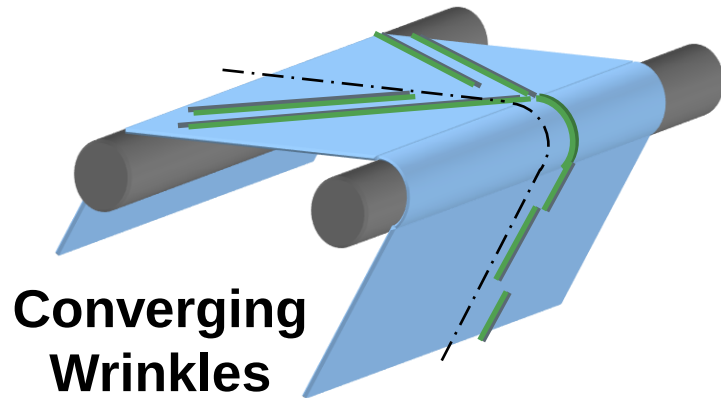
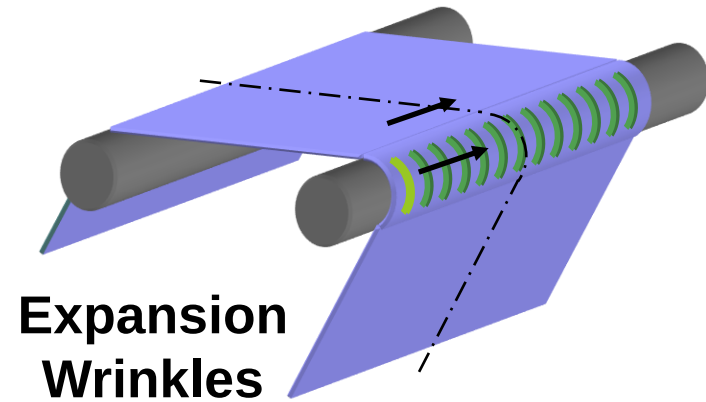
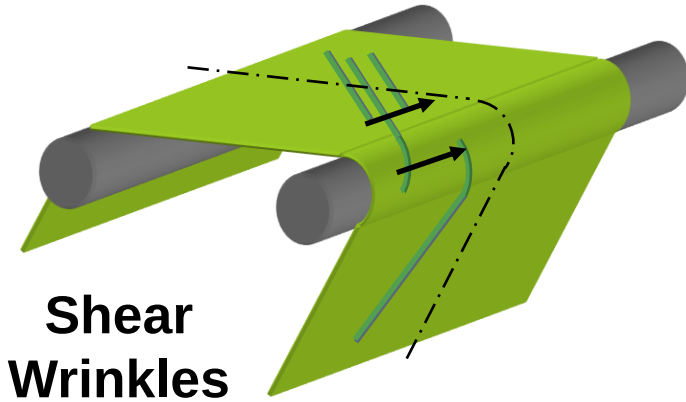


A web on a roller has a natural resistance to wrinkling.

Roadmap to Wrinkles



Identify Wrinkle by 4 Types



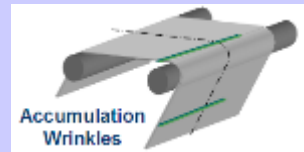
Definitions

- ▶ Shear Wrinkles (also known as Walking Wrinkles, Diagonal Wrinkles) – Wrinkles that form due to lateral bending in a span.
- ▶ Converging Wrinkles (also known as Stationary, Centering, Gathering, or Tracking Wrinkles) – Wrinkles that form due to opposite sides of the web converging to a position or lane.

4B1: Accumulation Wrinkles

Definition:

All or a portion of the web's width web is under machine direction compression and buckling as it passes over a roller, through a nip point, or enters a winding roll.

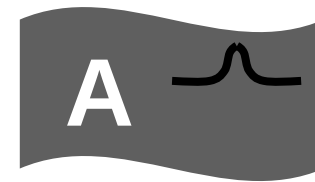


**CD Buckles
On Rollers**

Measuring Wrinkles:

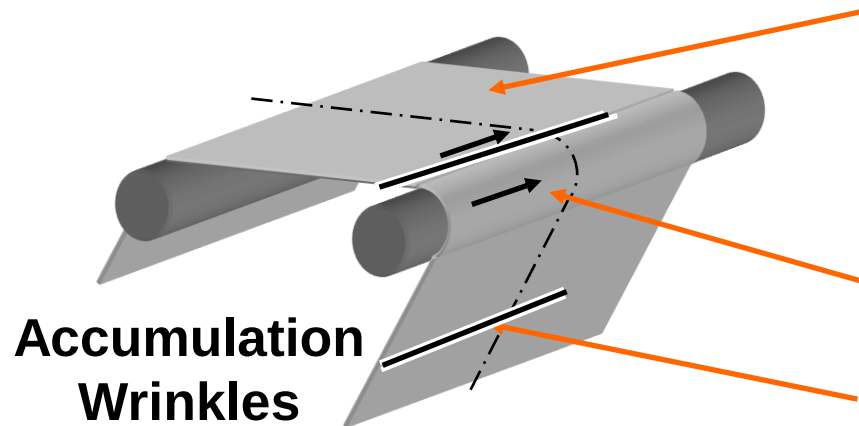
- Most wrinkles are measured by viewing or counting the creases in the web. Note these things:
 - ✓ The angle and position any troughs in the span immediately upstream of where the wrinkles forms,
 - ✓ The angle and crossweb position of the wrinkles.
 - ✓ Whether the wrinkle shifts or walks laterally.

Accumulation Wrinkles



Mechanism:

Part of the web accumulates upstream of a roller, then buckles and passes over the roller.



Troughs:

Crossweb, forming immediately upstream from roller.

Wrinkles:

Crossweb over part or all of web width. Often forming in regular, but not exact frequency.

Creases:

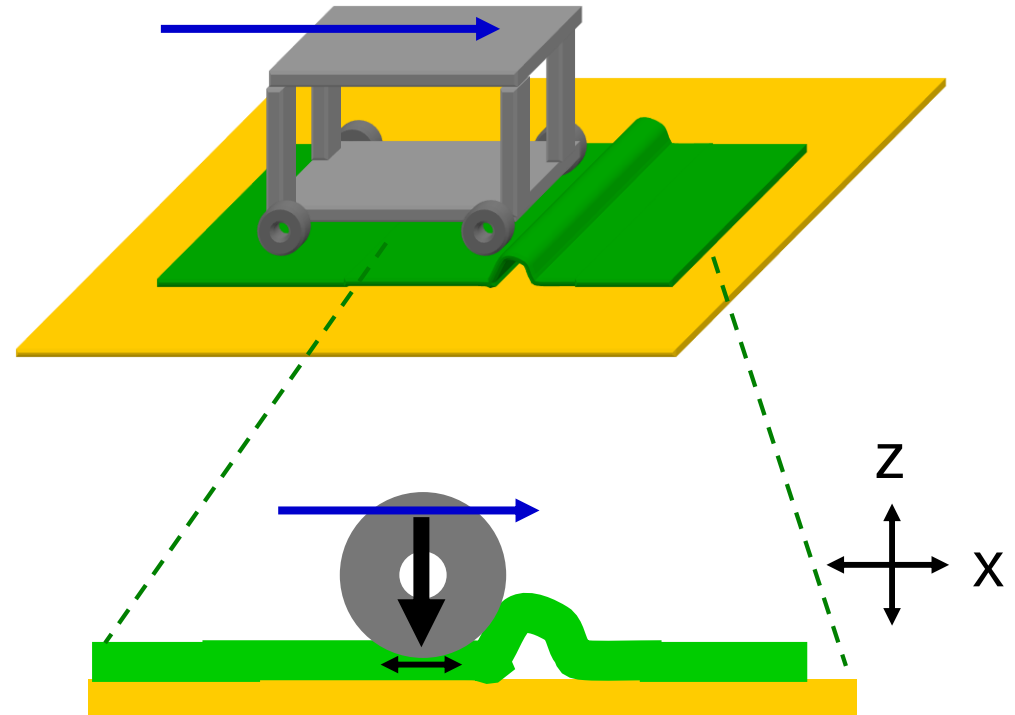
Same as wrinkles.

Nip Induced Tension

A cart rolled across a rug will form a wrinkle or buckle in front of it.

The nip force compresses the rug in the Z-direction and elongates it in the X-direction.

The wheel pins the elongated material to the ground, pushing the excess material forward..



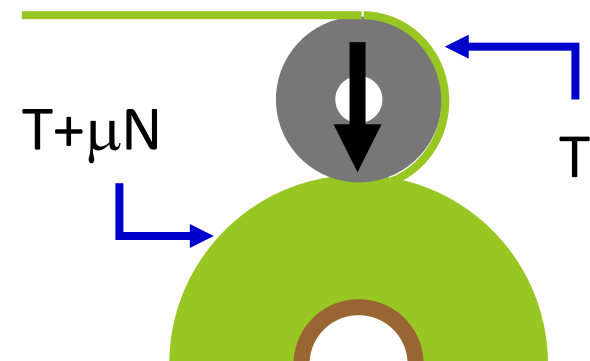
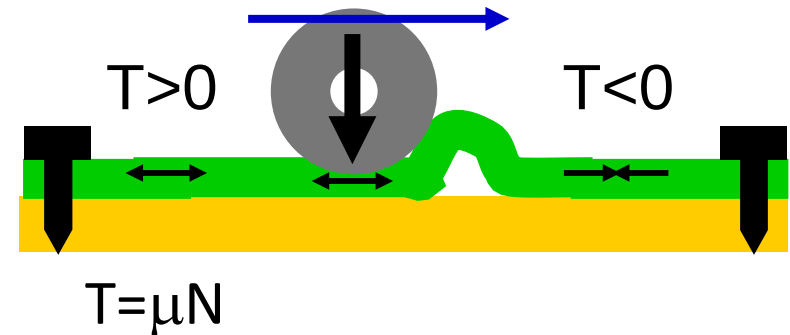
Nip Induced Tension

If the rug was fixed to the floor on both ends, its tension would increase after the nip rolled over it and decrease ahead of the nip.

The tires ability to tension the rug is limited by friction – the frictional force between the rug and floor (μN).

The winding nip has the same effect.

The web that passes under the nip, slips relative to the layer below it, increasing tension by approximately μN .



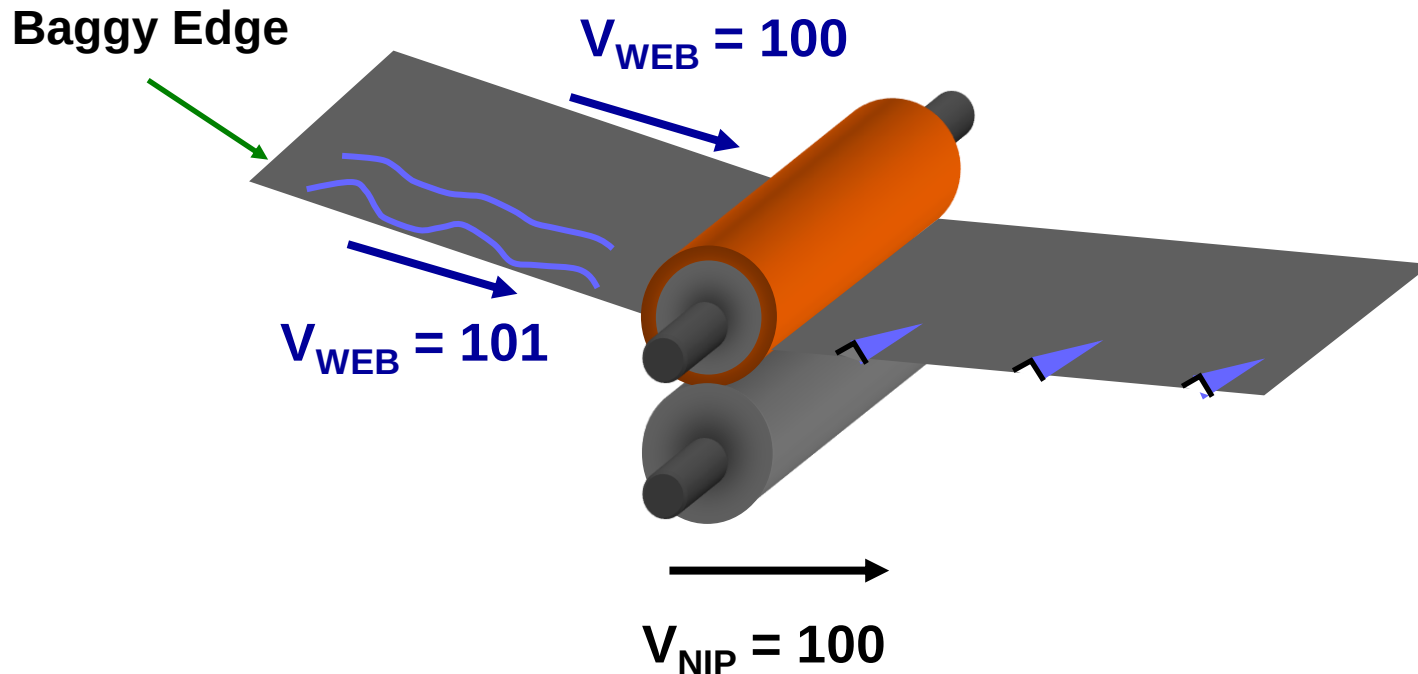
A1



Accumulation Wrinkles

Case 1:

Baggy Webs and Nipped Rollers

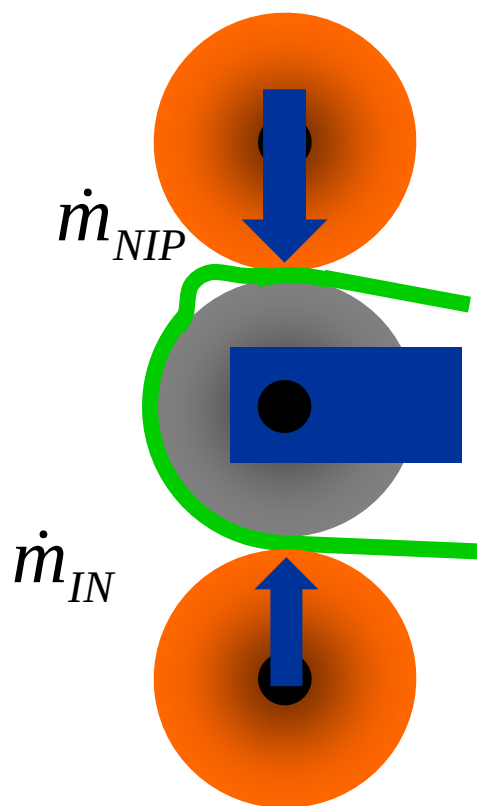


Accumulation Wrinkles

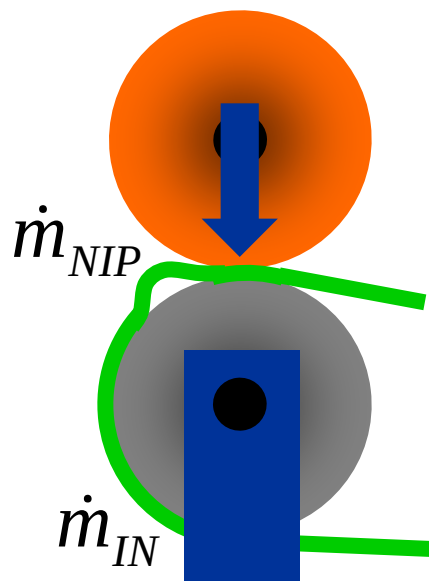
A2

Case 2: Overfed Nips

As nip load goes up, the mass flow through the nip may drop, creating excess web ahead of the nip.

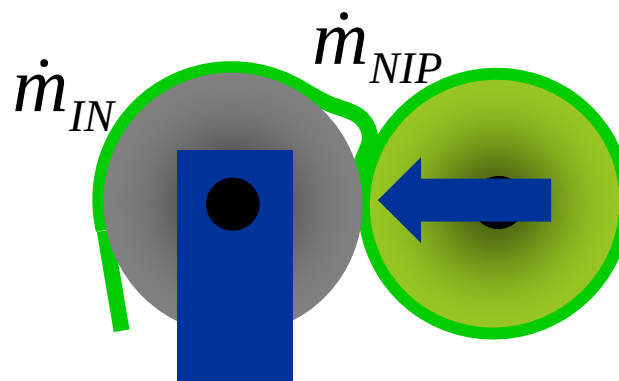


Calendar Stack



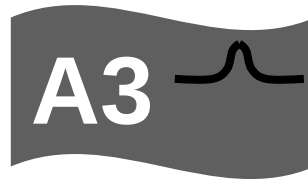
Nipped Rollers

$$\dot{m}_{NIP} < \dot{m}_{IN}$$

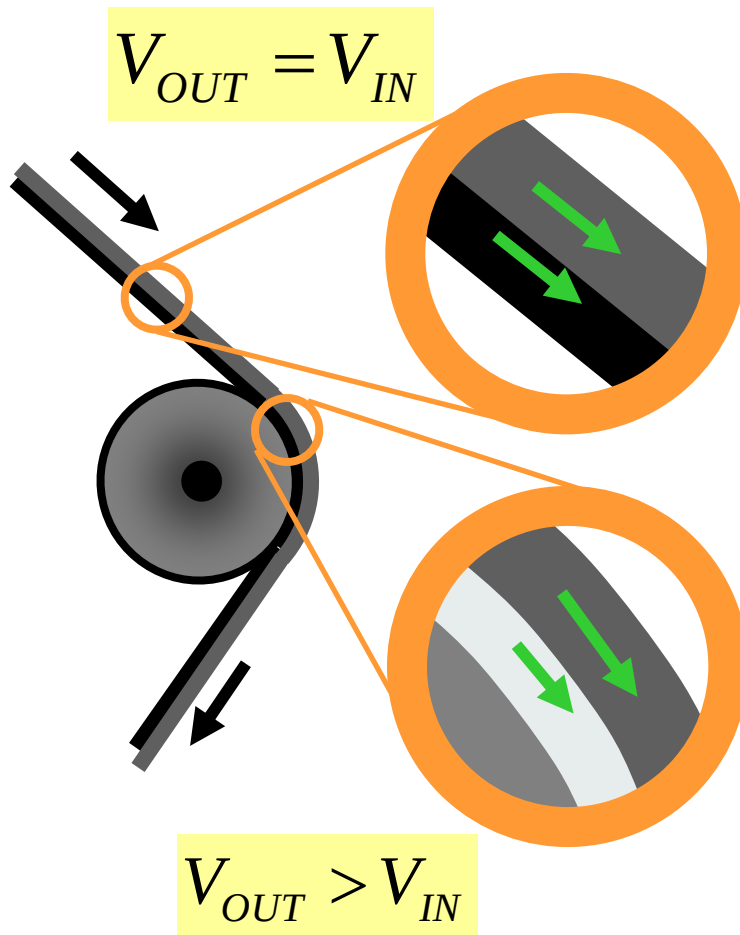


Surface Winder

Accumulation Wrinkles



Case 3: Laminates on Small Radius Rollers



As a laminate bends around a roller, the outer layer rotates at a larger radius and therefore a higher speed.

$$V_X = 2\pi r_X \omega$$

$$r_{OUT} > r_{IN}$$

The laminate interface must carry the load created by the differential strains and stresses.

Accumulation Wrinkles

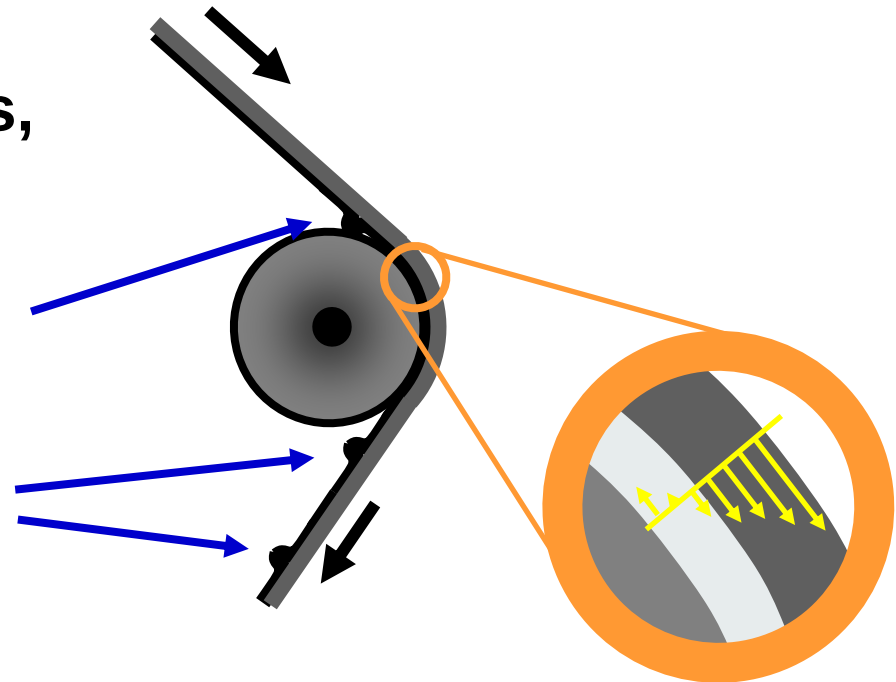
A3

Cause:

Laminate Bond < Shear Force

If the laminate bond breaks, the inner ply will slip relative to the outer ply, accumulating upstream of the roller...

...until a crossweb wrinkle passes over the roller.



Remedies: High laminate bond, larger rollers, or lateral roller ridges.

Accumulation Wrinkles

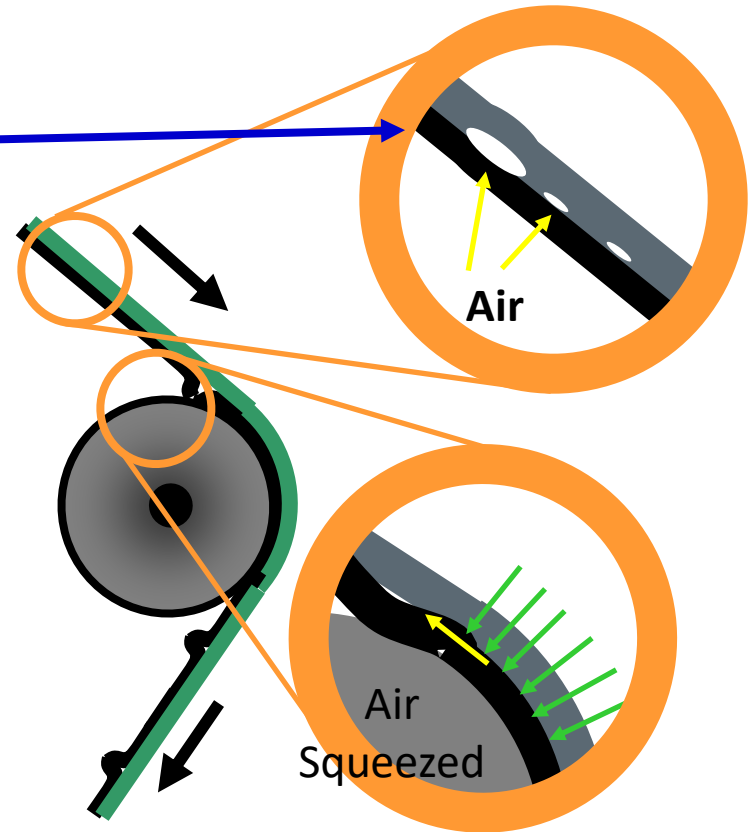


Case 4: Air Bubble Accumulation

Many laminates have air bubbles at their interface.

At a roller, the top layer's pressure may force the air bubble upstream.

If the laminate bond is low, the delaminated inner layer may accumulate until a wrinkle forms.



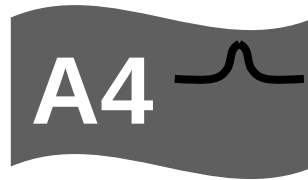
Remedies: Eliminate bubbles, higher laminate bond, larger rollers, lower tension, or lateral roller ridges.

Eliminate Bubble-Related Delamination

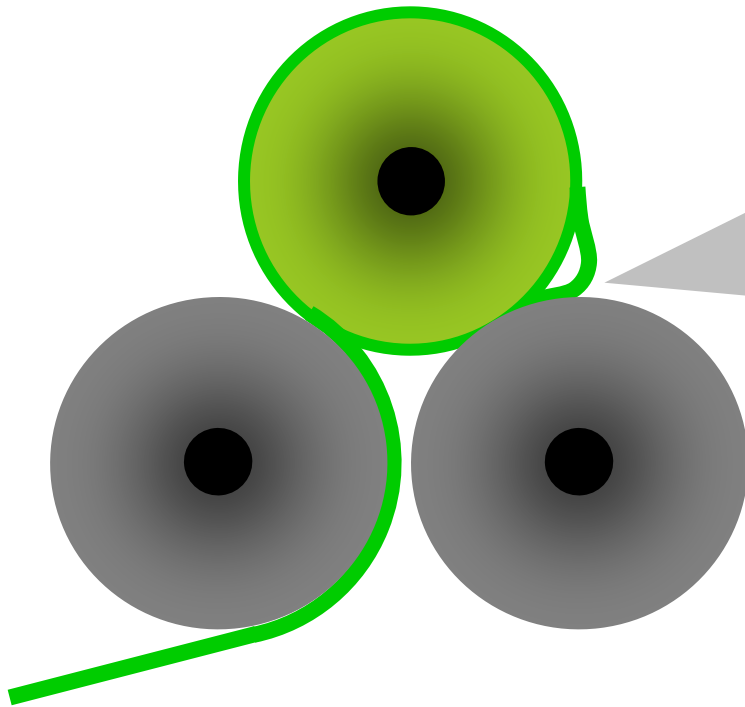
- ▶ Reduce air at laminating (low speed, uniform nipping, sufficient nipping, in vacuum/absence of air).
- ▶ Increase adhesive bond between layers.
- ▶ Reduce pressure of top web to bottom web, allowing air to pass over rollers (low tension, large radius)
- ▶ Design the product with porosity, holes, or perforations to allow air through the product thickness.
- ▶ Use lateral ridges on rollers to pass small bubbles downstream without accumulating to defect-causing size.
- ▶ Pierce the bubbles manually with a blade or automatically with a close proximity blade or needle.
- ▶ Wipe bubbles to the side manually or use a high bond, full width tape to push bubbles through to the winder or end of the process.

Google :
“PFFC bubbles”

Accumulation Wrinkles

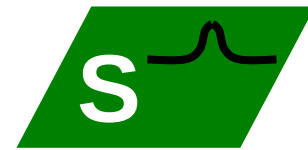


Air Entrained in Wound Rolls



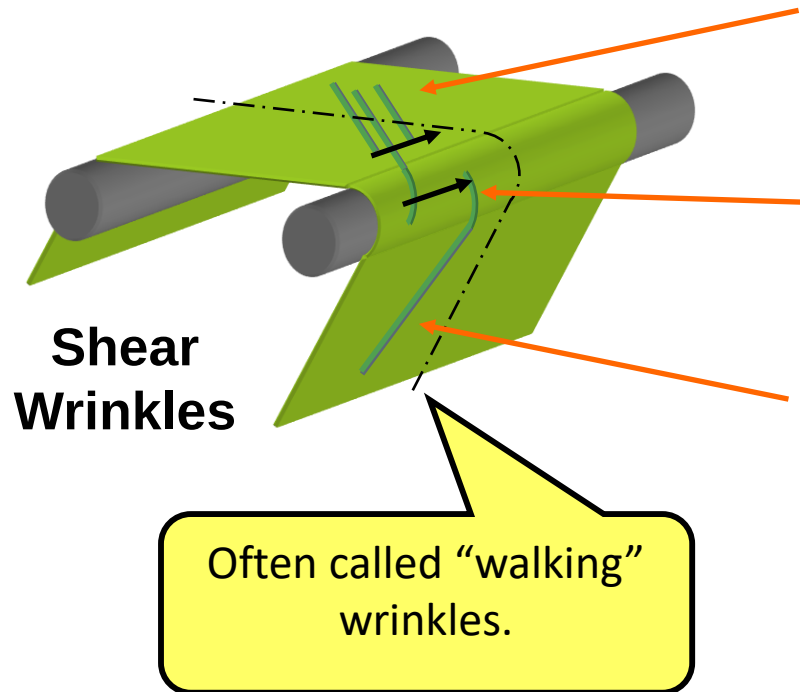
In two drum surface winders, the second nip point may squeegee and accumulate entrained air and web leading to a wrinkle.

Shear Wrinkles



Mechanism:

Web pulled laterally to one side.

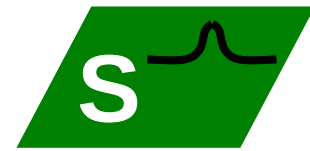


Troughs:
Asymmetrical and angled.

Wrinkles:
Form and “walk” laterally.

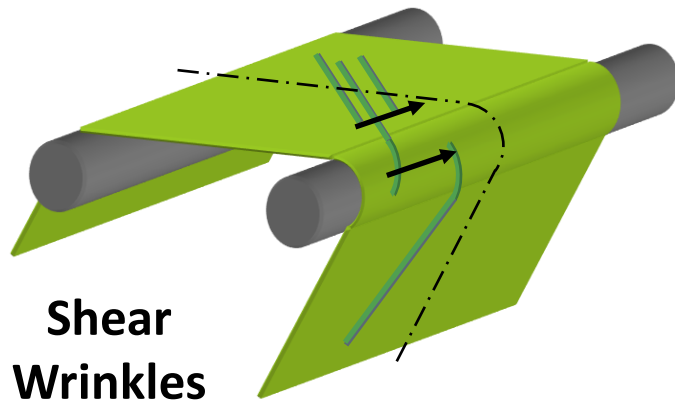
Creases:
Angled 5-30 degrees from machine direction.

Shear Wrinkles



Mechanism:

Web pulled laterally to one side.

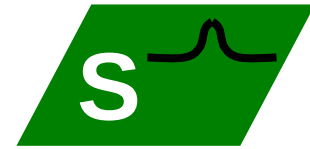


Eliminate any one of these and wrinkles will not form.

Prerequisites:

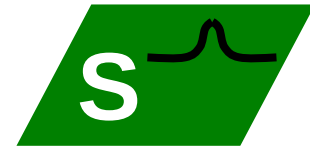
1. Mechanism to induce lateral bending.
2. Tractional force to overcome span's bending stiffness.
3. Tractional force on roller to exceed buckling criteria.

Shear Wrinkles Requirements



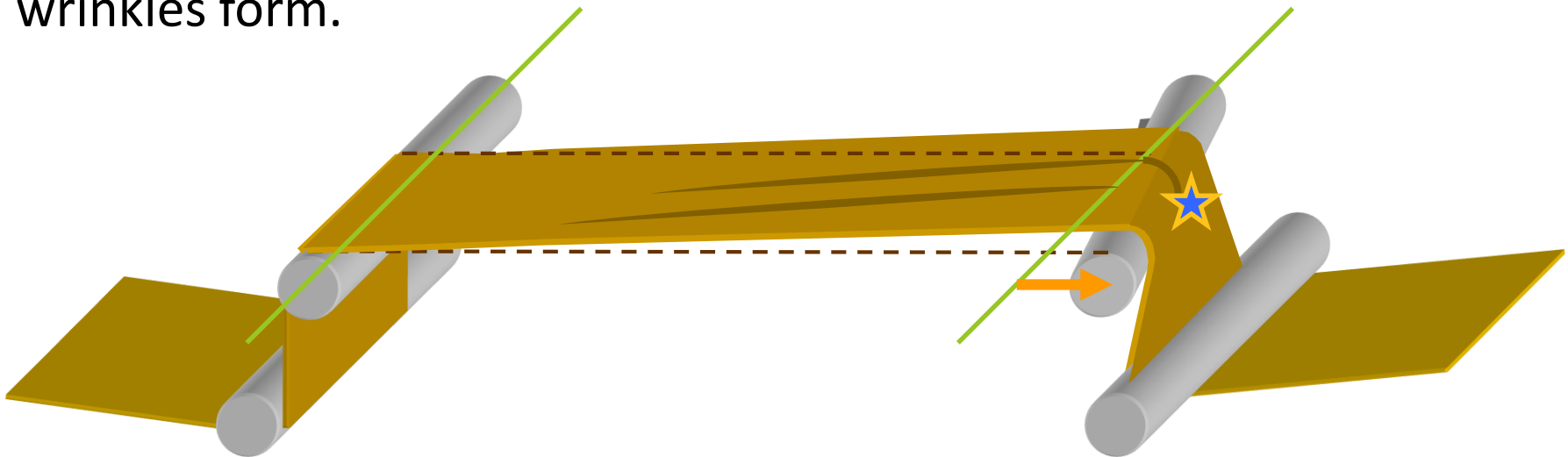
1. Potential for Critical Bending
2. Traction to Bend
3. Traction to Wrinkle
4. Freedom to Buckle
5. Time to Form

Shear Wrinkles Requirements



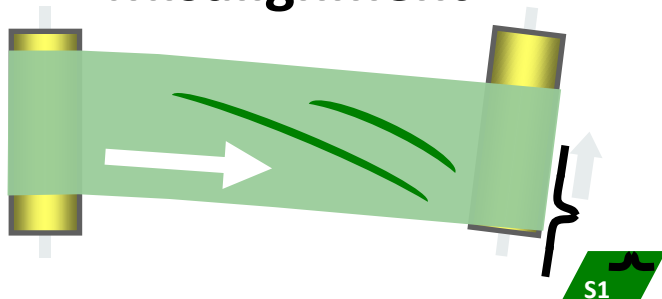
1. Potential for Wrinkle-Critical Bending

Any web span has a roller angle or amount of lateral offset that will trigger troughing. If the angle or offset is increased, wrinkles form. If the errors in angle or offset are kept below the critical value, no wrinkles form.

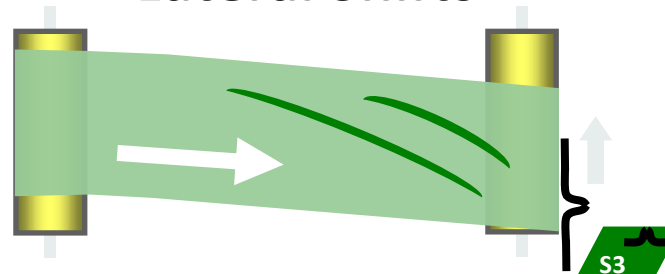


Shear Wrinkle Causes

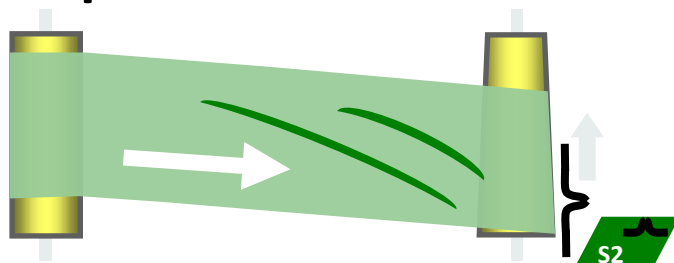
Misalignment



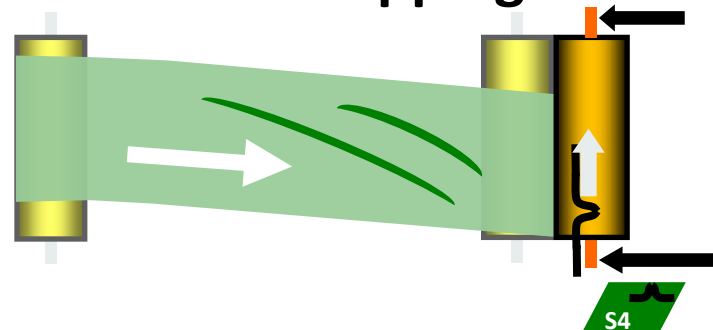
Lateral Shifts



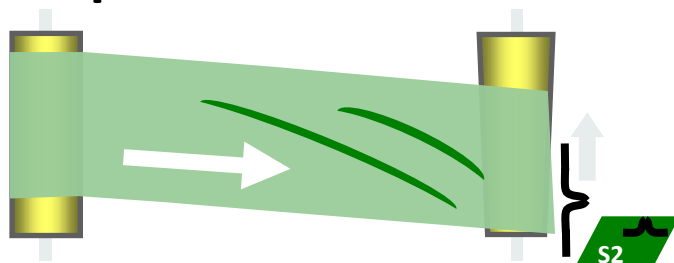
Taper + Good Traction



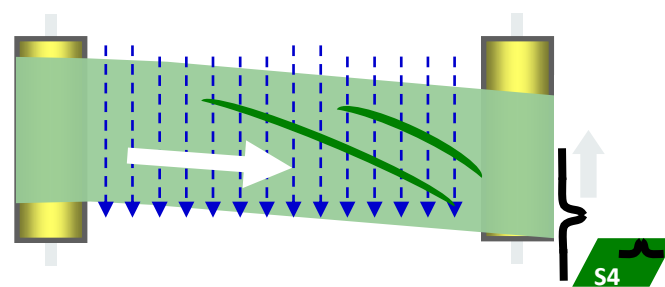
Uneven Nipping



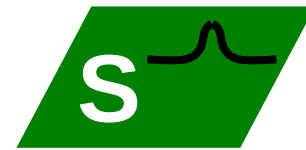
Taper + Poor Traction



External Forces



Shear Wrinkles Requirements



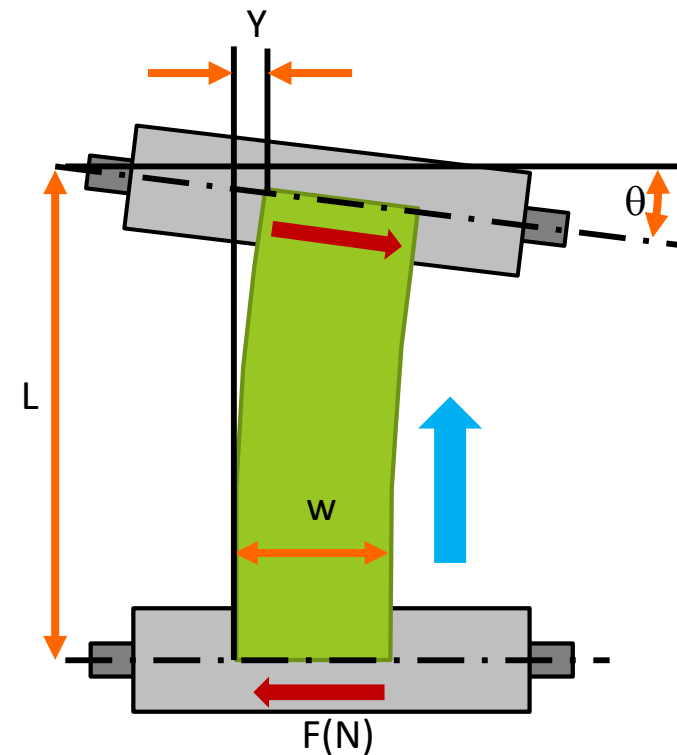
2. Traction to Shear

Bending mechanisms, such as misaligned or tapered rollers may attempt to bend the web, but they are limited by web-roller traction.

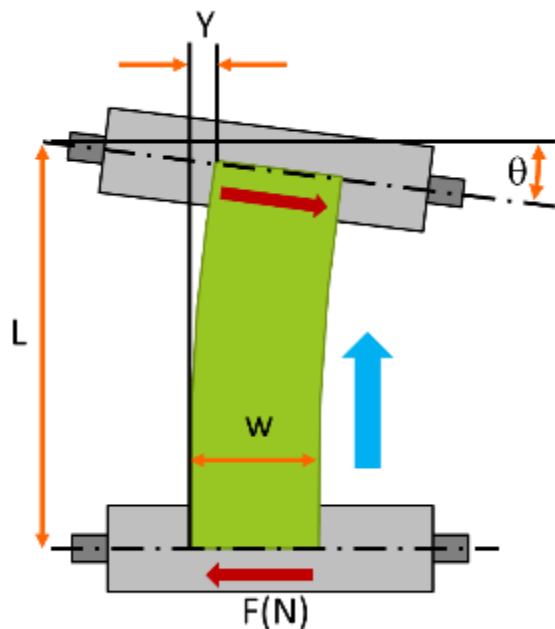
If the traction is high, the mechanism will control the web bending.

However, if the traction is below what is required to bend the web, then the bending will be traction limited and wrinkles will not form.

$$F = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

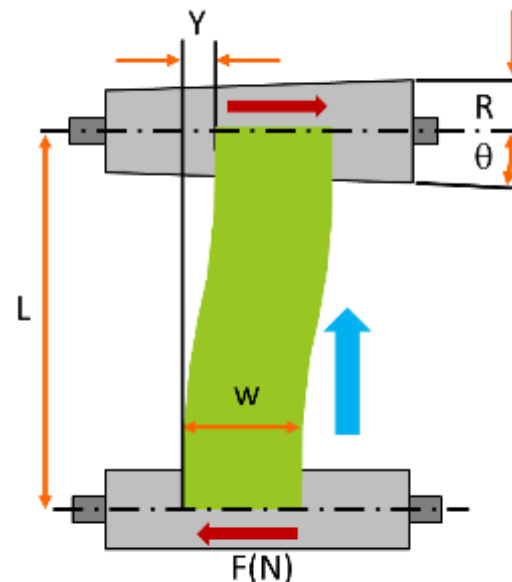


Lateral Tracking Comparison



$$F = \frac{2EI\theta}{L^2} = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

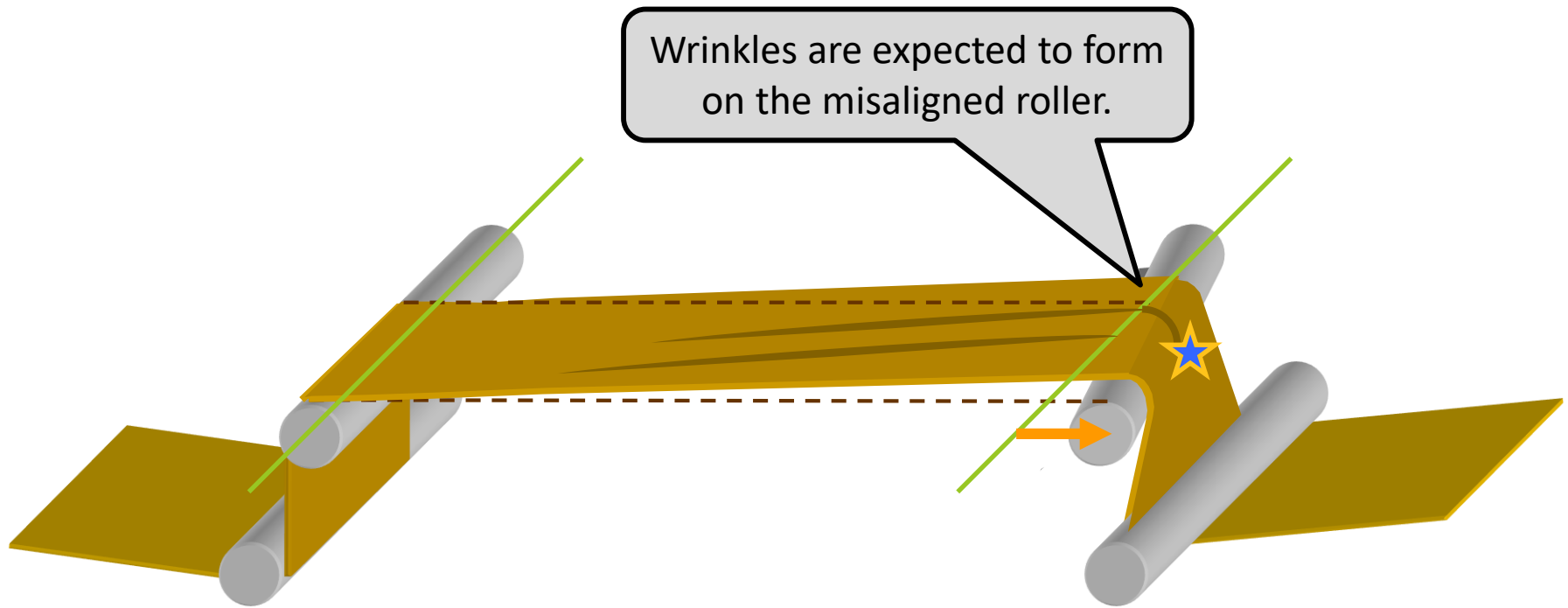
$$Y(L) = \frac{2}{3} L \theta$$



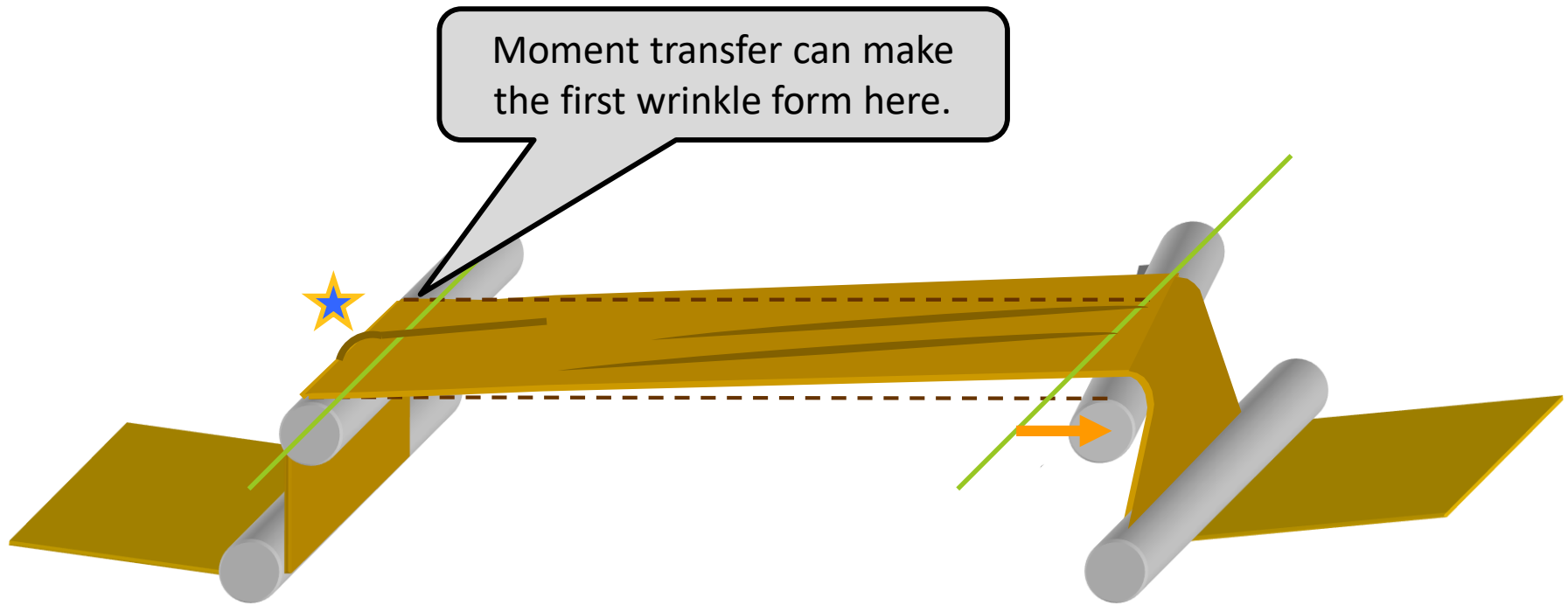
$$F = \frac{2EI\theta_c}{RL} = EtY \left(\frac{w}{L} \right)^3$$

$$Y(L) = \frac{\theta_c L^2}{6R}$$

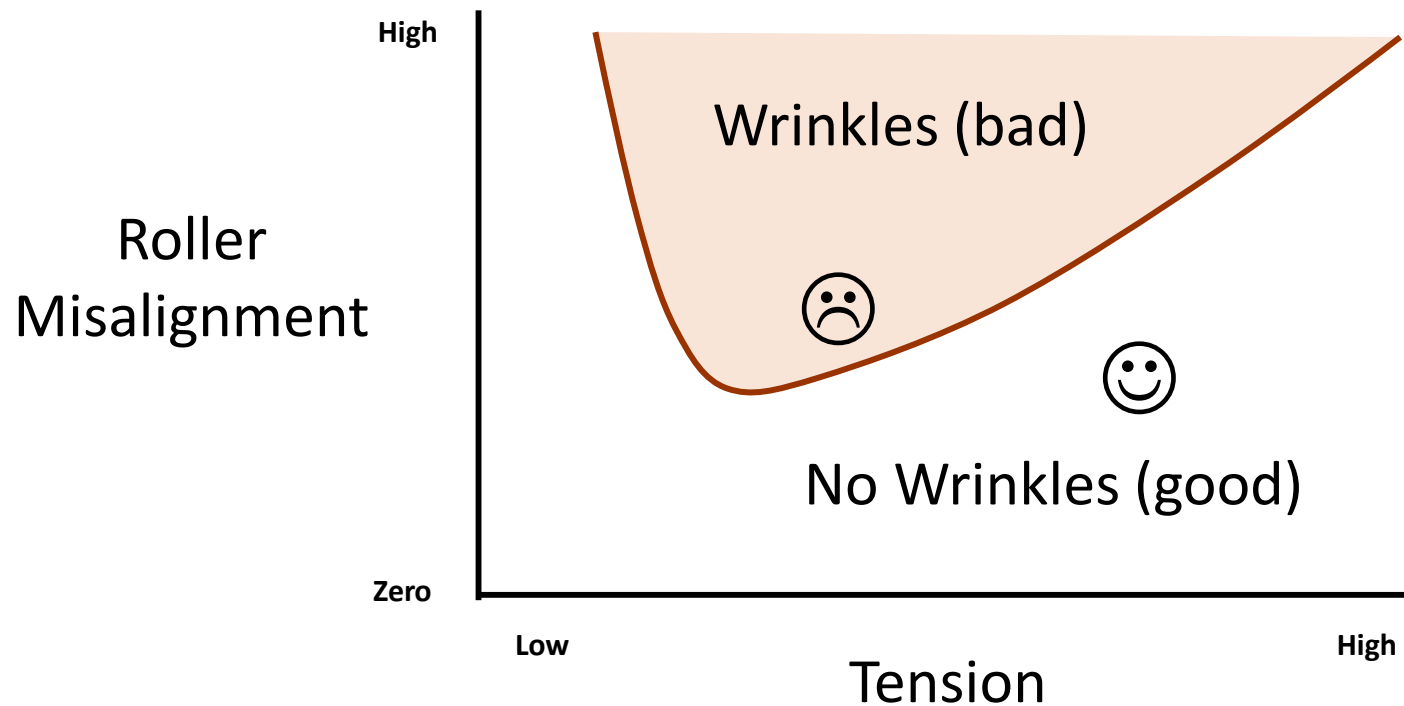
Shear Wrinkle Experiments



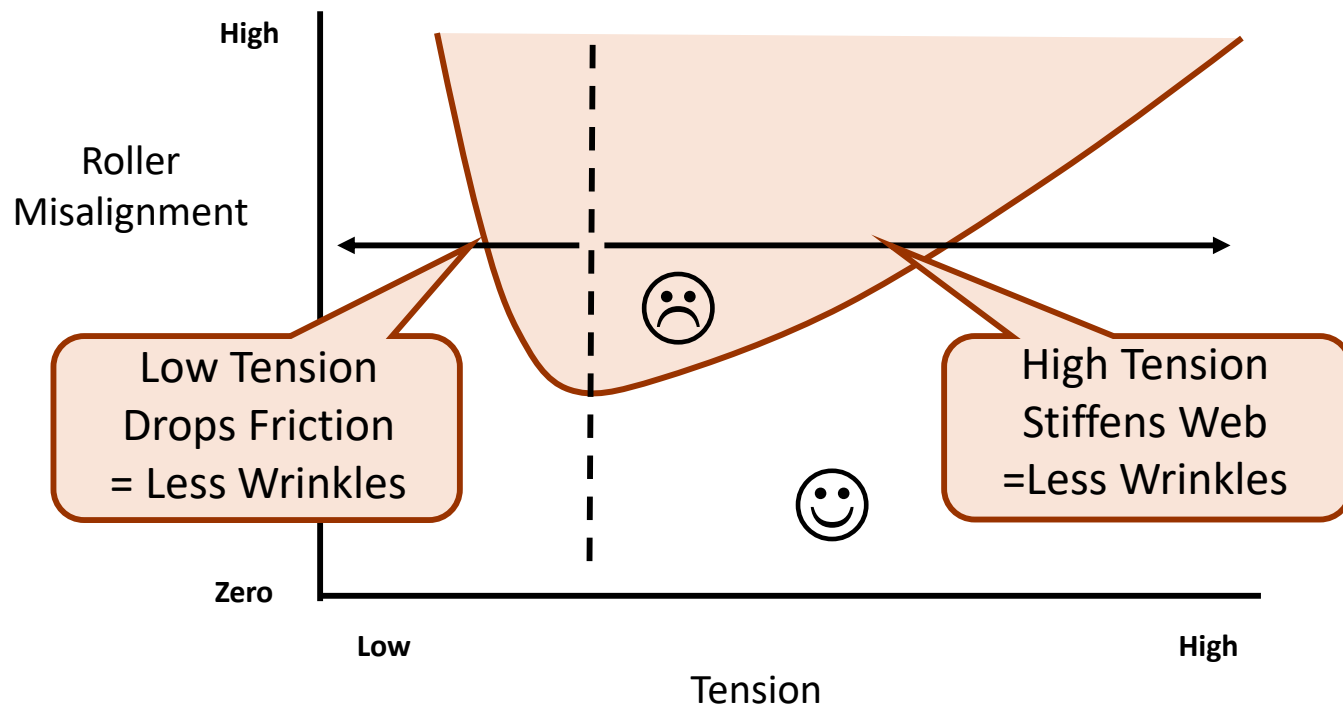
Shear Wrinkle Experiments



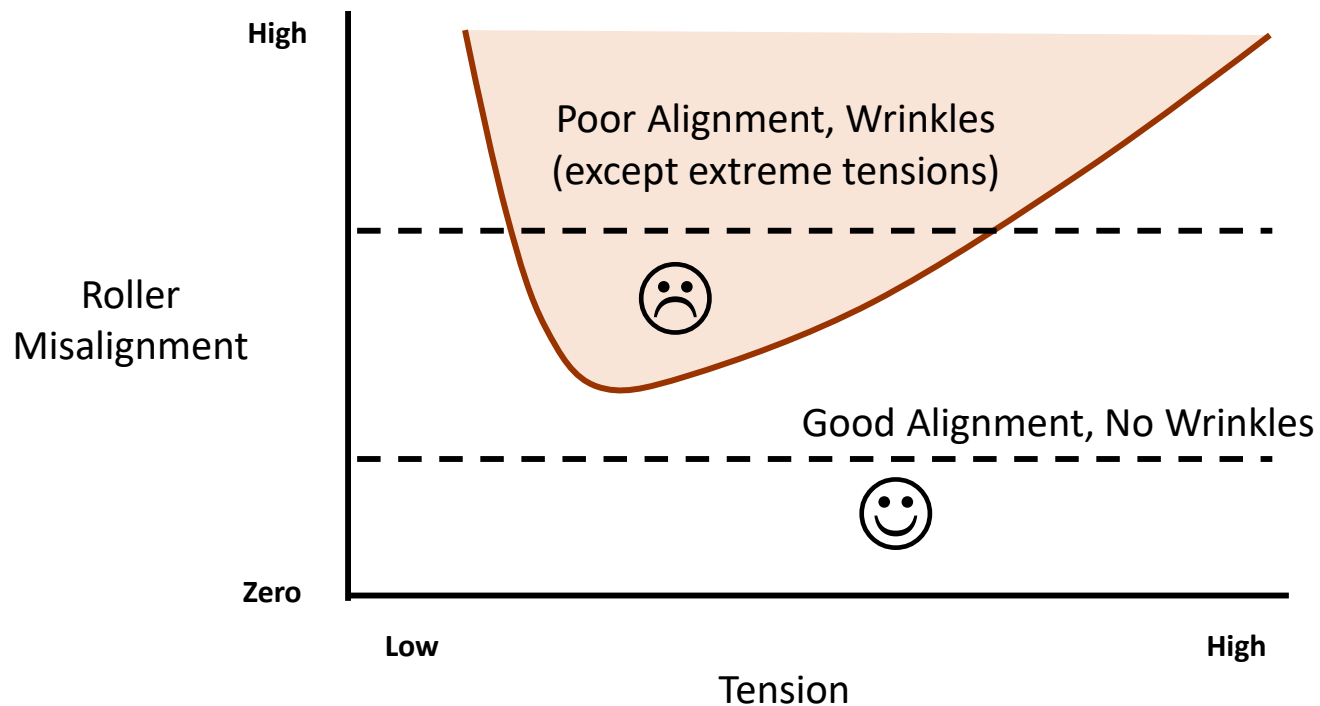
Wrinkle Behavior



Wrinkle Behavior

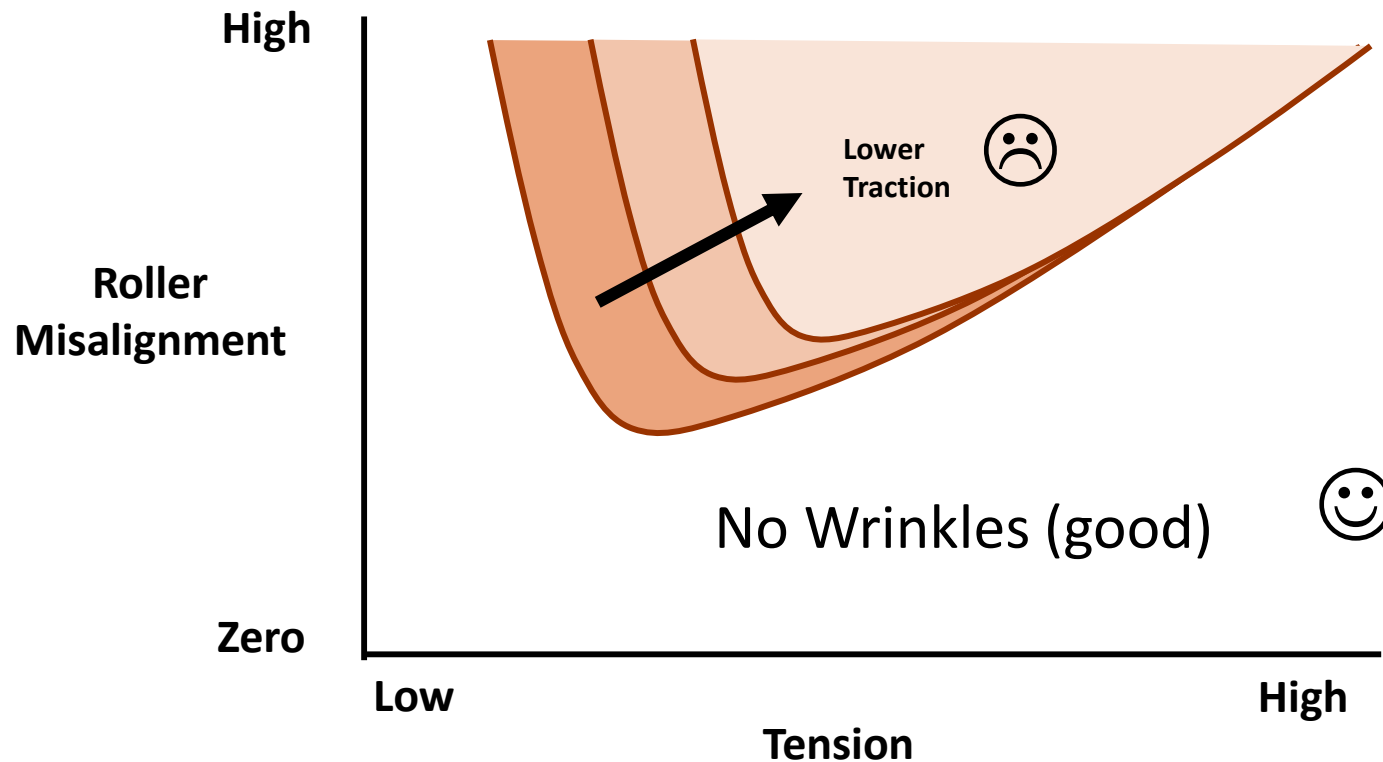


Wrinkle Behavior



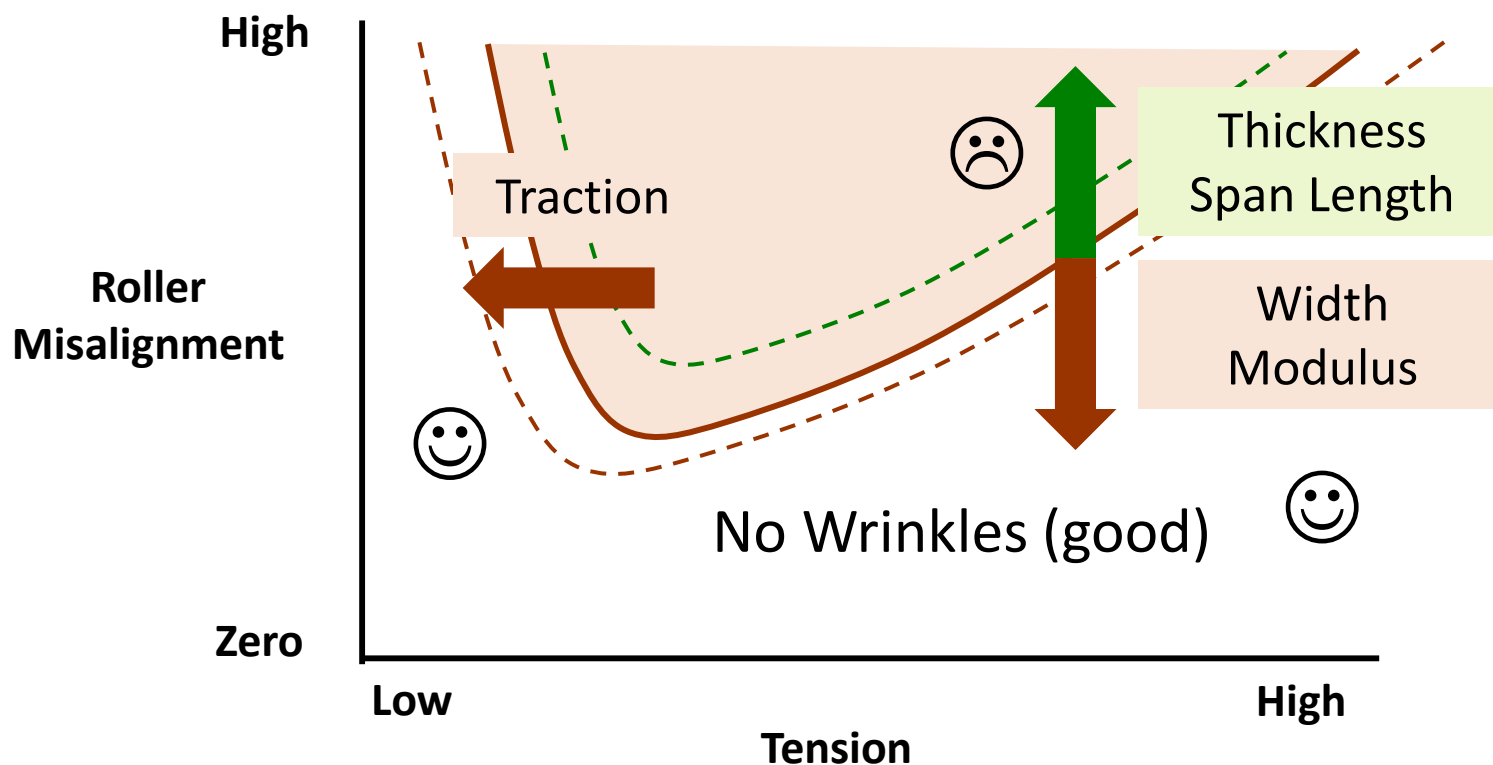
Wrinkles and Traction

Wrinkle-free conditions increase on low tension side with reduced traction coefficient.

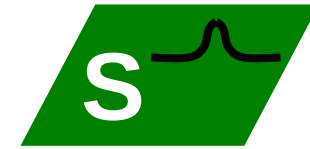


Wrinkles and Traction

Wrinkle-free conditions will shift with traction, thickness, width, span length, and modulus.



Shear Wrinkles Requirements



3. Traction to Wrinkle

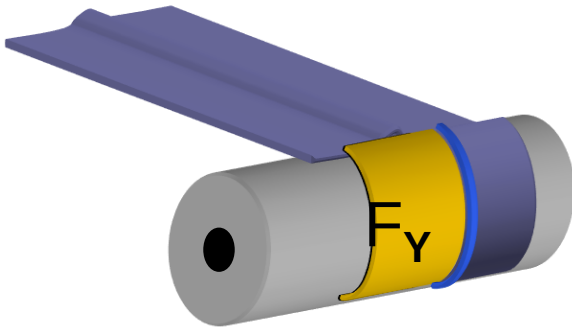
Force must be applied from two directions to hold the web in a buckled shape. If either side of an attempted wrinkle has insufficient traction, the wrinkle can slide out.

Traction develops over contact width. If the web-roller contact is disrupted into bands where no individual band can achieve the force, then again, wrinkles will not form.

Traction and Wrinkling

Traction is required to form and hold a wrinkle.

The most wrinkle-holding traction is at the roller's center.



$$F_Y = \frac{\mu T \theta w}{2}$$

Eliminate wrinkles by:

Reducing Coefficient of Traction

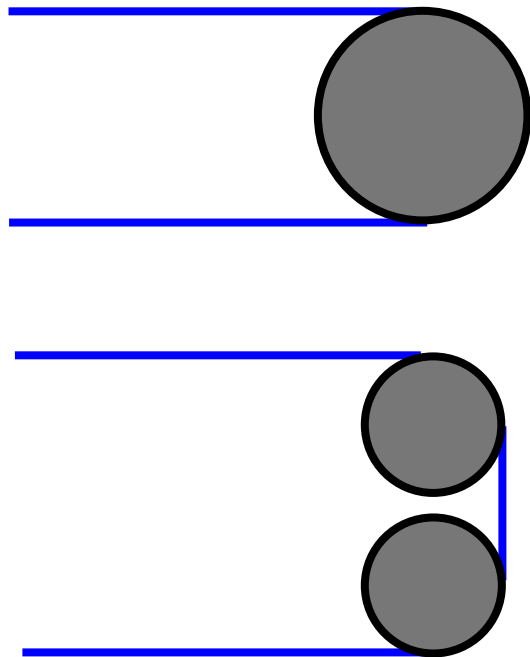
Reducing Tension / Width

Reducing Wrap

Reducing Width

Wrap Angle & Wrinkles

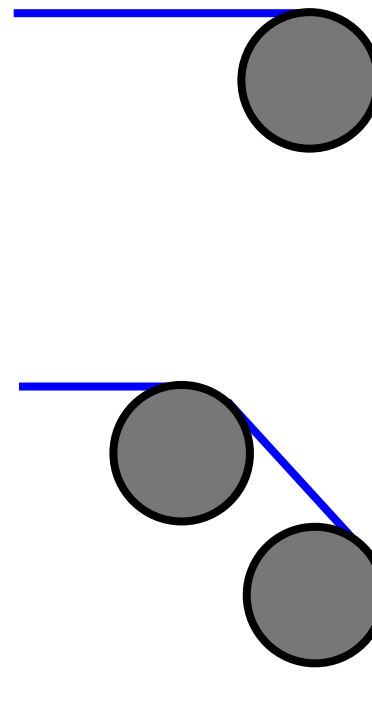
Avoid 180° wrap on rollers (except dancers).



Worse

Better

Reducing wrap will often decrease wrinkle sensitivity.

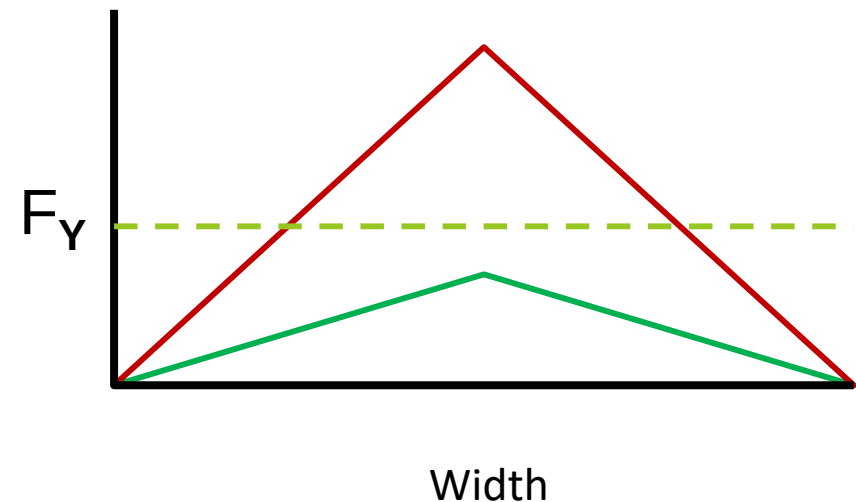
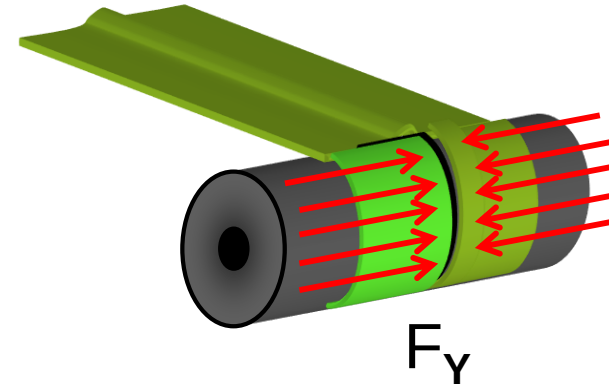


Traction and Buckling Stress

The compressive stress required to hold the web in a buckled shape in opposition to shape stiffening is delivered by web-roller traction forces.

If a roller can apply enough tractional force, wrinkles are possible.

If a roller cannot apply enough tractional force, wrinkles can't form.

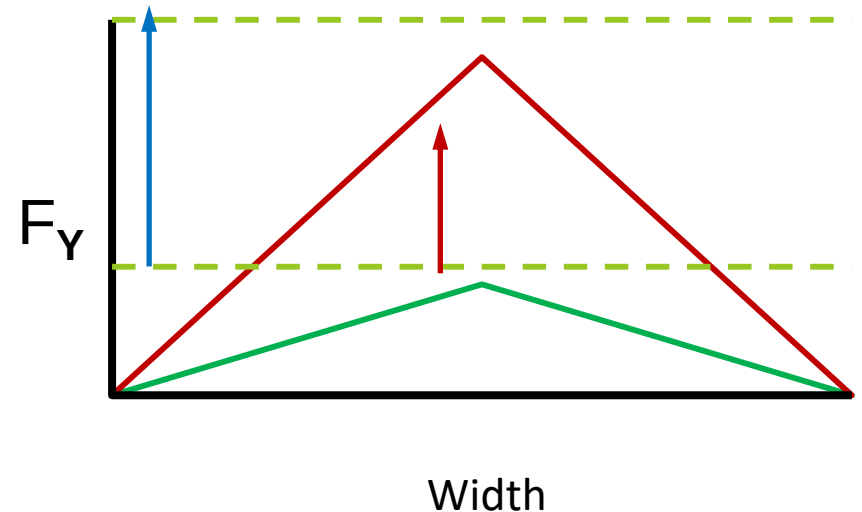
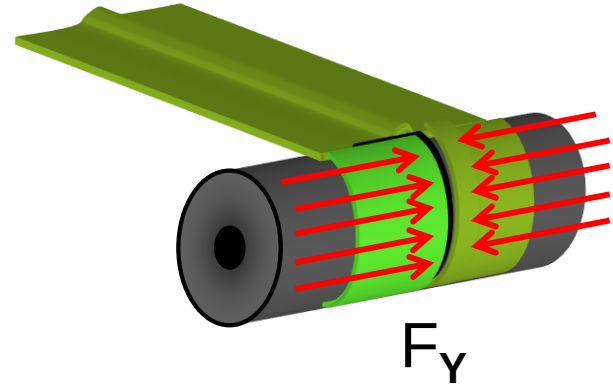


Traction and Buckling Stress

The buckling stress increases with thickness, modulus, and inversely with radius.



The traction force to enable wrinkles increases with coefficient of traction, tension, and width.

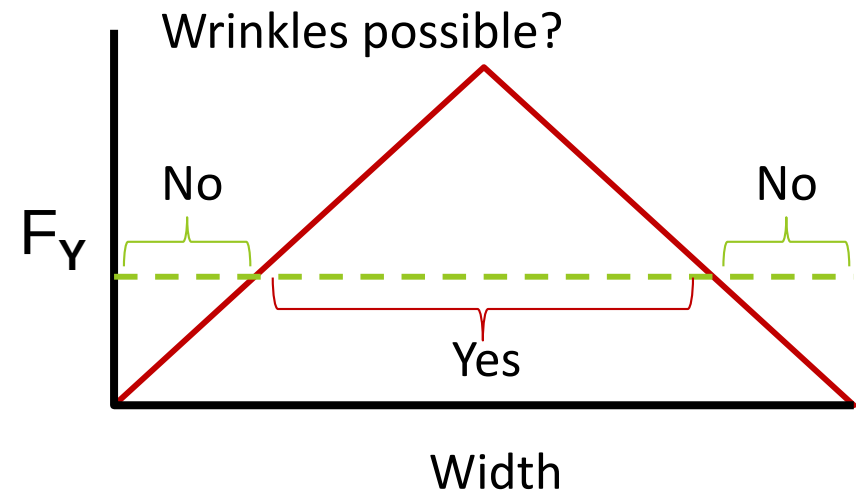
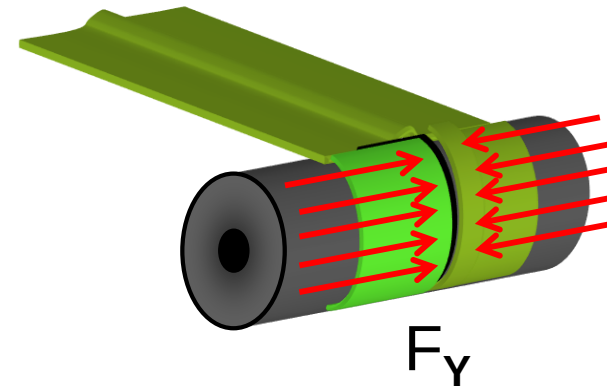


Traction and Buckling Stress

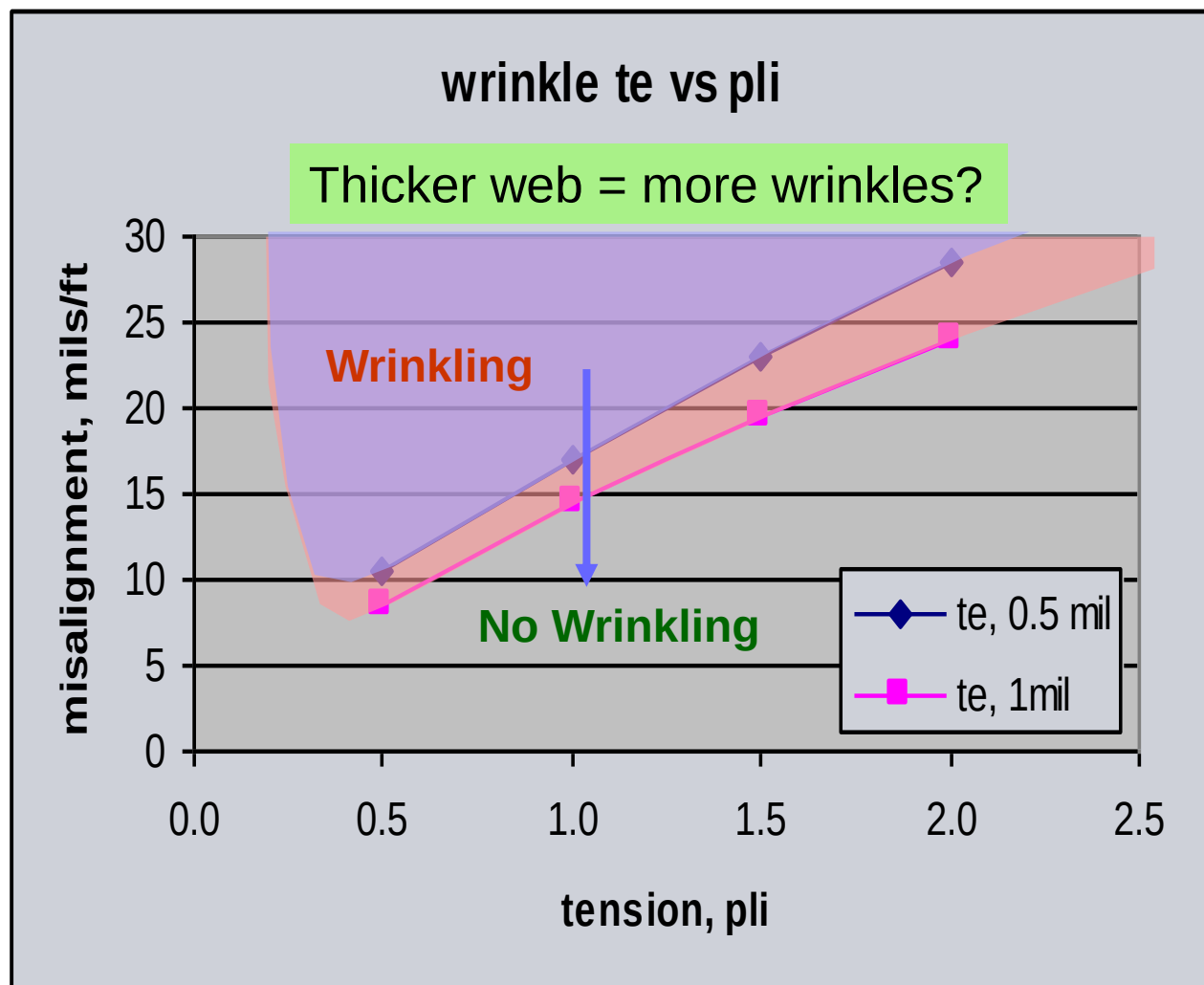
To hold a wrinkle, traction must be available from both direction laterally.

The center of the web-roller contact has the highest traction greatest ability to hold a wrinkle.

The edges have the least traction and often can't sustain a wrinkle.



Shear Wrinkles vs. Tension (PLI)



On a PLI basis,
you may have
less wrinkles with
thinner webs,
since the strain is
higher.

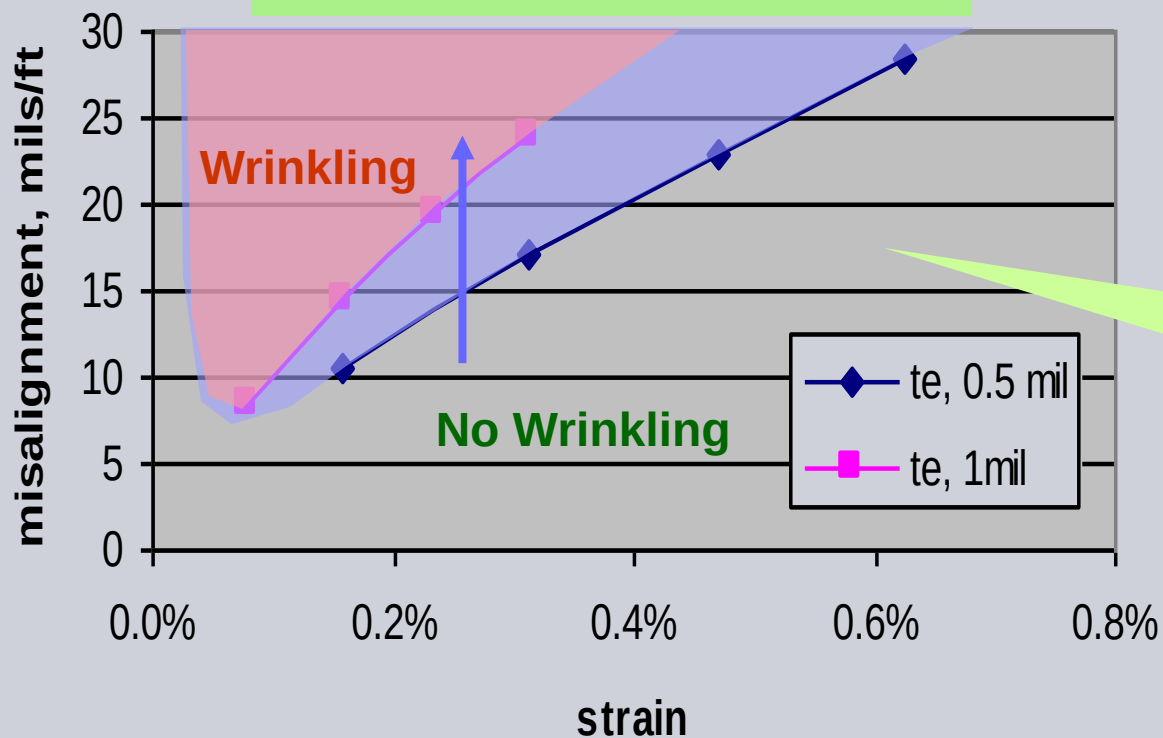
This is not
necessarily
intuitive!

w	24	in
E	640000	psi
COF	0.3	--
radius	2.5	in
L	48	in

Shear Wrinkles vs. Strain

wrinkle te vs strain

Thicker web = less wrinkles!



But viewed
versus equal
strains or
stresses, thicker
webs are better.

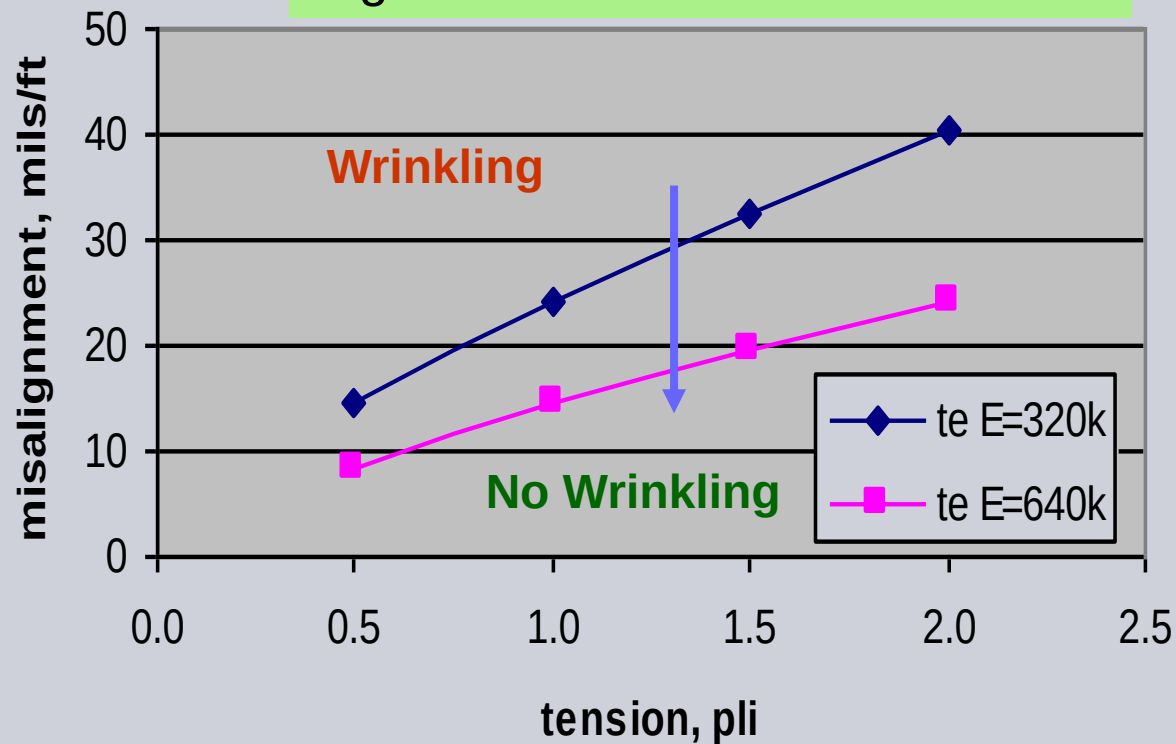
This is the exact same
data as the last graph.

w	24	in
E	640000	psi
COF	0.3	--
radius	2.5	in
L	48	in

Shear Wrinkles vs. Tension (PLI)

wrinkle te vs pli

Higher modulus = more wrinkles?



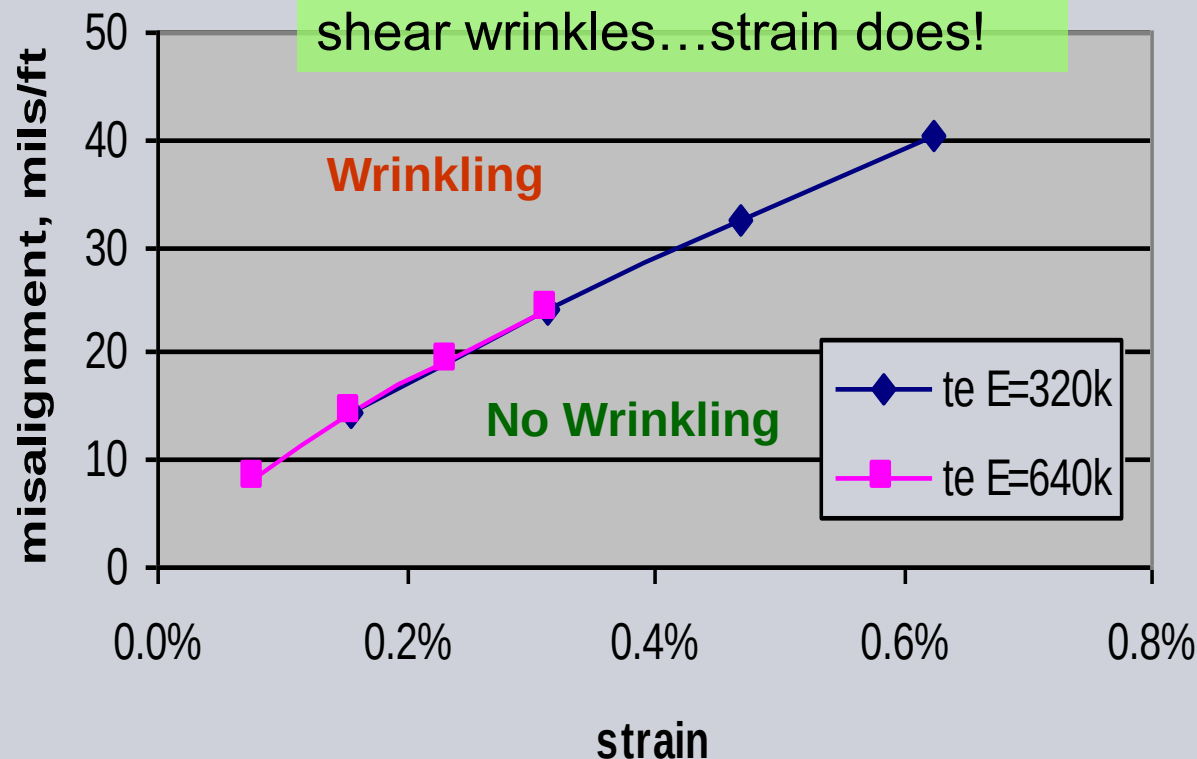
On a PLI basis,
you may have
less wrinkles with
lower modulus
web.

The web stiffness
drops, but the
sensitivity to
misalignment
drops faster.

Shear Wrinkles vs. Strain

wrinkle te vs strain

Higher modulus doesn't affect shear wrinkles...strain does!

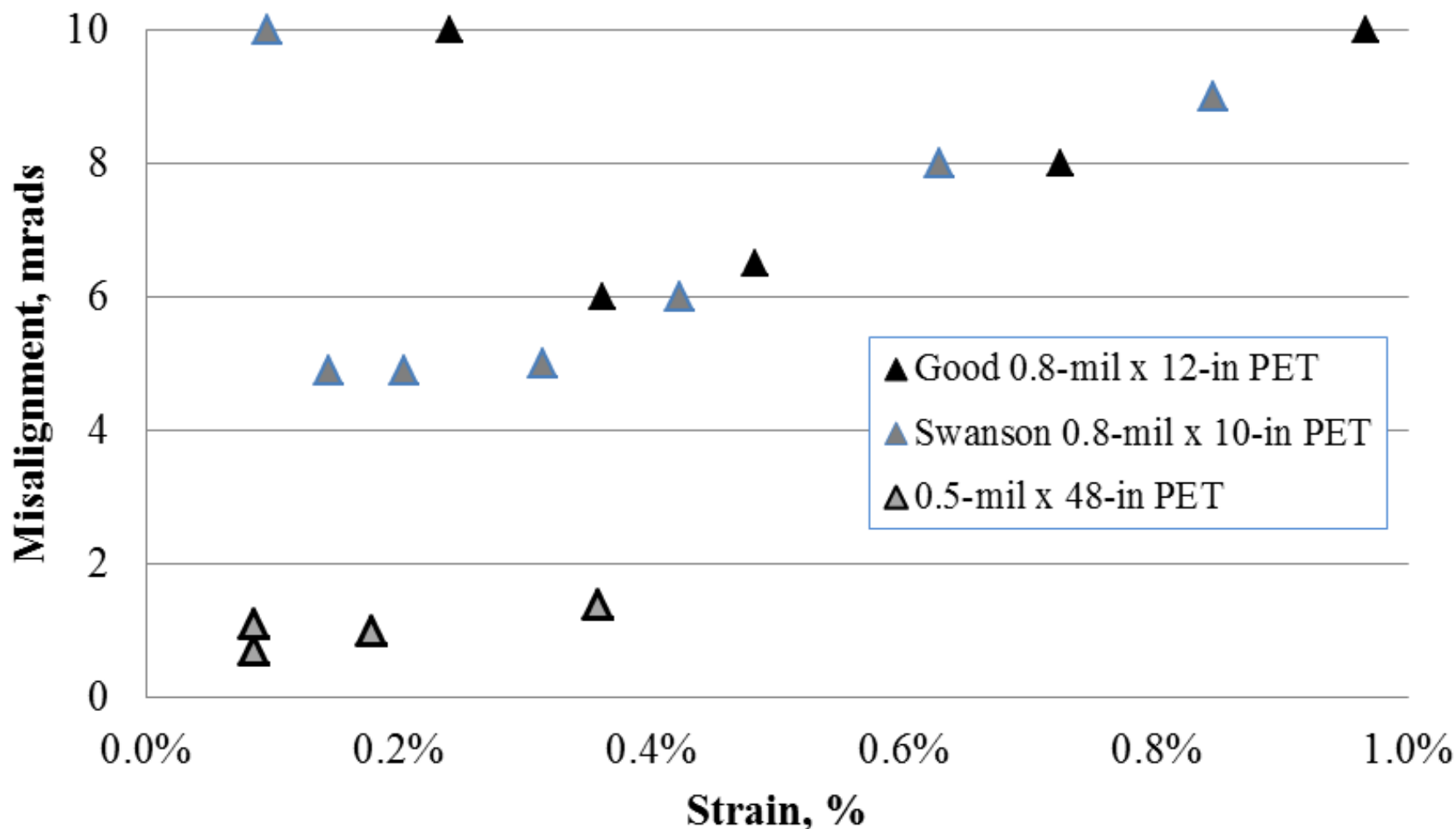


On a STRAIN basis, there is no difference between a high and low modulus web.

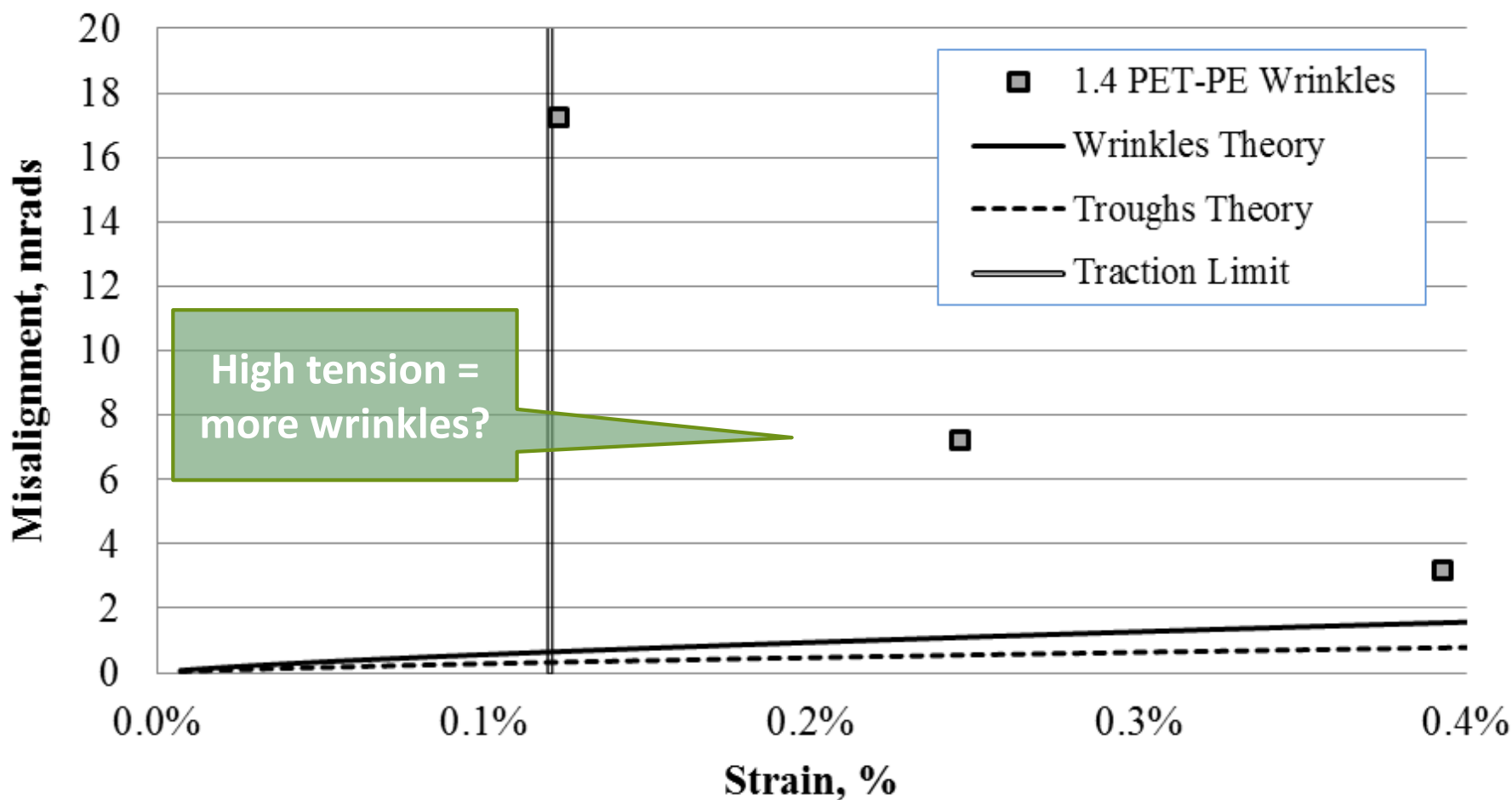
Remember, the secret to web handling is STRAIN.

This also, is not necessarily intuitive.

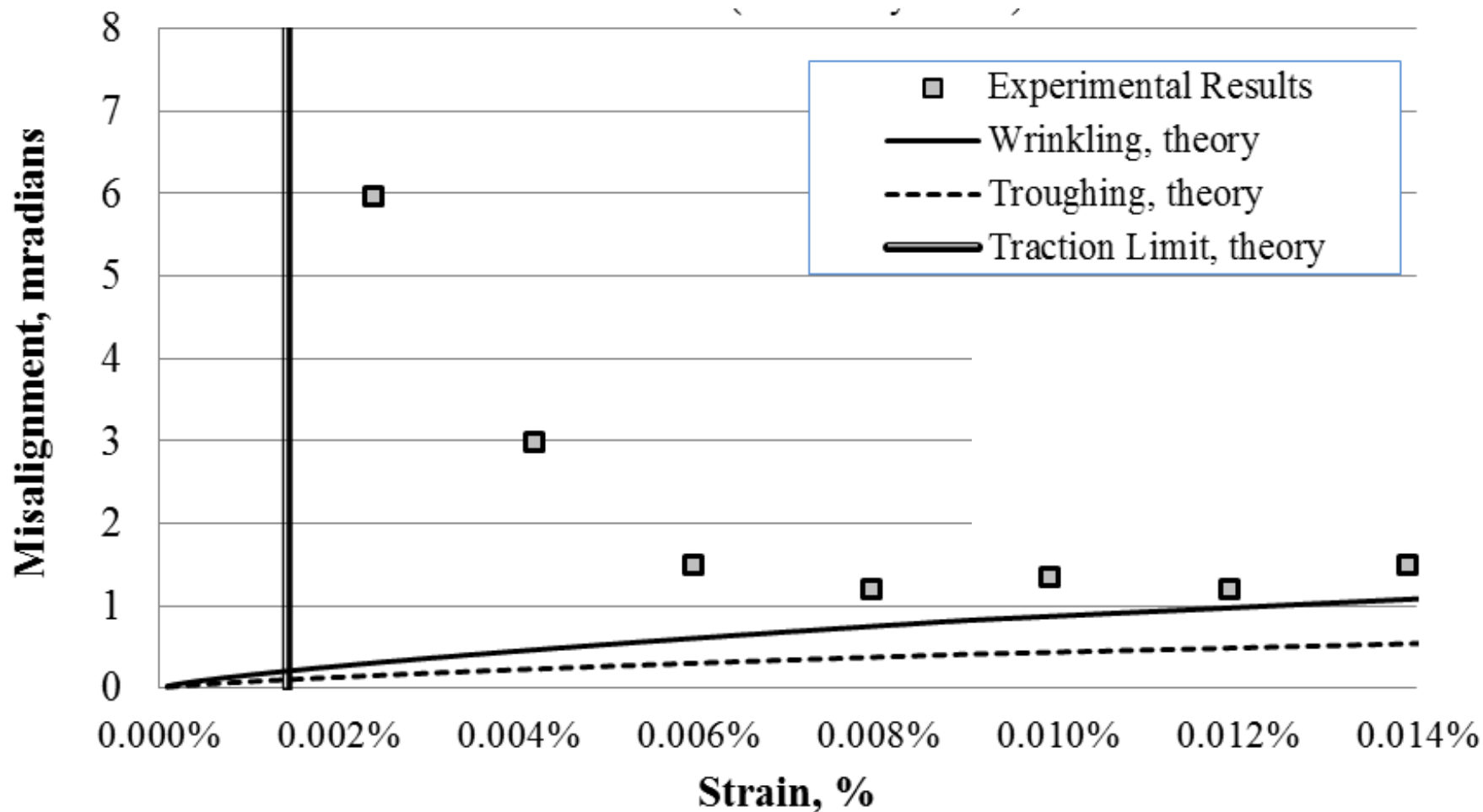
Comparison of Narrow and Wide: Polyester Shear Wrinkles



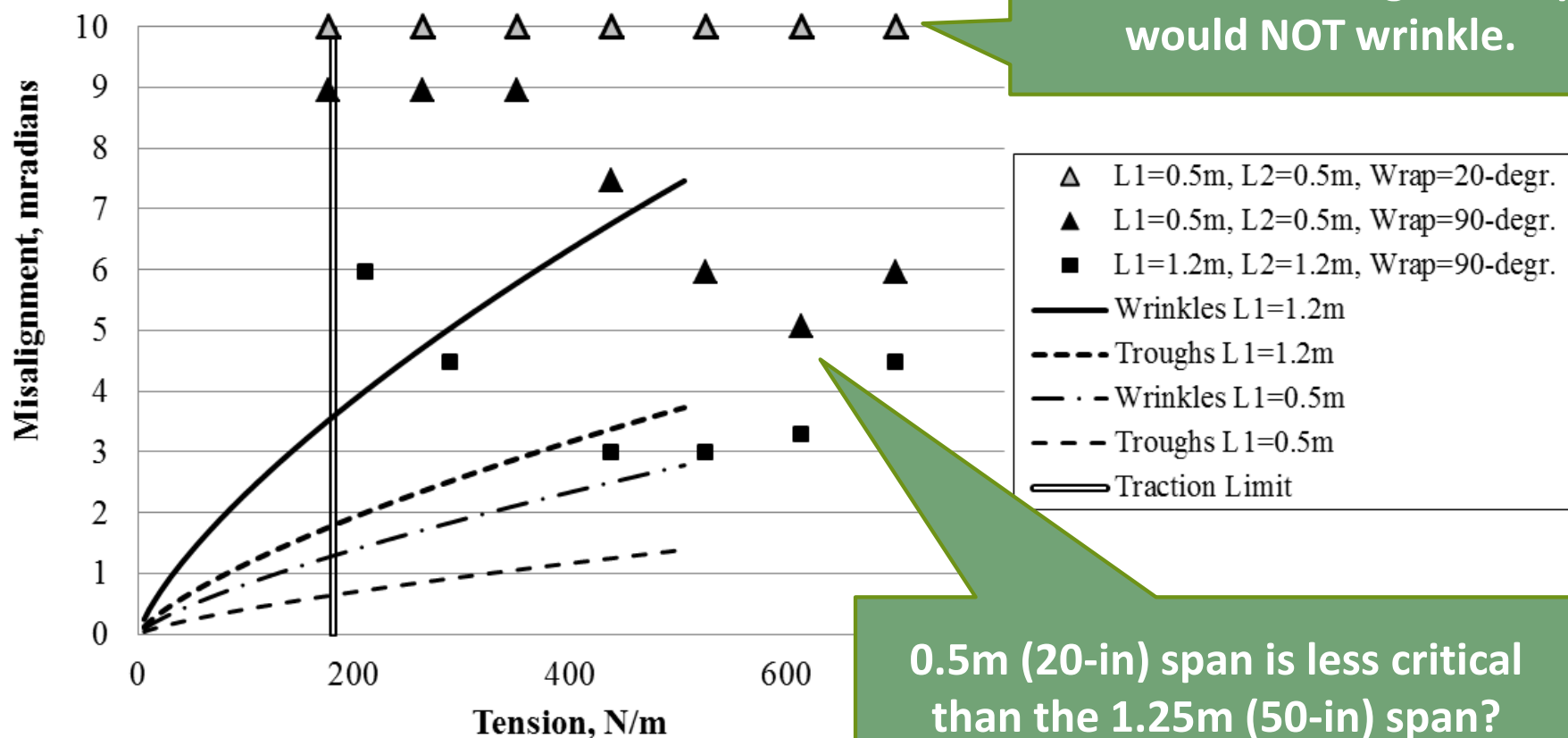
Shear Wrinkles in 35 micron: Polyethylene-Polyester Web



Shear Wrinkles in Copper Foils - Results vs. Theory



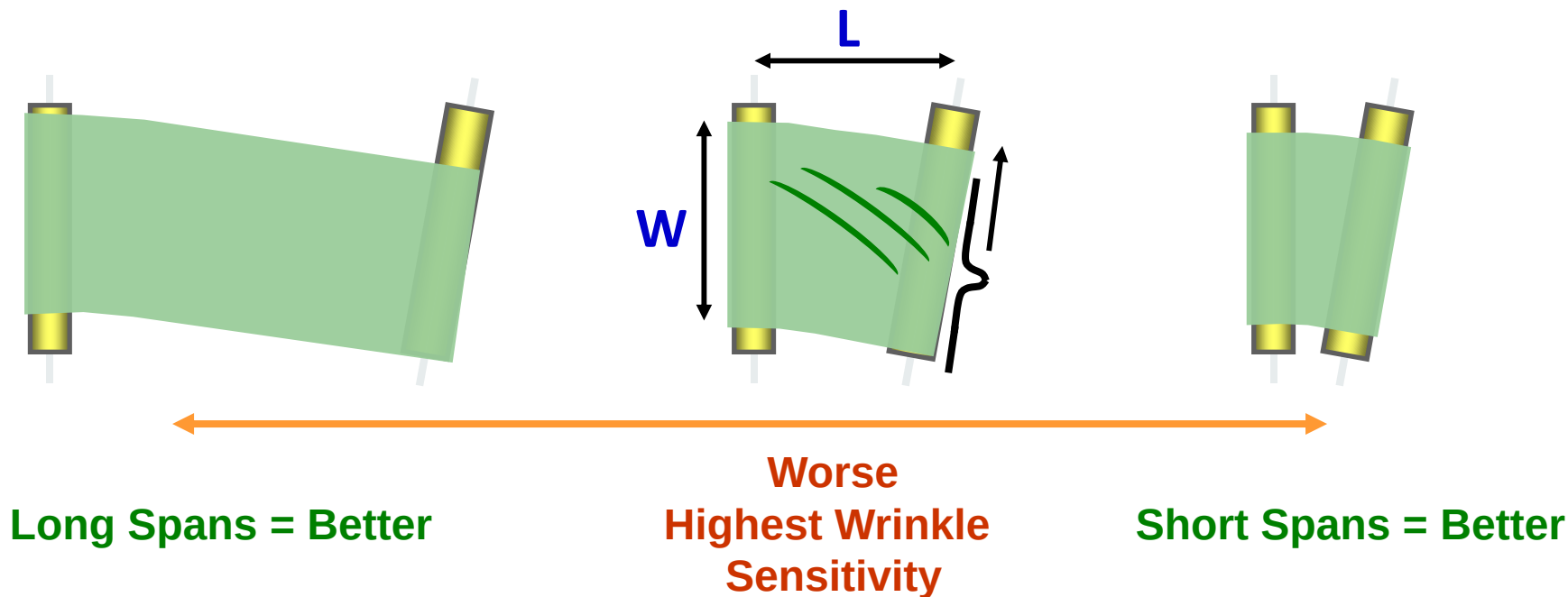
Shear Wrinkles in 75 micron Paper: Results vs. Theory



Shear Wrinkles vs. Span Length

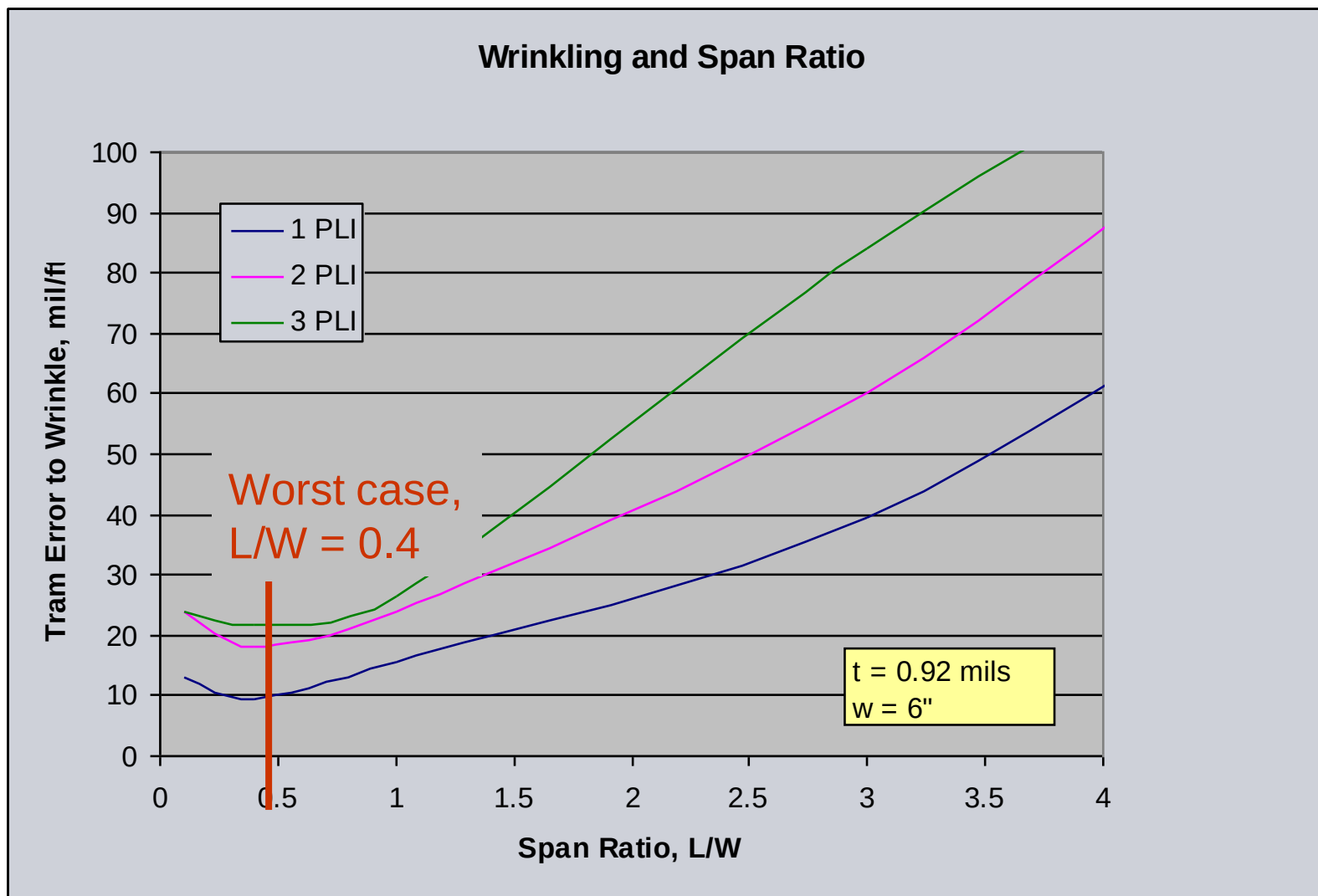
Shear Wrinkles:

The “worse case” span for shear wrinkles is $L/W = 0.4$

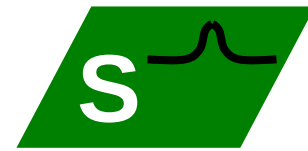


Wrinkle sensitivity decreases with longer and shorter spans.

Shear Wrinkles and Span Ratio



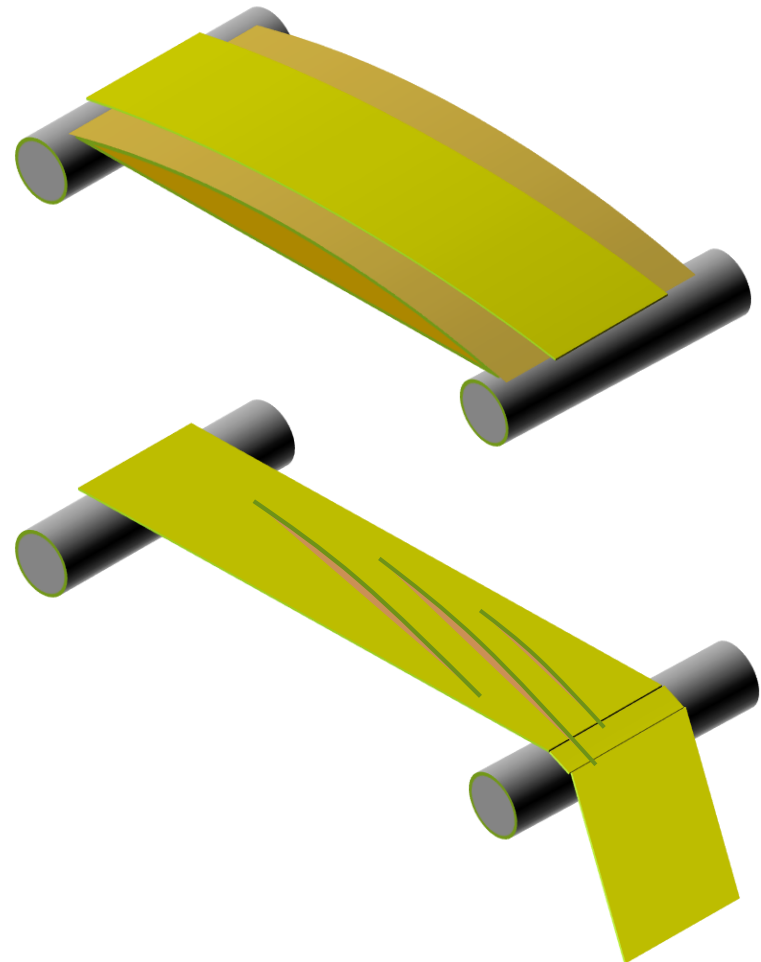
Shear Wrinkles Requirements



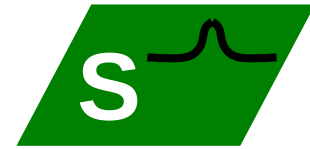
4. Freedom to Buckle

Troughs are the seeds of shear wrinkles. If the web is restricted from forming troughs by in-plane or lateral constraints (drag bars, drag plates, air flotation bars/plates, or carrier webs/belts, tenters) then shear wrinkles will not form.

Web imperfections (bagginess) and external forces (uneven air or liquid flow) promote out-of-plane shape, troughs, and wrinkles forming.



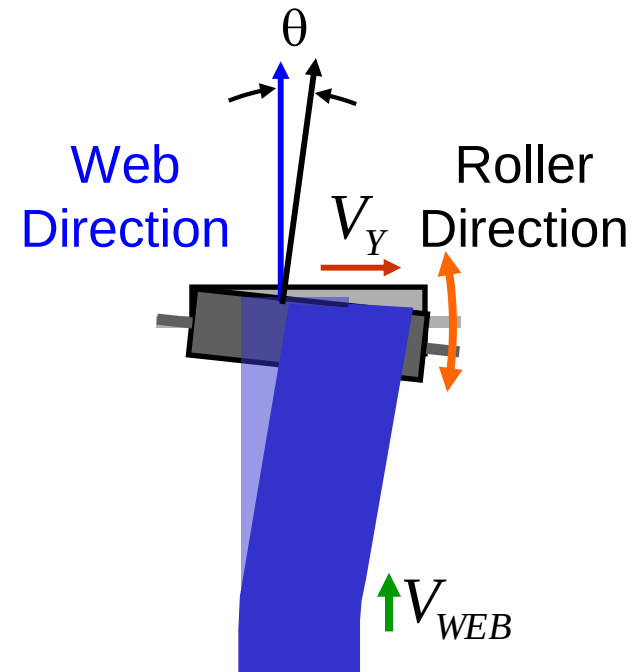
Shear Wrinkles Requirements



5. Time to Form

A web bending cause, such as a misaligned roller or diameter variation requires a finite time to redirect the web to the bent shape and position.

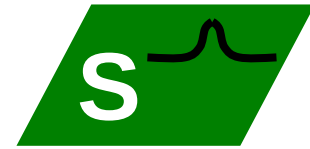
If the bending cause is transient and doesn't last long enough for the web to bend to the critical lateral displacement, then the wrinkle will not form.



$$V_Y = V_{WEB} \sin \theta$$

θ = angle error
between web and
roller directions.

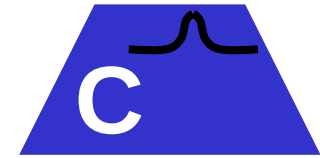
Shear Wrinkles Requirements



1. Potential for Critical Bending
2. Traction to Bend
3. Traction to Wrinkle
4. Freedom to Buckle
5. Time to Form

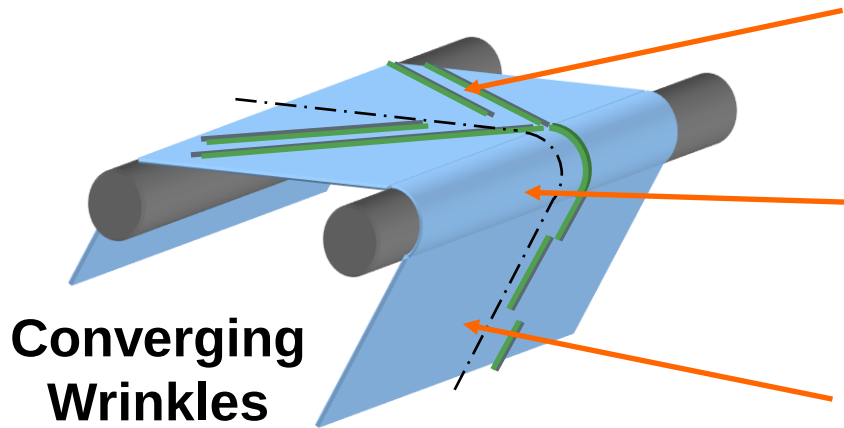
Each of these wrinkle requirements provide clues to wrinkle cures.

Converging Wrinkles



Mechanism:

Both sides of web tracking inward.



Troughs:
Symmetrical and angled.

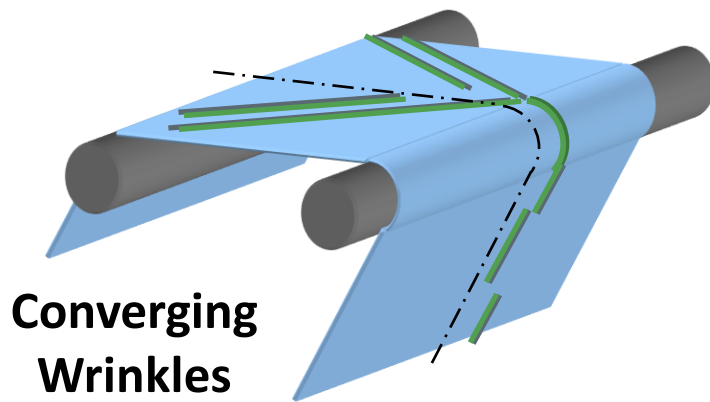
Wrinkles:
Form in one position.

Creases:
Generally machine direction with small amount of wander.

Converging Wrinkles

Mechanism:

Web pulled laterally to one side.

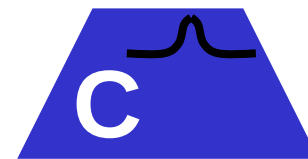


Eliminate any one of these and wrinkles will not form.

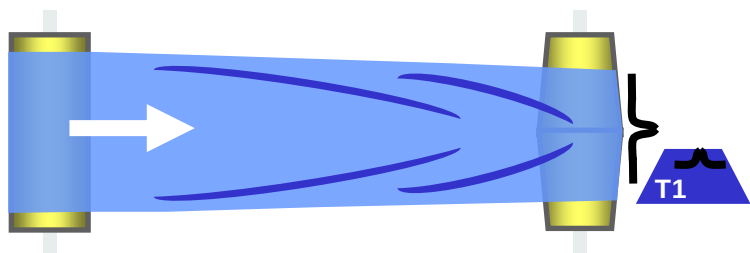
Prerequisites:

1. Mechanism to induce tracking to center.
2. Tractional force to overcome span's bending stiffness.
3. Tractional force on roller to exceed buckling criteria.

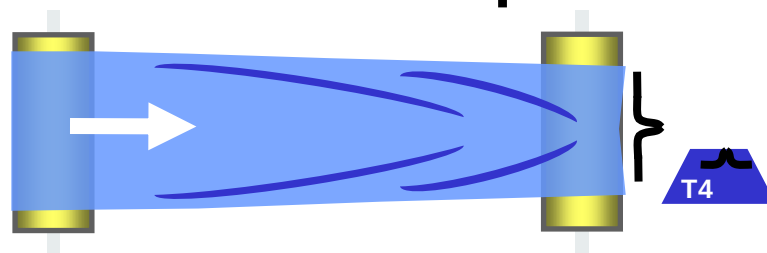
Converging Wrinkle Causes



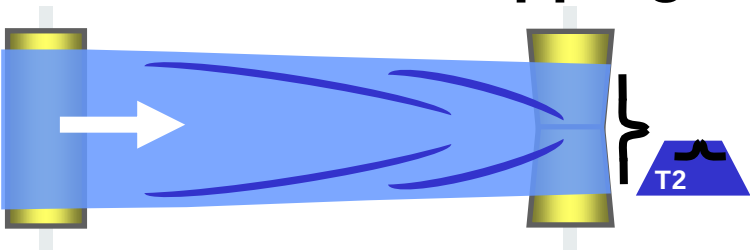
Convex Rollers in Traction



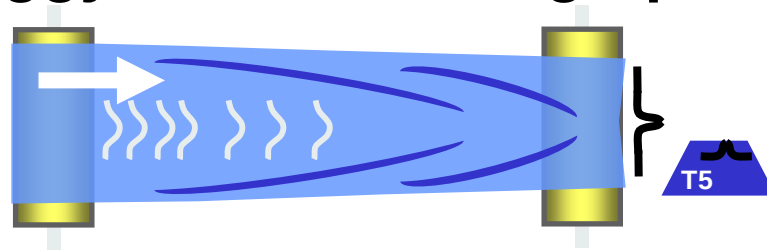
Pre-Roller Web Expansion



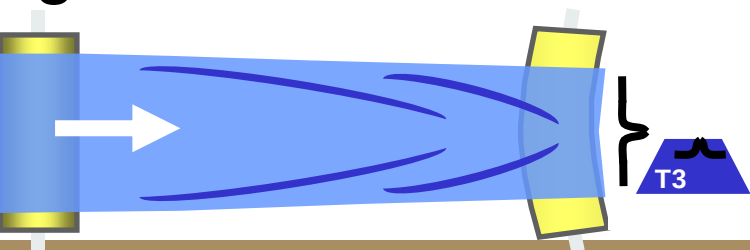
Concave Rollers Slipping



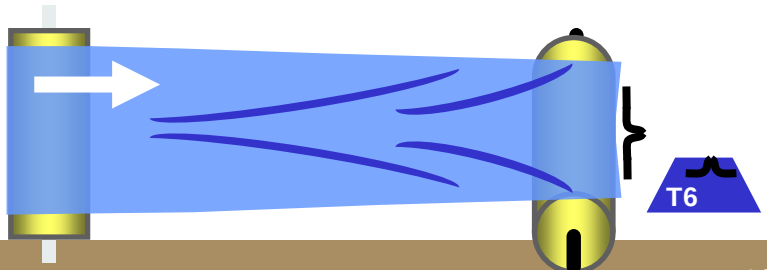
Baggy Center in Long Span



Negative Bowed Roller

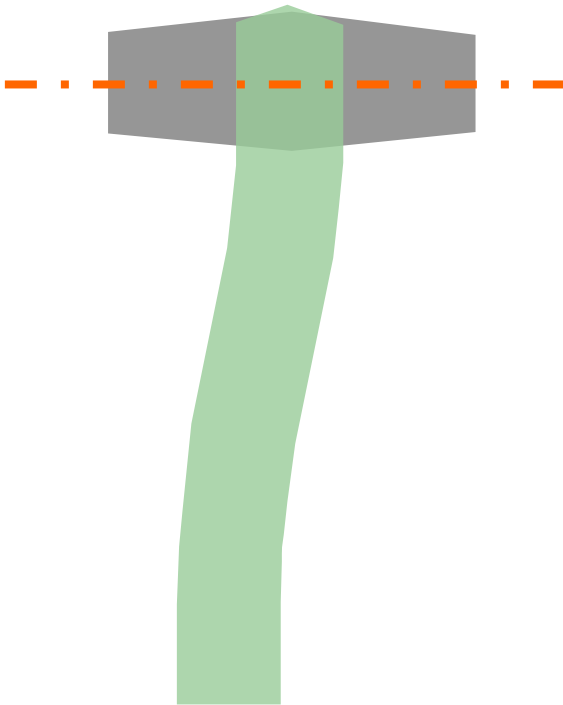


Twist

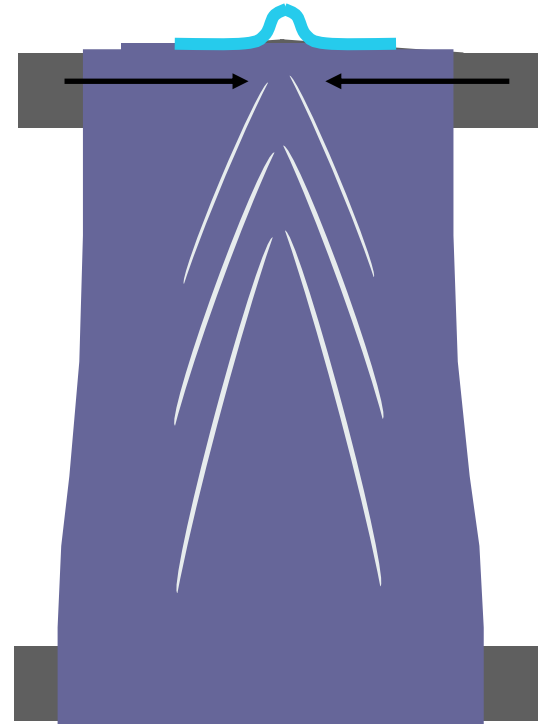


Crowned Rollers: Centering vs Wrinkling

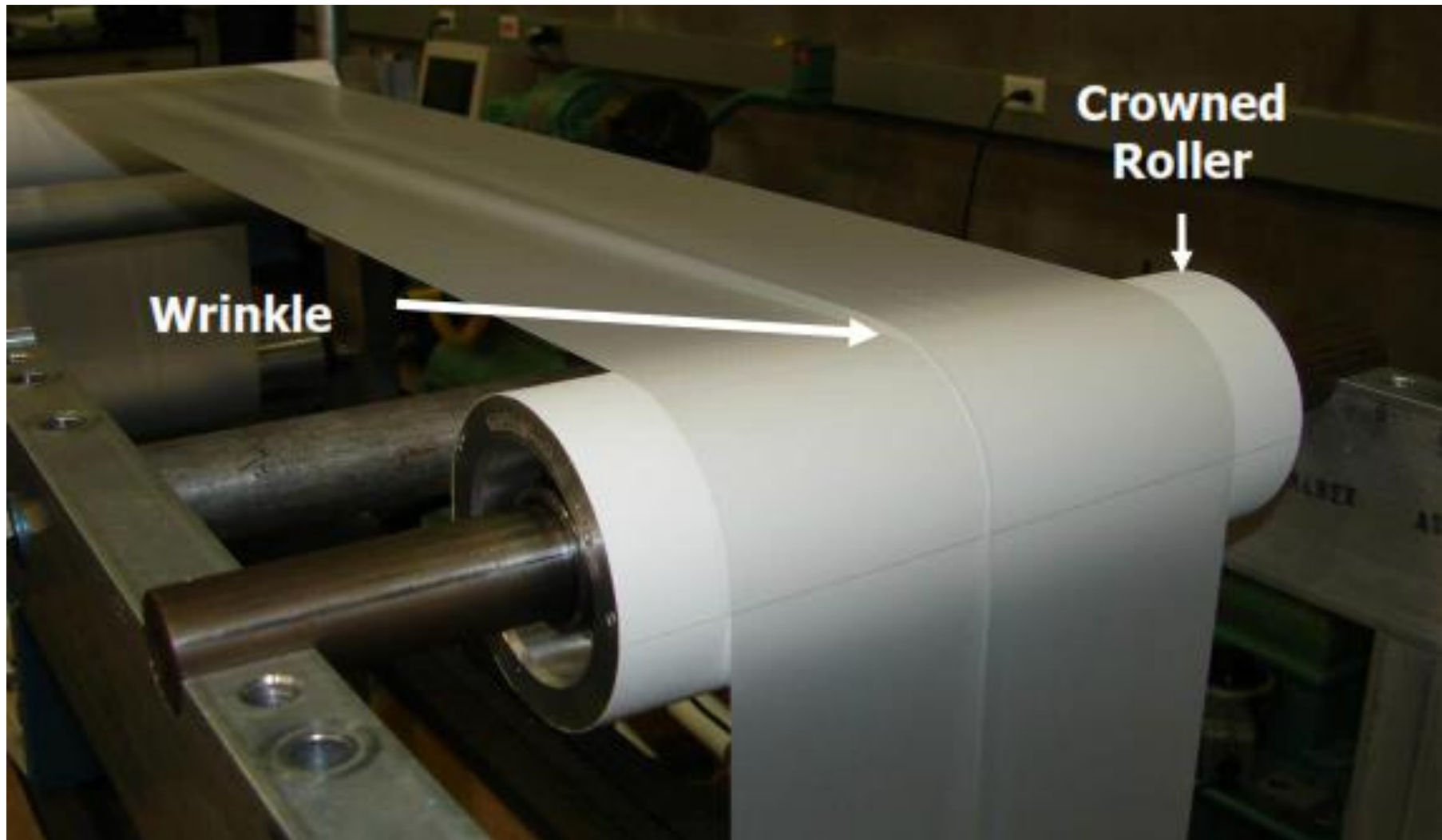
A crowned roller is an effective passive centerline web guide for narrow, stiff webs.



A crowned roller is an undesired wrinkle generator for wide, thin webs.



Converging Wrinkle on Crowned Roller



Single Span Web Buckling Due to Roller Imperfection in Web Process Machinery, Biesel, J. A., Oklahoma State University, PhD Thesis 2006

Pre-Roller Expansion

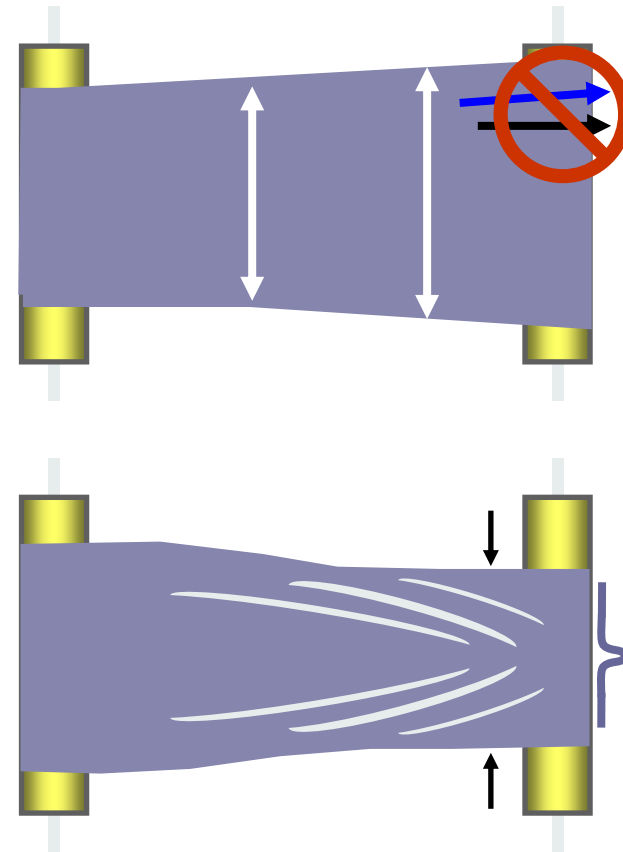


Causes:

Tension Change, Thermal, Viscoelastic Recovery, Moisture (paper)

A web that laterally expands in the upstream span will not have a “parallel” entry.

The parallel entry rule will displace the non-parallel web laterally, each edge to the center, creating wrinkles.

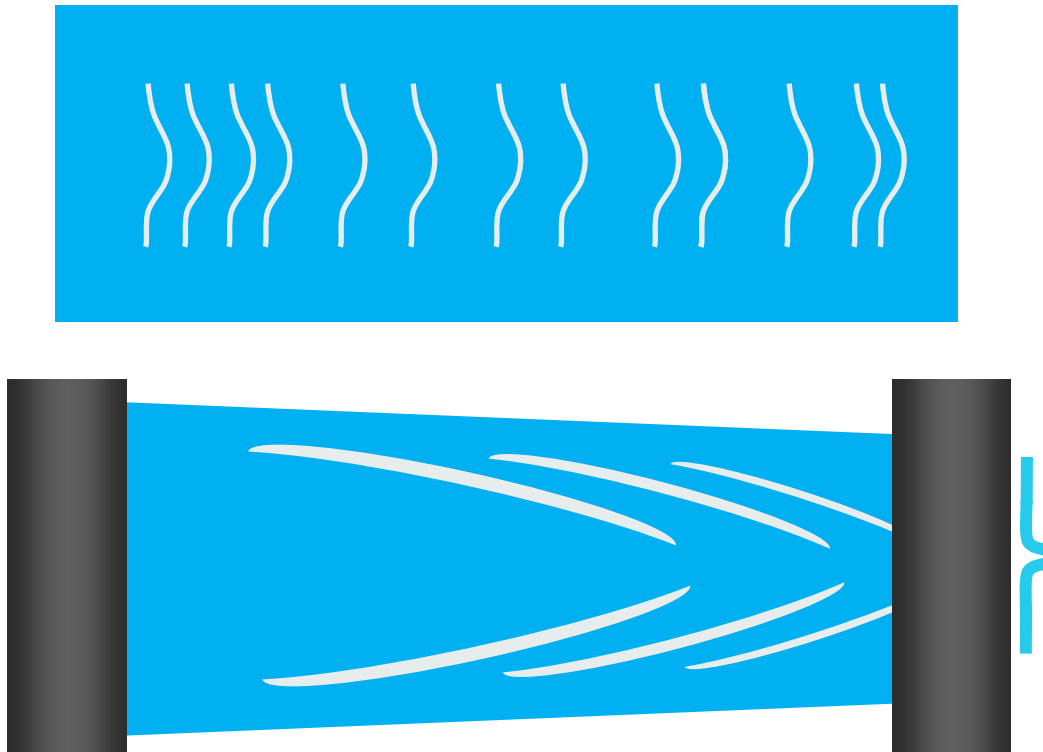


Converging: Baggy Center



Web Non-Uniformity

A “baggy” center web will track towards the long lanes



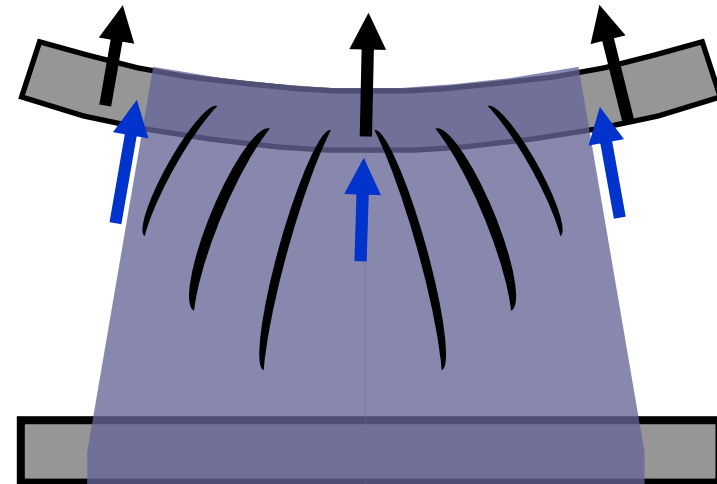
Negative Bowed Roller



Negative bowed rollers have a uniform diameter, but have a curved axis of rotation.

The curved axis creates varying parallel entry vectors across the roller's face.

The center lanes of web track straight, but the edges will track towards the center.

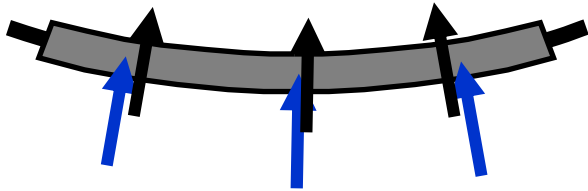


The result is a gathered, wrinkling web.

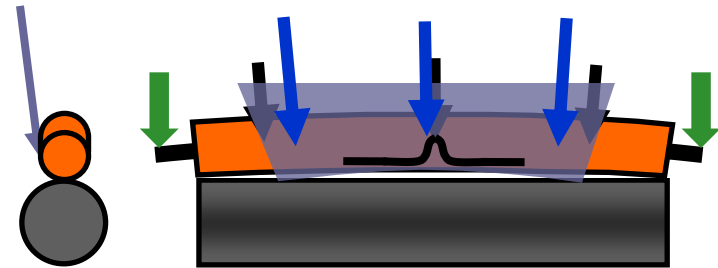
Negative Bowed Roller



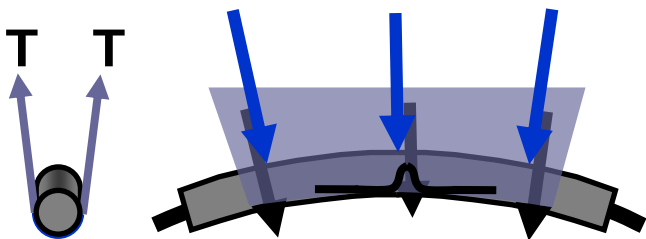
Gravity



End-Loaded Nipped Rollers



Web Tension



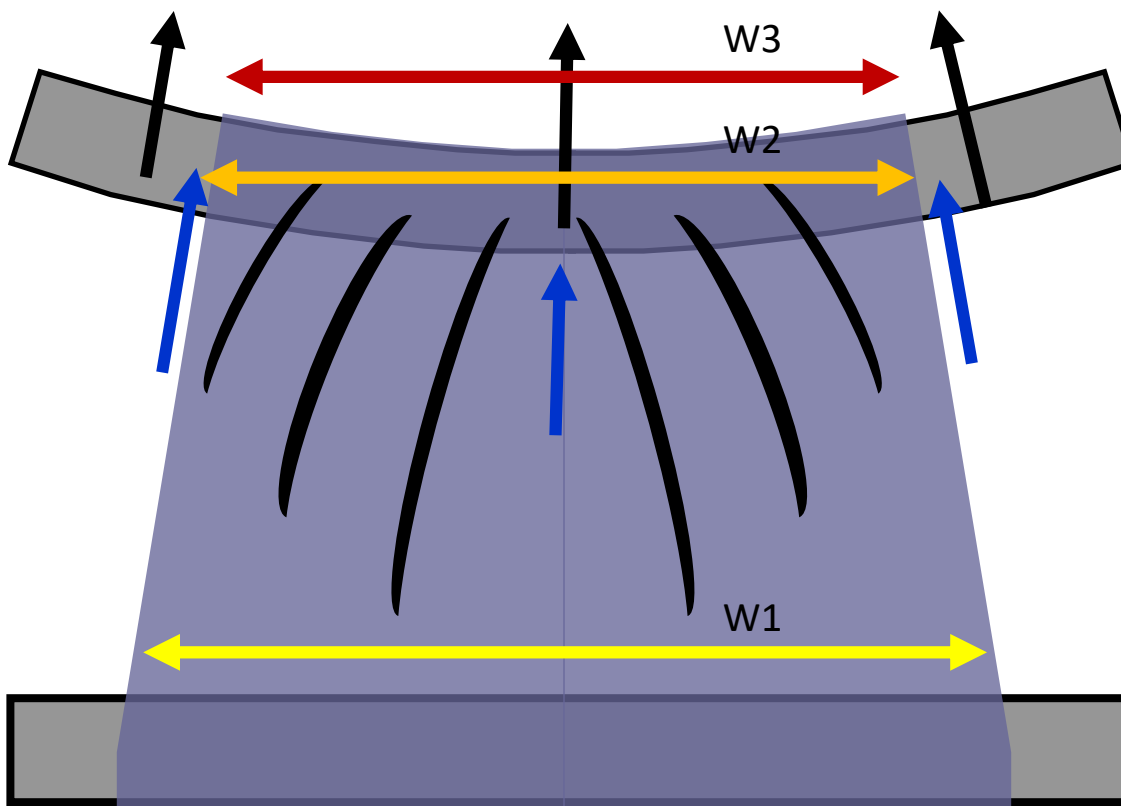
Reversed Bowed Rollers



-  Local roller rotation vector
-  Local parallel entry web tracking path

Lateral Strains Induced by Deflection

$$\epsilon_{y,critical} = -\frac{0.605t}{R} \quad -\epsilon_{d,c} - \epsilon_{s,c} > \frac{0.605t}{R}$$



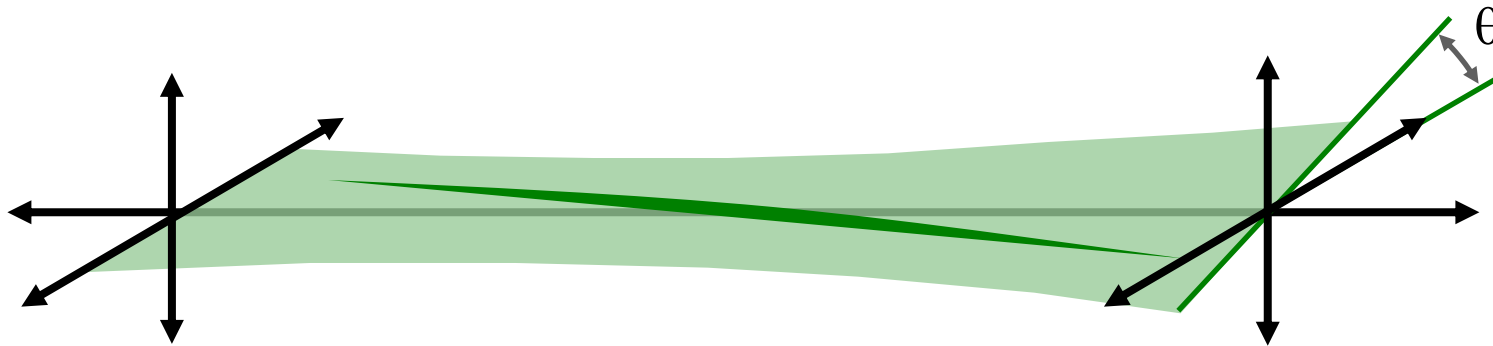
ϵ_d
On-Roller
CD Strain

ϵ_s
Pre-Roller CD
Strain



Stresses from Twisting

Web twisting is stable and non-deforming when applied to narrow webs and long spans.



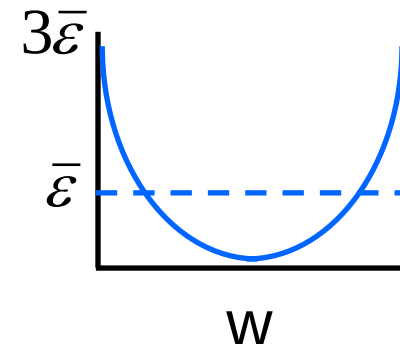
Assuming a parabolic tension distribution, the center web tension is zero at the minimum twisting span, L_{min} , or maximum twisting angle, θ_{crit} .

$$L_{MIN} = w\theta \sqrt{\frac{twE}{24F_T}} = \frac{0.2w\theta}{\varepsilon^{0.5}}$$

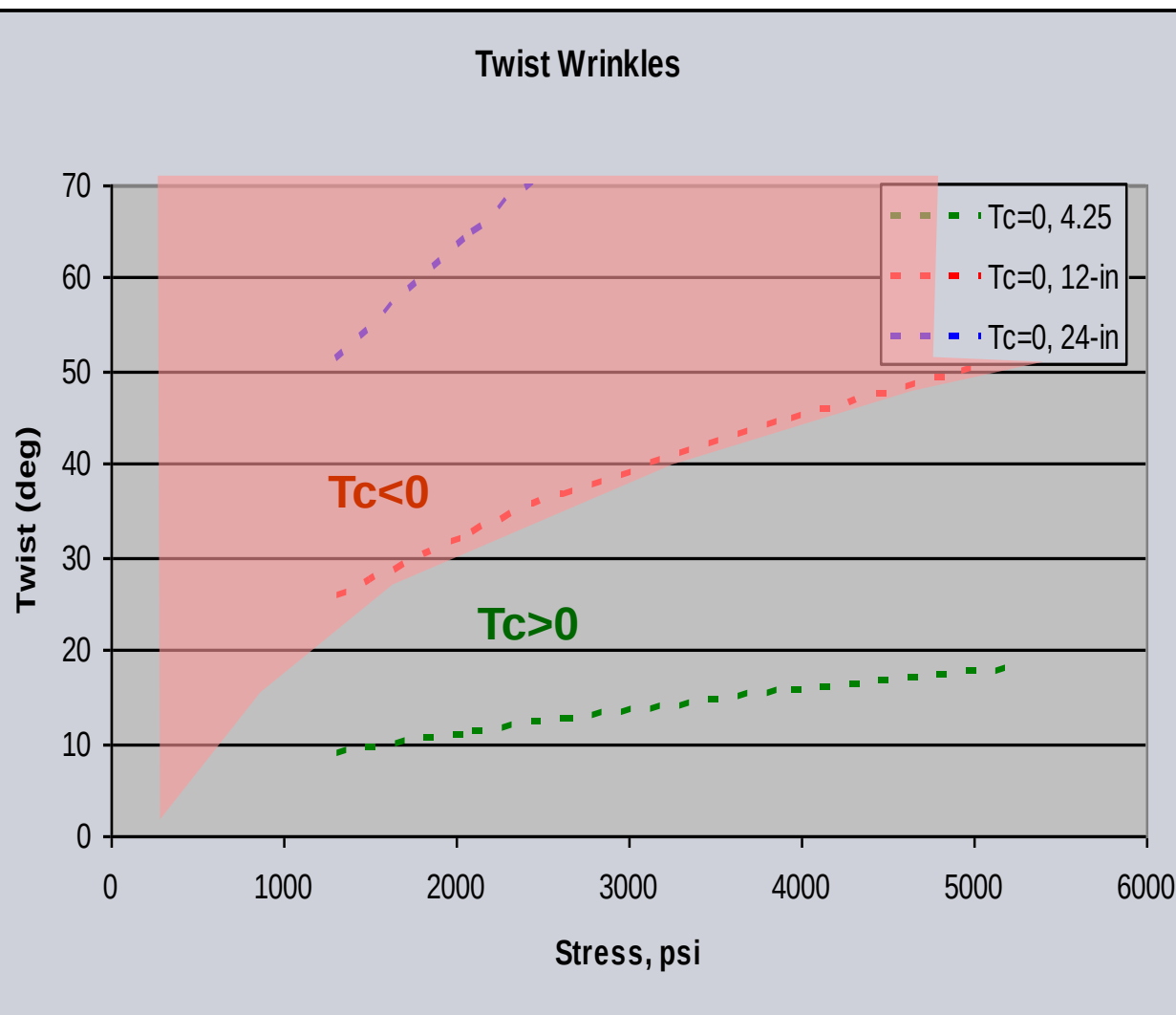
$$\theta_{crit} = \frac{L\varepsilon^{0.5}}{0.2w} = \frac{(40)\sqrt{0.002}}{(0.2)(20)}$$

$$= 0.44 \text{ radians}$$

$$= 25 \text{ degrees}$$



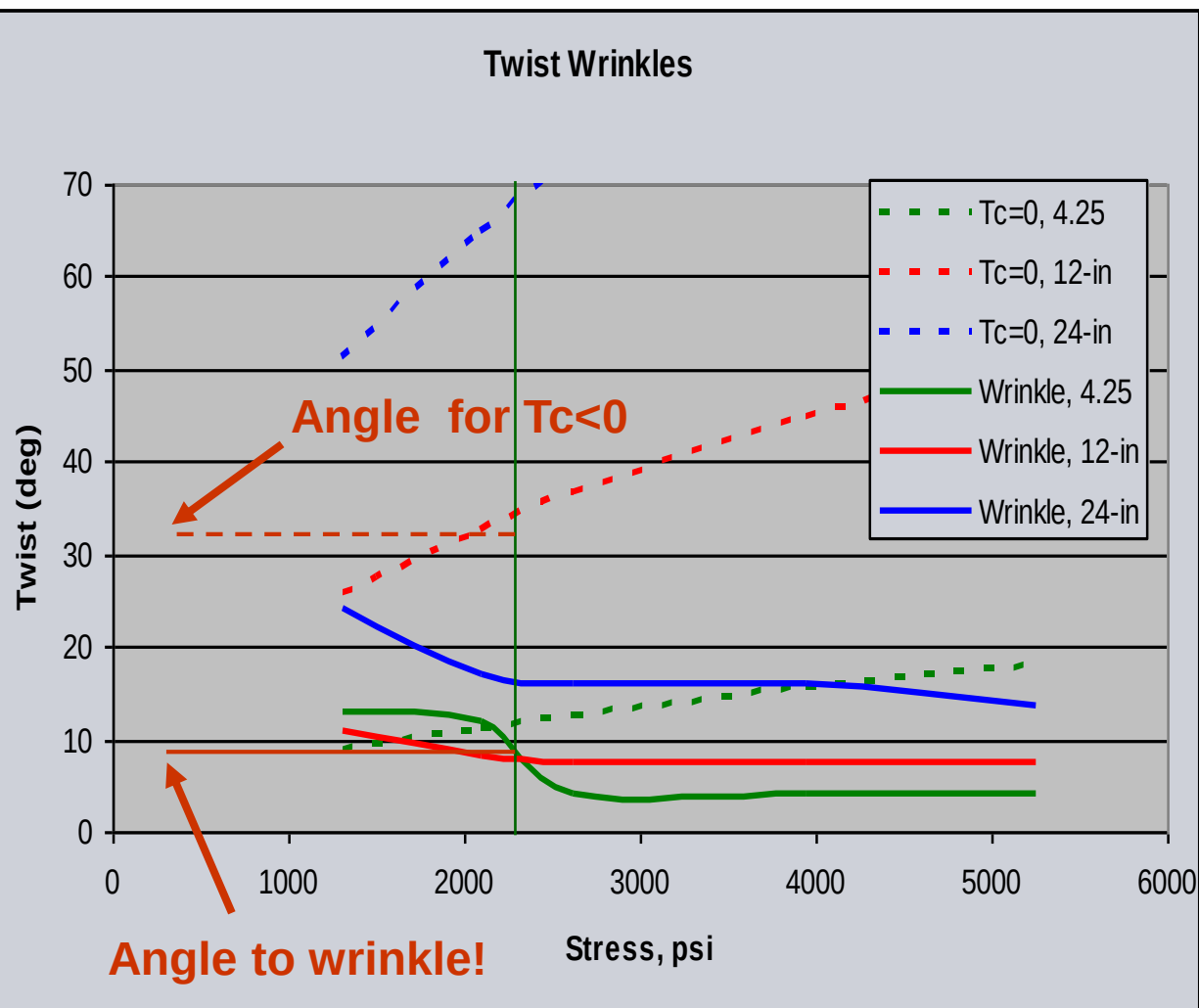
Shear Wrinkles vs. Twist



Twisting spans have traditionally assumed a parabolic tension profile and specified to avoid slackness at the web's center ($T_{\text{center}} > 0$).

t	0.92	mils
w	15	in
E	640000	psi
COF	0.3	--
radius	2.5	in

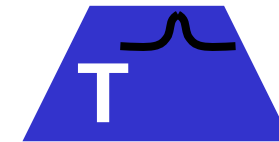
Shear Wrinkles vs. Twist



In wrinkle experiments, the slack web center ($T_c > 0$) criteria is too liberal.

Wrinkles form well before $T_c < 0$.

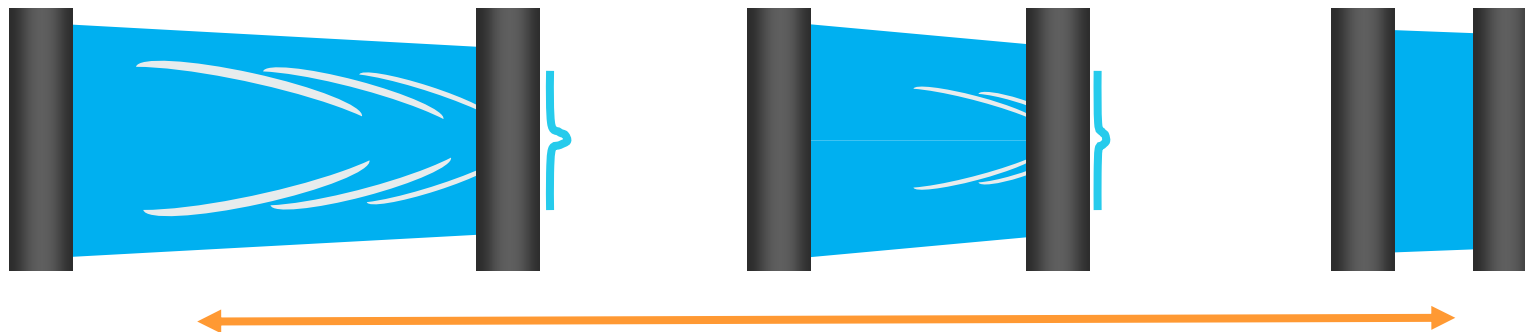
t	0.92	mils
w	15	in
E	640000	psi
COF	0.3	--
radius	2.5	in



Converging Wrinkles vs. Span Length

Converging Wrinkles:

Since converging wrinkles are dependent on flexible lateral bending, they are promoted by long spans.



Long Spans = Worst
More Wrinkles

Short Spans = Best
Less Wrinkles

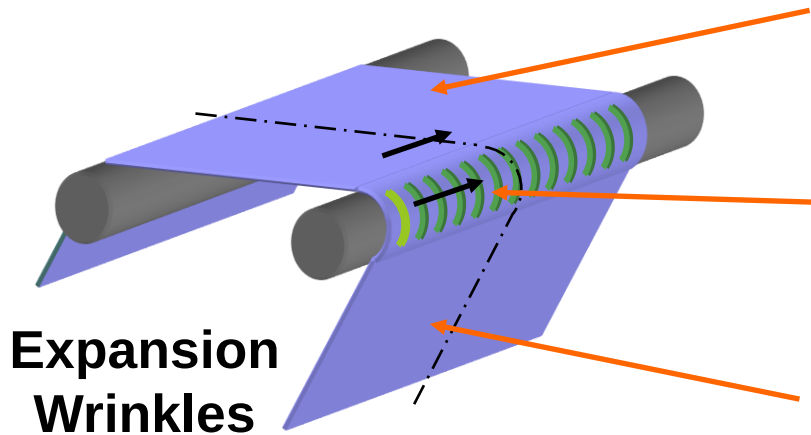
Short spans rarely have converging wrinkles.

Expansion Wrinkles



Mechanism:

Traction constrains expanding web during roller contact.



Troughs:
Minimal or no troughs.

Wrinkles:
Uniformly spaced, but no wrinkles near the edges.

Creases:
May not cause creases.

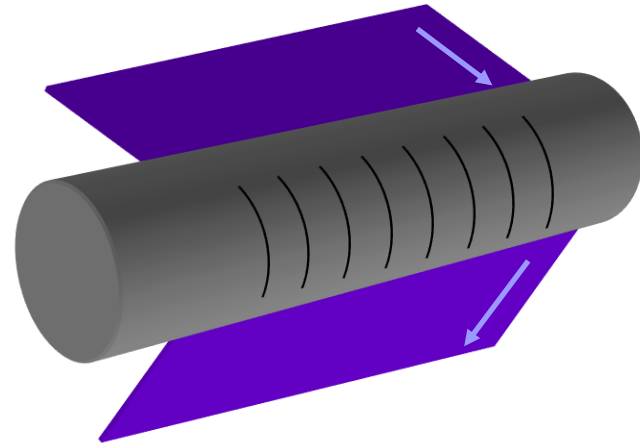
Expansion Wrinkle Causes



Web Width Expansion: On-Roller

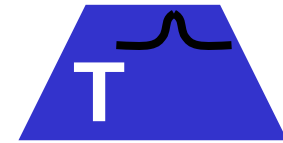
Thermal, Tension Change, Moisture (paper)

Web width expands while constrained by roller friction...on-roller expansion wrinkles form.

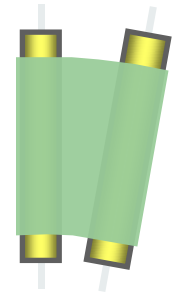
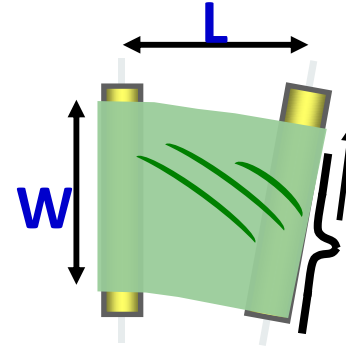
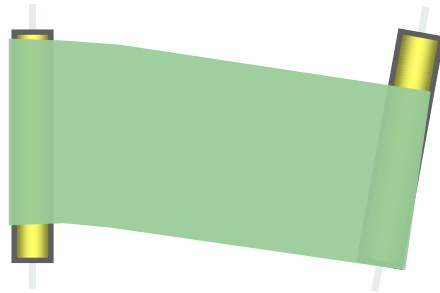


Since expansion takes time, these wrinkles are more common at slow or zero speed.

Wrinkles vs. Span Ratio



**Shear
Wrinkles**



Long Spans = Better

**Worse
Highest Wrinkle
Sensitivity**

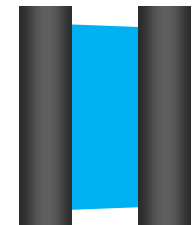
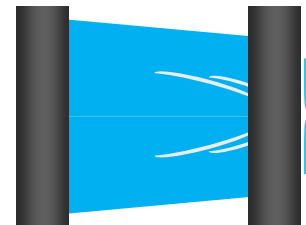
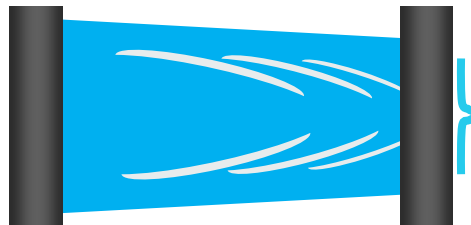
Short Spans = Better

What span ratio
is best?

**Long Spans = Worst
More Wrinkles**

**Short Spans = Best
Less Wrinkles**

**Converging
Wrinkles**

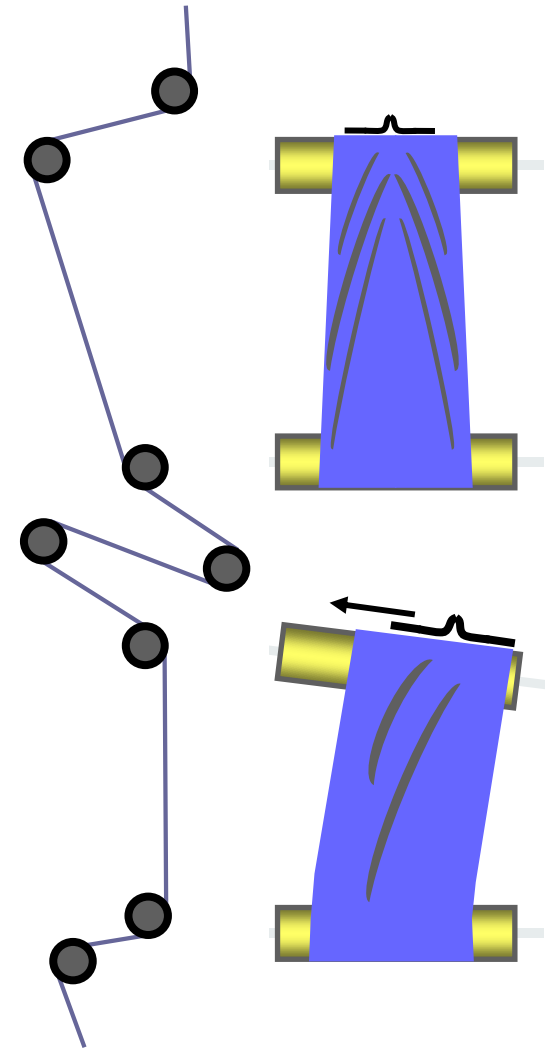


Magnifying Wrinkles

Making a defect worse can be the first step to making it better.

Running a thin, high modulus web will reveal a web line's wrinkle sensitivity.

This simple test will show misalignments and diameter variations better and faster than indirect precision measurements.



Wrinkle Remedies

Immediate

1. Change the input roll
2. Change the local process tension
3. Check local alignment or diameter variations
4. Try tape collars at both web edges
5. Modify the friction coefficient
6. Stop the roller
7. Try a new web path

Long-Term

1. Understand and eliminate the source
2. Adjust span lengths
3. Adjust wrap angles
4. Try gap or nip rollers at winding
5. Try an adjustable roller
6. Install spreader or anti-wrinkle rollers



5.3 Spreading

Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

5.3 Learning Objectives

- ▶ Where are spreading rollers used?
- ▶ What is a spreader roller? What is an anti-wrinkle roller?
- ▶ How much spreading is required to prevent wrinkling?
- ▶ What is the best way to spread slit strands?
- ▶ How do the following spreading / anti-wrinkle rollers work:
 - ▶ Bowed roller? Expanding surface roller? Edge nips?
 - ▶ Flex expander? Concave roller? Hyper crowned roller?
- ▶ Do spiraling tape, grooves, or ridges spread a web?
- ▶ What is the best anti-wrinkle roller?

Guidelines for Spreading

Spreaders are used to:

- 1) Prevent wrinkling at the downstream end of a span with a persistent wrinkle cause (e.g. heating foil, moisturizing paper)
- 2) Present a laterally taut web to a critical process that may create wrinkles from non-uniform tensioning or misalignment (e.g. nips, slitting, winding)

Winding Roll

Air

Flotation

Oven

Slitting

Unwinding

Roll

Nipped Process

Guidelines for Spreading

Spreading occurs mostly upstream of the spreader roller.

- Some spreaders have additional on-roller spreading, but this is a much lesser effect.
- On-roller spreading can lead to roller wear, as the roller surface stretches below the laterally taut web.

Most spreaders are more effective with long entry spans.

- Longer entry spans require less force to bend laterally.
- Longer entry spans will shift more for a given spreading angle.

Guidelines for Spreading

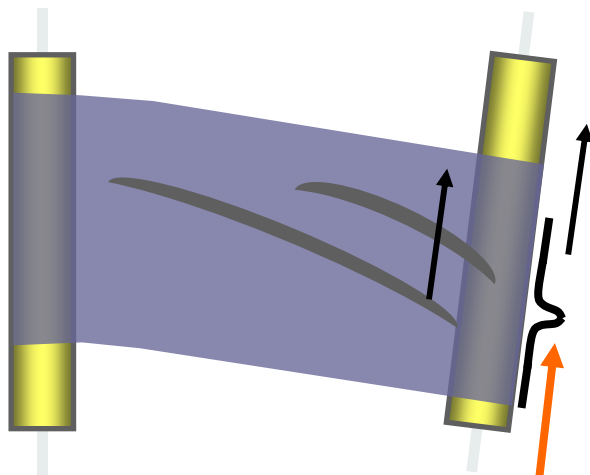
The spreading benefit is lost soon after the spreader roller. For spreading ahead of wrinkle cause (nips, slitting, winding), a short span reduces ability of wrinkle source to gather the web.

Spreading and anti-wrinkle rollers should be used only after cylindrical rollers with optimized traction, wrap angle, and span lengths have proven ineffective.

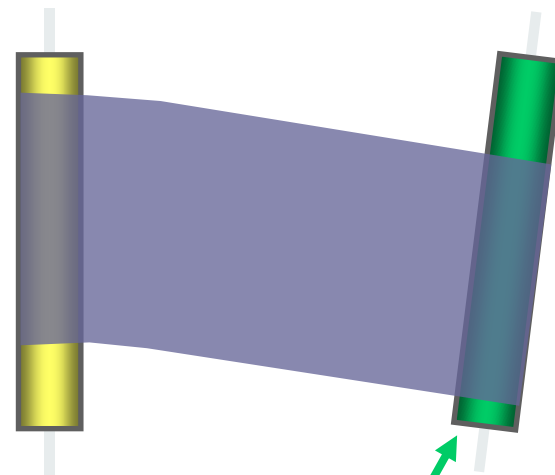
Definition: Anti-Wrinkle Roller

Anti-Wrinkle Roller:

A roller of special design that eliminates a wrinkle that would otherwise form.



Shear wrinkle from
misalignment

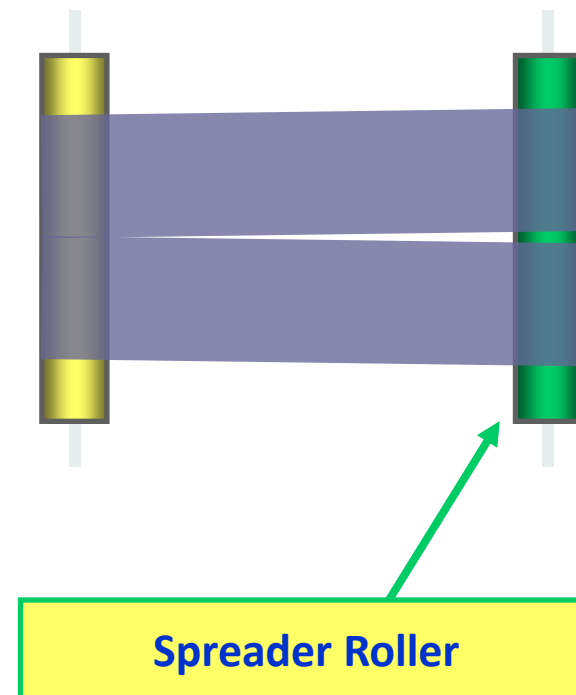
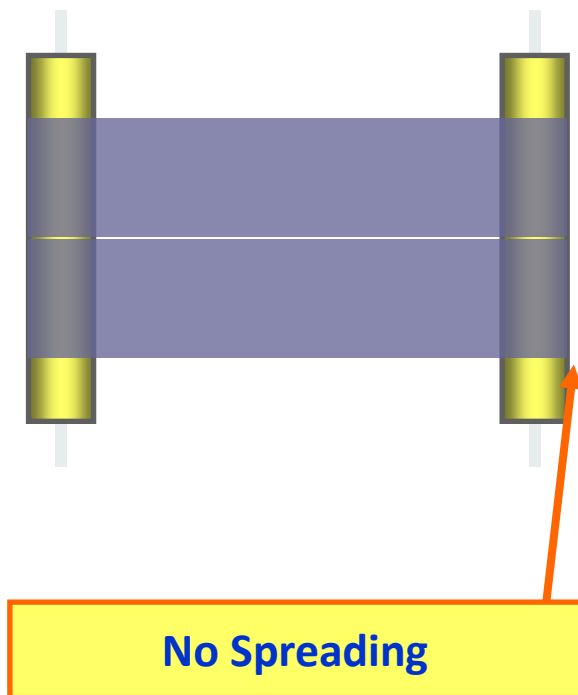


Anti-Wrinkle Roller

Definition: Spreader Roller

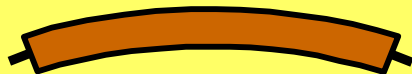
Spreader Roller:

A roller of special design that increases the web's width...or creates a gap between slit strands.



Spreader Options

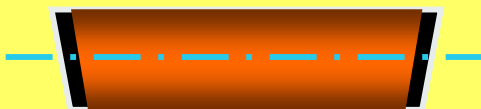
Bowed Roller



Skewed Edge Nips



Flat Expander Roller



Flex Expander Roller



Tape Collared Roller



Concave Roller



Also known as:
Reverse crowned roller

Curved Drag Bar



Hyper-Crowned Roller

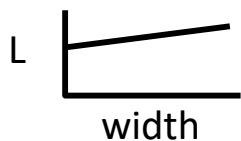


**Old Bowed Roller
used as Drag Bar**

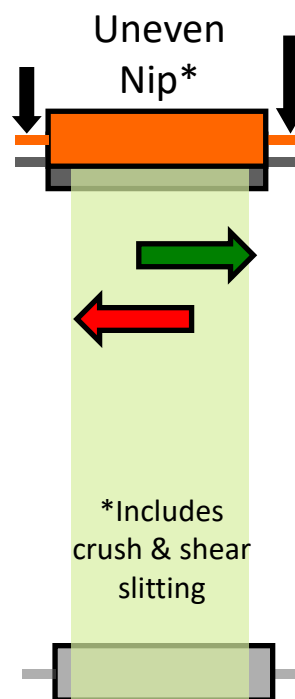
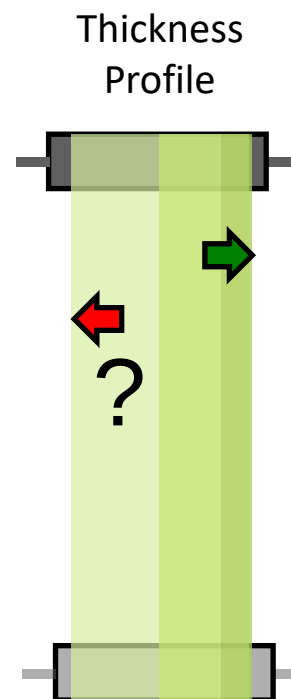
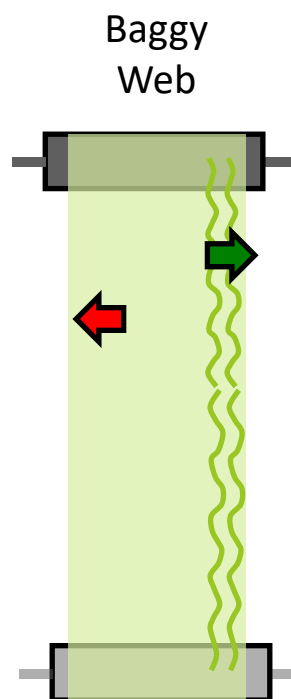
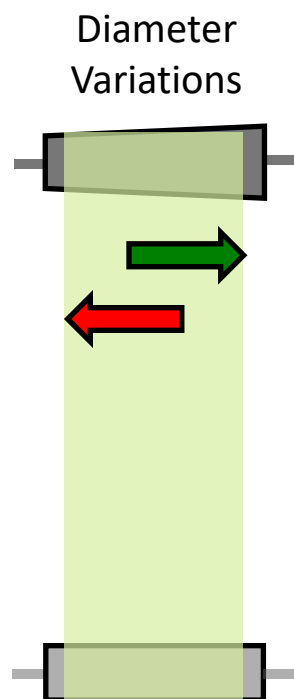
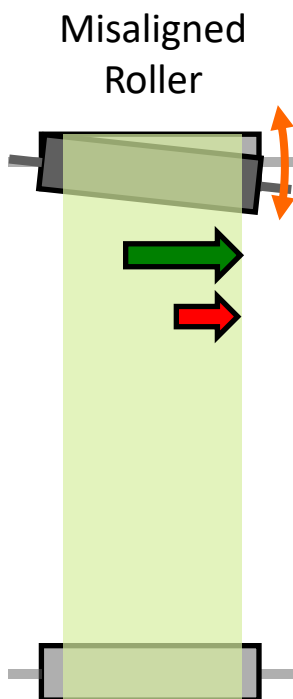
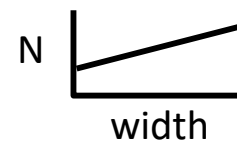
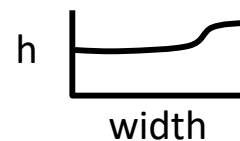
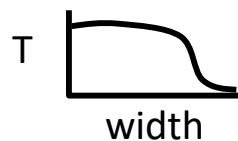


Summary - Five Tracking Rules

Top causes of web CD shifting.



$$\theta_{WEB} \neq \theta_{ROLLER}$$



*Includes crush & shear slitting

Good or Poor Traction

What is...?

Slip?

PER?


**Good
Traction
Tracking**

**Web bonds to
roller on
contact and
does not slip.**

**Frictional force
is sufficient to
enforce PER.**


**Poor
Traction
Tracking**

**Web forms
poor bond to
roller and may
slip.**

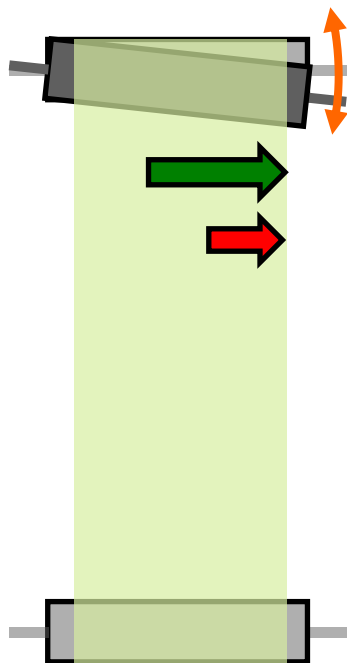
**Frictional force
is too low to
enforce PER.**

Summary - Five Tracking Rules

Rollers have **4** ways to shift a web.

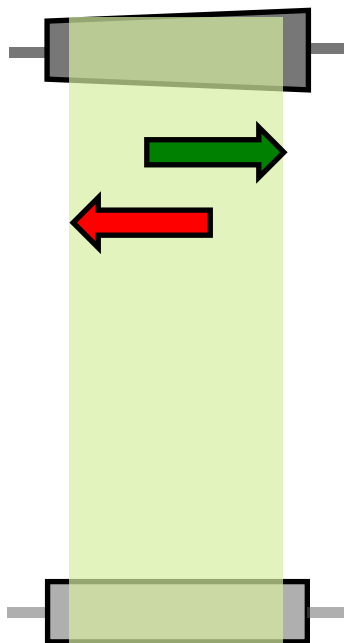
2 1

Misaligned
Roller



4 3

Diameter
Variations



1. Misalignment
+ Traction.

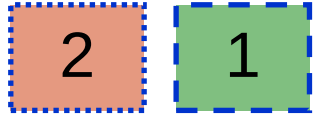
2. Misalignment
+ Slip

3. Diameter Variations
+ Traction

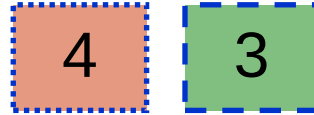
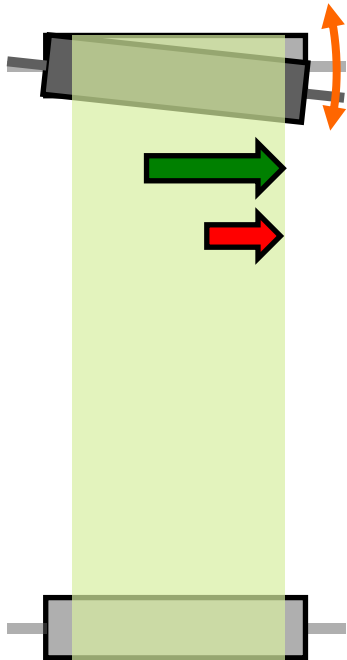
4. Diameter Variations
+ Slip

Summary - Five Tracking Rules

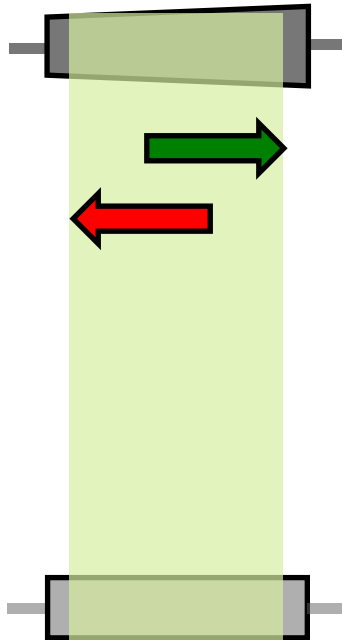
Rollers have **4** ways to shift a web.



Misaligned
Roller



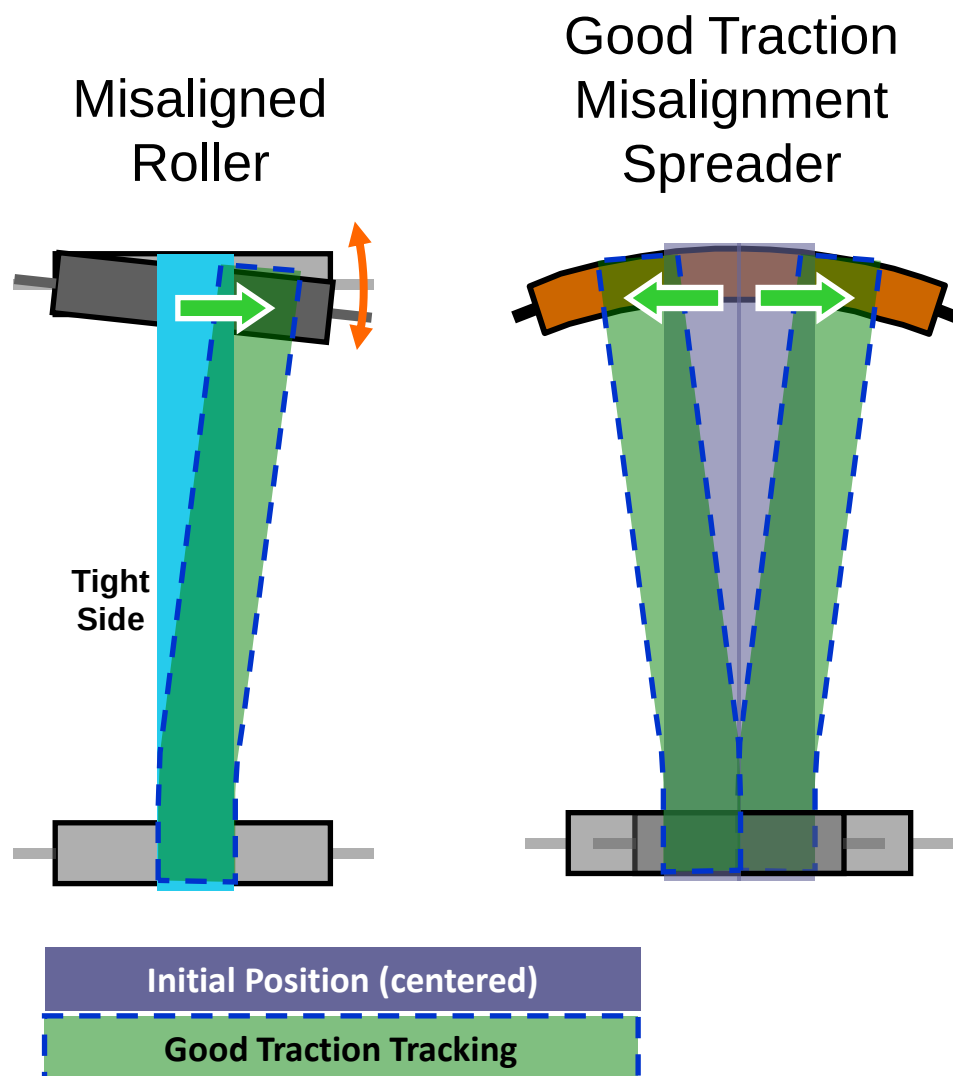
Diameter
Variations



1. Misalignment
+ Traction.

Most anti-wrinkle
and spreader rollers
use this mechanism.

Misalignment-Based Spreader



Bowed Roller



Skewed Edge Nips



Flat Expander Roller



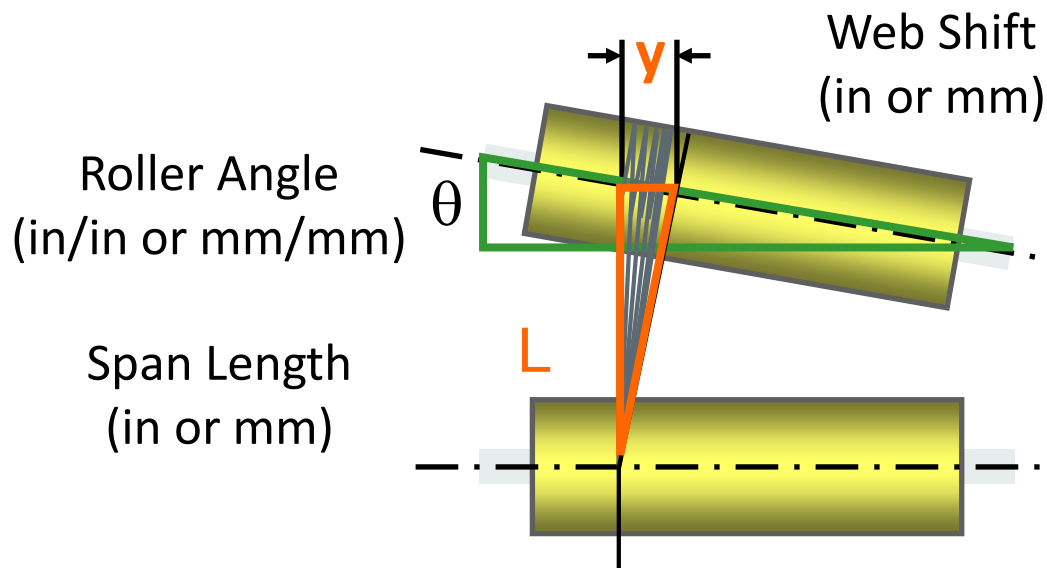
Flex Expander Roller



Tracking Result: Lateral Position

Two similar triangles estimate parallel entry tracking of a string.

Webs will shift as curves (not linear) and will shift 2/3 of strings.



String

$$\theta = \frac{y}{L}$$

$$y = L \theta$$

Web

$$\frac{2}{3} \theta = \frac{y}{L}$$

$$y = \frac{2}{3} L \theta$$

Bowed* Roller Supplier (USA)

*A.K.A. Banana Rollers, Curved Axis Rollers, Bent Rollers

Applied Roller Technology

www.appliedroller.com



Mount Hope

www.xerium.com/mountHope/



Double-E

www.doubleeusa.com



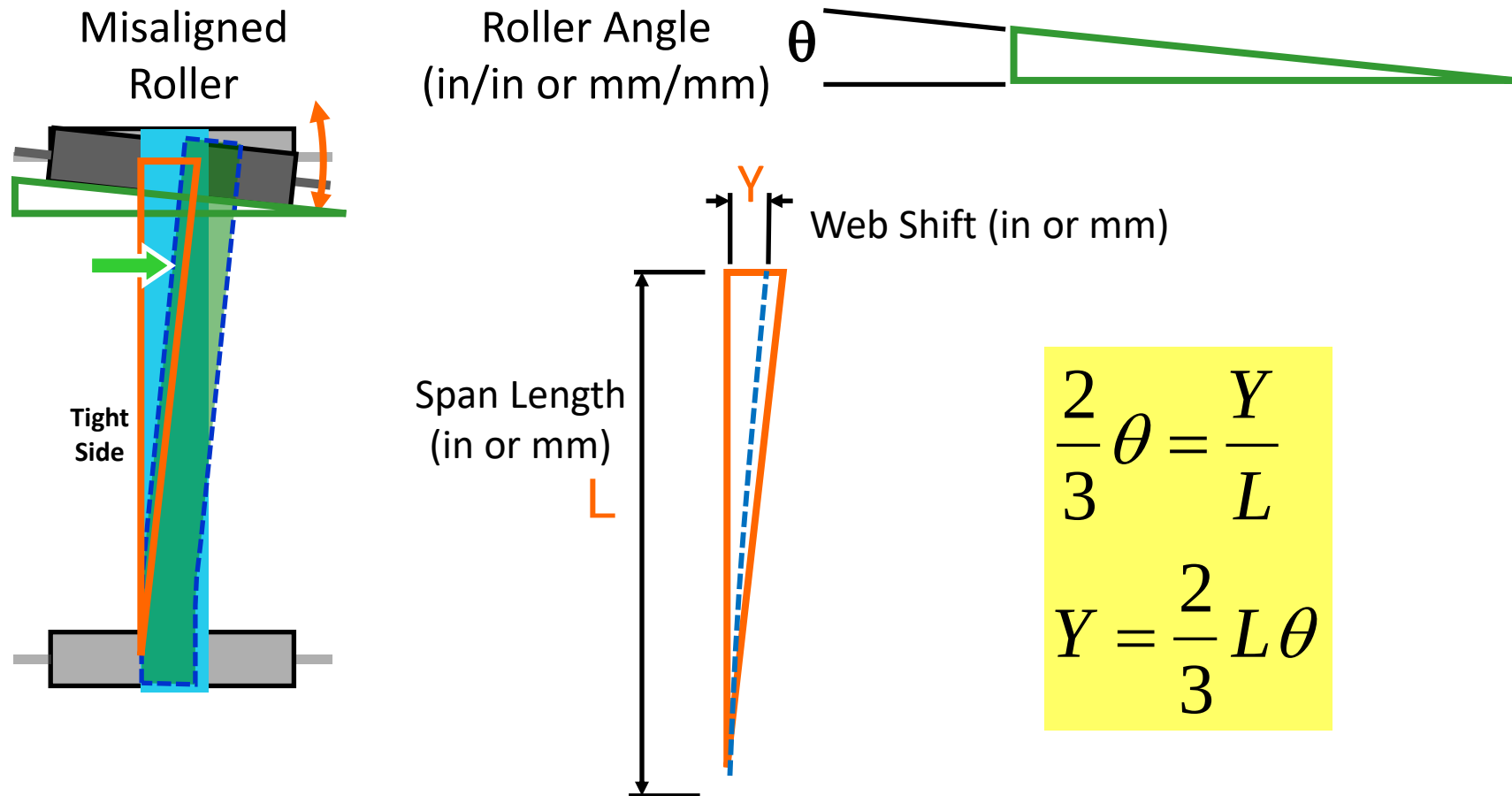
Spencer Johnston

www.spreader.spencerjohnston.com



Misalignment Web Shift

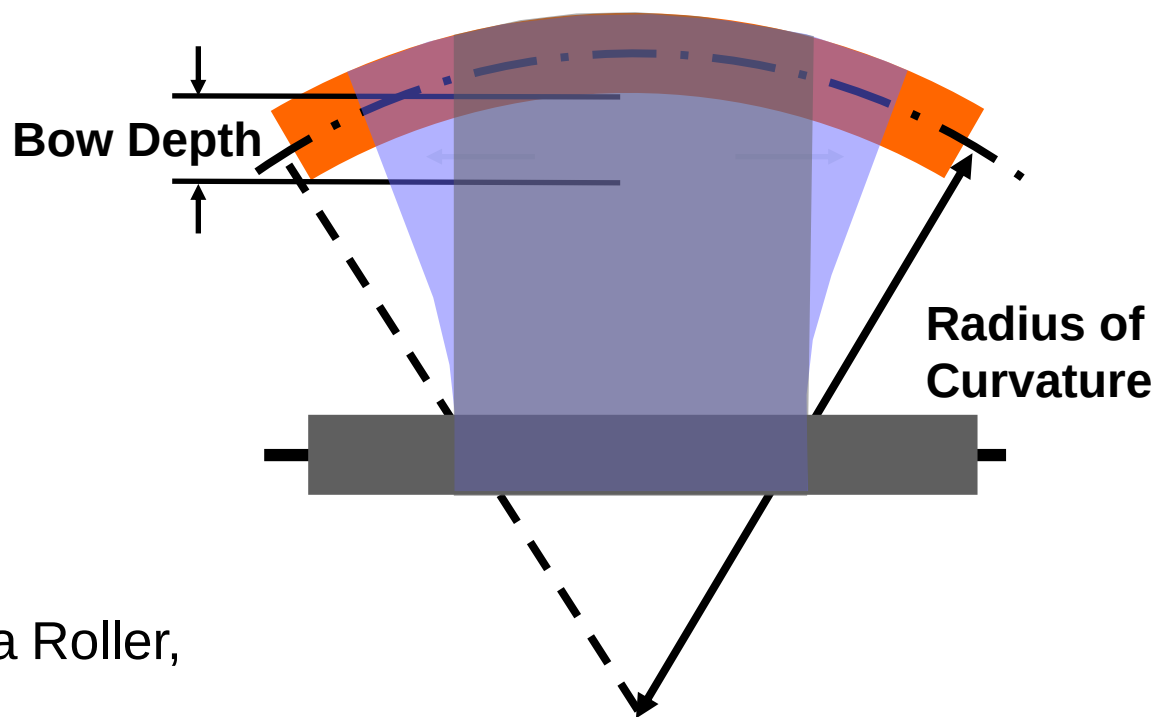
Misalignment spreader shift a web based on their angled surface motion and entry span length.



Bowed Roller

Bowed rollers have a varying axis of rotation.

The web will move laterally based on the parallel entry rule.



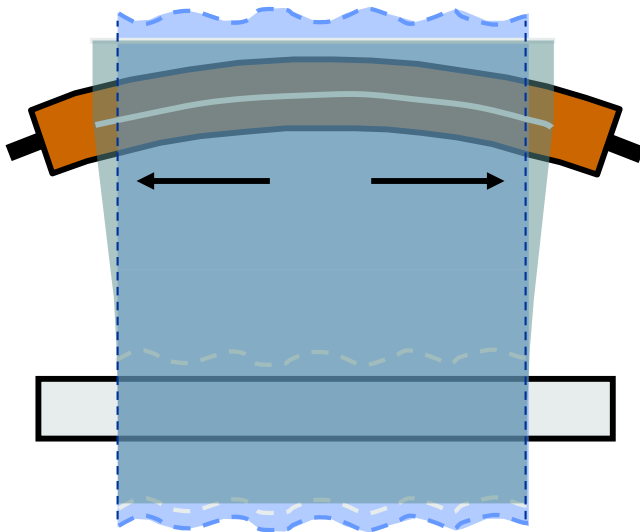
Also known as:
Curved-Axis Roller, Banana Roller,
Mount Hope™ Roller

Bowed Roller Key Features

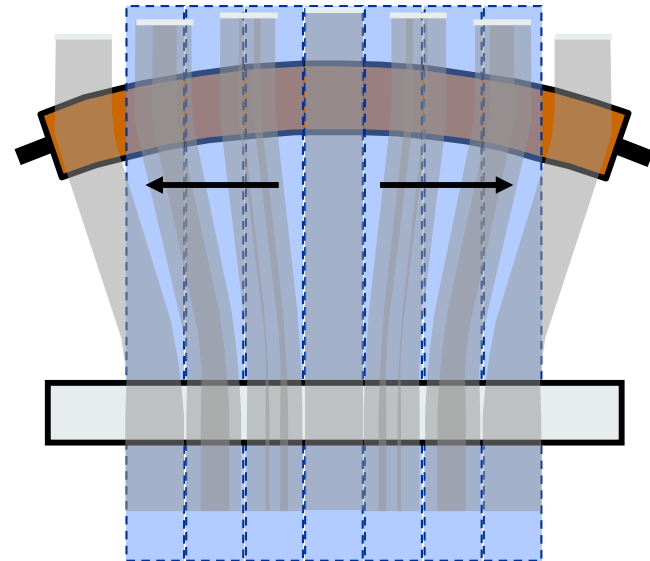
- ▶ Bowed roller purpose?
 - ▶ Prevent wrinkles
 - ▶ Separate slit webs.
- ▶ Bow depth
 - ▶ How much? Radius of curvature? Adjustable?
- ▶ Entry and exit span lengths
- ▶ Wrap angle and bow plane angle
 - ▶ Baggy web compensation
 - ▶ Effect on spreading
- ▶ Surface options and traction
- ▶ Turning torque
 - ▶ If idling, what traction is required to turn?
 - ▶ If driven, controlled by torque or speed ratio?

Bowed Roller Applications

1. To pull out any lateral looseness of a single web to eliminate wrinkles.



2. To spread multiple strands to a uniform spacing.

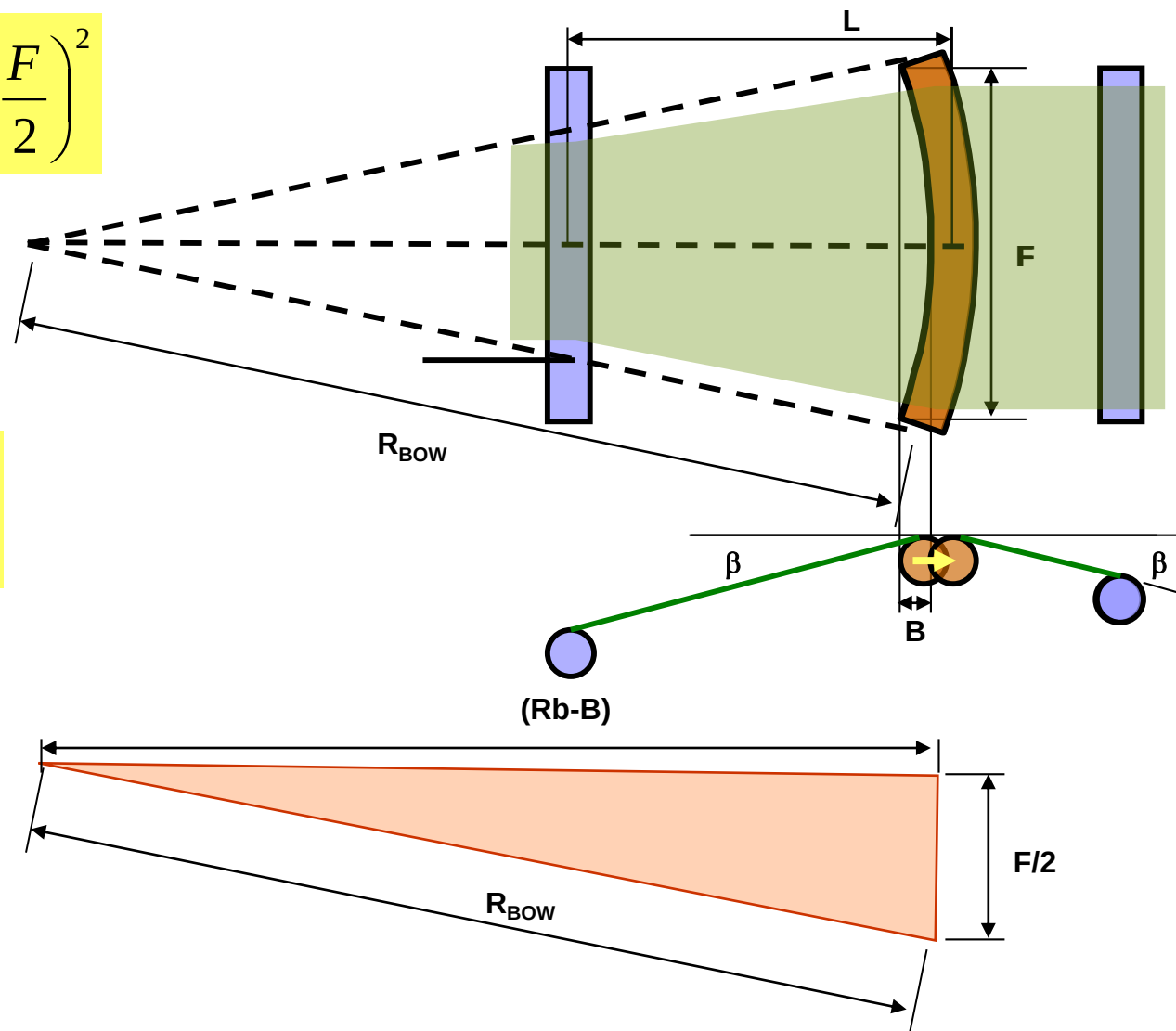


Bowed Roller Radius

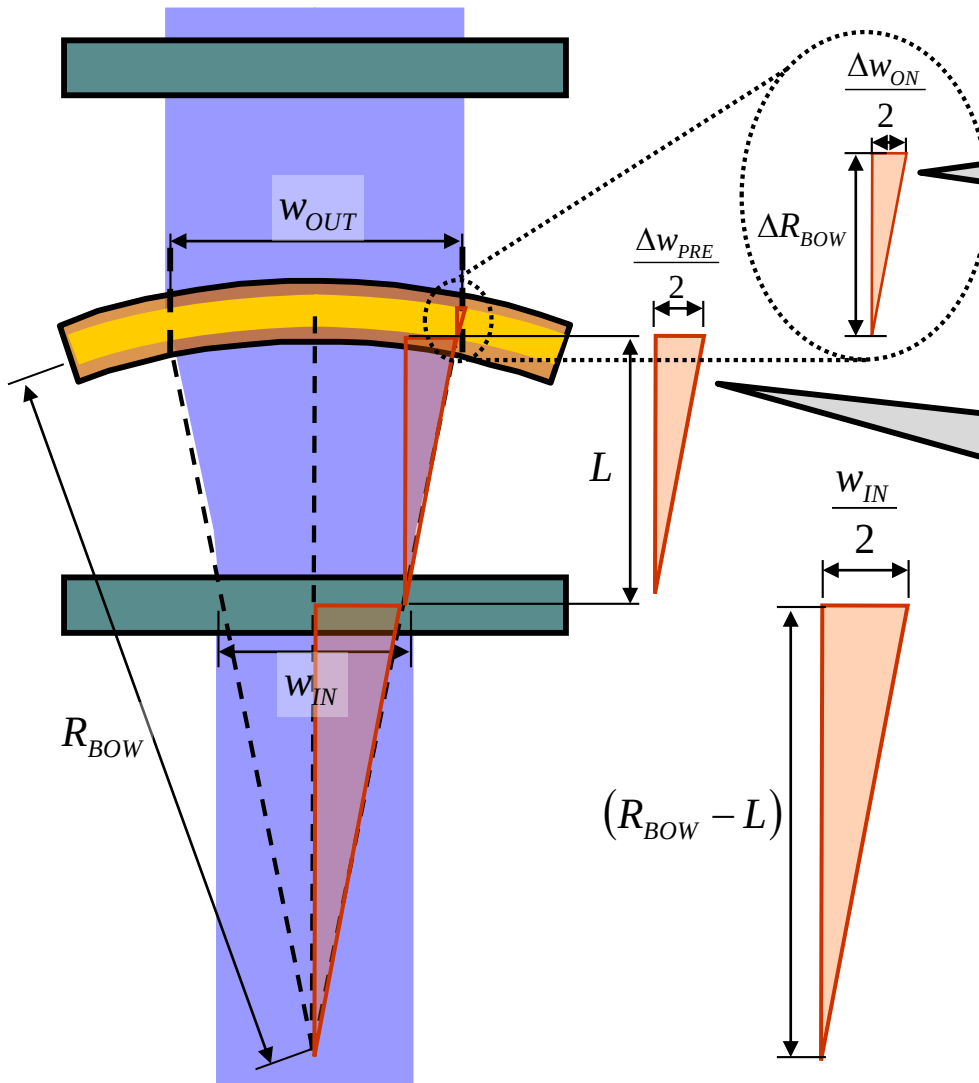
$$R_{BOW}^2 = (R_{BOW} - B)^2 + \left(\frac{F}{2}\right)^2$$

$$R_{BOW} = \frac{F^2}{8B} + \frac{B}{2} \approx \frac{F^2}{8B}$$

$$R_{BOW,eff} = \frac{R_{BOW}}{\cos \beta}$$



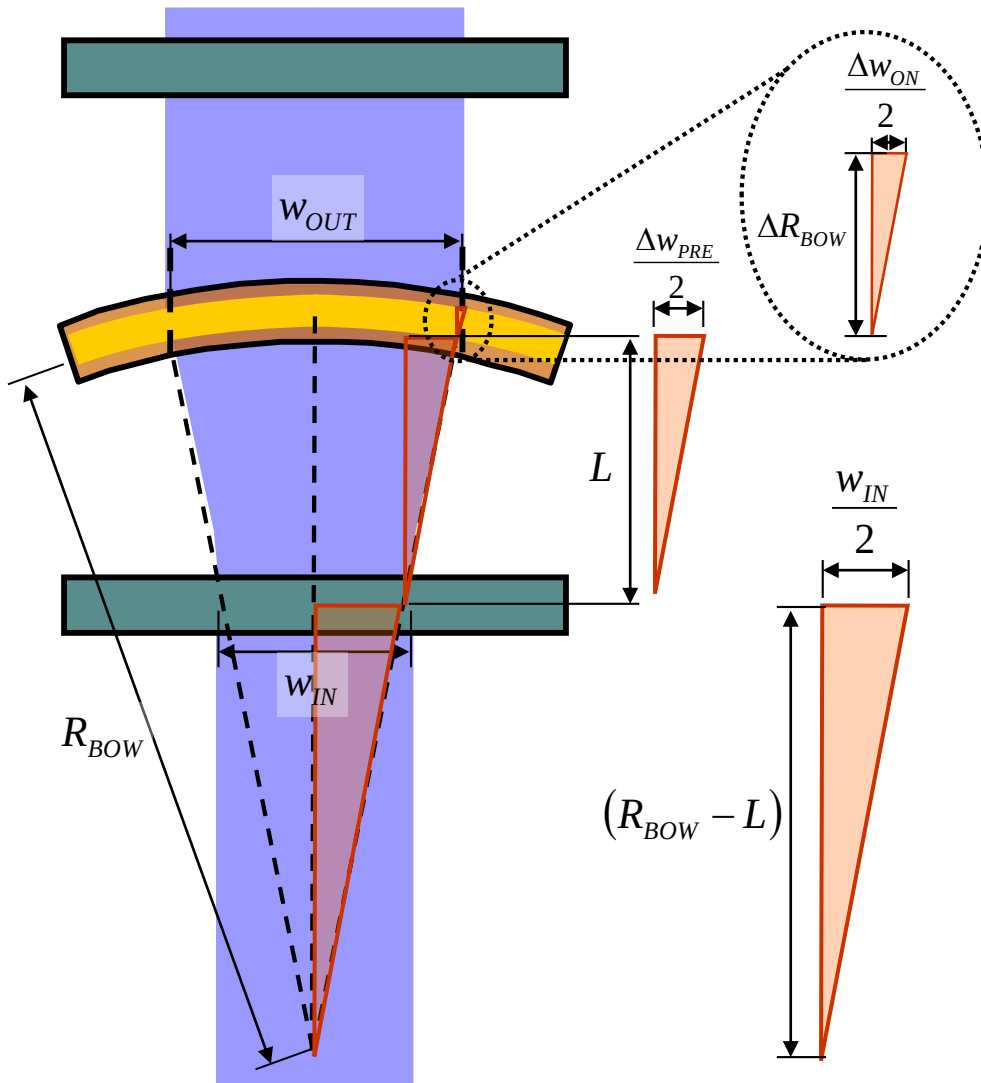
Bowed Pre-Roller Spreading



On-roller spreading is nearly insignificant, but can increase rate of rubber covering abrasion.

Pre-roller spreading based on the parallel entry rule is the primary mechanism of bowed roller spreading.

Bowed Pre-Roller Spreading



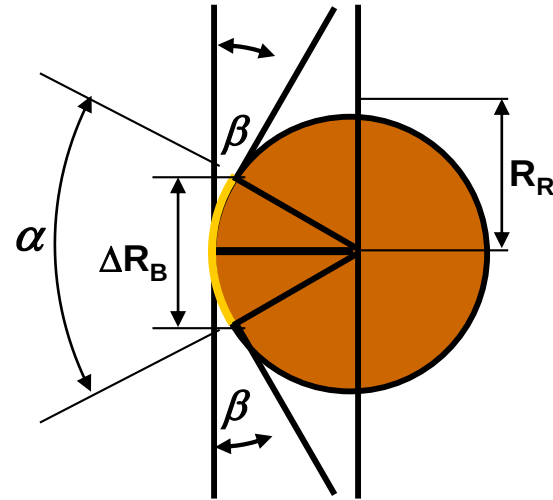
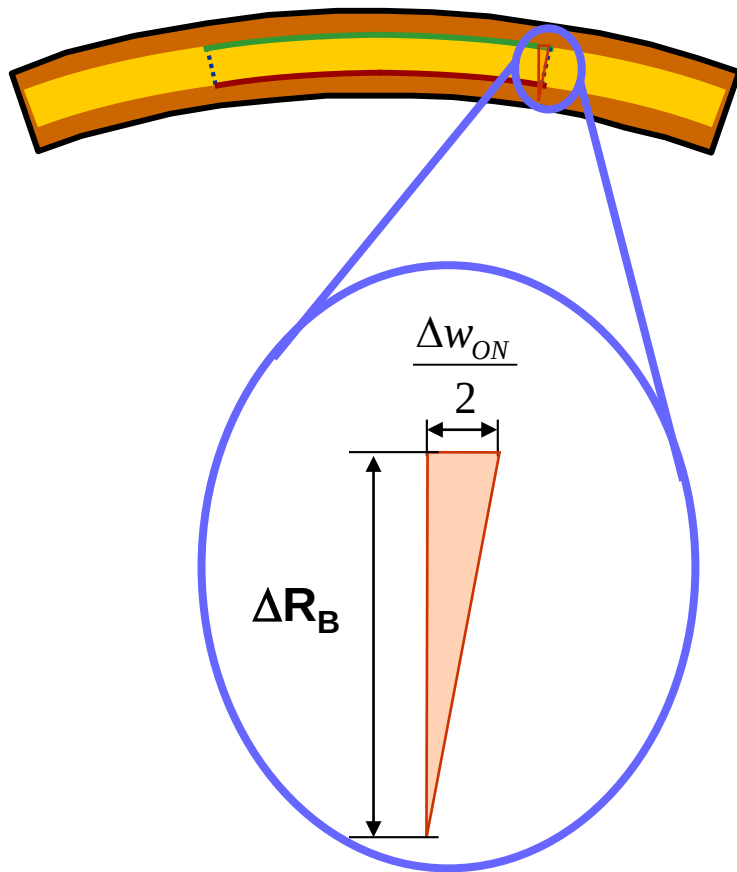
$$\frac{w_{IN}}{(R_{BOW} - L)} = \frac{\Delta w_{PRE}}{L} \approx \frac{\Delta w_{ON}}{\Delta R_{BOW}}$$

$$\varepsilon_{y \max, PRE} = \frac{\Delta w_{PRE}}{w_{IN}} = \frac{L}{(R_{BOW} - L)}$$

$$\varepsilon_{y \max, ON} = \frac{\Delta w_{ON}}{w_{IN}} = \frac{\Delta R_{BOW}}{(R_{BOW} - L)}$$

$$w_{OUT} = w_{IN} + \Delta w_{PRE} + \Delta w_{ON}$$

Bowed On-Roller Spreading



$$\frac{\Delta R_B}{2R_R} = \tan \beta$$

$$\Delta R_B = 2R_R \tan \beta$$

$$\varepsilon_{y,ON} = \frac{\Delta w_{ON}}{w_2} = \frac{\Delta R_B}{R_B}$$

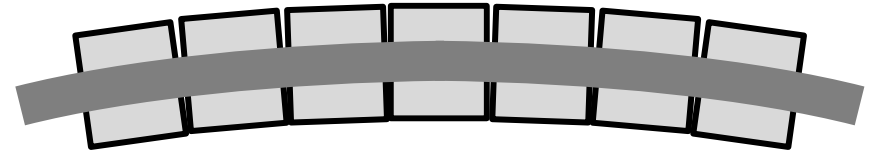
$$\varepsilon_{y,ON} = \frac{2R_R}{R_B} \tan \beta = \frac{R_R}{R_B} \tan \alpha$$

What is Inside a Bowed Roller?

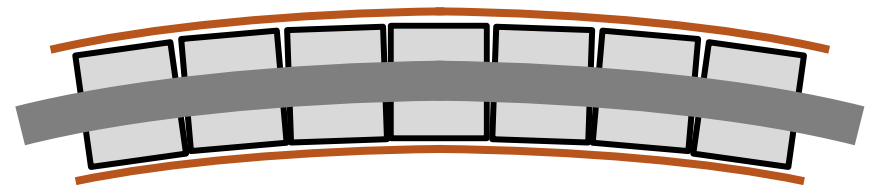
Most bowed roller have a fixed shaft curvature or depth.



Several narrow cylindrical roller segments are supported by the shaft (similar to a shish kabob).



A rubber sleeve covers the roller segments.



What is Inside a Bowed Roller?

Each internal roller segment has two bearings.



Bowed roller drag is created by the large number of internal bearings and the elastic hysteresis of the rubber sleeve stretching and relaxing.

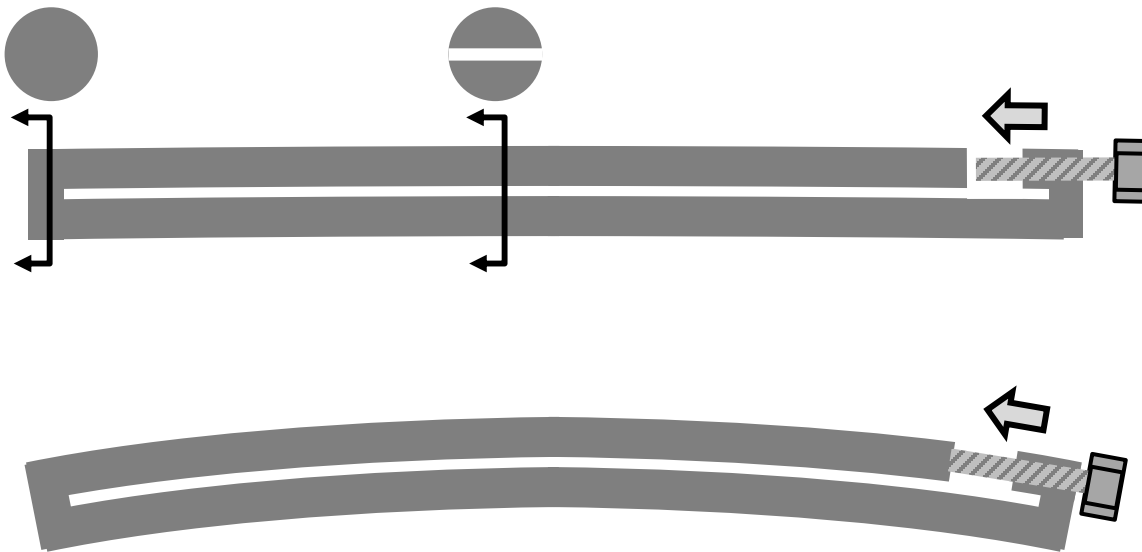


What is Inside a Bowed Roller?

Adjustable bowed roller have a split shaft.

- Stretchy webs use more bow for more spreading.
- Stiffer webs need less bow.

Adjusting a bolt on one end makes one side of the shaft longer than the other, inducing curvature.



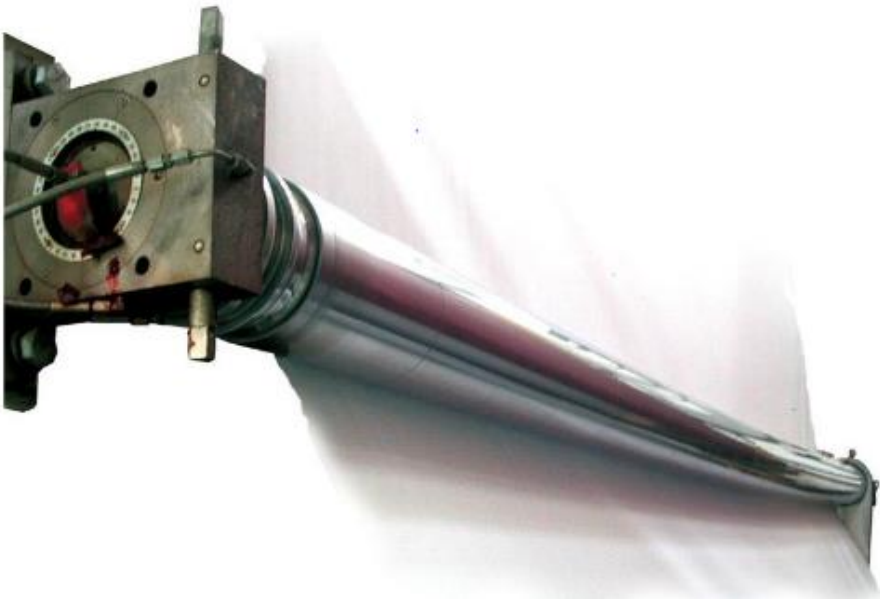
Options:

- Adjusted manually with a big wrench
- Adjusted by hydraulic cylinder
- Adjusted by motor

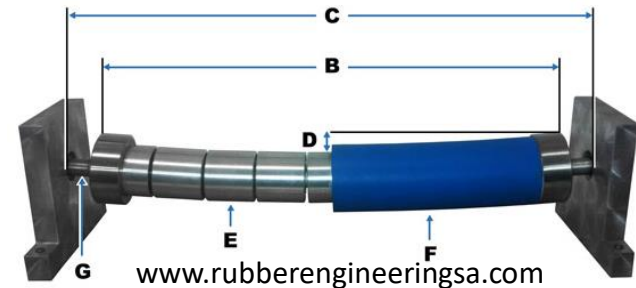
* Advanced systems provide position feedback for repeatable settings.

Bowed Rollers – Rubber or Metal Surfaces

Most bowed rollers have a continuous rubber surface or sleeve, but some have segmented metal (steel) surfaces for better wear and heat resistance.



[www.airexpandingshaft.com\(India\)](http://www.airexpandingshaft.com(India))



Thicker webs will have no problem with a segmented roller surface, but thin papers, films, and foils would likely wrinkle or fall into even small gaps in the roller surface.

Bowed Roller Spreading Requires Traction

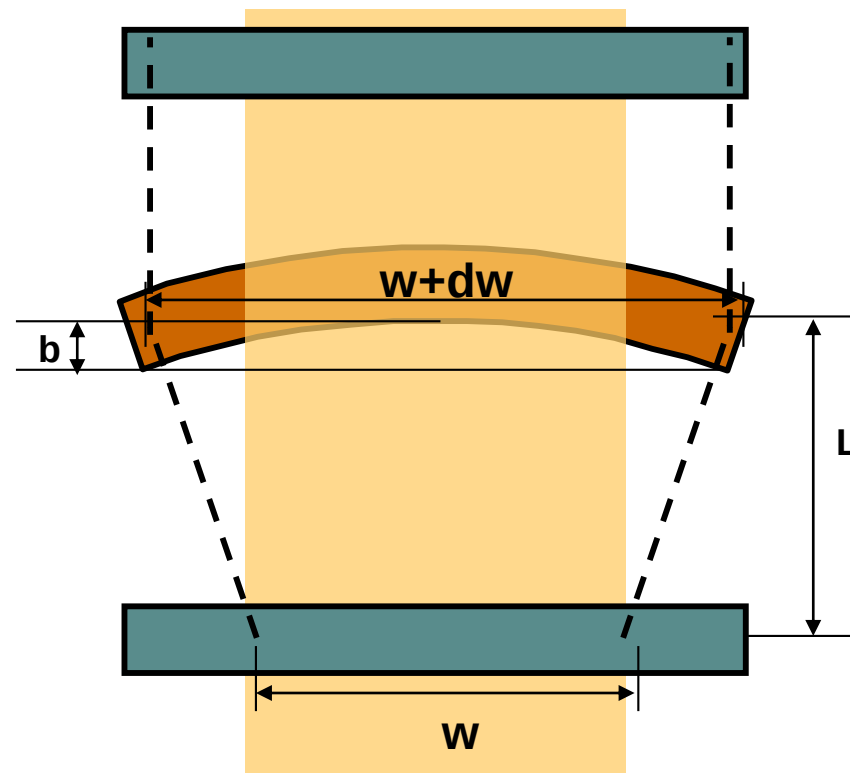
Maximum spreading = $f(\text{entry span length, effective bow})$

Actual spreading will be limited by:

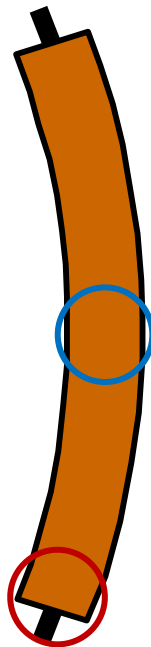
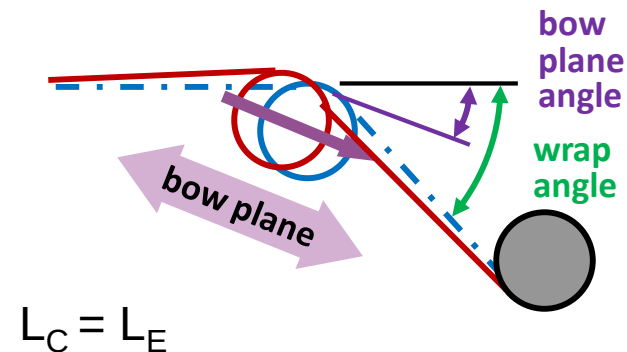
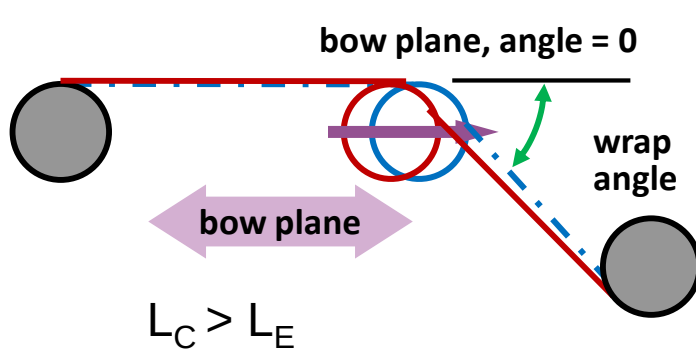
Traction – Without traction, the roller can't pull the web laterally.

Traction required by spreading function (most to least):

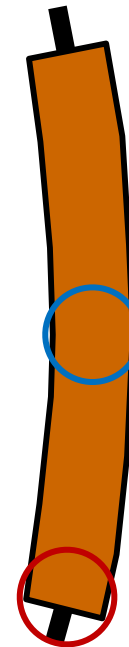
- Stretching a solid wide web.
- Stretching a wide porous film, tissue, or textile.
- Spreading two half width webs.
- Spreading multiple narrow slit strands.
- Pulling a gathered web to lateral tautness.



Bowed Roller Wrap Angle Effects: 45-deg wrap



With smaller wrap angles, the 'even tension' bow plan angle is smaller and spreading is stronger in this baggy 'neutral' position. (The bow plane angle reduce spreading by only 8%.)



- Edge Web Path
- . - Center Web Path

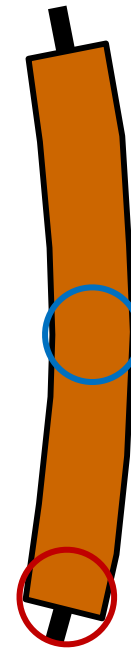
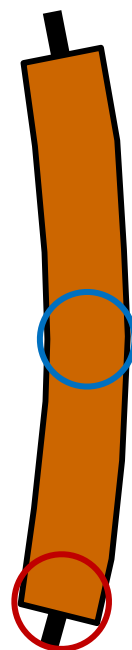
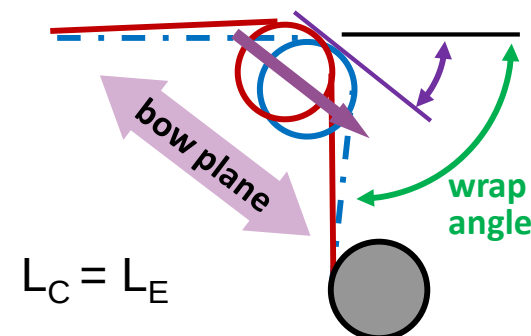
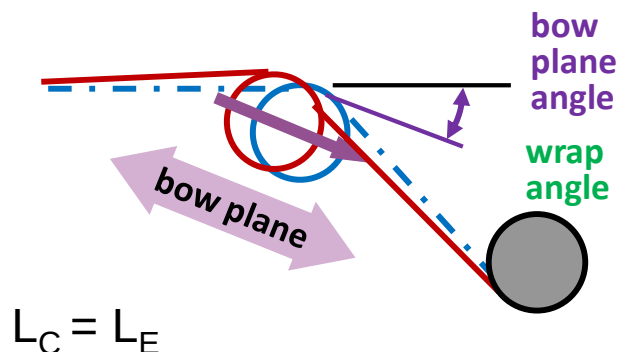
Bowed Roller Wrap Angle

The optimum wrap angle is 30-60 degrees.

Lower wrap angles may not have sufficient traction to laterally bend and spread the web or slit strands.

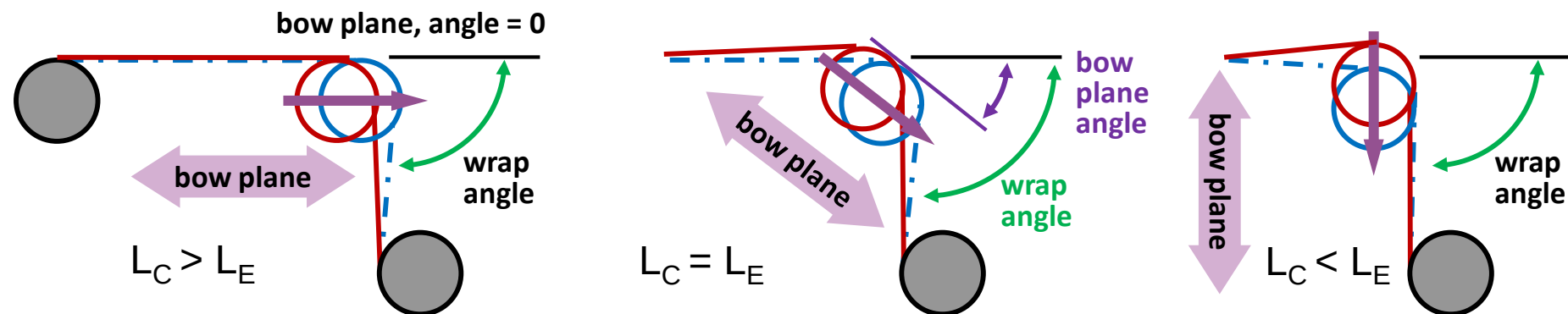
Larger wrap angle create several negative effects:

- More rubber wear from web-roller slippage.
- More difficult to adjust without crossweb tension variations.
- Less effective spreading in the default bow plane orientation.



— Edge Web Path
- - - Center Web Path

Bowed Roller Wrap Angle Effects: 90-deg wrap



Maximum spreading occurs when the bow plane is parallel to the entry, but this also can make edges loose in stiffer webs.

Top view: what the entry span "sees."

Edge and center web paths are equal and CD tension is even when bow plane angle is half of wrap angle, but spreading is reduced by the 30% (cosine of bow plane angle).

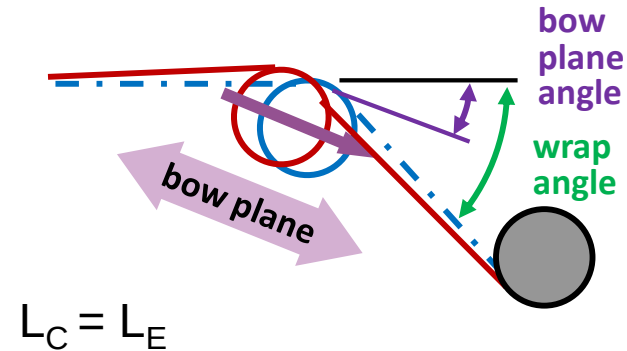
Top view

If the bow plane is perpendicular to the entry plane, there is no spreading in the entry span.

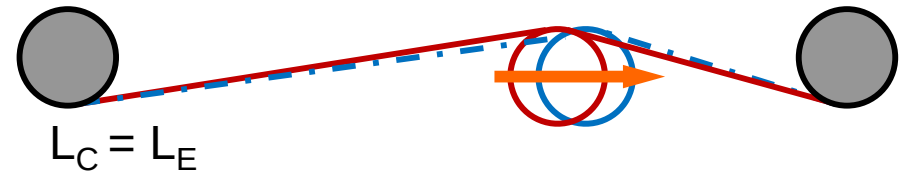
Top view

— Edge Web Path
- - - Center Web Path

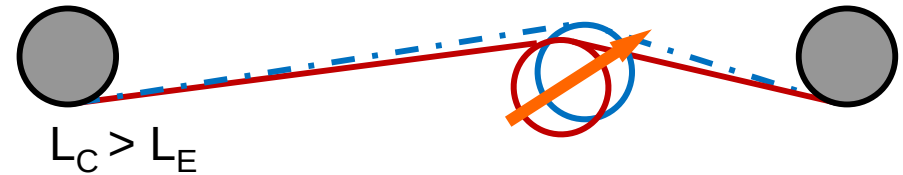
Baggy Web Compensation



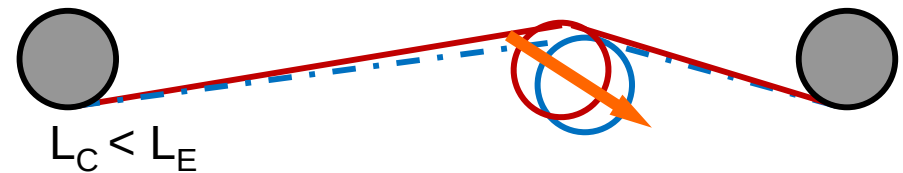
The default orientation creates equal edge and center web paths.



To tighten a baggy center, the bow plane may be turned into the web.



To tighten baggy edges, the bow plane may be turned away from the web.

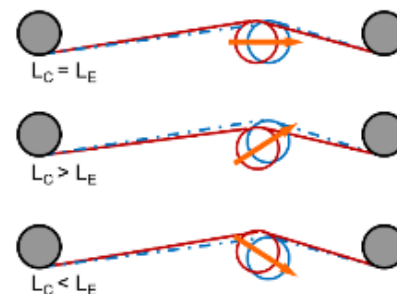


— Edge Web Path

- - - Center Web Path

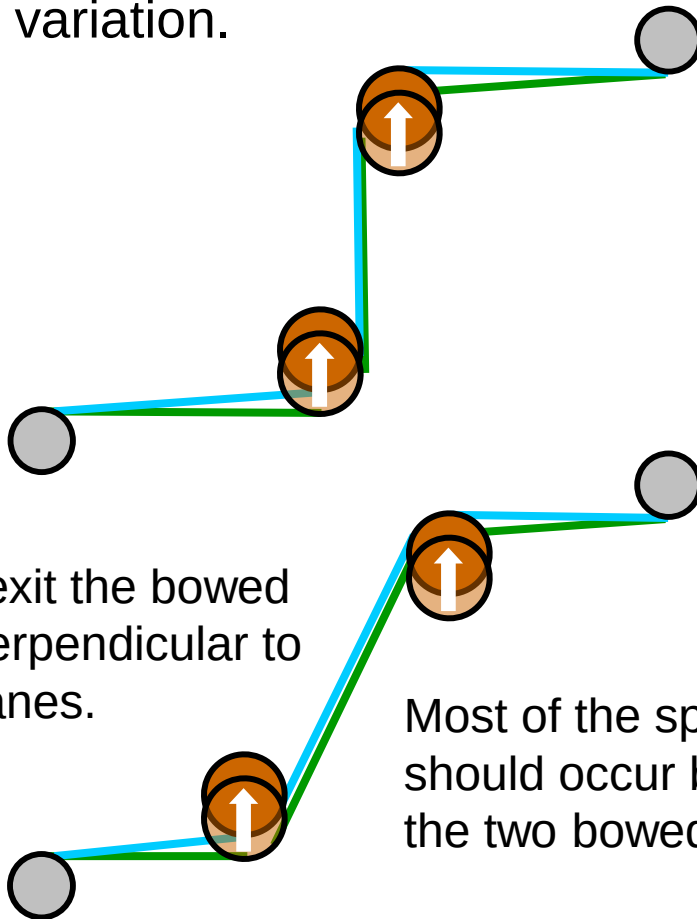
Keys to Bowed Roller Success

- ▶ Bow radius (set by bow depth) and entry span length determine spreading maximum.
- ▶ Bow depth will be lower for high modulus webs.
- ▶ Spreading may be limited by traction, especially in high speed/tension conditions, non-grooved rollers, and smooth and non-porous web.
- ▶ Bow plane orientation can:
 - ▶ tighten baggy center web,
 - ▶ tighten baggy edge web,
 - ▶ gather the web (if pointing upstream), or
 - ▶ make wrinkles due to excessive web path variations (usually when center-edge web path variations exceed web strain).
- ▶ Bowed roller used as 'D' bars (pointing up into the wrap angle), likely have low traction or slip problems.
- ▶ Wrap angles of 20-45 degrees work best. Smaller wrap angles may not have the required traction. Larger wrap angles reduce spreading effect in baggy-neutral bow plane orientation.



Dual-Bowed Roller Spreading

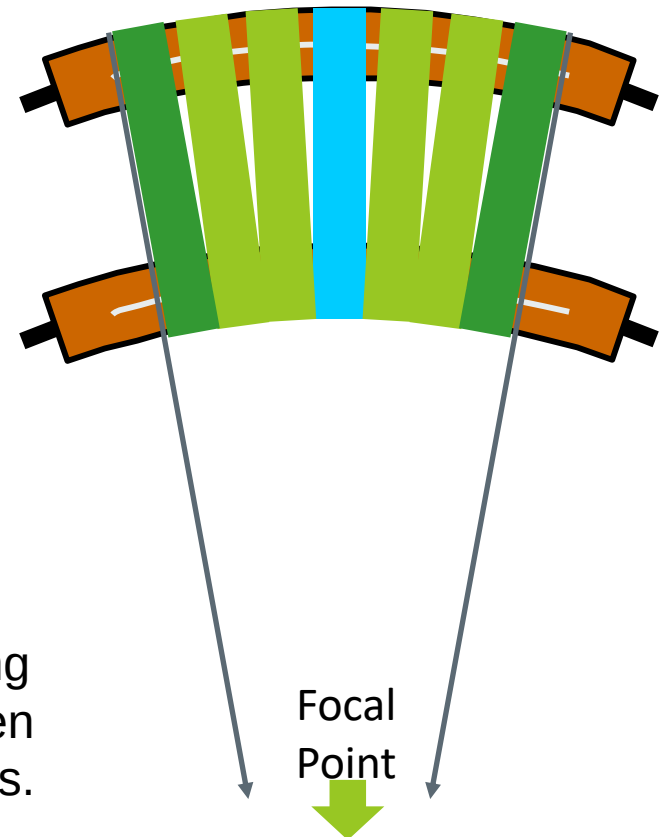
Two bowed rollers in series will provide spreading with minimal web path variation.



Enter and exit the bowed roller set perpendicular to the bow planes.

Most of the spreading should occur between the two bowed rollers.

The two bowed rollers will point to the same focal point.



Flat Expander Rollers

A flat expander roller's surface is made of polymer bands or a rubber sleeve.

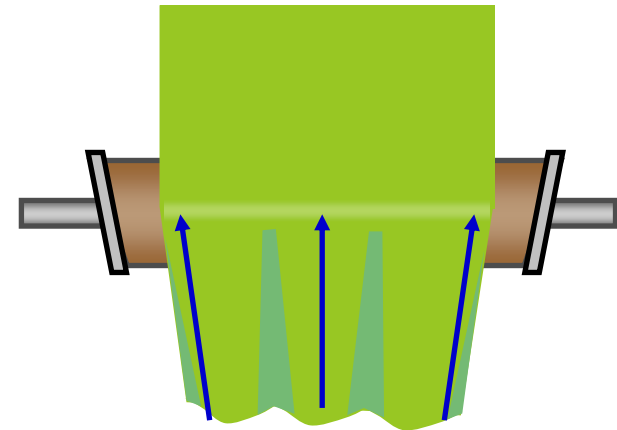
The surface expands laterally under the wrapping web.

The result is diverging surface vectors and two spreading mechanisms – pre-roller and on-roller web spreading.

Images courtesy of Izumi Int'l.



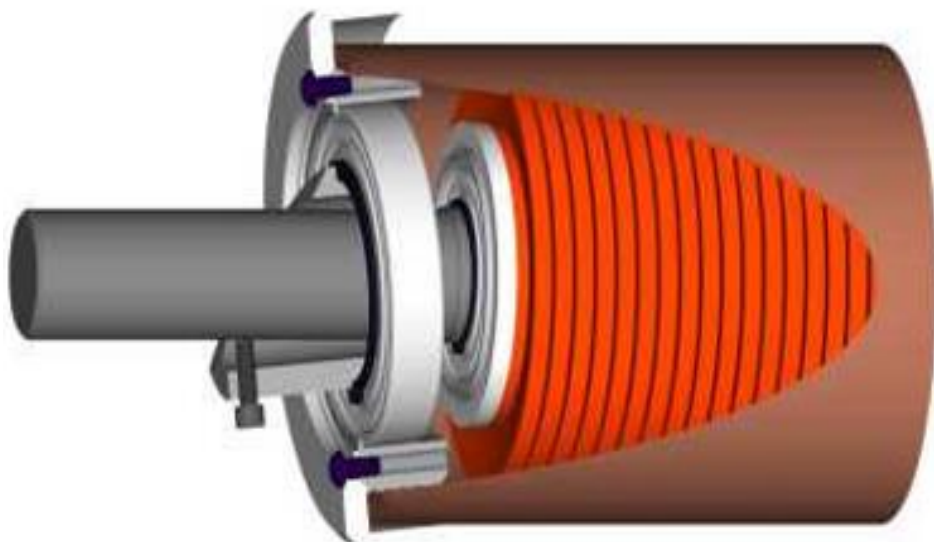
Images courtesy of Converter Accessory Corp.



Detail on WrinkleStop™ Roller

The set screw on each roller end tilts the rotating end disc of the rubber sleeve, causing the roller face width to expand and contract through the roller's full rotation.

Wrapping the roller on with initial contact on the expanding side creates a roller surface angle that directs the web outward.



Rubber Cover Spreading

Test: Apply tape on fully expanded side, rotate 180 degrees, observe.

CAC WrinkleStop Roller



Bowed Roller



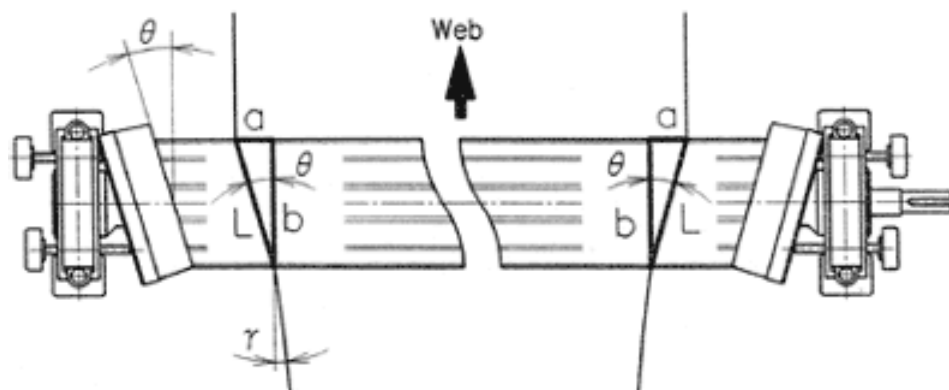
Flat Expander Rollers

**Mitsubishi Corp.
UE Miravo™ Roller**

Osaka, Japan

**Applied up to
200 m/min normal speeds,
300 m/min maximum**

www.mitsubishi-corp.co.jp/



Miravo™ roller images courtesy of U Engineering

Flat Expander Rollers

Izumi Int'l

Flat Expander No-Bow Roller

Osaka, Japan

www.izumiinternational.com



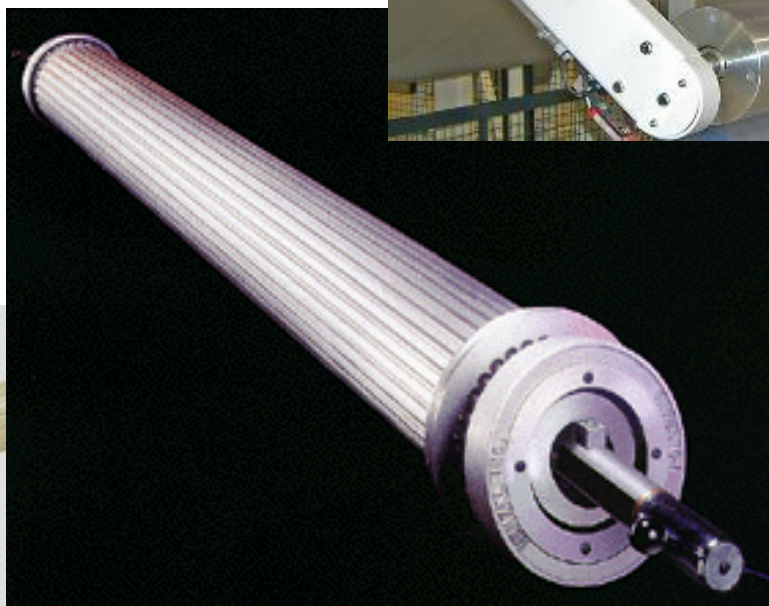
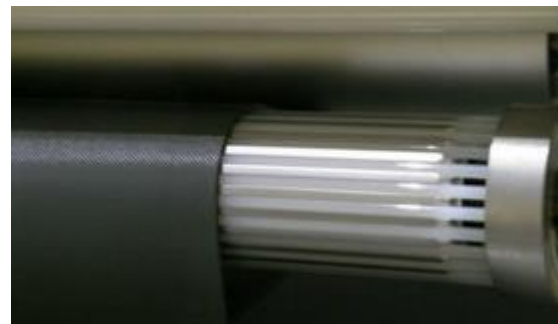
Courtesy of Izumi Int'l

Menzel USA

Polyband Roller

Flat Expander

No-Bow Roller



Courtesy of Menzel USA

Expander Roller Spreading

An expanding surface roller has two strong spreading mechanisms:

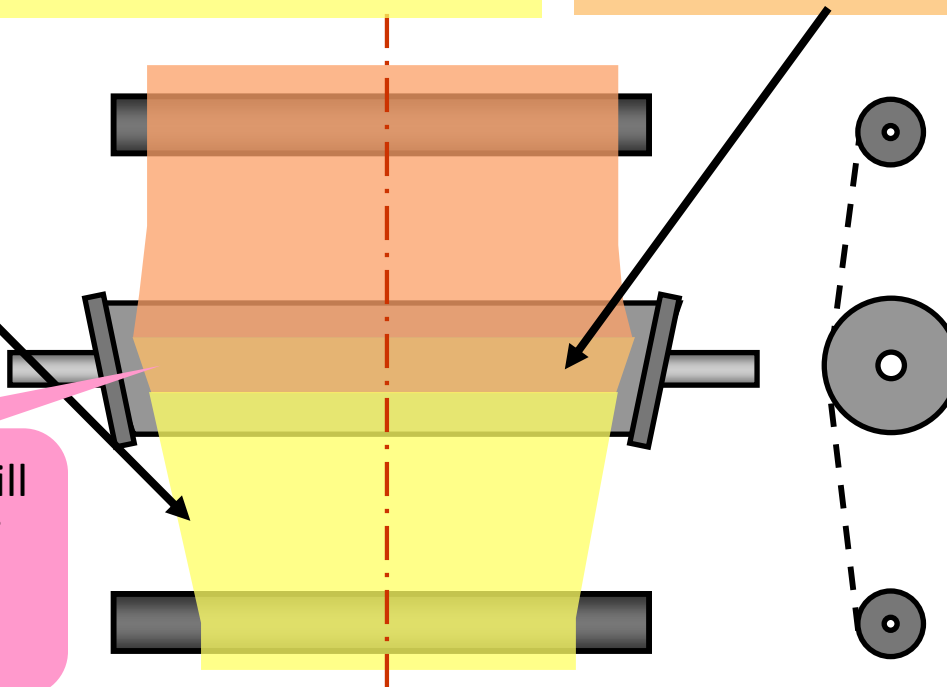
1. Pre-Roller Spreading

The web will track outward to align its edges to enter parallel to surface vector of the roller's initial contact point (parallel entry rule).

2. On-Roller Spreading

As the roller's surface expands, it will attempt to laterally expand the web, limited by the available friction.

Too much spreading will increase wear, loss of traction, and poor performance.



Flex Expander Rollers

A flex expander has a soft durometer elastomeric covering with spiral, bias-cut grooves.

Image
courtesy of
American
Roller

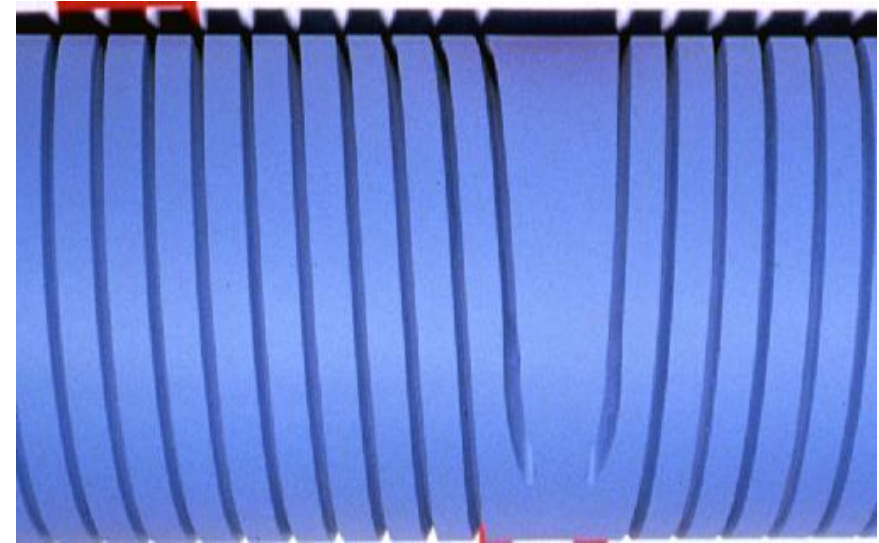


Image courtesy of Davis Standard

The flex expander works on the same principle as the bowed roller!

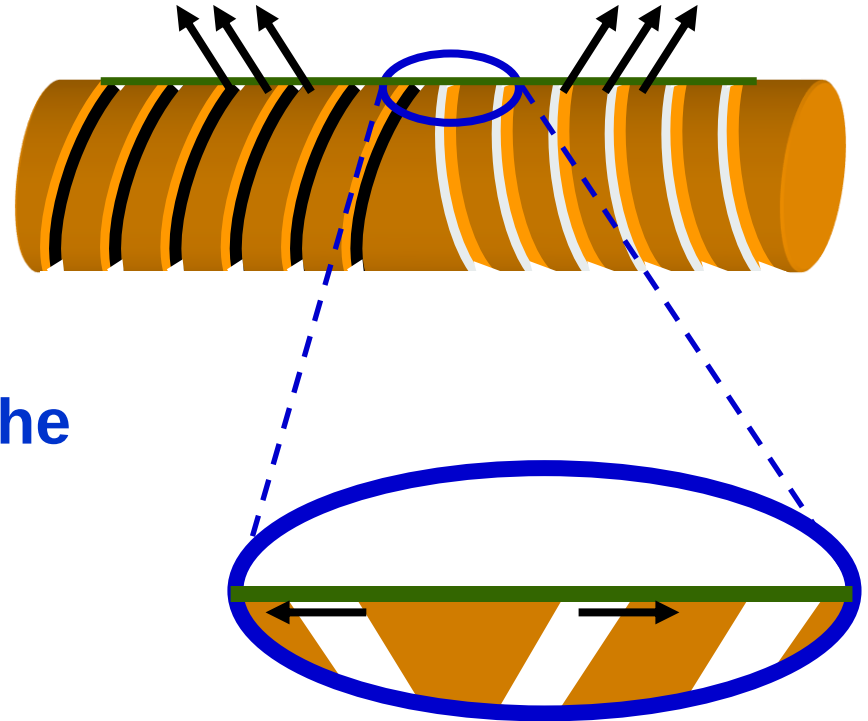


Flex Expander Rollers

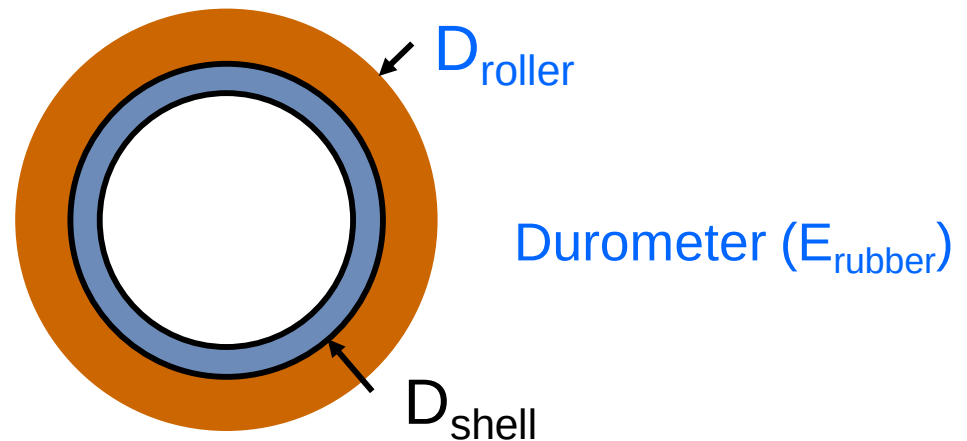
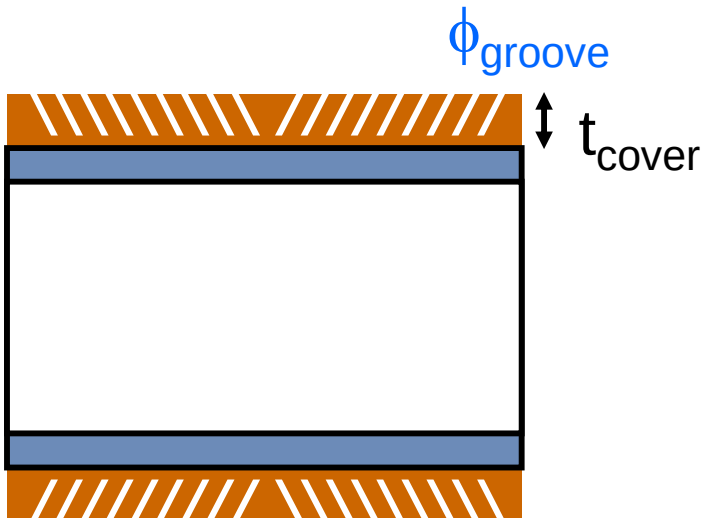
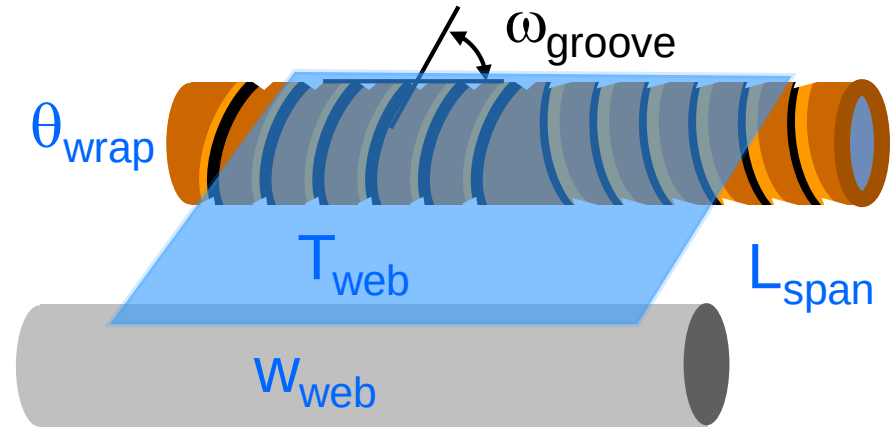
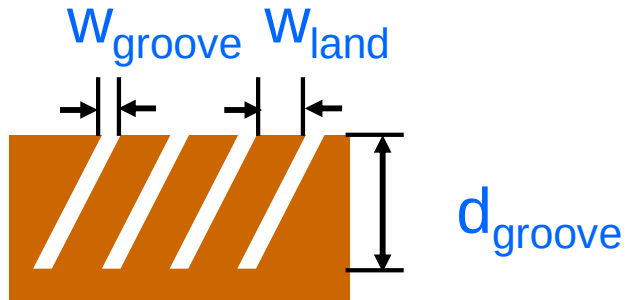
The tensioned web presses down on the angled elastic flutes.

In the initial contact zone, the roller surface is angling outward.

The outward surface motion causes web tracking due to the parallel entry rule, like all other misalignment-based spreaders.



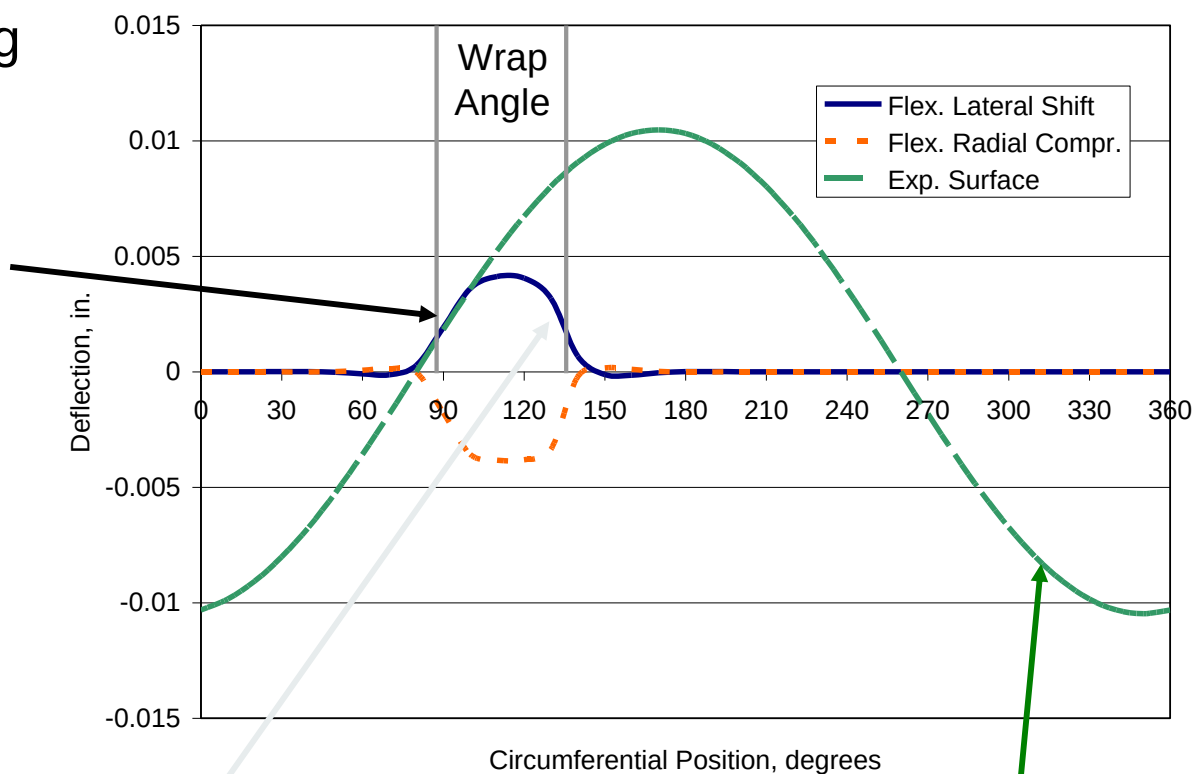
Flex Expander Rollers



Surface Vectors of Expander Rollers

The effective spreading of misalignment spreaders is directly related to the surface vector angle at initial contact.

Displacements of Expander Rollers



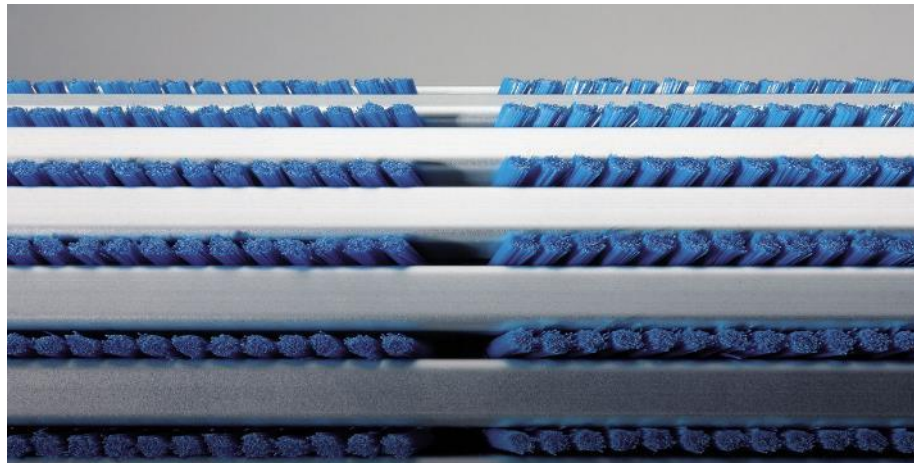
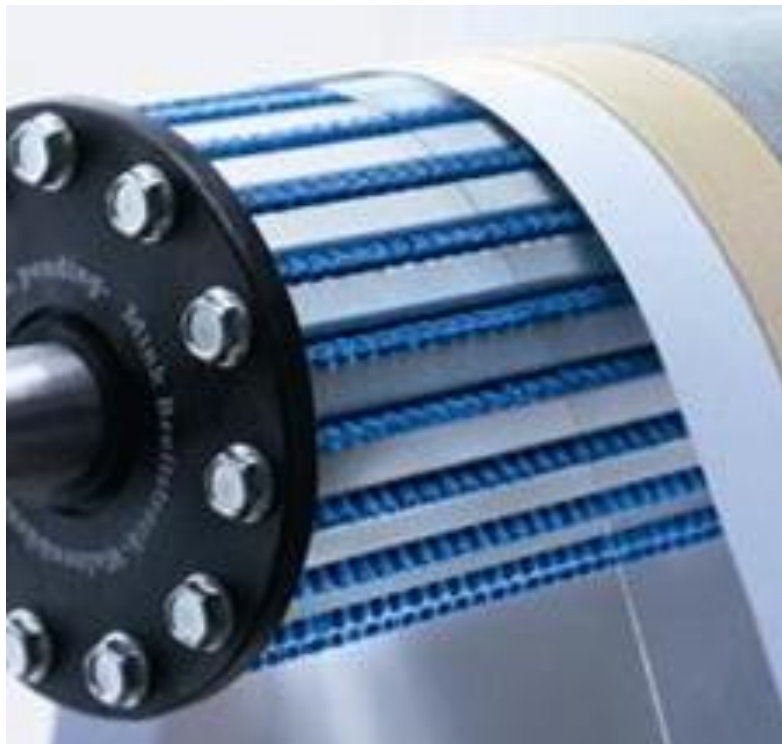
Flexible Expander



Straight Expander



Brush Spreaders (another flex expander)



The bristles on this roller are angled away from machine centerline similar to the rubber sections of a traditional flexible expander.

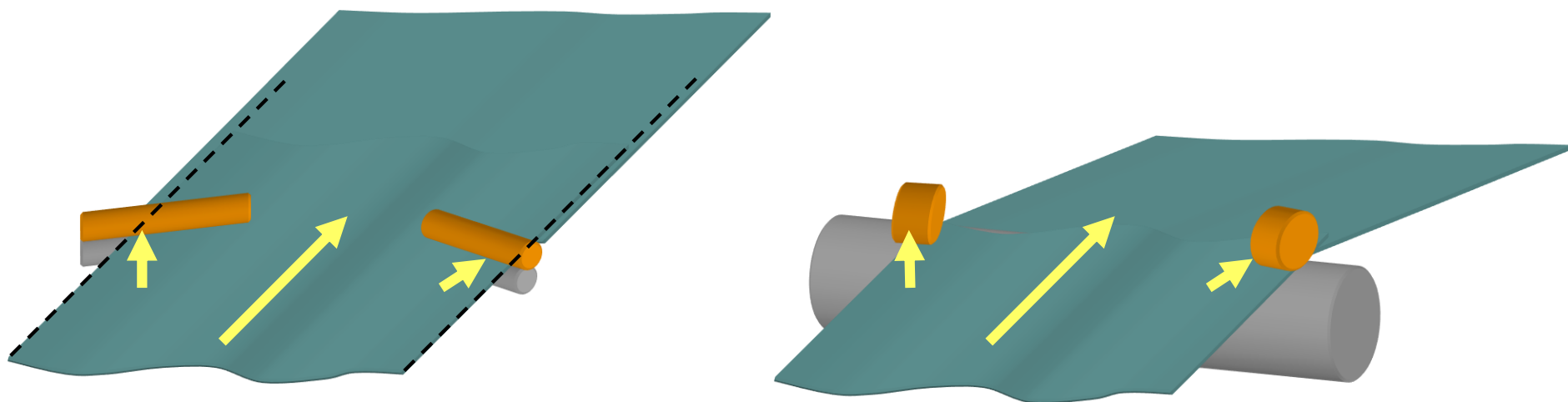
www.mink-buersten.de



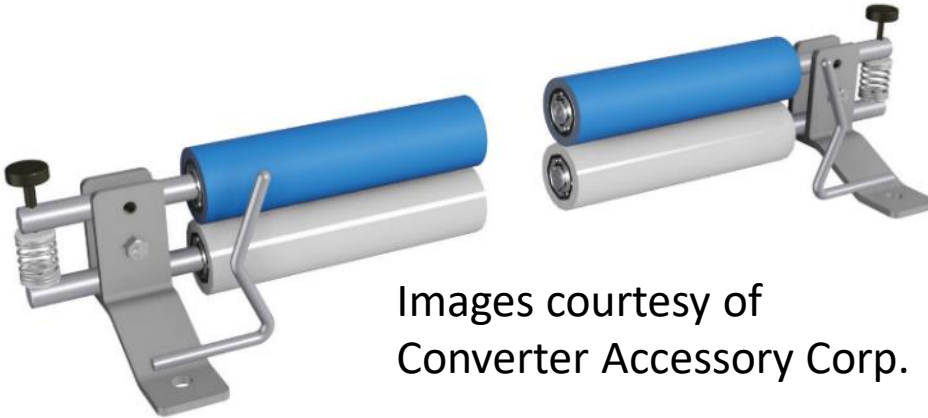
Skewed Edge Nips

Two rollers are nipped together, one rubber covered. Typically, the nips are spring loaded.

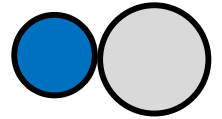
The roller nips are skewed at an angle (less than 2°) to pull the web laterally outward on each side by the parallel entry rule.



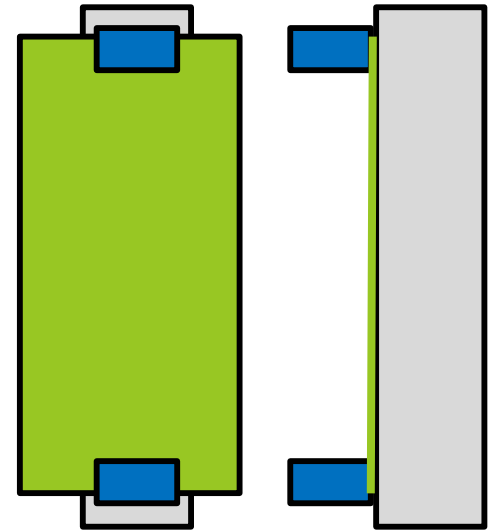
Skewed Edge Nips



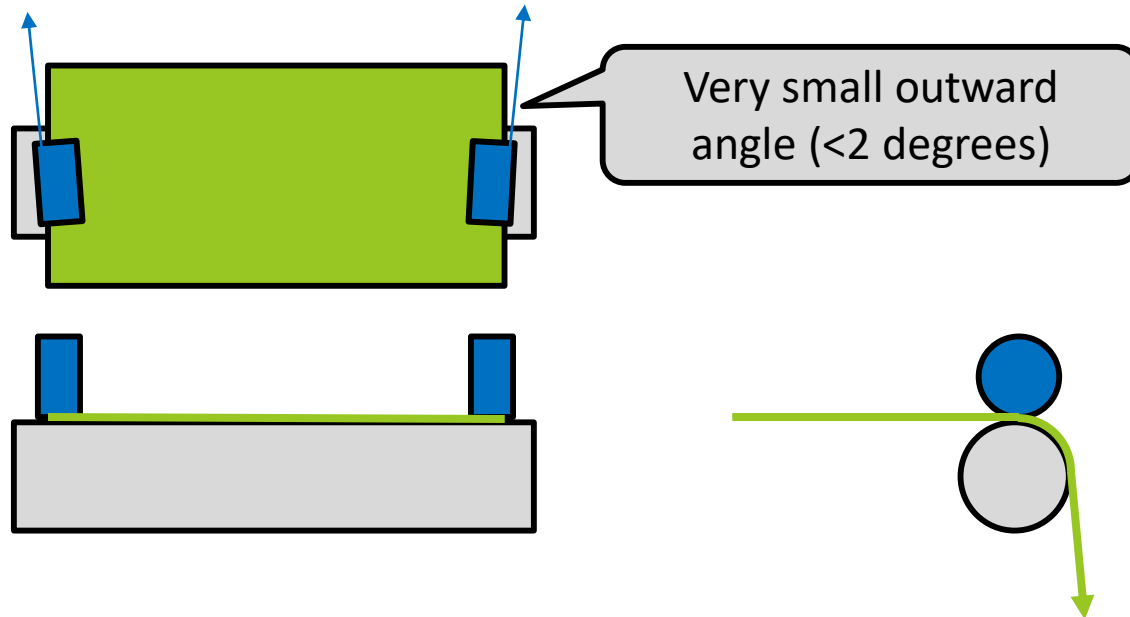
Images courtesy of
Converter Accessory Corp.



Courtesy of Bruckner
Machinery

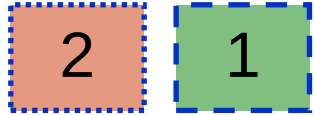


Skewed Edge Nips

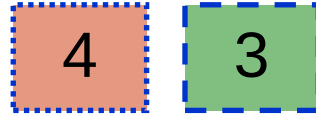
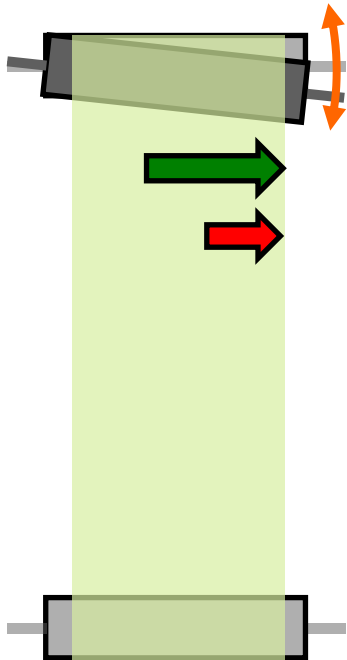


Summary - Five Tracking Rules

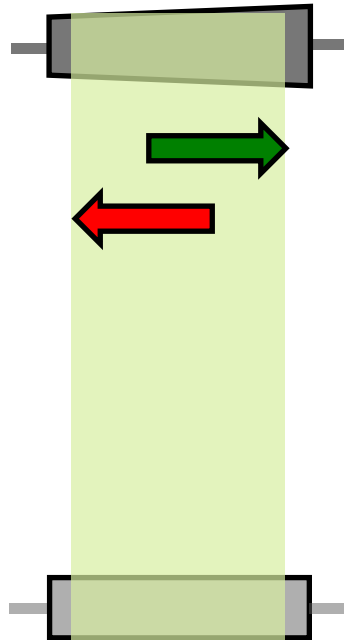
Rollers have **4** ways to shift a web.



Misaligned
Roller

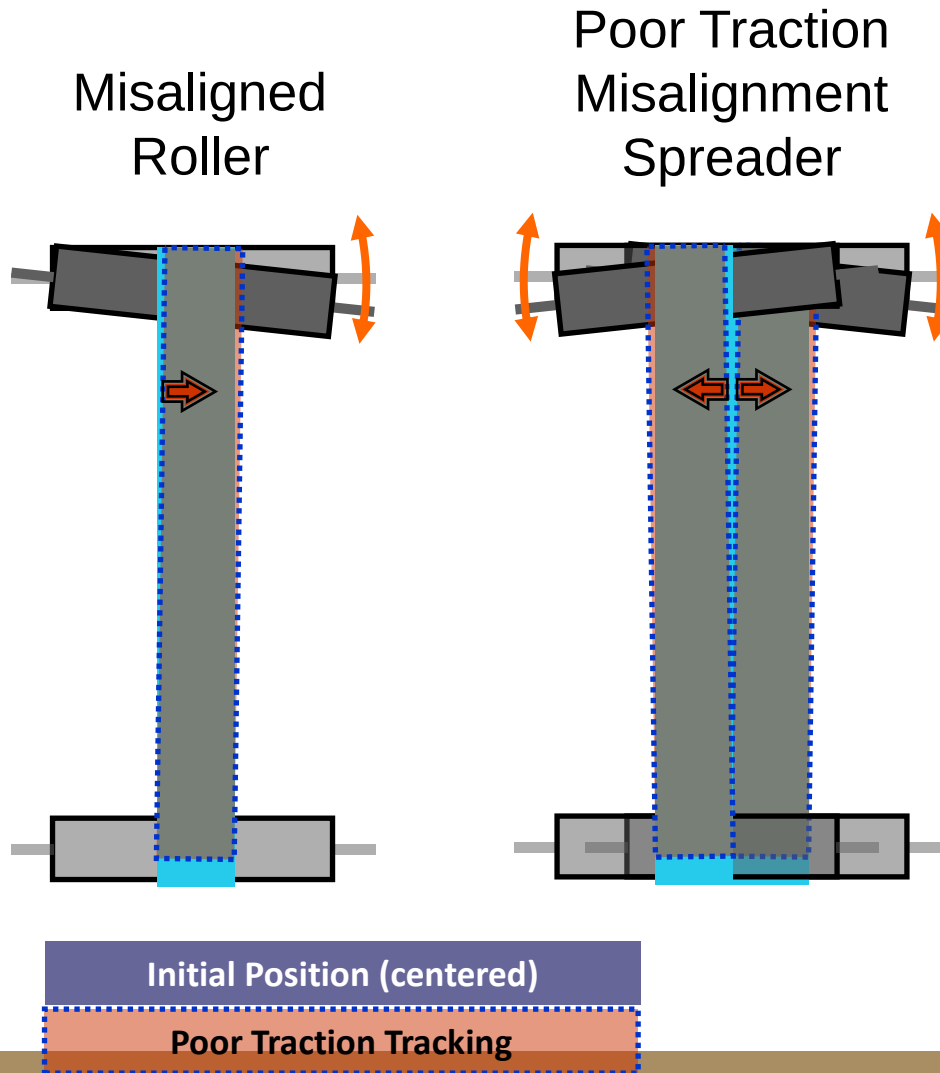


Diameter
Variations



2. Misalignment
+ Slip

Misalignment-Based Spreader



No spreaders are based on misalignment and poor traction.

Summary - Five Tracking Rules

Rollers have **4** ways to shift a web.

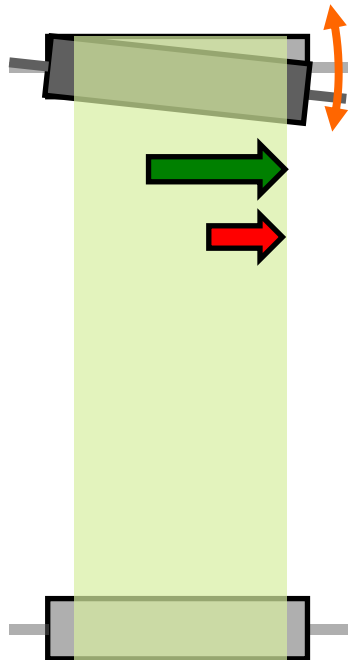
2

1

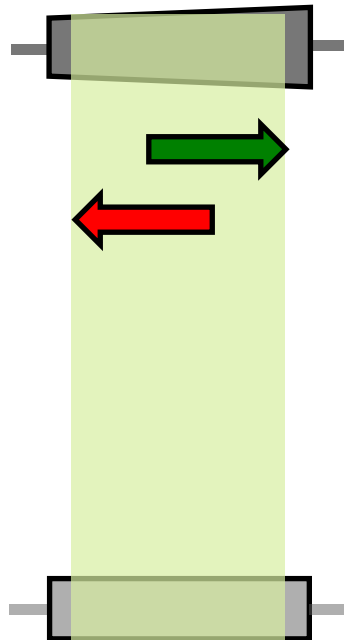
4

3

Misaligned
Roller



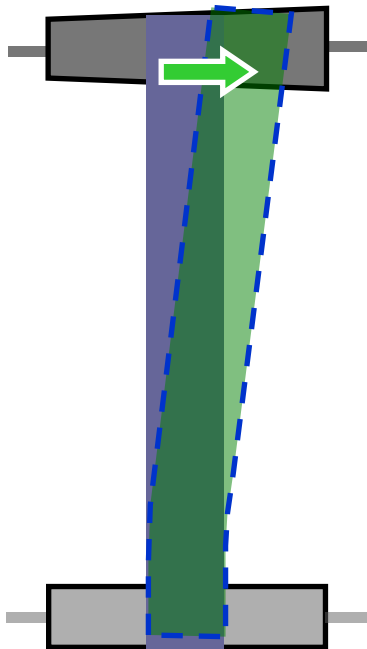
Diameter
Variations



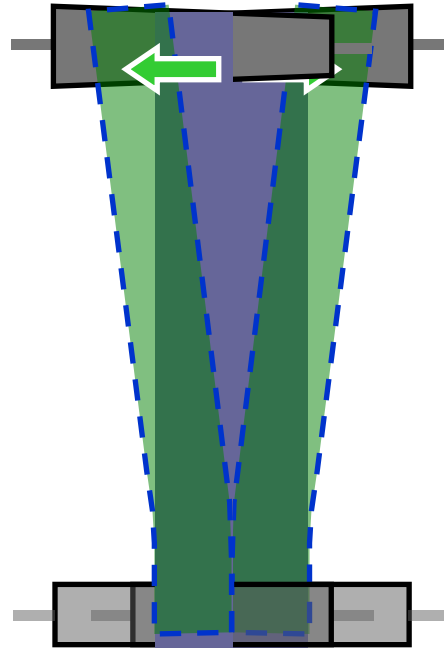
3. Diameter Variations
+ Traction

Diameter-Based Spreaders

Diameter
Variation Roller



Good Traction
Diameter
Spreader



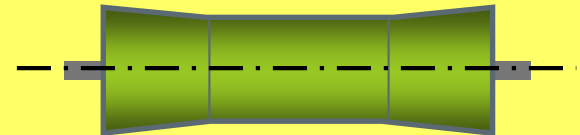
Initial Position (centered)

Good Traction Tracking

Tape Collared
Roller



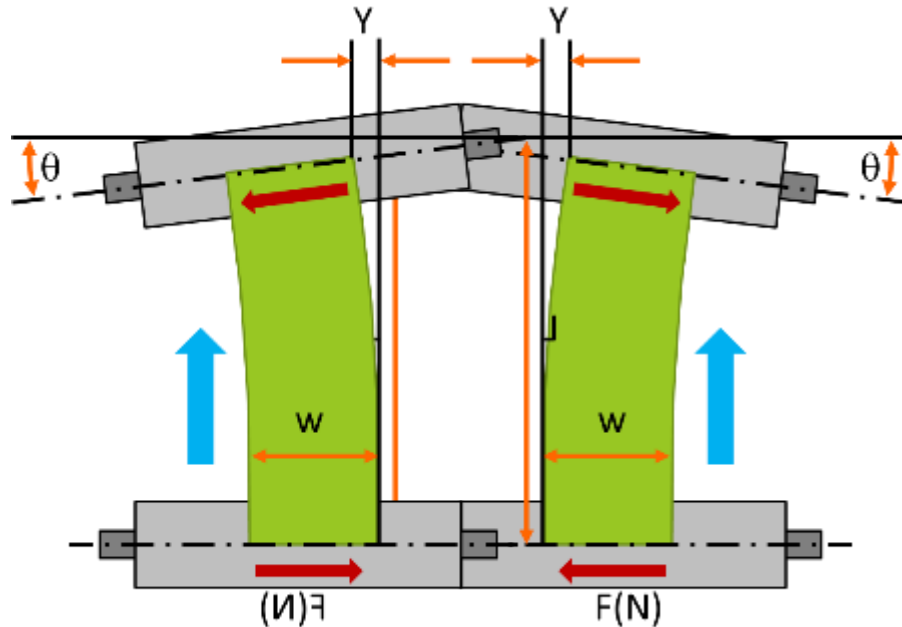
Concave Roller



Also known as:
Reverse crowned roller

Lateral Tracking Comparison

Mechanics of Bowed Rollers

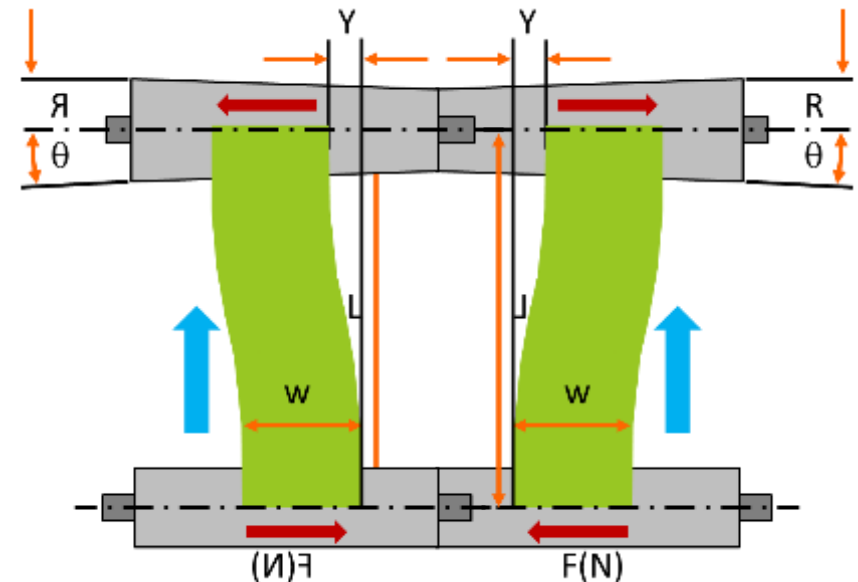


$$F = \frac{2EI\theta}{L^2} = \left(\frac{EtY}{4} \right) \left(\frac{w}{L} \right)^3$$

$$Y(L) = \frac{2}{3} L \theta$$

Concave roller need
4X friction
of bowed
rollers.

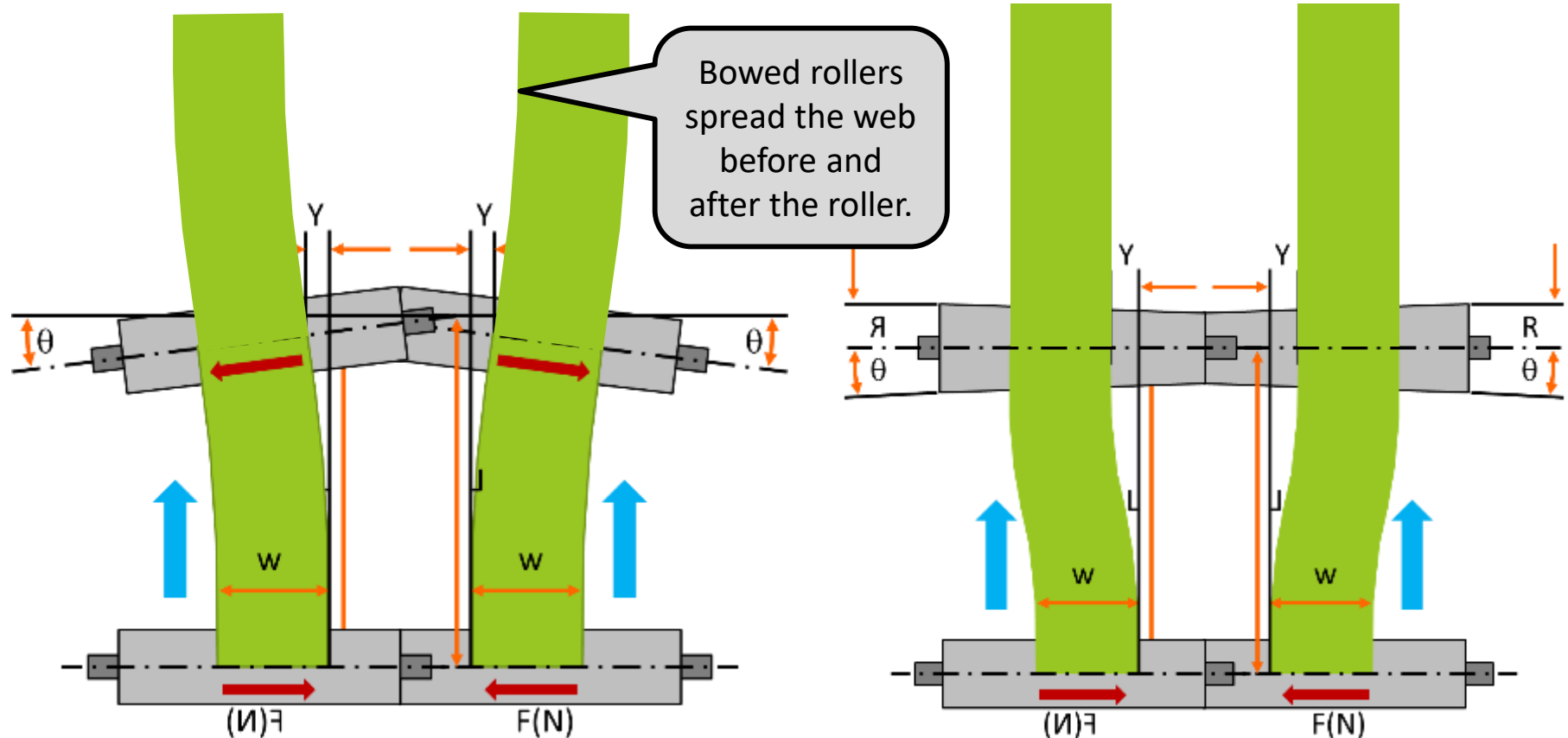
Mechanics of Concave Rollers



$$F = \frac{2EI\theta_c}{RL} = EtY \left(\frac{w}{L} \right)^3$$

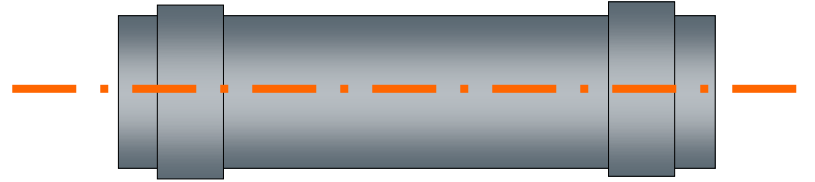
$$Y(L) = \frac{\theta_c L^2}{6R}$$

Lateral Tracking Comparison

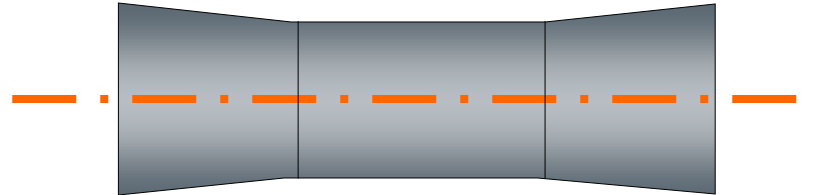


Concave Roller Profile Options

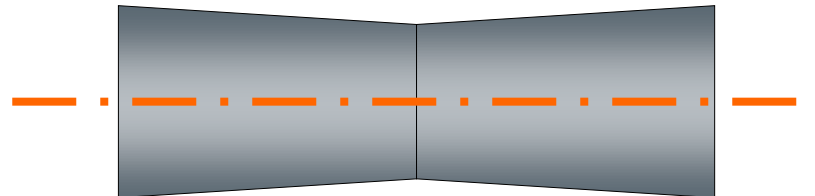
Stepped or Taped



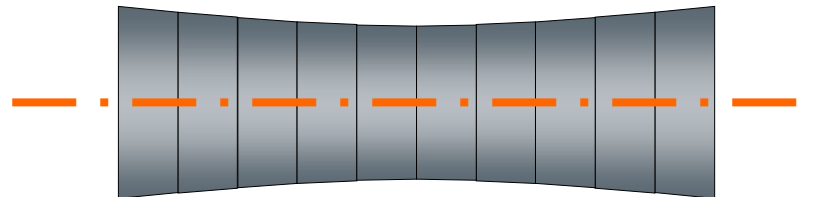
Bow Tie



Hourglass



Parabolic

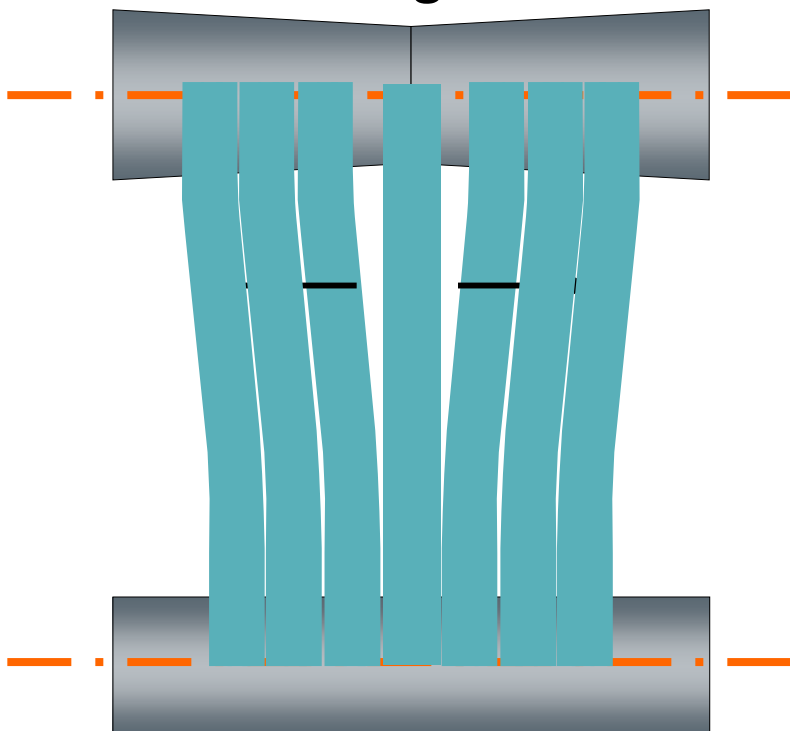


Concave Roller Profile Options

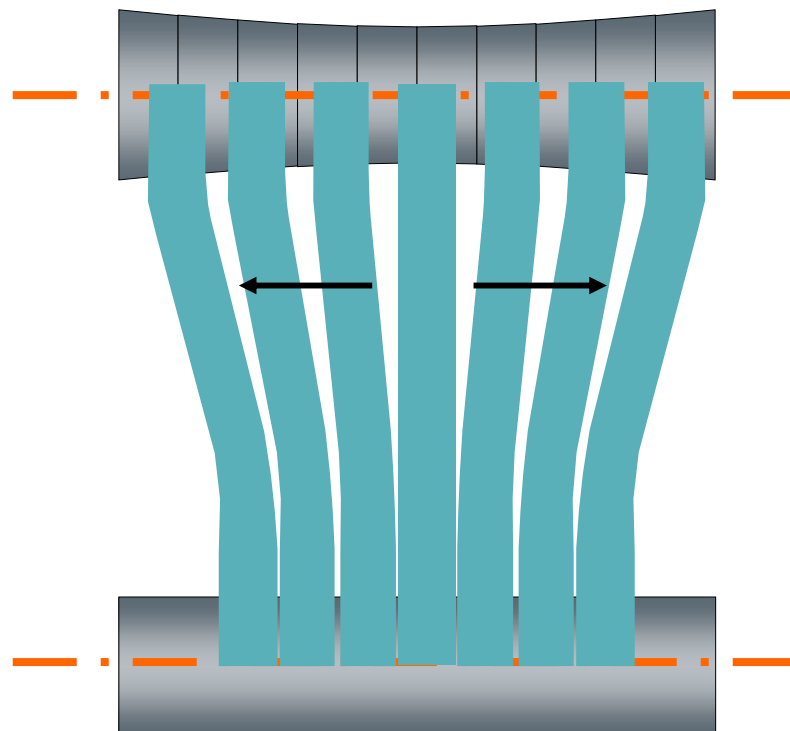
The constant taper angle of an hourglass profile shifts all lanes equally: strong central spreading, but weak spreading between lanes or slit strands.

The progressively changing taper angle of a parabolic profile shifts all lanes nearly equally: good spreading at all widthwise positions, spreading all lanes or slit strands.

Hourglass



Parabolic



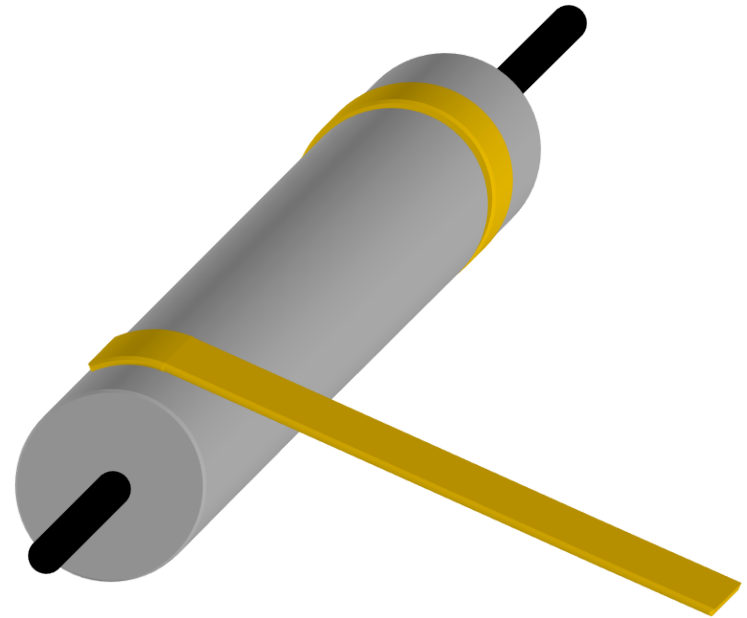
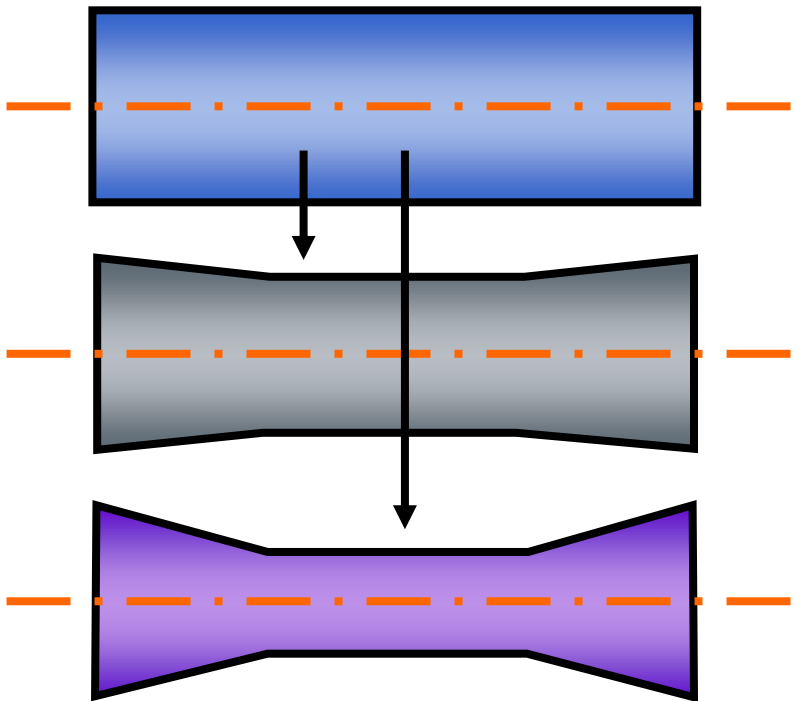
Concave Profile Details

Profile	Spreading	Radial Profile	Advantages/Disadvantages
Stepped / Taped	Spreads edges, hopefully pulling entire width taut.	3-10x web strain	Creates highest edge strain. May wrinkle at shear of abrupt high to low tension step change. Does not adapt to width changes.
Bow Tie	Spreads edges, hopefully pulling entire width taut.	3-10x web strain	Avoids wrinkling and shear of stepped/taped rollers. Taper may run 20-30% of roller width, allowing effectiveness over moderate width changes.
Hourglass	Each lane on each web half shifts laterally equal amounts.	1.5-2x web strain	Proven more effective than bow tie profile in some tests. Positive spreading for all web widths. Web halves would spread apart, but multiple strands side of center may not spread relative to each other.
Parabolic	Lanes of web spread progressive more from center to edges.	2-2.5x web strain	All lanes spread and move away from each other. Multiple slit strands would each open gaps between them.

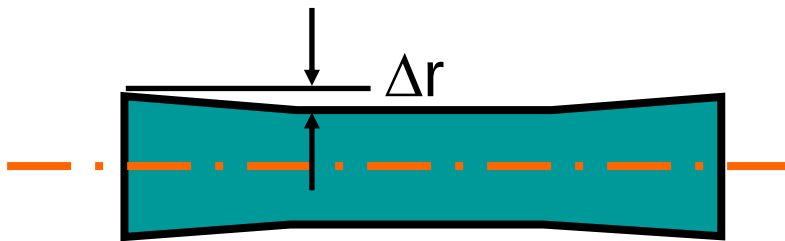
Concave Rollers

A reverse crowned roller can be machined into the roller cylinder...

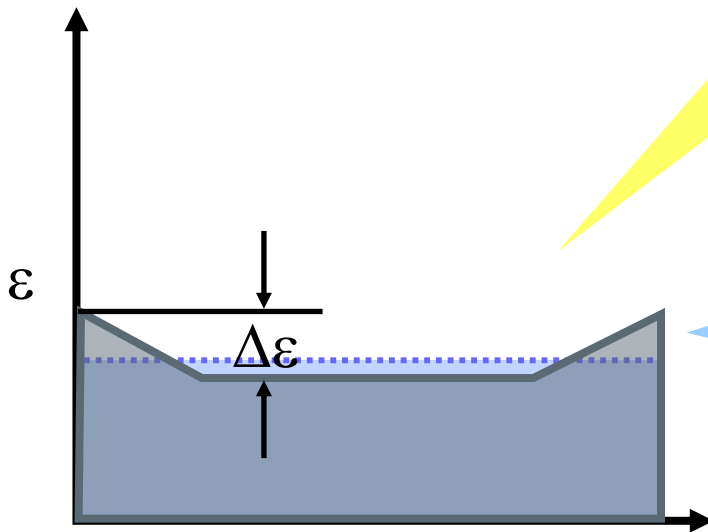
...or created by applying tape to a standard cylindrical roller.



Concave Rollers



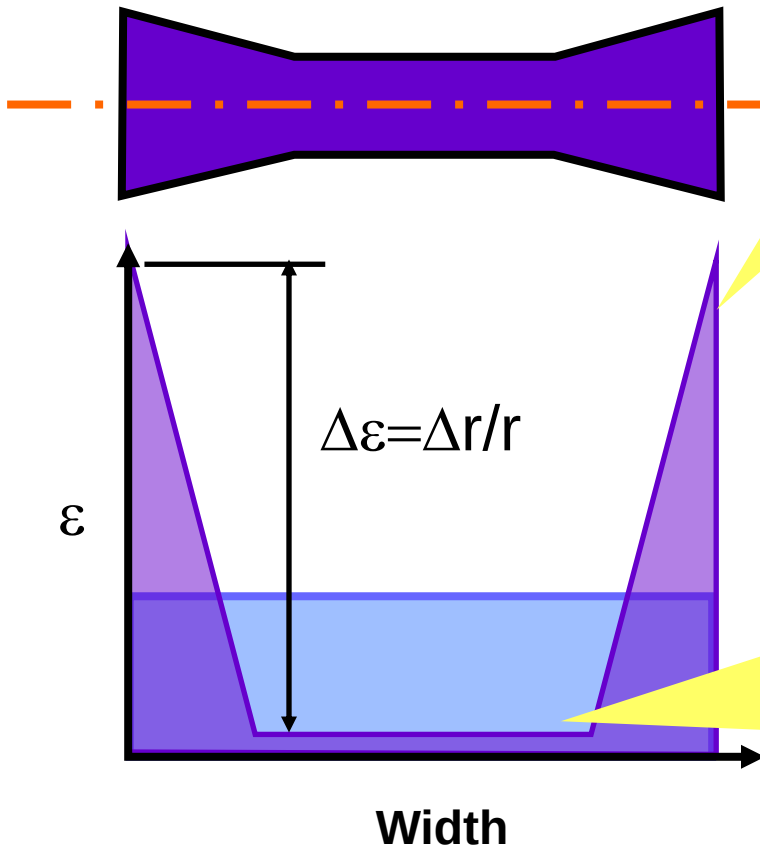
A cylindrical roller pulls with uniform strain and tension.



A concave roller pulls with strain and tension varied proportional to the diameter variations.

Width

Reverse Crowned Rollers



Greater percent diameter variations create greater strain and tension variations.

Be careful not to drive the center to slackness or the edge to **break** or **deformation** stresses.

Summary - Five Tracking Rules

Rollers have **4** ways to shift a web.

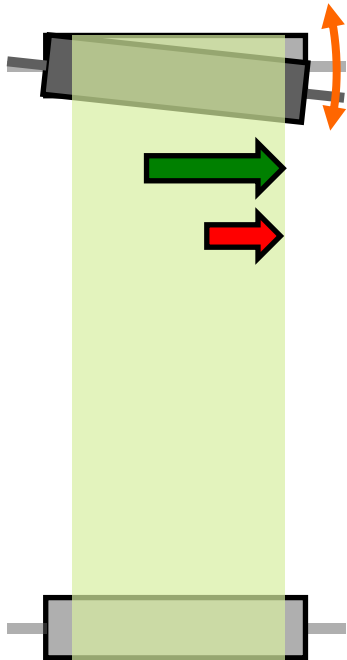
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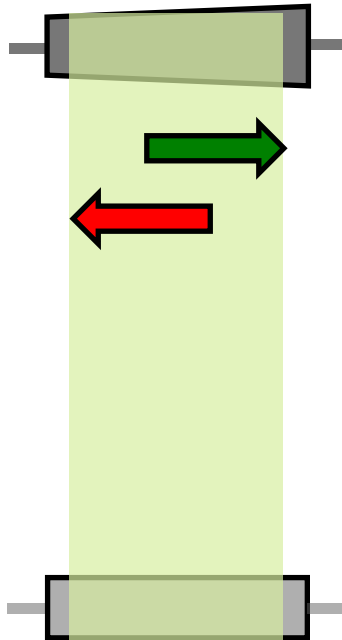
4

3

Misaligned
Roller



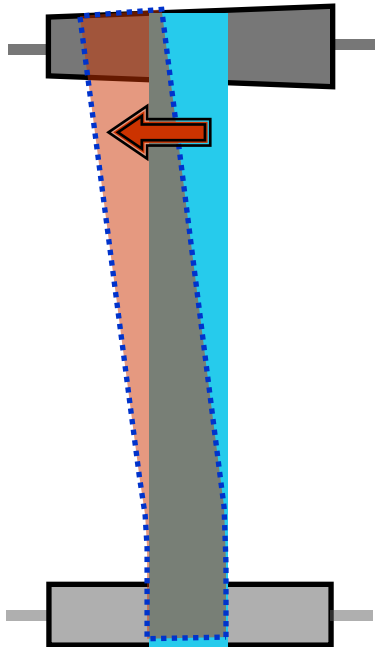
Diameter
Variations



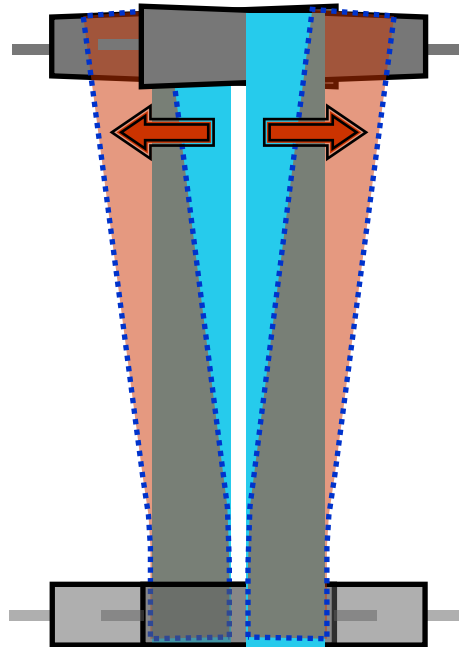
4. Diameter Variations
+ Slip

Misalignment-Based Spreader

Diameter
Variation Roller



Poor Traction
Diameter
Spreader



Initial Position (centered)

Poor Traction Tracking

Curved Drag Bar



Hyper-Crowned
Roller



Old Bowed Roller
used as Drag Bar



“D” Bar

D-bars create a tension variation due to the web path deflection.

The induced moment bends the web outward from the center.

Since the bar is not rotating, the web does not follow the parallel entry rule.

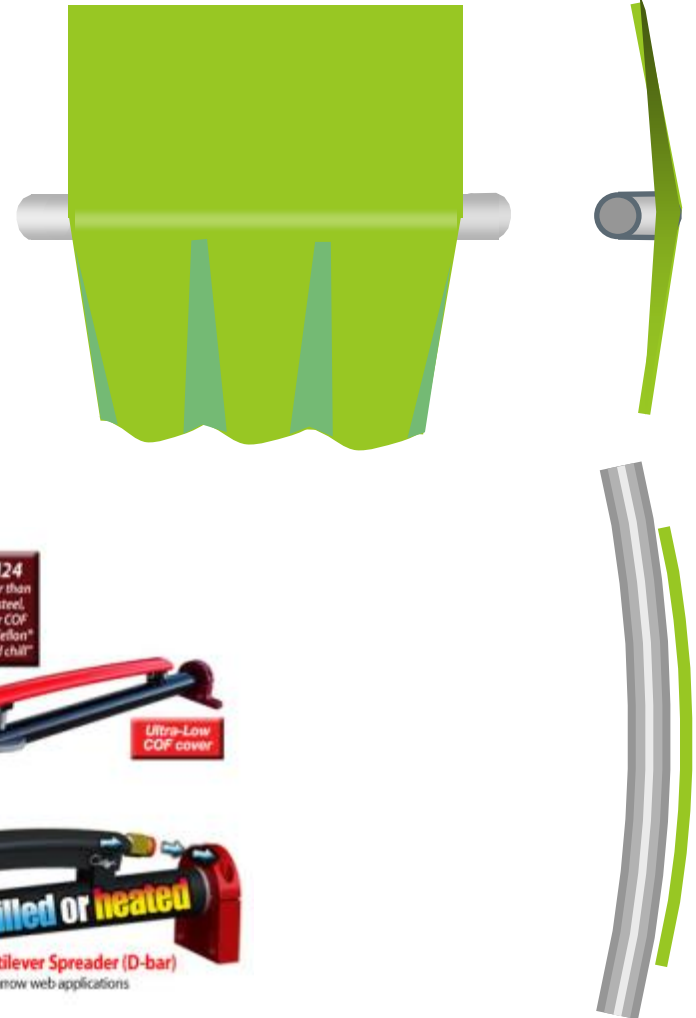


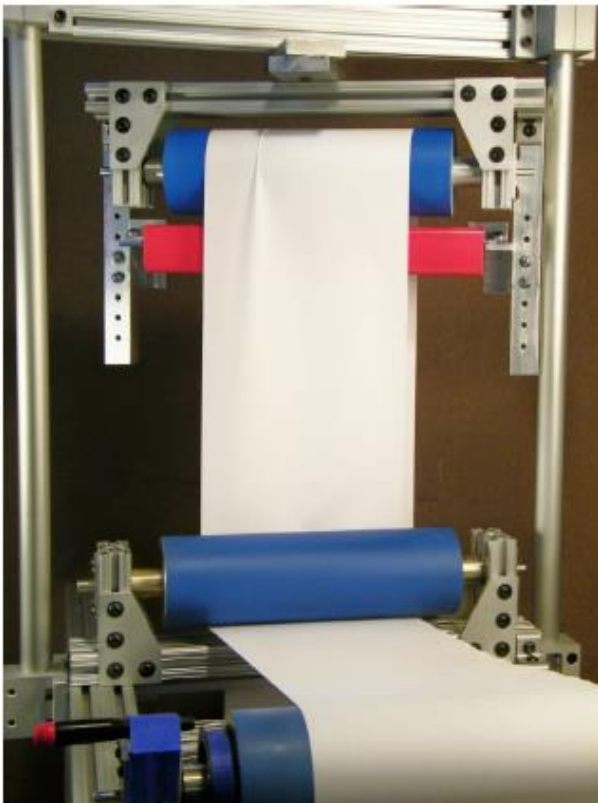
Image courtesy of Pearl Technologies

Is There Any Science for D-bars and Bent Pipes

Jerry Brown
Essex Systems

APPLIED WEB HANDLING
CONFERENCE 2010

Test setup



- Twist angle = 21°
- Span = $13 \frac{3}{4}$ in
- Width = 7 in
- Tension = 1 pli
- Modulus = "high" in the neighborhood of 0.5M psi
- Roller radius = 1.5 in
- Bar $3 \frac{5}{16}$ below roller centerline
- Bar height = For curved bar, $\frac{3}{4}$ in on right end and 1 in on left (to get full contact with curve). For straight bar, $\frac{3}{4}$ in at both ends.

Spreading with the bowed bar

Bowed bar disengaged



Bowed bar engaged



Curved bar engaged.jpg

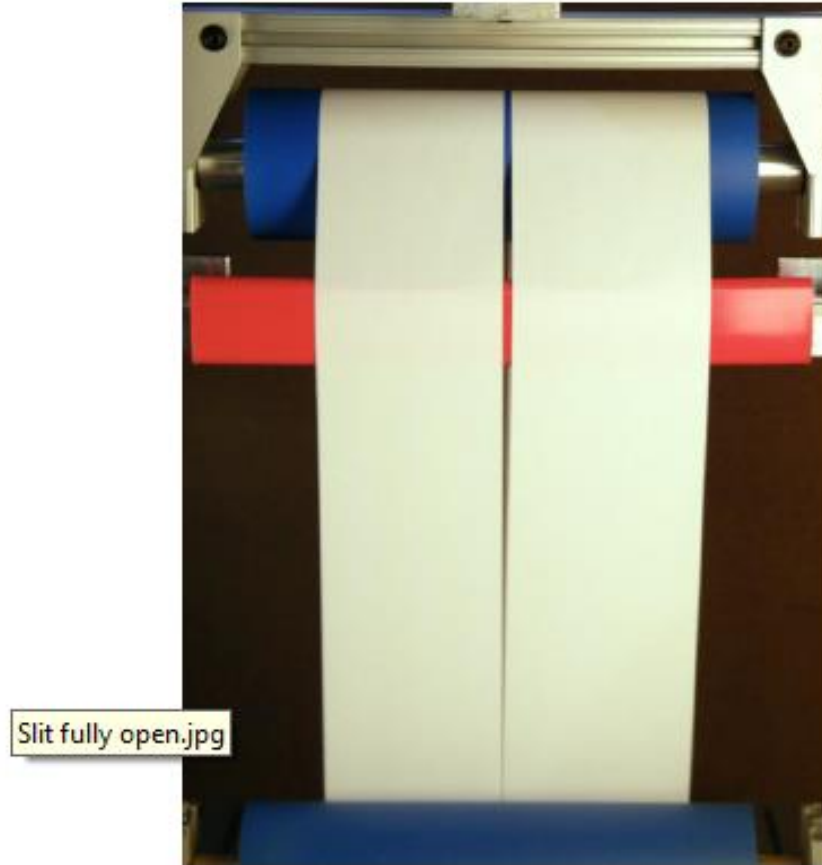
Is the bar producing positive lateral stress?

The Swanson slit test

Slit approaching bowed bar

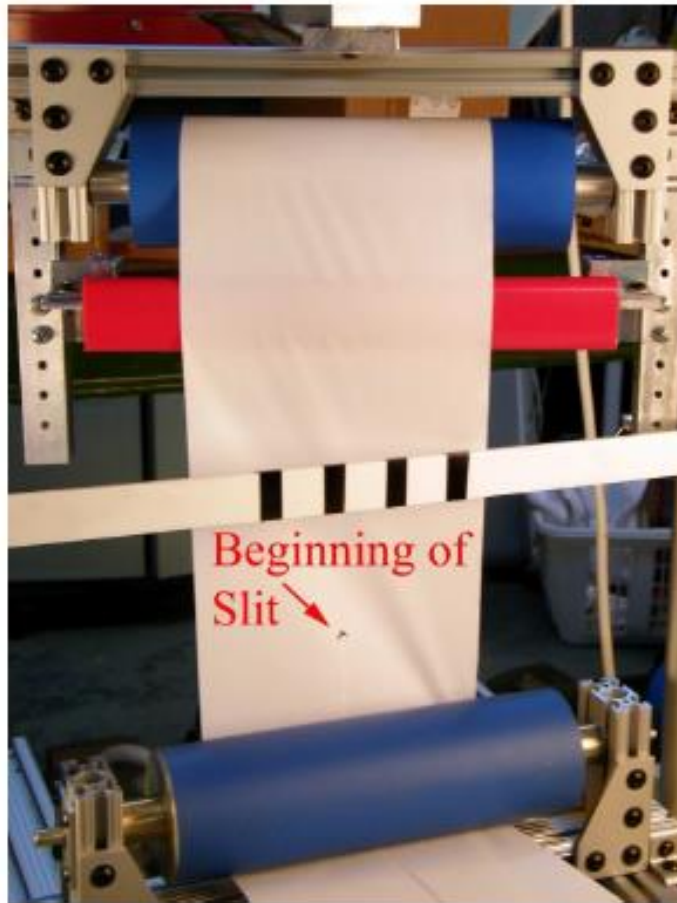


Slit fully opened after running a while

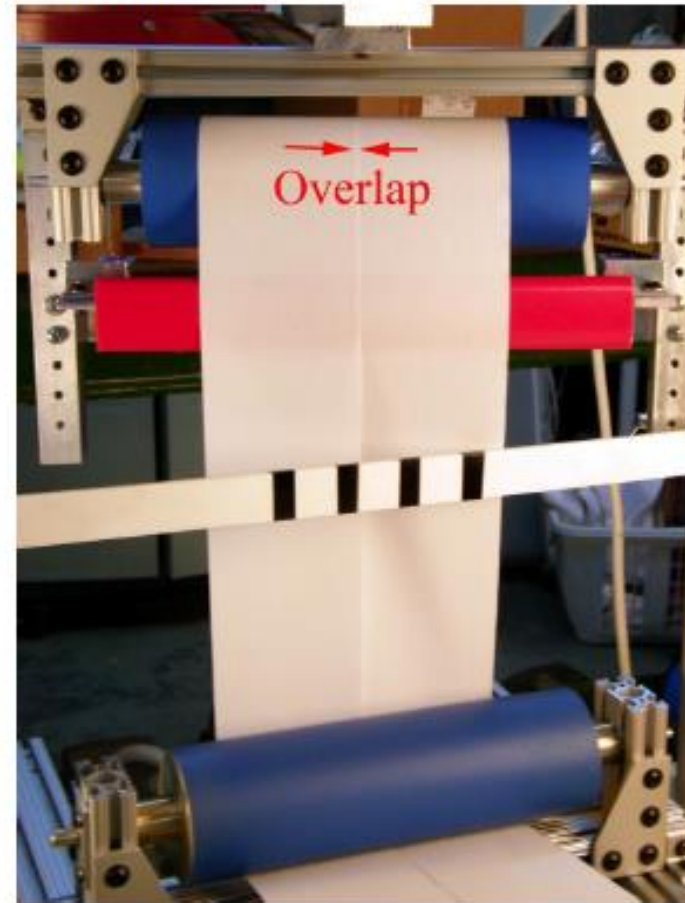


Does a straight bar separate a slit?

Slit approaching straight bar



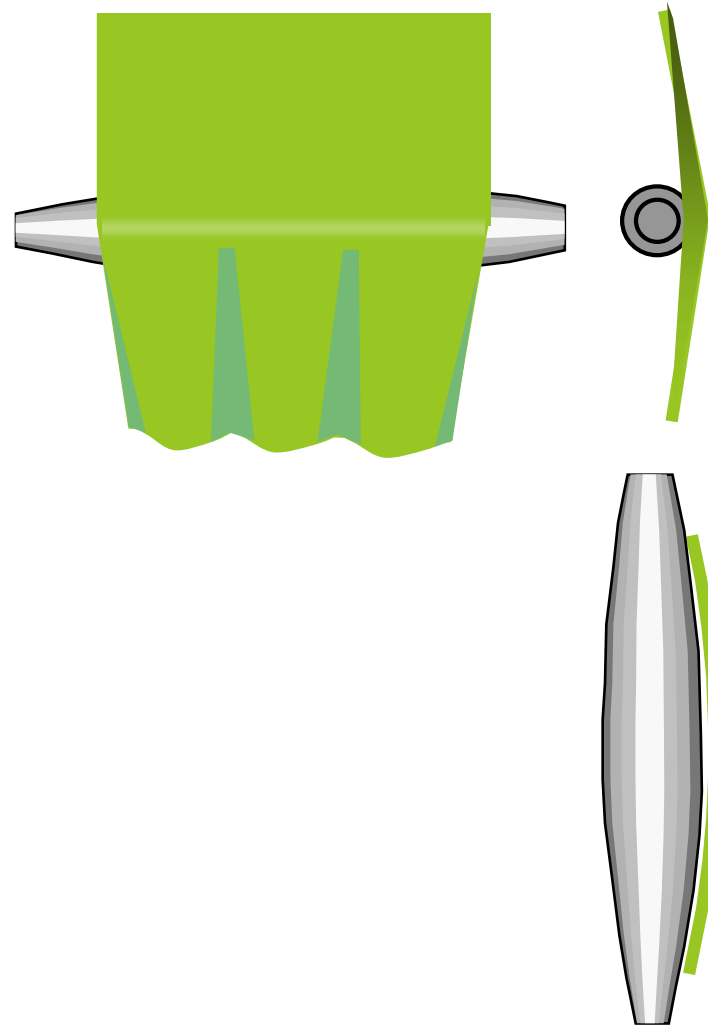
Slit overlapped after running several span lengths.



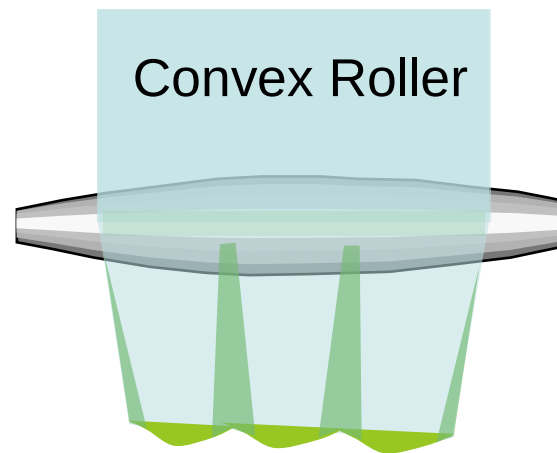
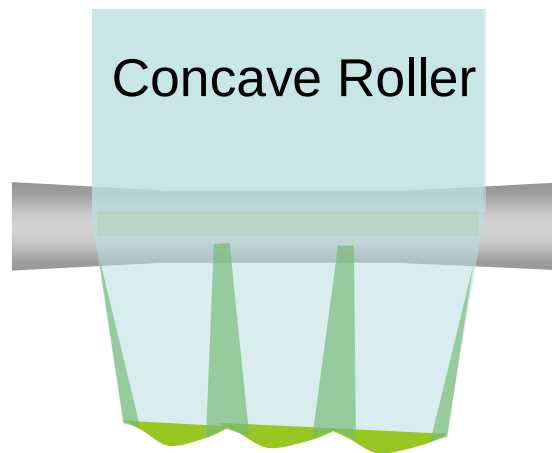
Hyper-Crowned Rollers

A hyper-crowned roller has so much diameter variation, the web is forced into slip.

The hyper-crowned (or barrel roller) works like a d-bar, spreading the web via induced moment bending.



Concave and Convex Rollers



Mechanism

Induced bending
and parallel entry

Induced bending

Diameter
Variation

2-10 times web
strain

10-100 times web
strain

Preferred
Wrap Angle

Low to high, but ensure
no web-roller slip

Low

Preferred
Traction

Moderate to high

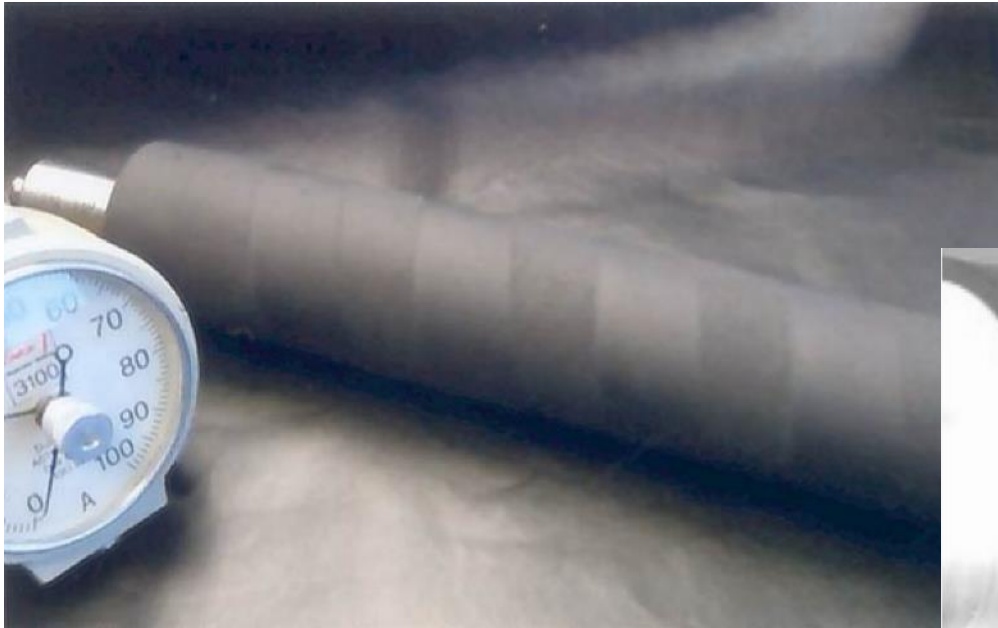
Low to moderate

Question

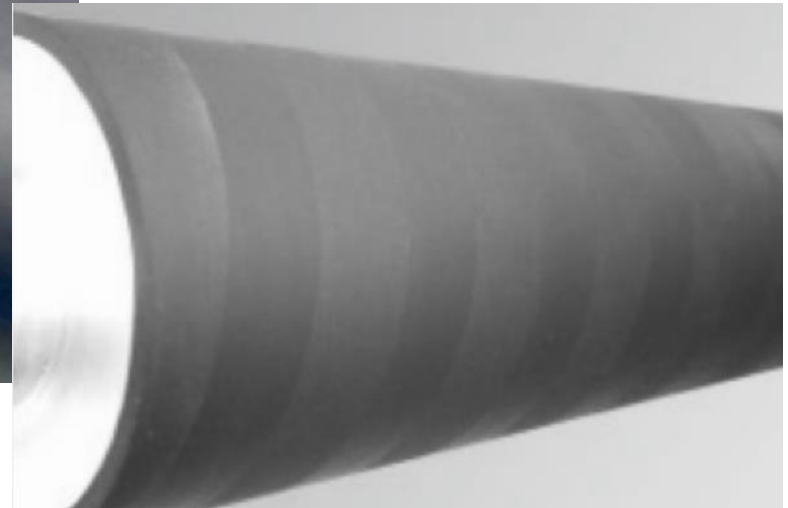
Are there other
anti-wrinkle roller
options?



Variable Traction Roller



I have not tested this roller to prove effectiveness.



The roller's rubber surface is modified to create a spiral pattern of high and low traction bands. The anti-wrinkle or spreading mechanism is unidentified.

Images courtesy of Mitex (Germany)

Taped Rollers

◆ Standard Rollers (Cylindrical)



◆ Tape Collars on Standard Rollers



◆ Spiral Tape on Standard Rollers

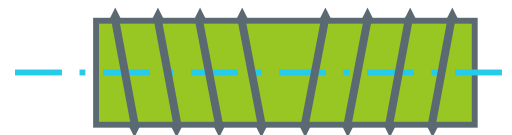


Spiral and Ridged Rollers

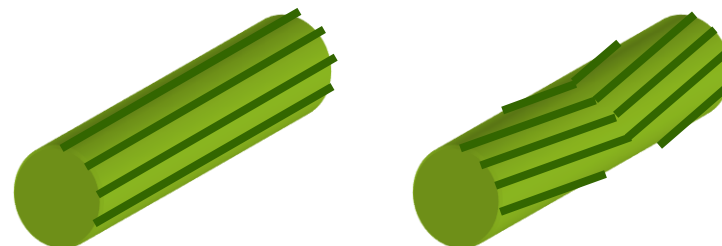
◆ Spiral Tape Rollers



◆ Spiral-Ridged Roller



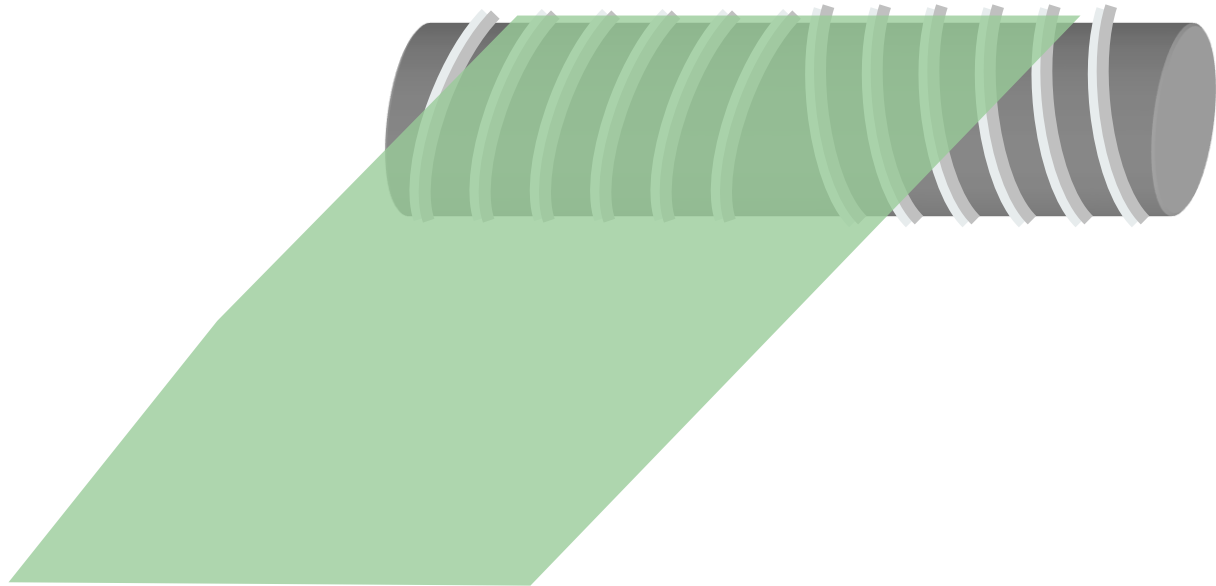
◆ Axially-Ridged Roller



Spiral Ridged Roller

Spiral rollers feature hard machined ridges in a helical pattern from the center outward.

The exact mechanism of spiral rollers is debatable.

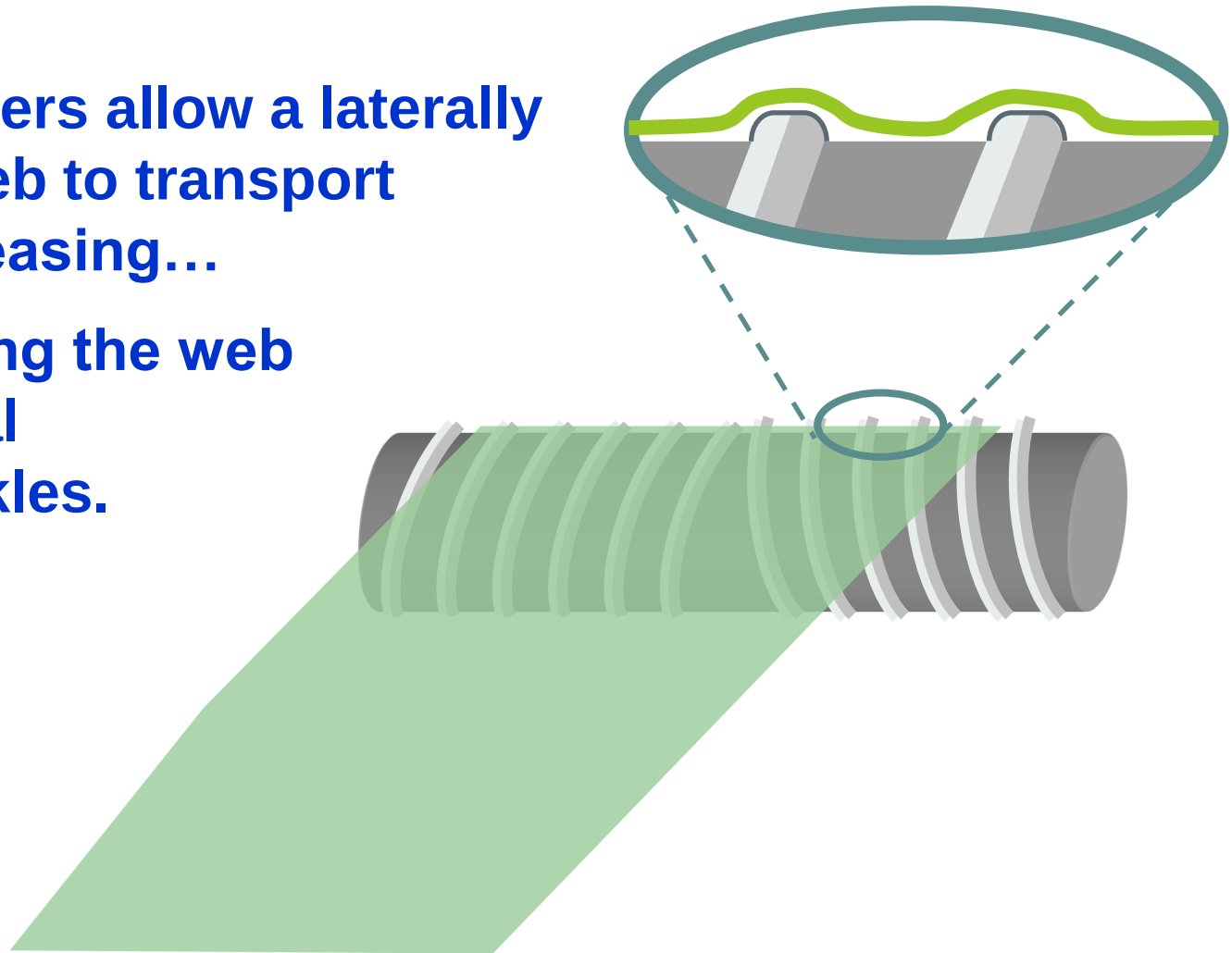


Spiral Ridged Roller

Benefit:

Ridged rollers allow a laterally buckled web to transport without creasing...

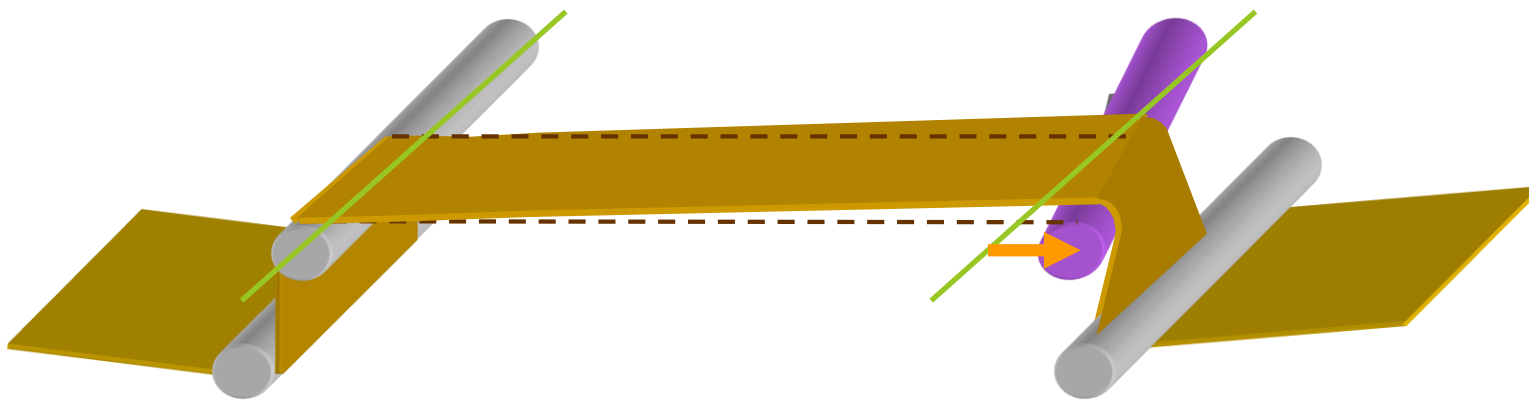
...by forming the web into several micro-buckles.



Testing Anti-Wrinkle Rollers

Test #1

Compare the misalignment required to create a shear wrinkle of standard vs. “anti-wrinkle” roller.

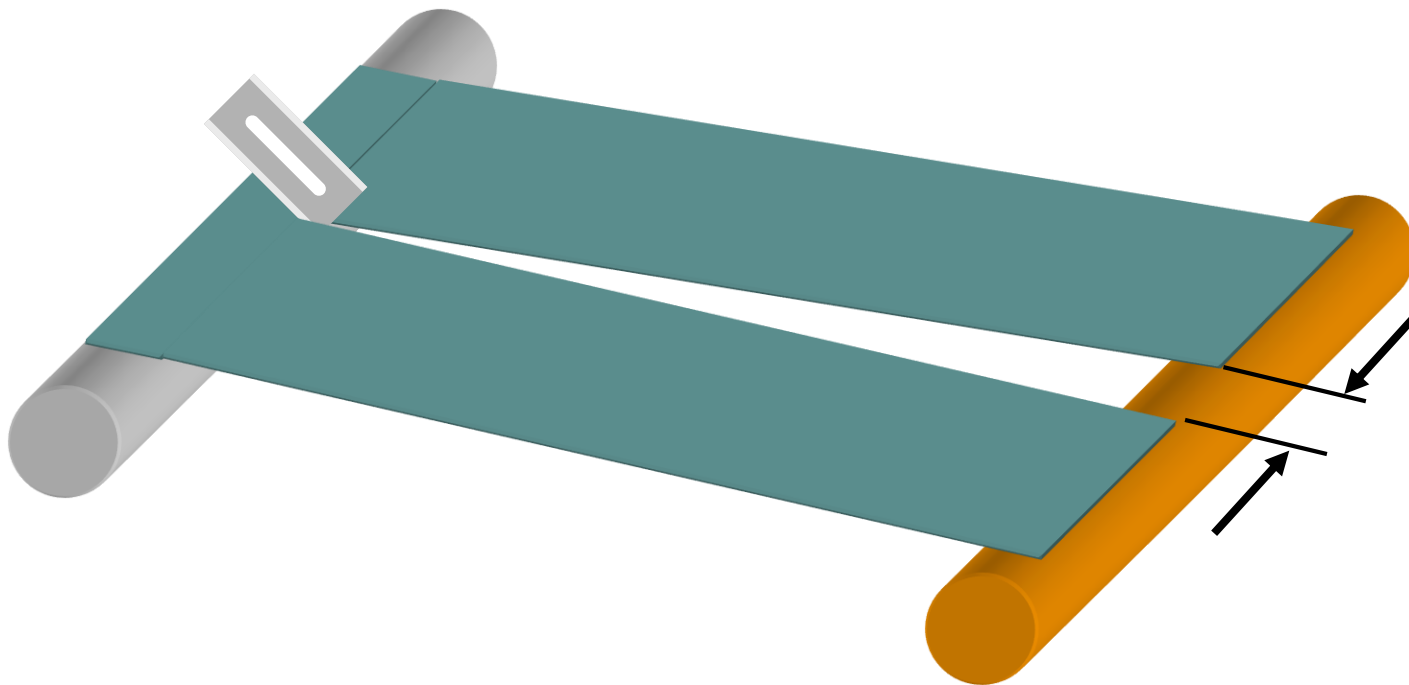


R. P. Swanson, *Testing and Analysis of Web Spreading and Anti-Wrinkle Devices*, Proceeding of the Fourth International Conference on Web Handling, pp. 414-428, 1997

Testing Spreading Rollers

Test #2

Slit the web at the beginning of the upstream span and measure the slit gap at the target roller.



Rollers Evaluated in "Swanson 1997"

- ▶ Standard Idler



- ▶ Outward Spiral Tape



- ▶ Inward Spiral Tape



- ▶ Low Friction Tape



- ▶ Tape Collars



- ▶ Reverse Crown



- ▶ Bowed Roller



- ▶ Expanding Surface



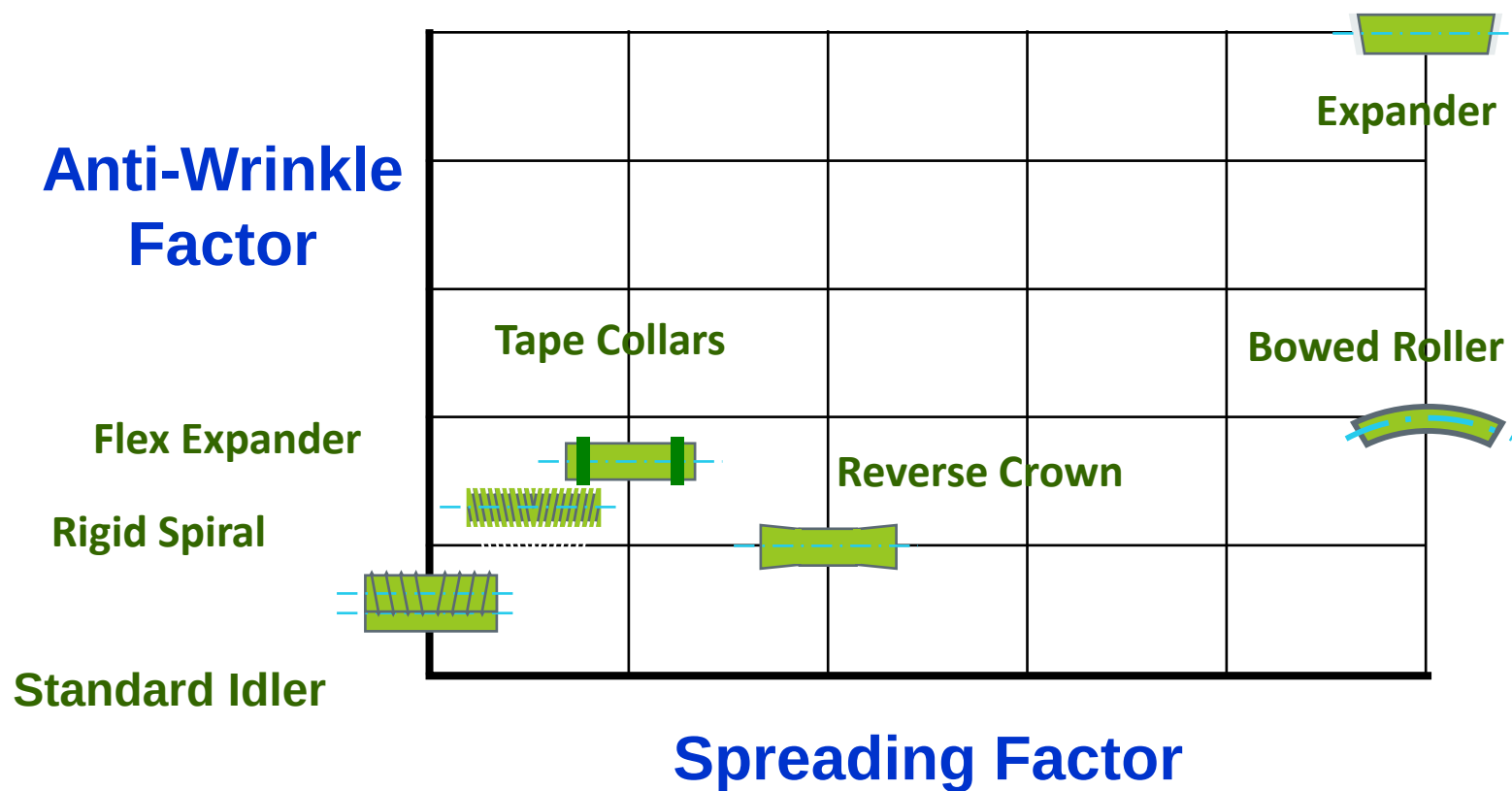
- ▶ Flex Expander



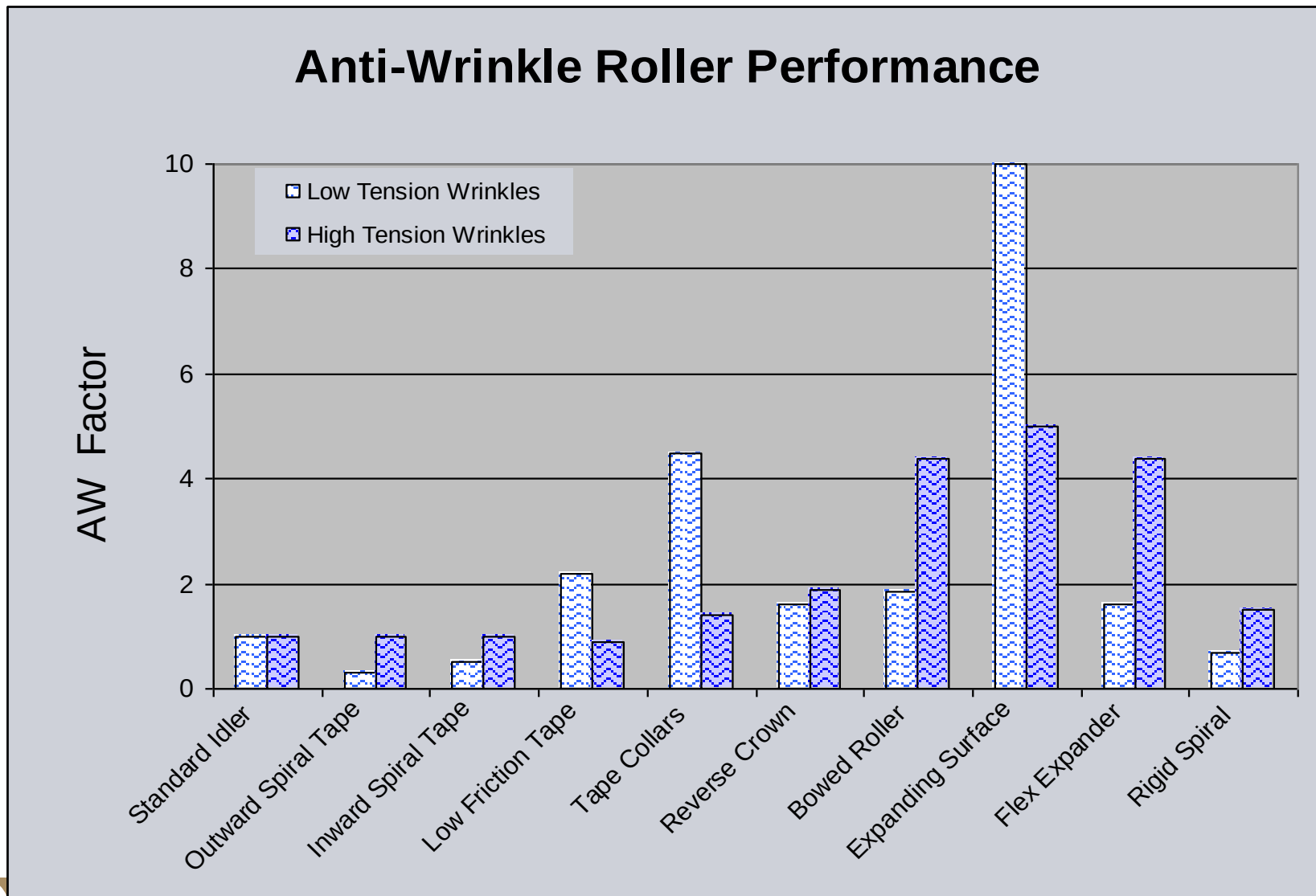
- ▶ Rigid Spiral



Anti-Wrinkle? Spreader?



Anti-Wrinkle Performance



Anti-Wrinkle Performance

Reverse crown had positive effect at low and high tensions.

Bowed roller were better at high tensions.

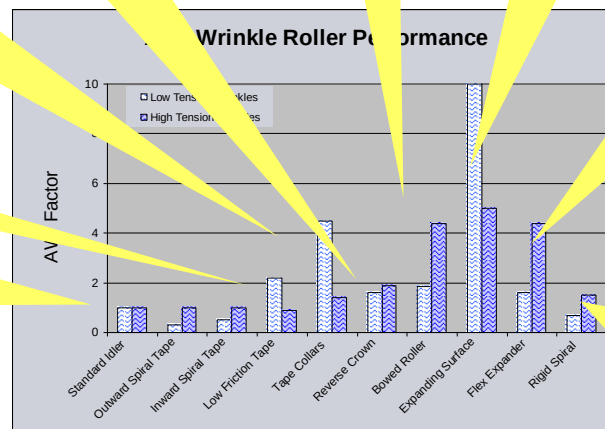
Expanding surface rollers had the top anti-wrinkle performance!

Tape collars had a strong benefit for low tension wrinkles.

Flexible expanders were as good as bowed rollers.

Low friction tape helped in low tension wrinkle relief.

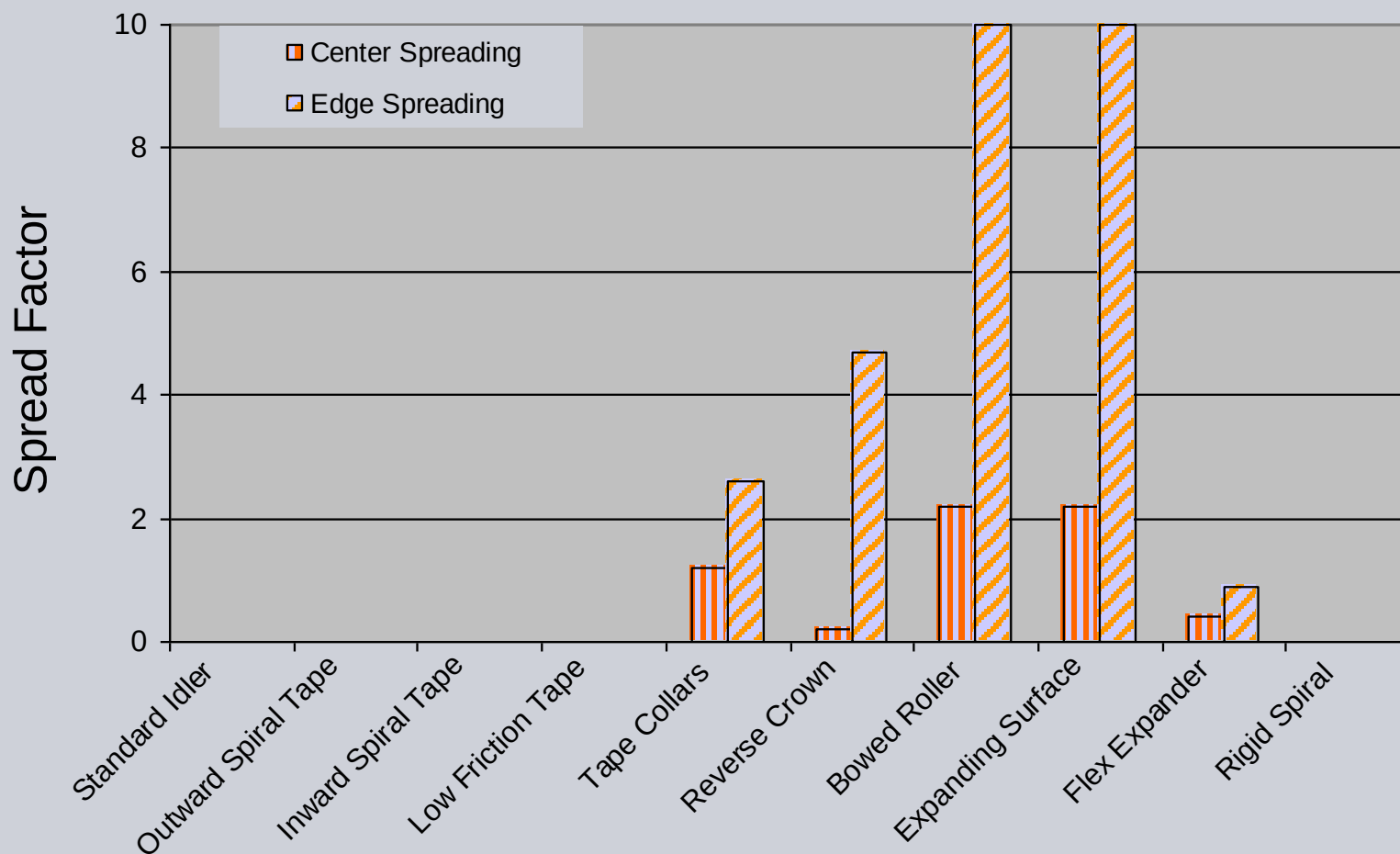
Spiral tape had a negative or zero effect, regardless of direction.



Rigid spiral rollers had small anti-wrinkle benefit for high tensions.

Spreading Performance

Anti-Wrinkle Roller Performance

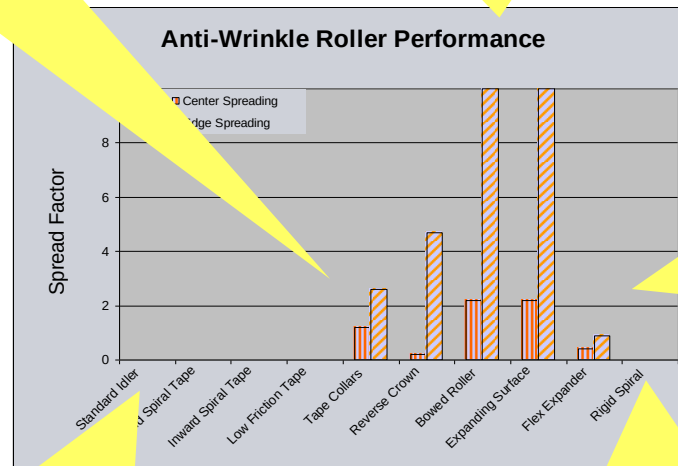


Spreading Performance

Tape collars and reverse crown have more spreading at edges.

Bowed and expanding surface rollers had similar powerful spreading action.

The flex expander was a weak spreader (though a strong anti-wrinkle roller).



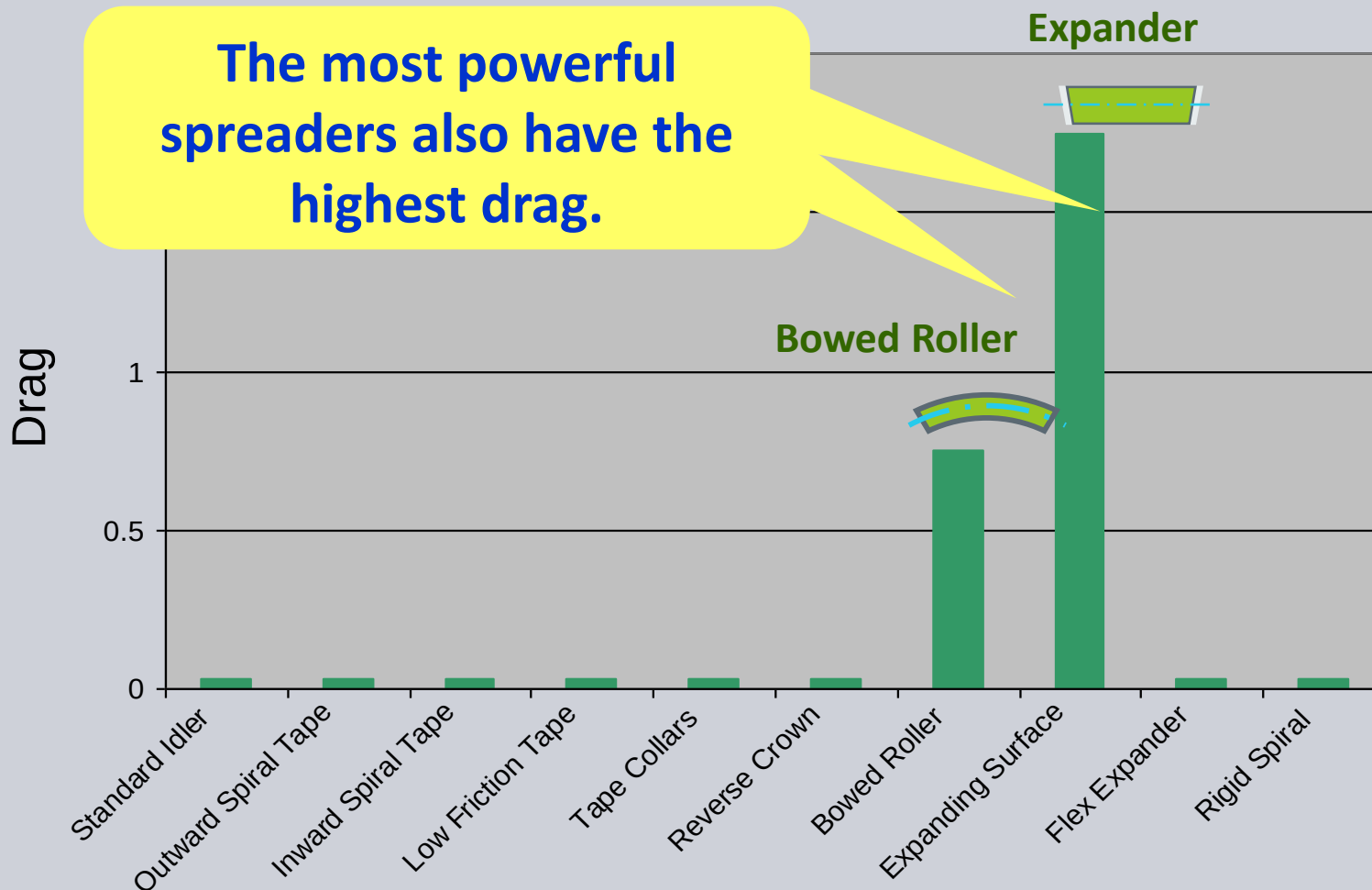
Spiral tape, either direction, and low friction tape did not *spread* the web.

The rigid spiral had zero spreading effect.

Spreader Roller Drag

Anti-Wrinkle Roller Performance

The most powerful
spreaders also have the
highest drag.



AW-Spreaders 2009



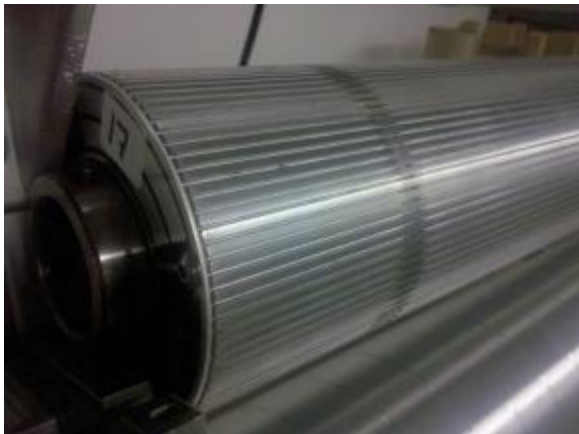
Cylindrical



Flex Expander #1



Flex Expander #2



Concave, Grooved

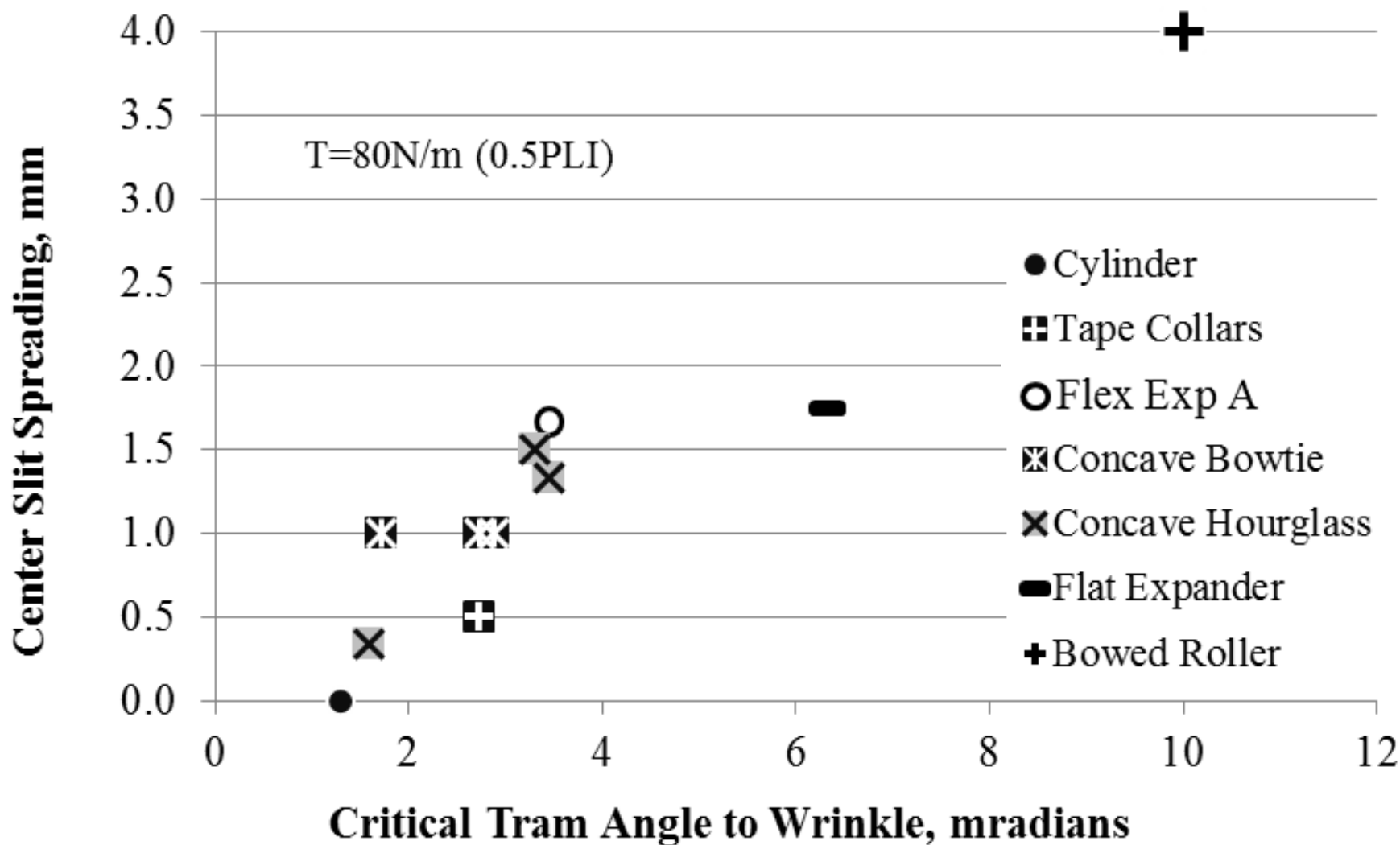


Flat Expander

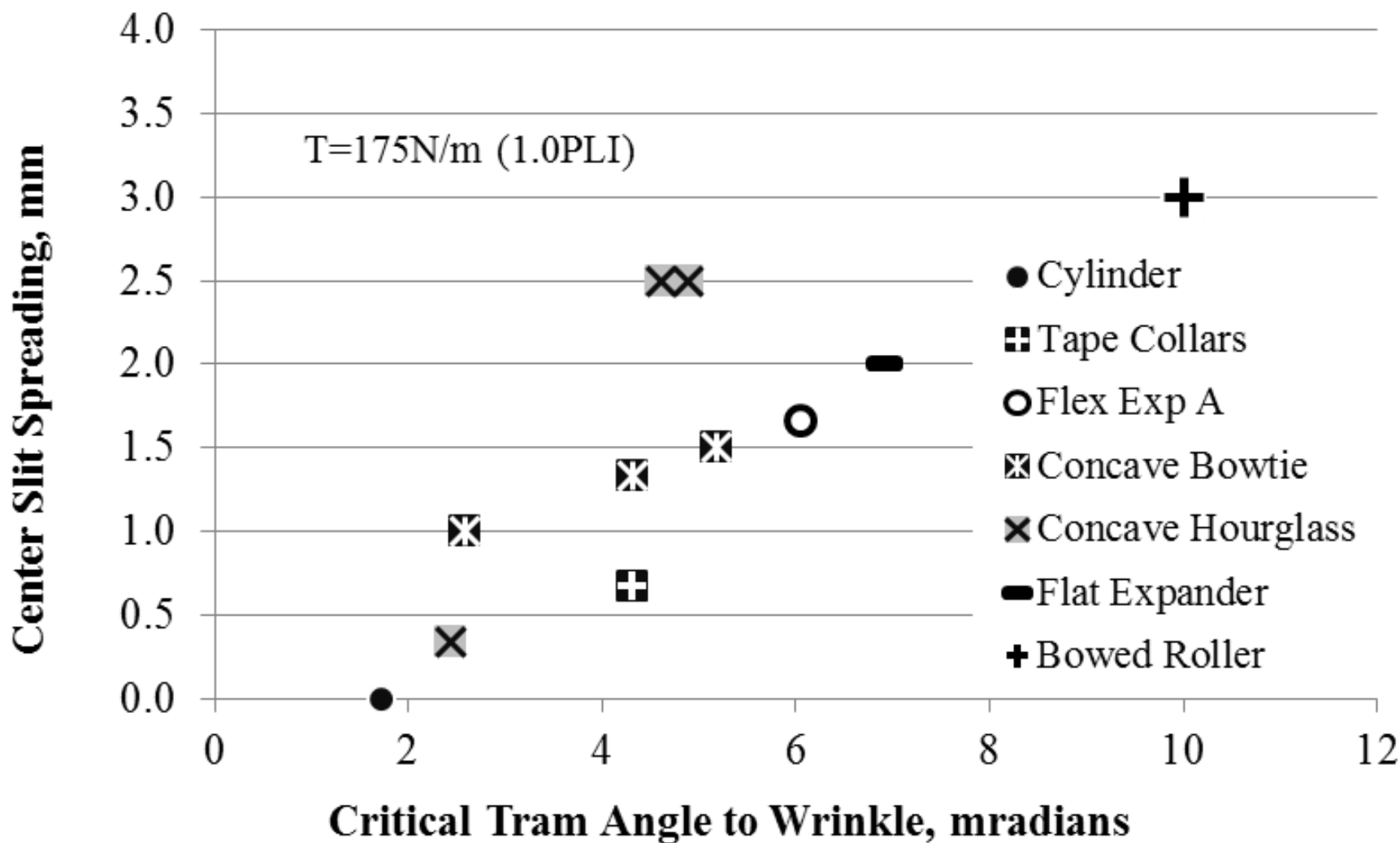


Bowed

Anti-Wrinkle vs. Spreading Test Results by Roller, T = 80 N/m

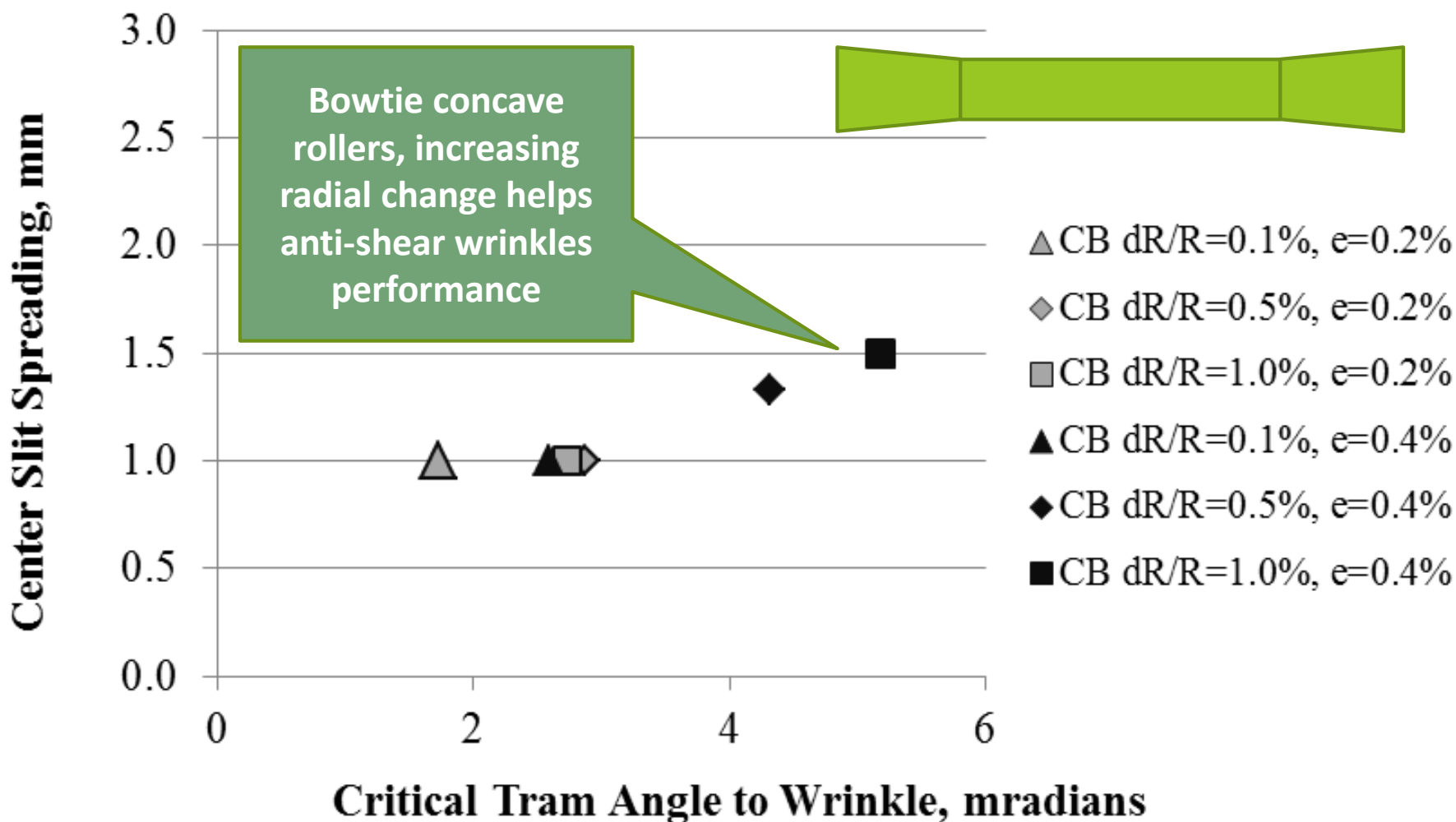


Anti-Wrinkle vs. Spreading Test Results by Roller, $T = 175 \text{ N/m}$



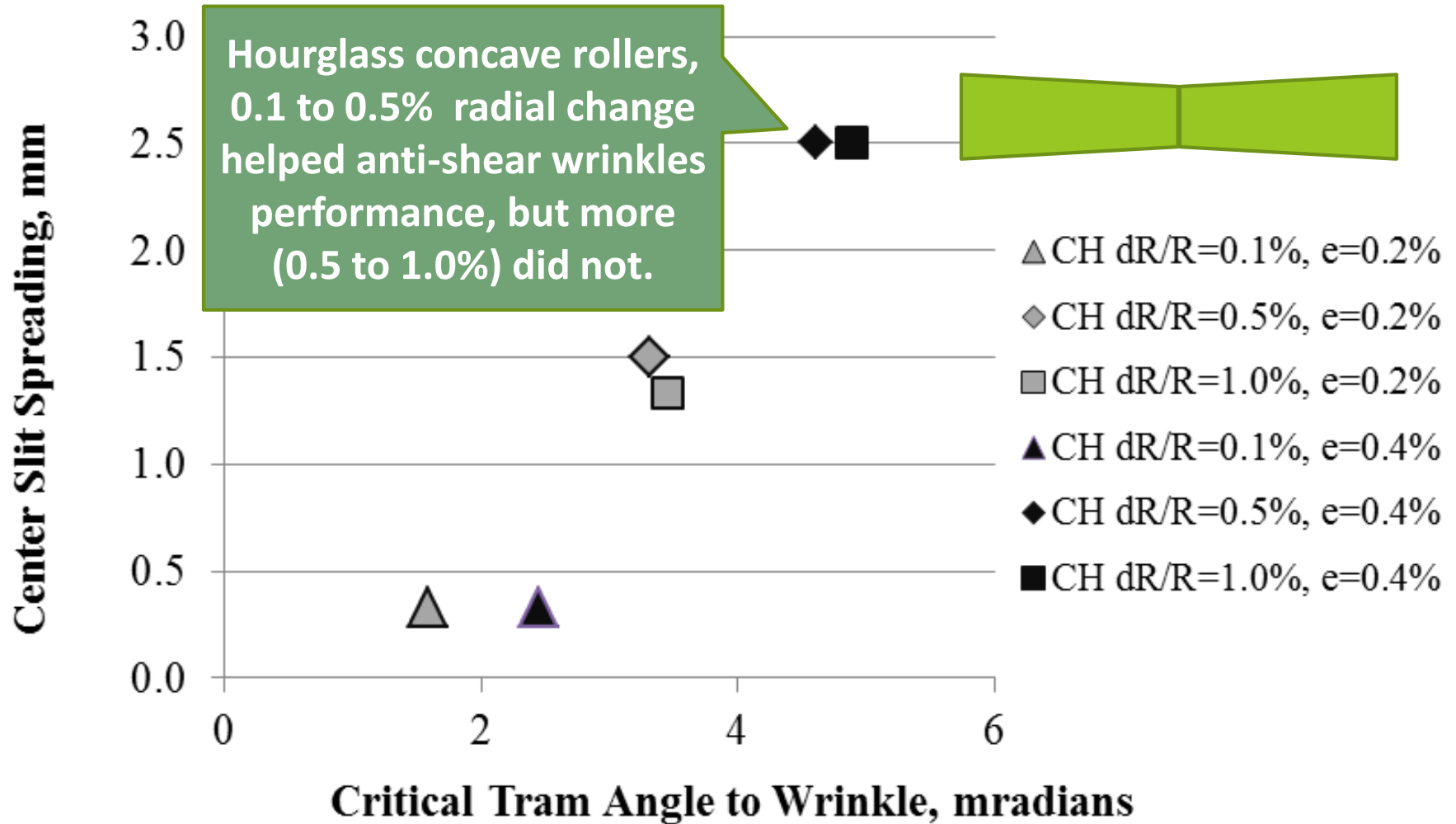
Bowtie Concave Roller

Anti-Wrinkles vs. Spreading Performance



Hourglass Concave Roller

Anti-Wrinkles vs. Spreading Performance



Question

How do I use this information to select the best AW roller for my process?

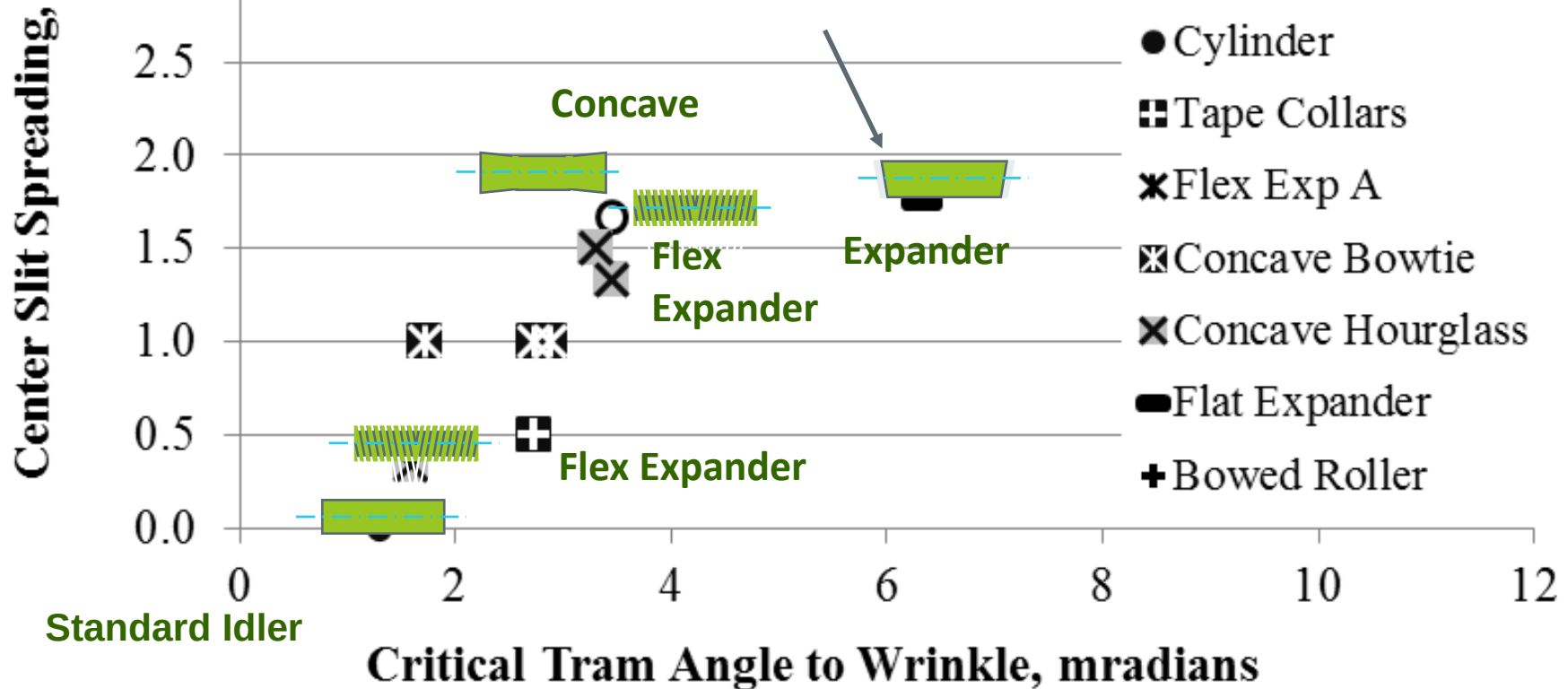


Anti-Wrinkle vs. Spreading Test Results by Roller,

$T = 80 \text{ N/m}$

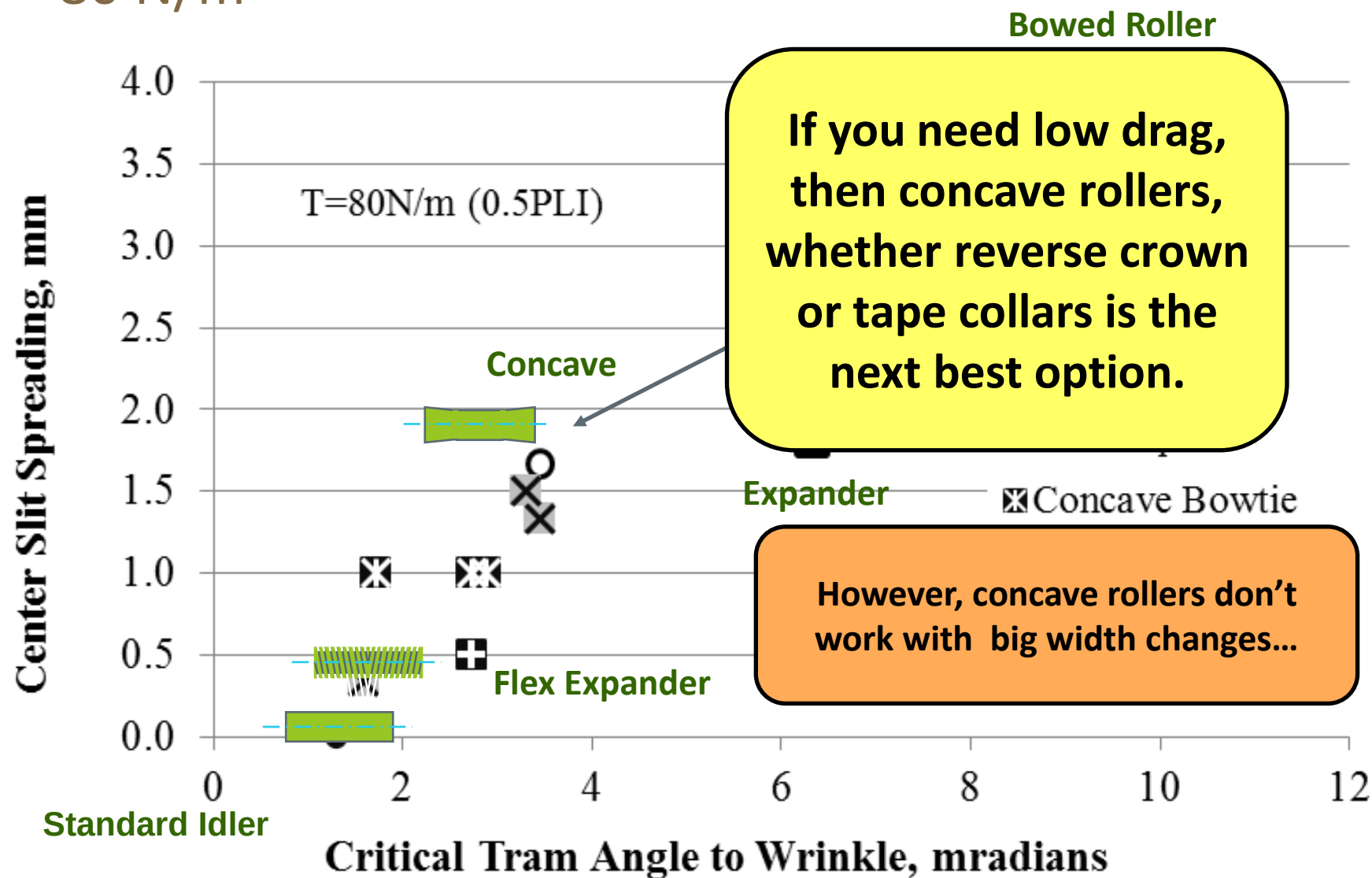
Expanding surface and bowed rollers are by far the most powerful spreaders, but also the highest drag, may need to be driven...

Bowed Roller

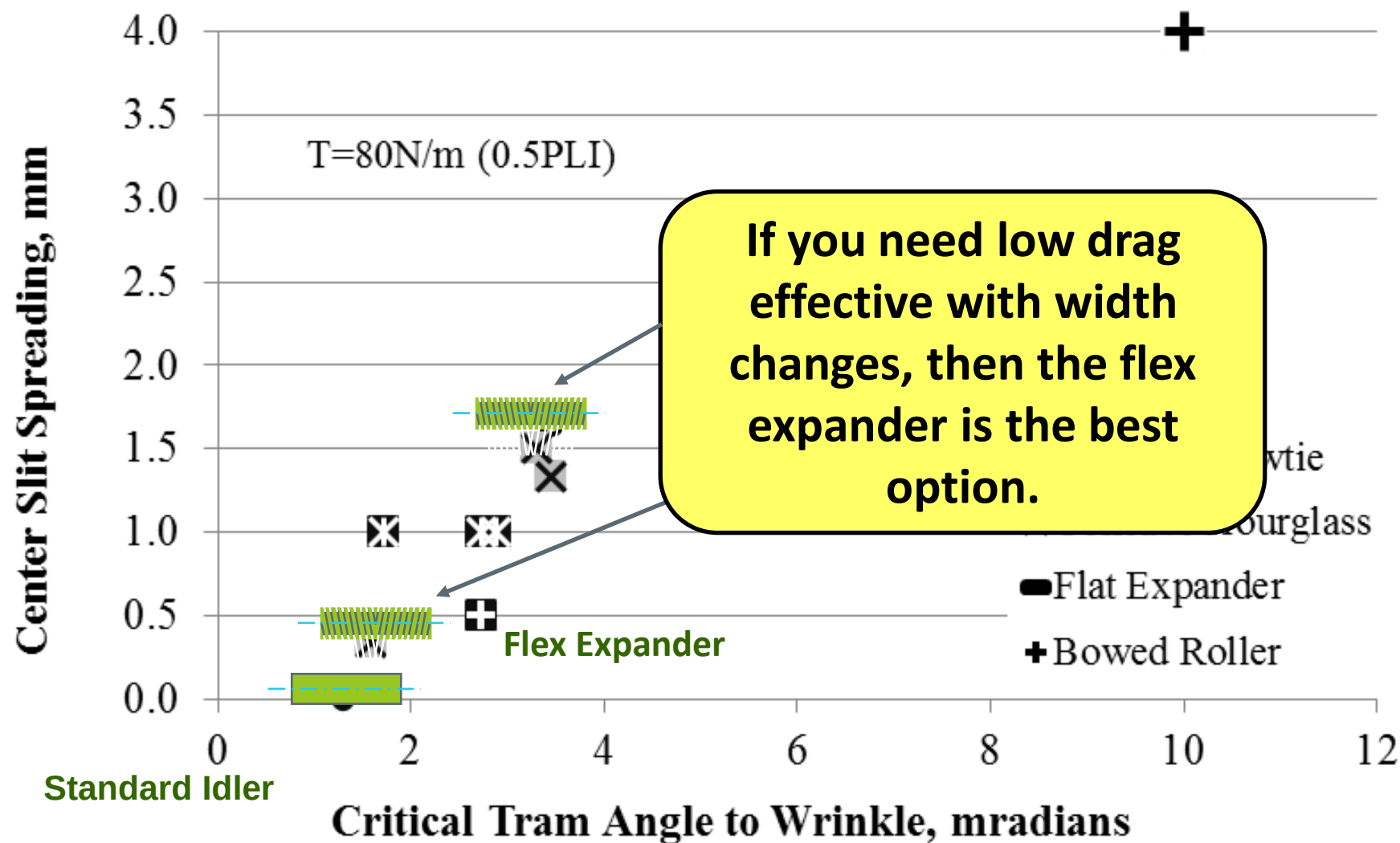


Anti-Wrinkle vs. Spreading Test Results by Roller,

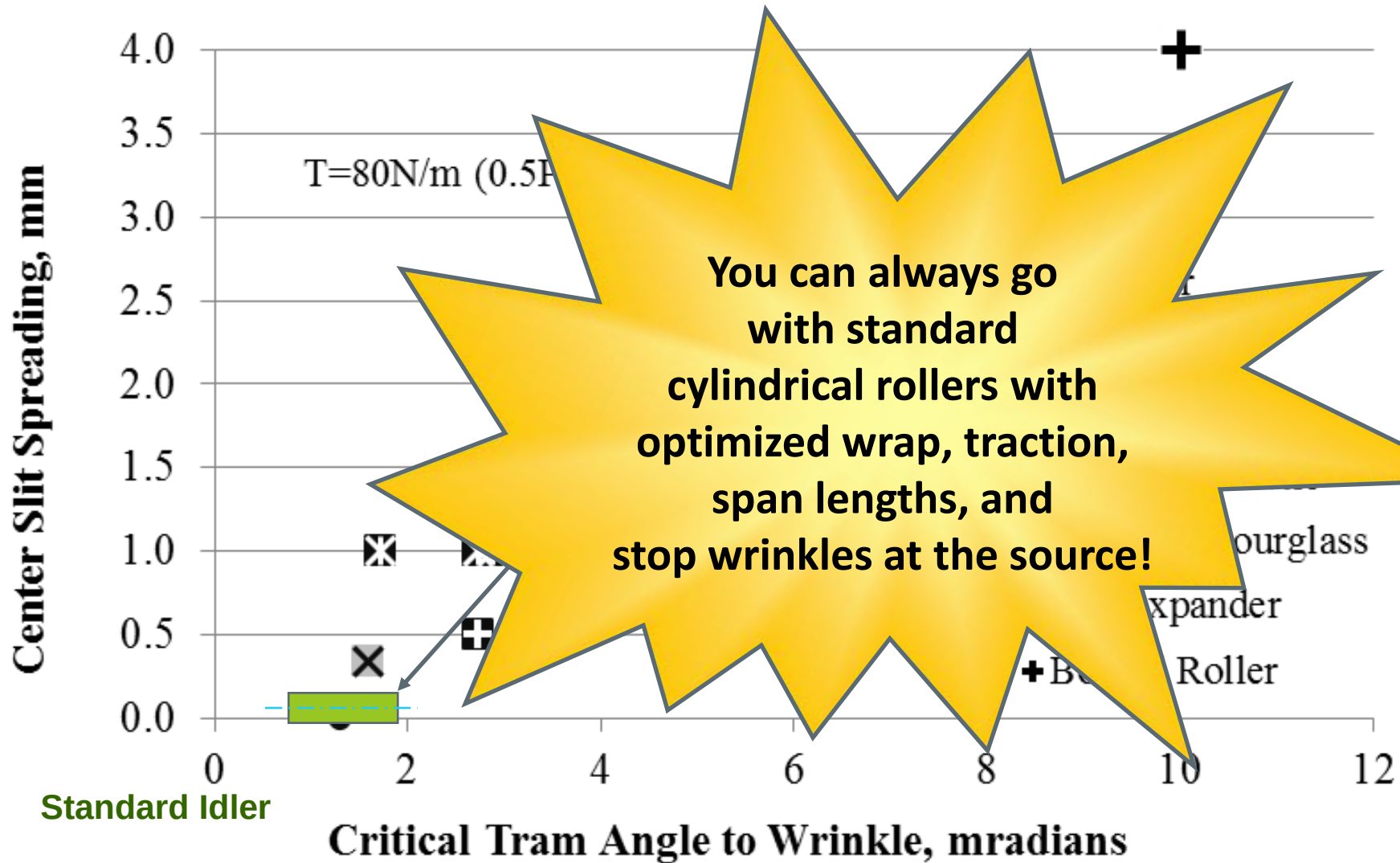
$T = 80 \text{ N/m}$



Anti-Wrinkle vs. Spreading Test Results by Roller, $T = 80 \text{ N/m}$

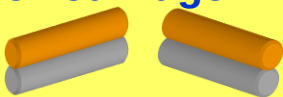


Anti-Wrinkle vs. Spreading Test Results by Roller, T = 80 N/m



Spreader Options

All need sufficient traction

	Entry Span	Exit Span	Wrap Angle	Spreading?
Bowed Roller 	Long ($L > w$)	Short ($L < w/2$)	15-45 degr.	Entry Span 95%, On Roller 5%
Skewed Edge Nips 	Long	Short	0 degr.	Entry Span
Flat Expander Roller 	Long	Short	15-45 degr.	Entry Span 95%, On Roller 5%
Flex Expander Roller 	Long	Short	15-180 degr.	Entry Span
Tape Collared Roller 	Longer better	Short	As needed for traction	Entry Span
Concave Roller 	Longer better	Short	As needed for traction	Entry Span

Spreader Options

All need sufficient traction

Bowed Roller



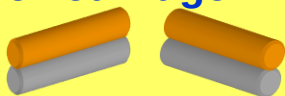
Biggest Advantages

Powerful spreader, baggy edge/center compensation, spread slit strands, adjustable*

Biggest Disadvantages

Often poorly applied, rubber* surface wears, difficult to align, high drag, overly aggressive for stiff web (e.g. foils)

Skewed Edge Nips



Powerful spreader, nip creates good traction

Unsafe, difficult to thread, step to unnipped web damages lighter webs

Flat Expander Roller



Powerful spreader, adjustable, easy to align

Rubber* surface wears, some designs don't expand away from edges, high drag, overly aggressive for stiff web (e.g. foils)

Flex Expander Roller



Surprisingly powerful spreader

Thin webs buckle if grooves too wide, moderate spreading, rubber* cover ages
(*All rubber rollers have temperature limits and will lose performance with age.)

Tape Collared Roller



Least expensive, quickest to try

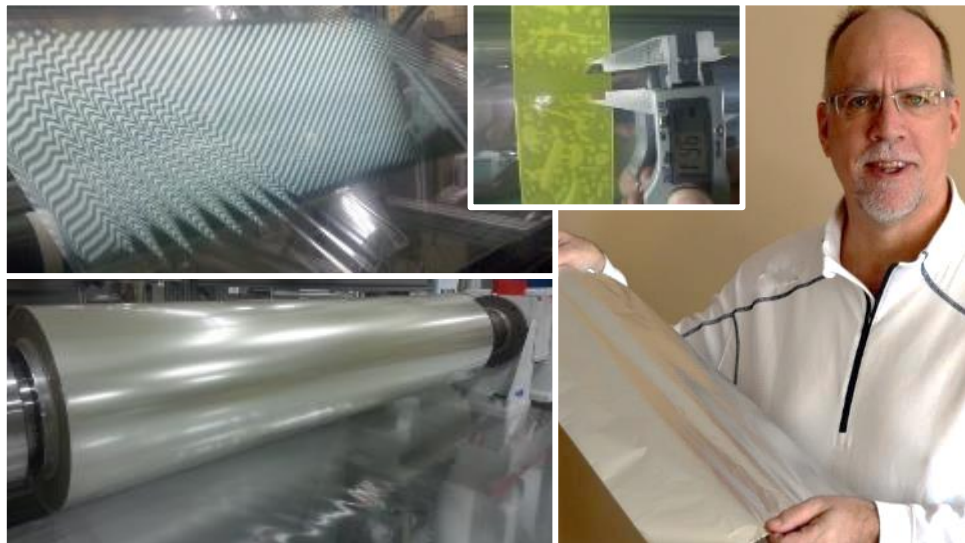
Must be optimized for web width, strain, and roller diameter, tape may shed, difficult to apply to stiff webs (e.g. foils)

Concave Roller



Low expense, long term solution if tape collars are effective

(See Tape Collars), won't work with big width or strain changes.



5.4 Guiding

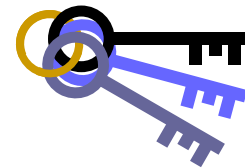
Timothy J. Walker, TJWalker + Associates Inc.

1620 Edgcumbe Road, Saint Paul, MN 55116

tjwalker@tjwa.com www.webhandling.com 651.686.5400

5.4 Learning Objectives

- ▶ What is an example of passive web guiding?
- ▶ What are the components of a automatic web guiding system?
- ▶ What are the limits to automatic guiding accuracy?
- ▶ What is the optimal position of a web guide sensor?
- ▶ What are the basic rules of the following guide systems:
 - Sidelay unwind guiding
 - Sidelay rewind guiding
 - Displacement guiding
 - Steering guiding
 - Chase guiding



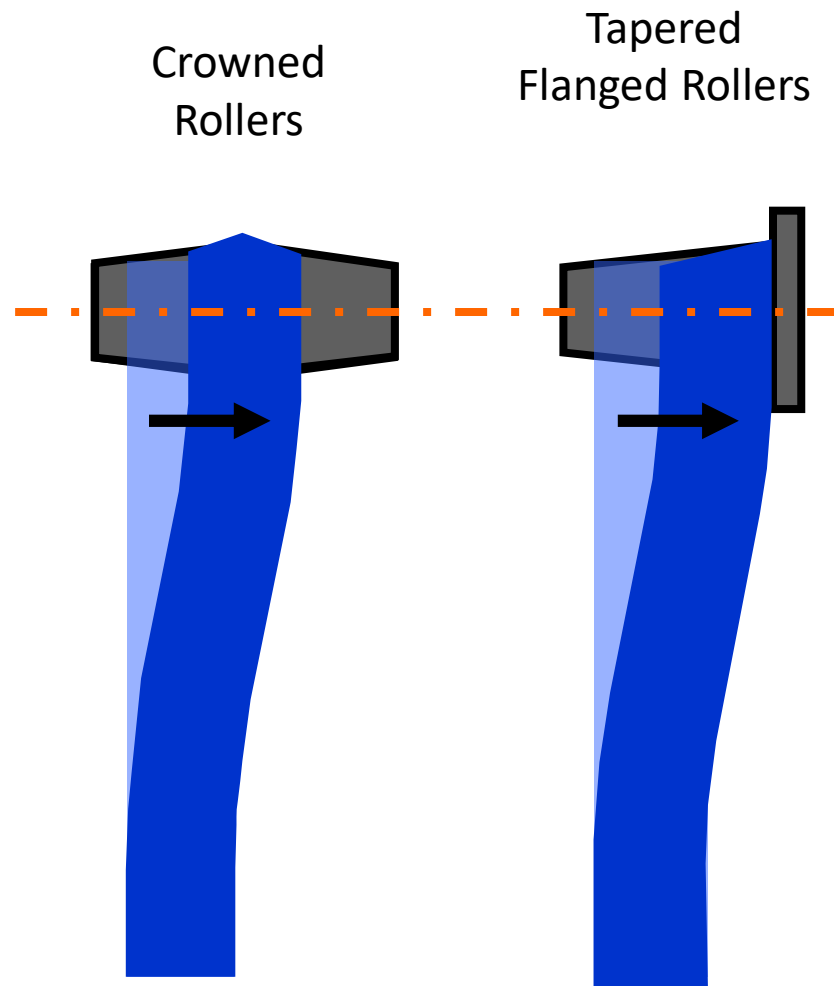
Tracking on Diameter Variations

How a web tracks on diameter variations:

1. The diameter variations create crossweb roller surface speed and web tension variation.
2. The crossweb tension variation exerts a moment (torque) on the web.
3. The moment induces bending in the web.
4. The web bending curvature violates the parallel entry rule.
5. The web tracks laterally in order to satisfy the parallel entry rule.

Passive guiding works best with long spans and narrow webs. (Good traction required)

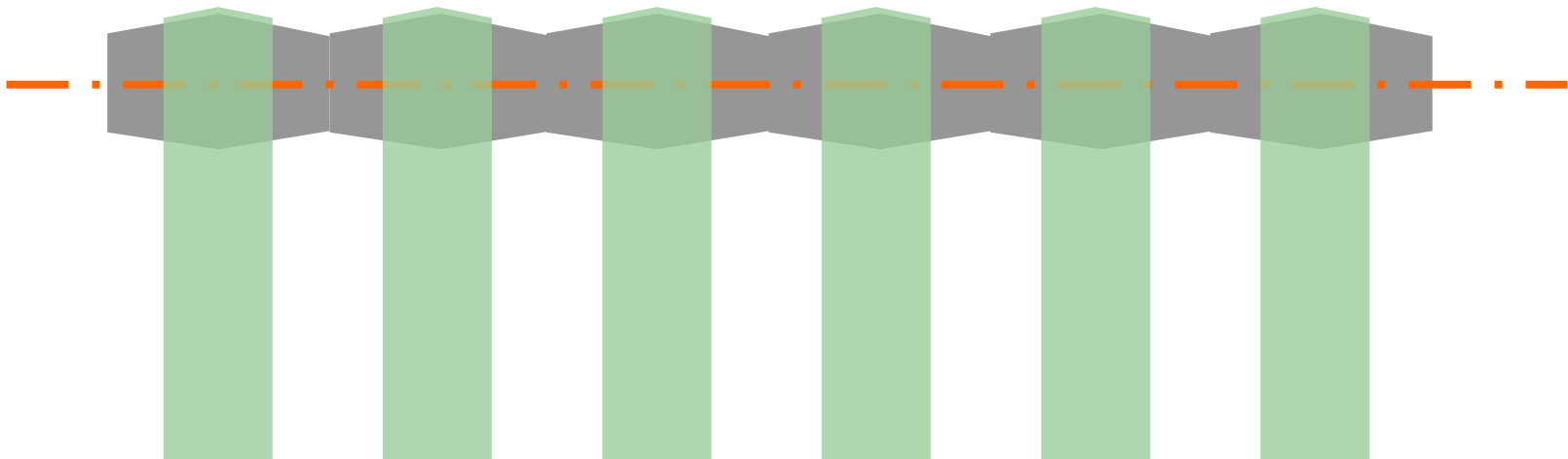
May create wrinkles in wide webs.



Passive Guiding

Multi-Crowned Rollers

Used to keep multiple strands in their lanes after slitting.



Passive Guiding at Winding

A tapered roller will improve narrow web wind edge alignment.

**Tapered,
Flanged Roller**

Gap Winding

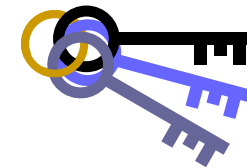
(Note: Do not use tapered roller as pack roller.)

**Tapered,
Flanged Roller**

**Cylindrical Pack
Roller**

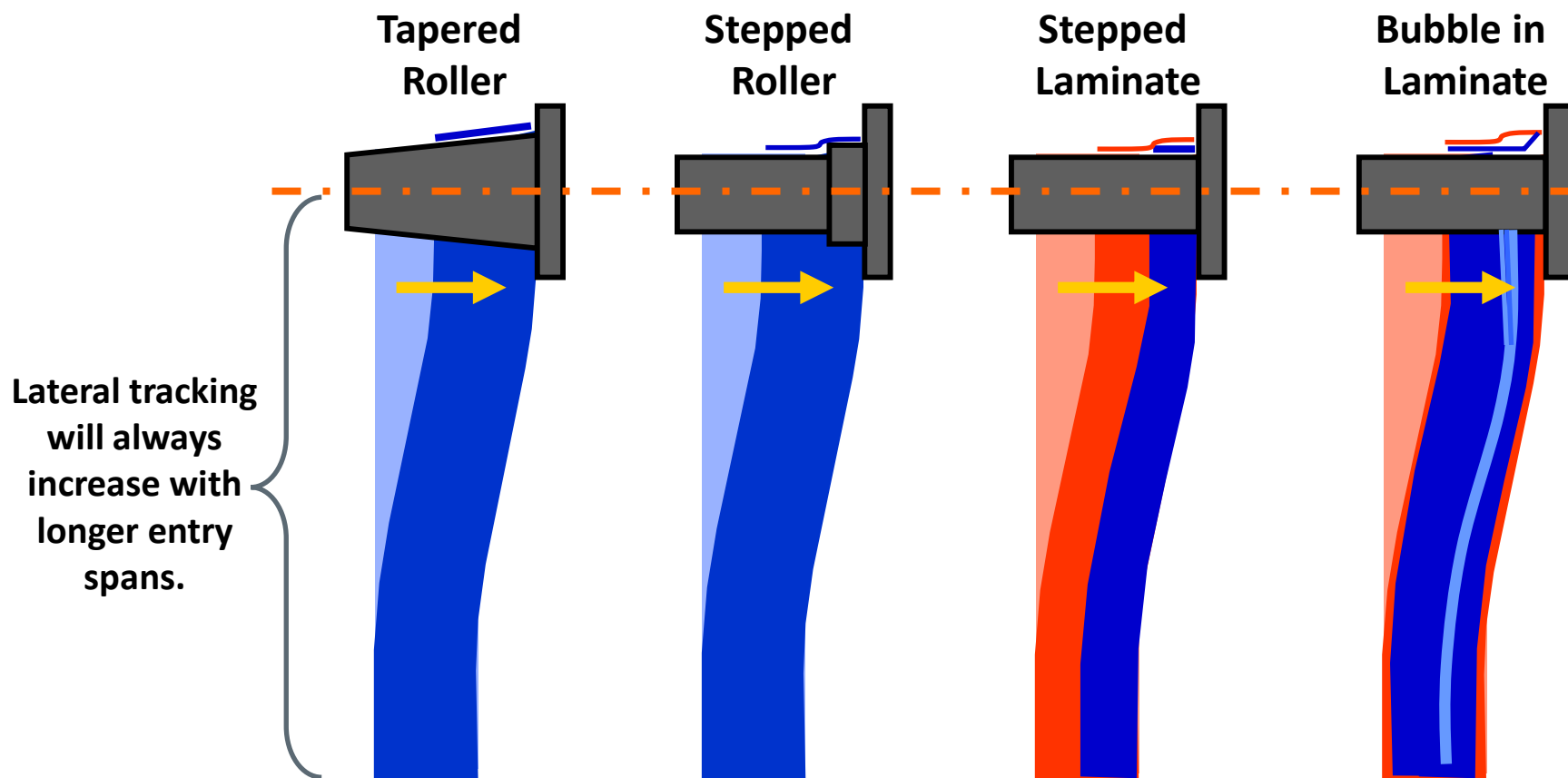
Contact Winding

A tapered, flange wind can have a sidewall equal to post-slit wind.



Diameter Variations From Web

Lateral web shifting can come from many “diameter” variations sources:

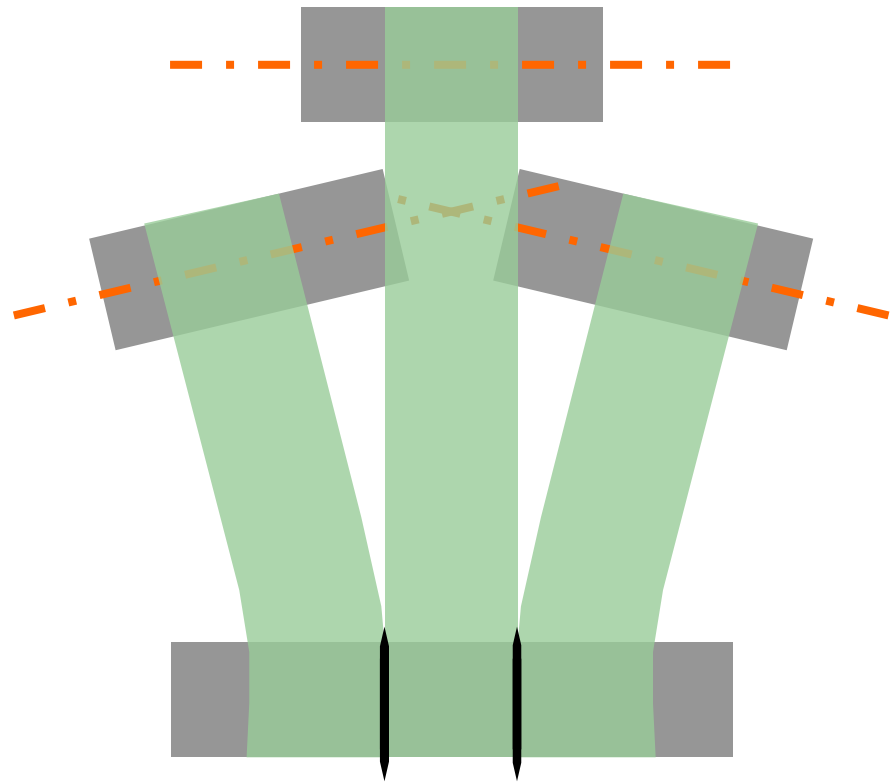


Skewed Roller Spreading

Used to create a small separation in a limited number of fixed width strands.

Advantage:

Better than a bowed roller for products with consistent number and width of strands.

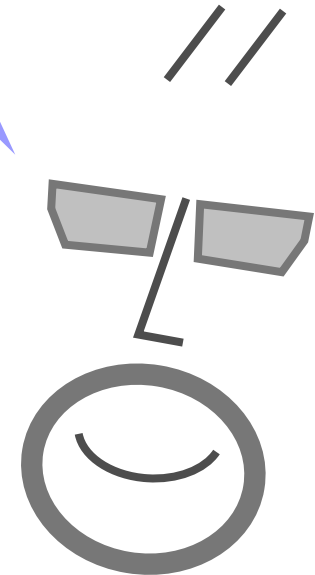


Automatic Guide Qs

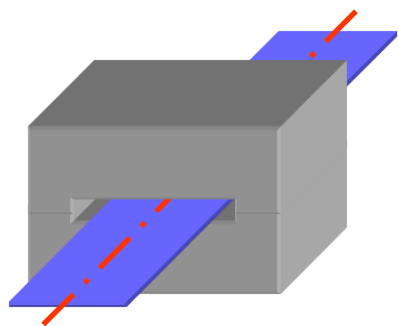
Where are web guides used?

Upstream of any lateral position sensitive process, including:

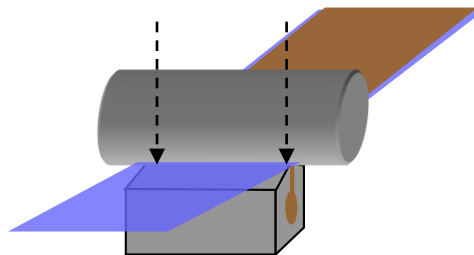
At unwinding to align to machine centerline, and ahead of coating, laminating, printing, slitting, and winding...



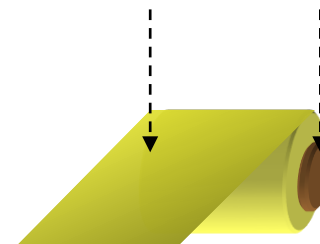
Problems of CD Shifting and Slipping



Web to equipment
($\pm 0.5''$)



Coating to web
($\pm 0.2''$)



Web to core
($\pm 0.2''$)

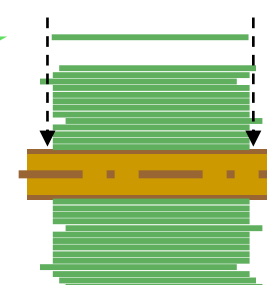
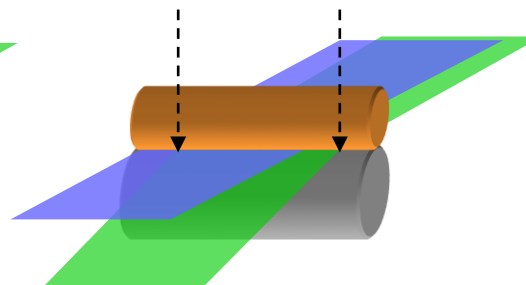
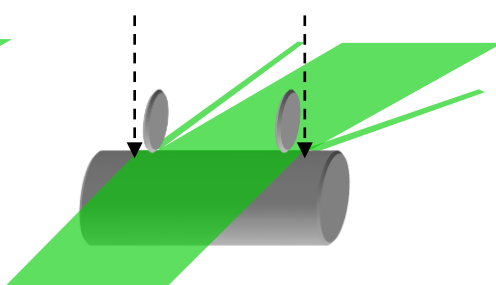
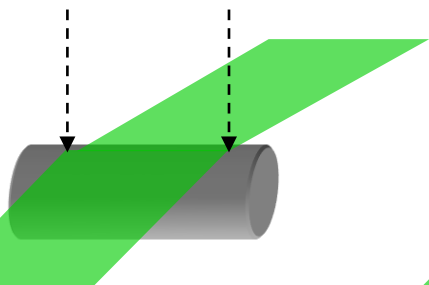
Easy - Lateral Alignment Requirements - Difficult

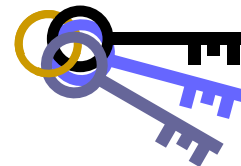
Web to roller
($\pm 2.0''$)

Web to knives
($\pm 0.5''$)

Web to web
($\pm 0.2''$)

Wound roll edges
($< \pm 0.030''$!!!)

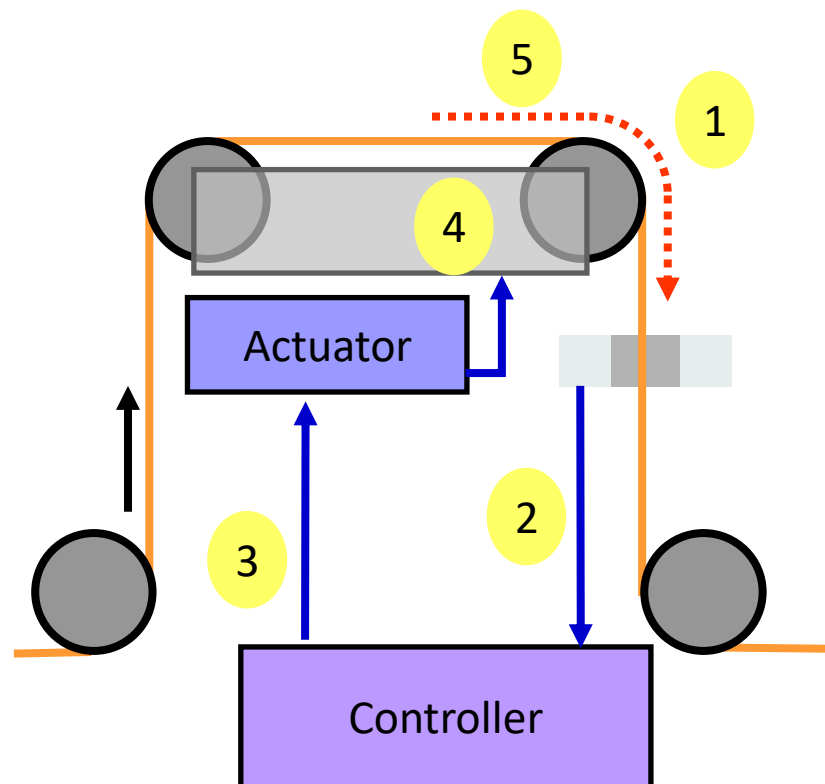




Closed Loop Web Guiding

Control Loop Cycle:

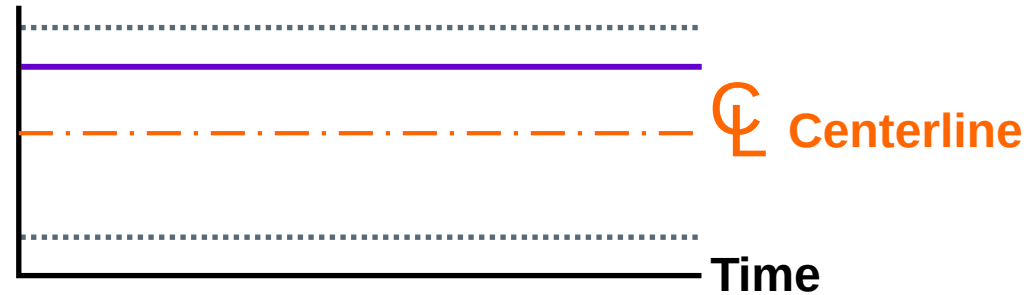
1. Web travels through the web guide.
2. Web position sensor detects the web and sends signal to the controller.
3. Controller compares web position to set point, requests correct motion from the actuator.
4. Actuator shifts.
5. Web travels on corrected path where sensor rechecks it position...



Types of Lateral Position Error

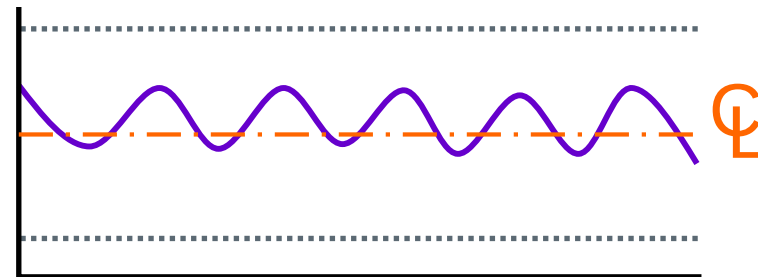
Offset

Lateral
Position



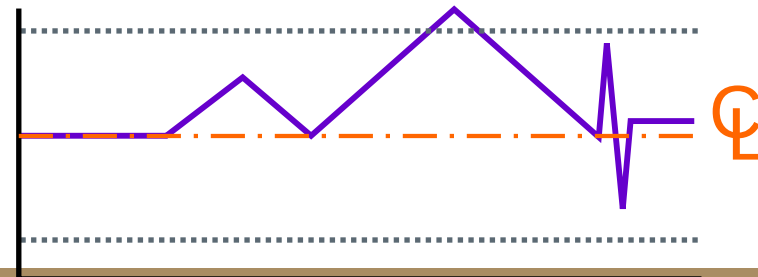
Sinusoidal

Lateral
Position



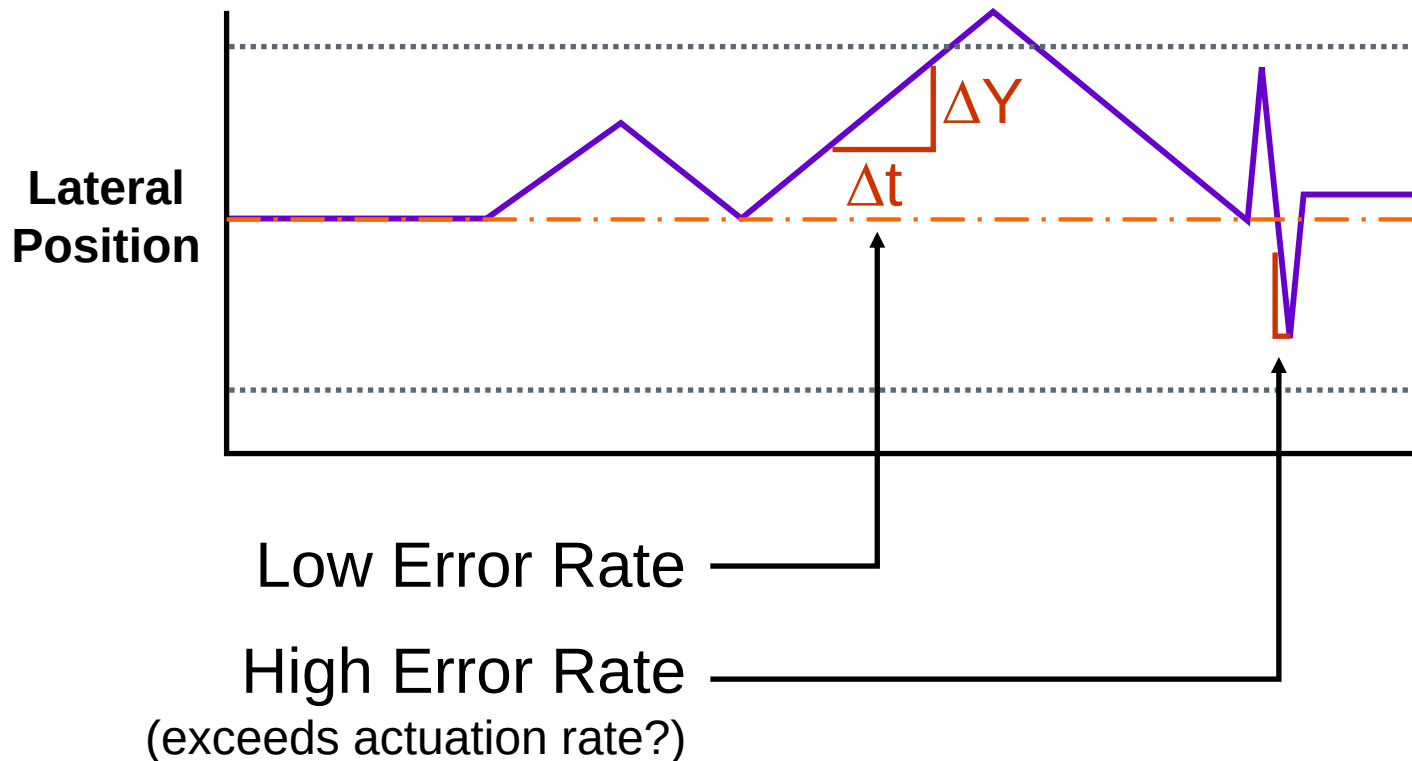
Linear Shift

Lateral
Position



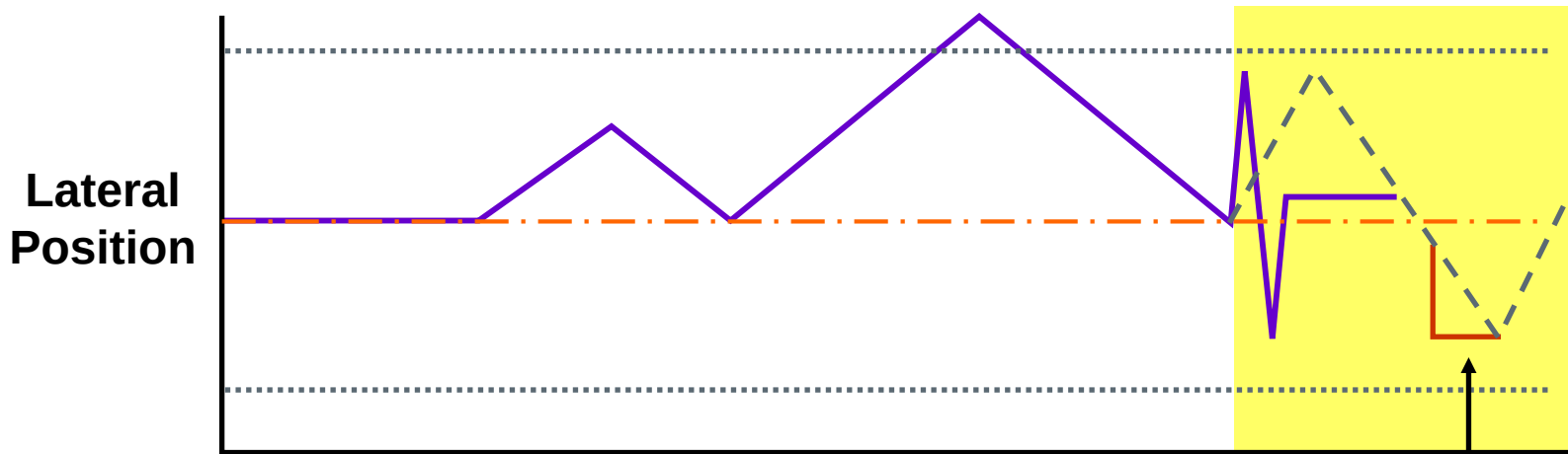
Guiding Limits: Actuation Rate

$$\text{Error Rate} = \frac{\Delta Y}{\Delta t} = V_Y$$



Guiding Limits: Actuation Rate

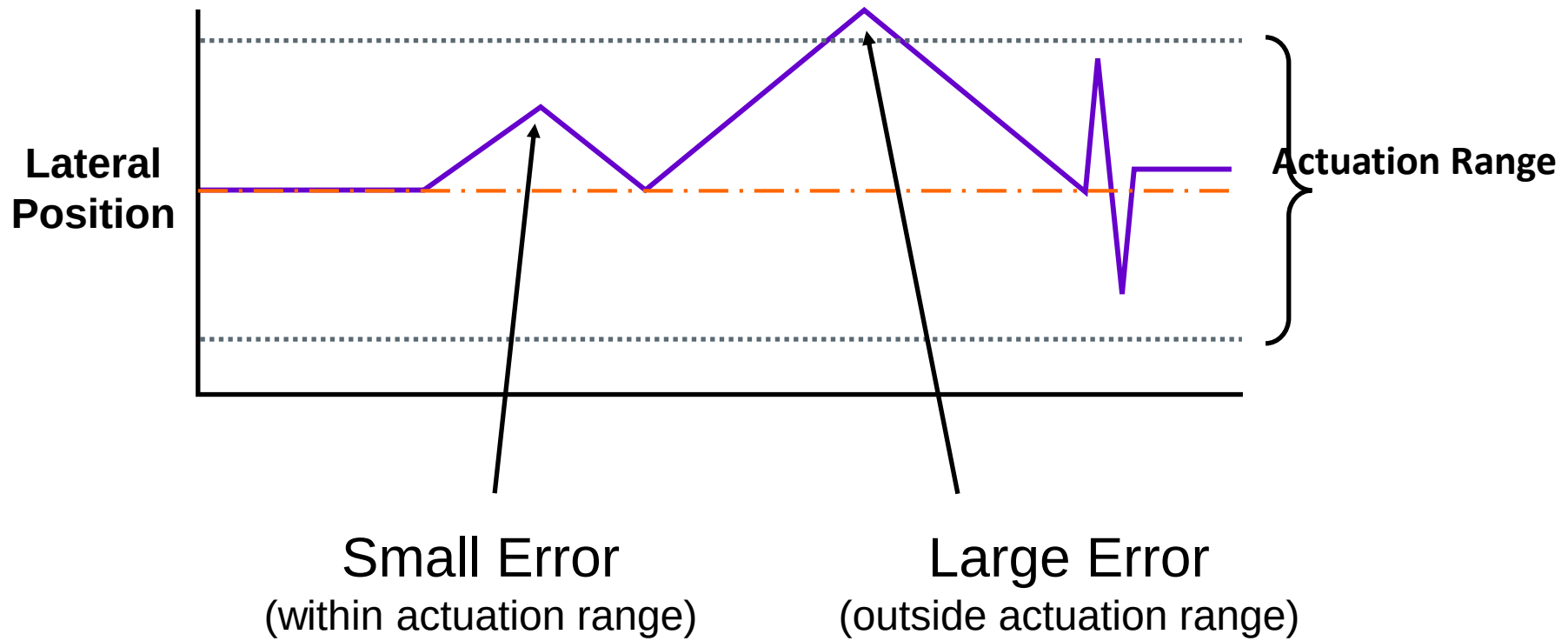
$$\text{Error Rate} = \frac{\Delta Y}{\Delta t} = V_Y$$



Error rate is reduce by
slower web speed
(within actuation rate!)

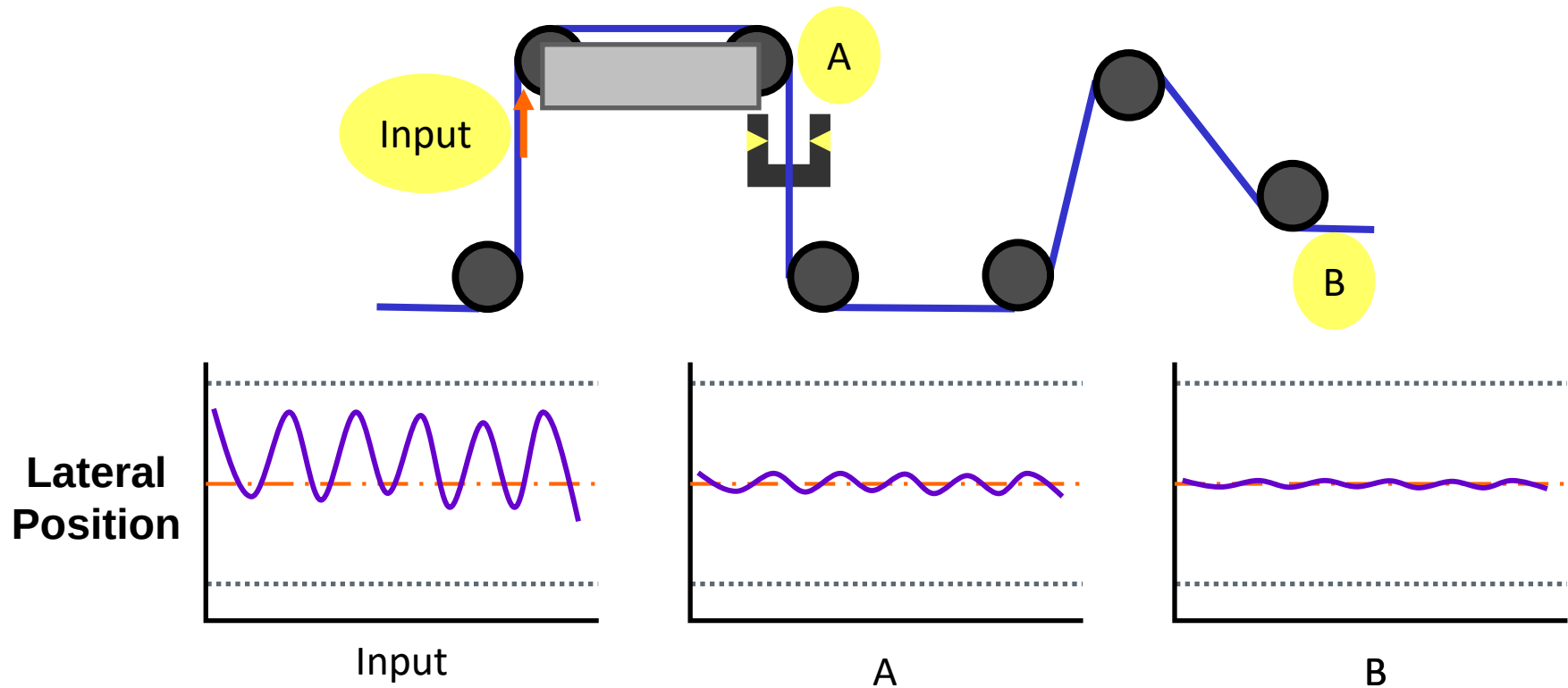
Guiding Limits: Actuation Range

$$\text{Error Range} = Y_{WEB} - Y_{CL} = \Delta Y$$



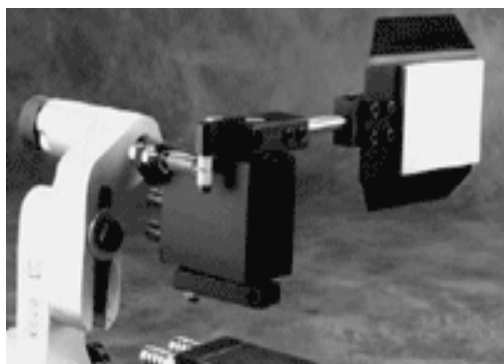
Rollers Dampen Lateral Motion

Rollers and spans naturally dampen sinusoidal web motion, with the best position accuracy usually a few rollers downstream of the web guide.



Edge Sensor Options

Mechanical



Images courtesy of Fife

Optical Array



Images courtesy of Fife

Ultrasonic



Images courtesy of Accuweb

Pneumatic



Image courtesy of Nireco

Line Scan



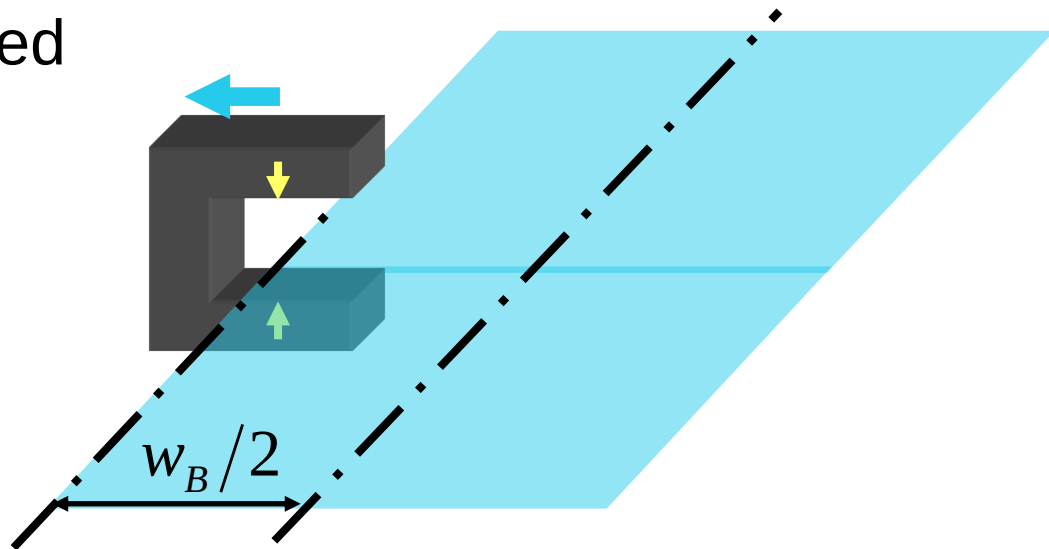
Image courtesy of Nireco

Line scan sensors are used to follow printing or coating lines.

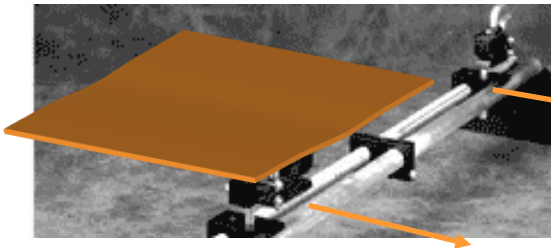
Edge vs. Width Sensors

Edge sensors are placed $1/2$ web width from machine centerline.

Edge sensors can be moved as width changes to stay on machine centerline.



Width sensing system adjust automatically to web width changes.



Courtesy of Fife Corp.



Courtesy of Accuweb

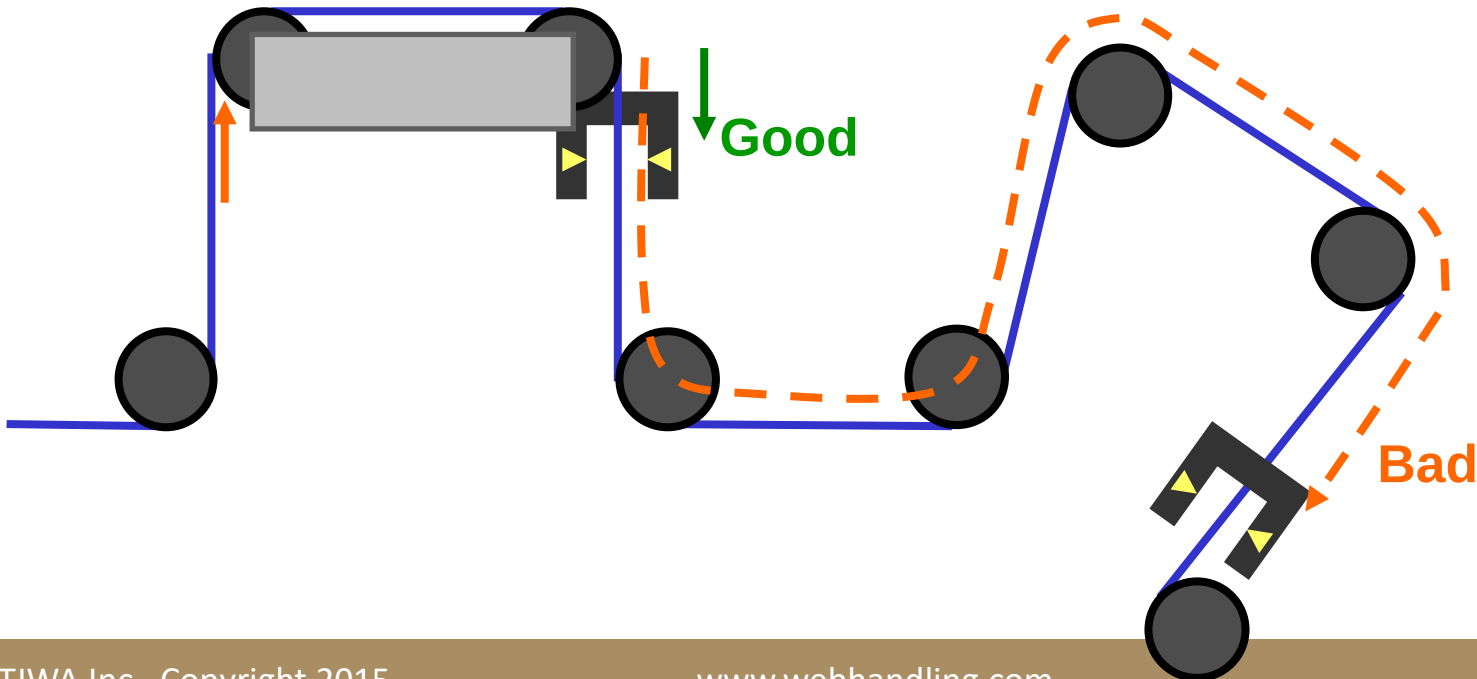
Web Position Detection Methods

Method	Pros	Cons
Light Transmission or Reflectance	Good accuracy with small, discrete sensing elements	Sensitive to dust, ambient light.
Ultrasonic Absorption	Good for thin, transparent webs.	Sensitive to temperature and humidity changes
Pneumatic	Good for solvent applications, simple electronics.	May upset thin, low tension webs.
Mechanical	Simple	Requires stiff web with low sensitivity to edge abrasion.

Place Sensor Near Web Guide Exit

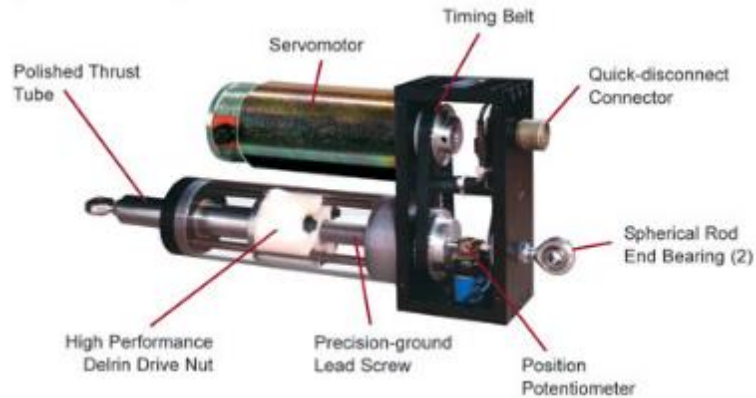
Web sensors should be as close as possible to the web guide exit.

Any web path between the guide and sensor is a time lag, leading to over-steering and web oscillation.



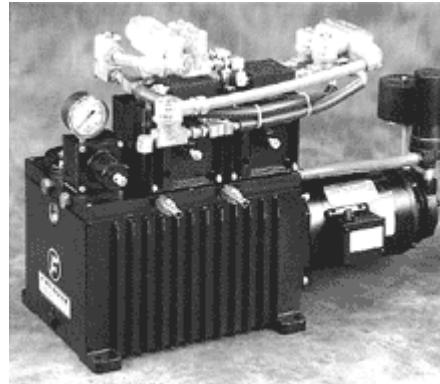
Web Guide Actuators

Electro-Mechanical



Courtesy of Accuweb

Pneumo-Hydraulic



Courtesy of Fife Corp.

Pneumatic

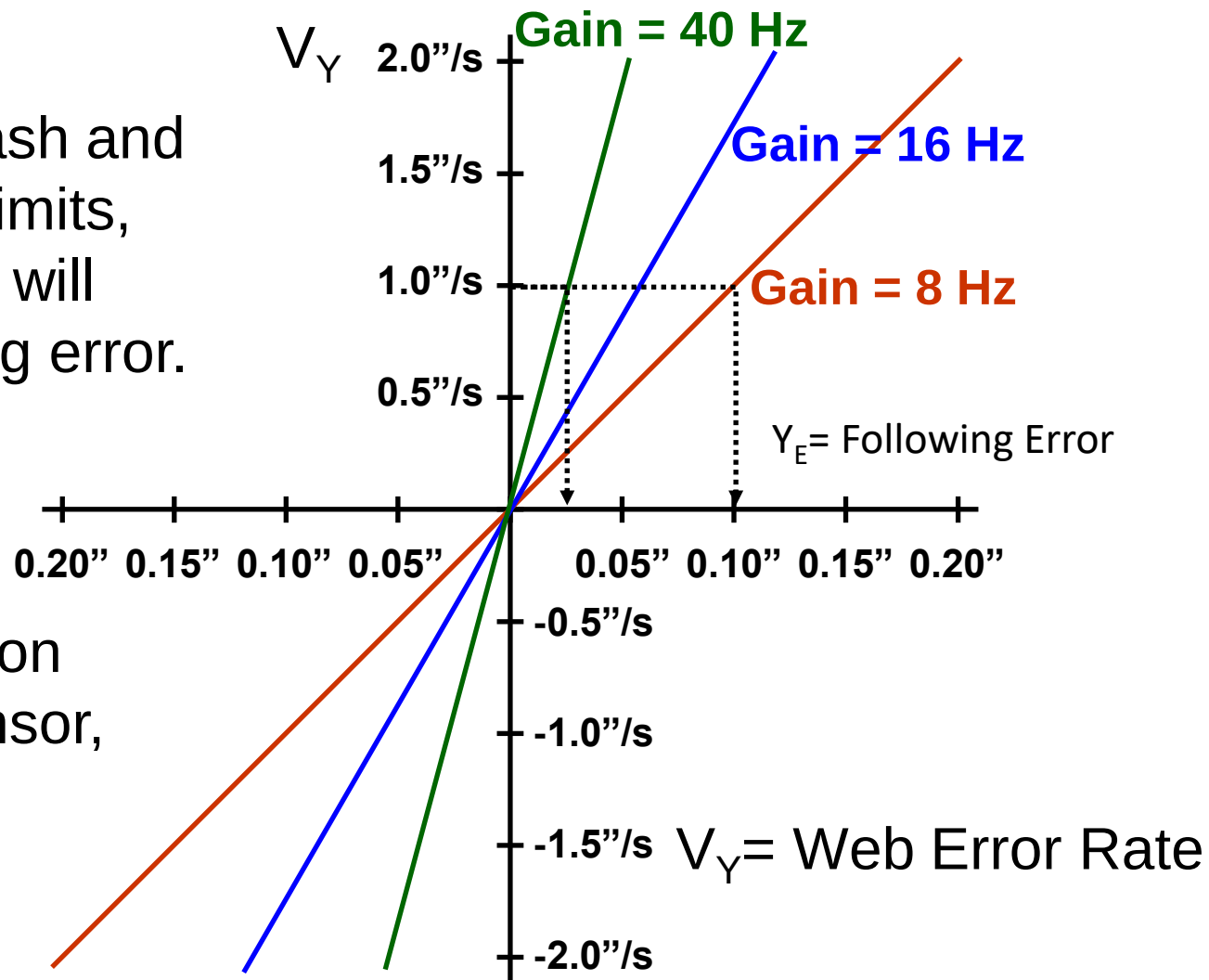


Courtesy of Coast Controls

Higher Gain Reduce Guiding Error

Ignoring backlash and actuation rate limits, the higher gain will reduce following error.

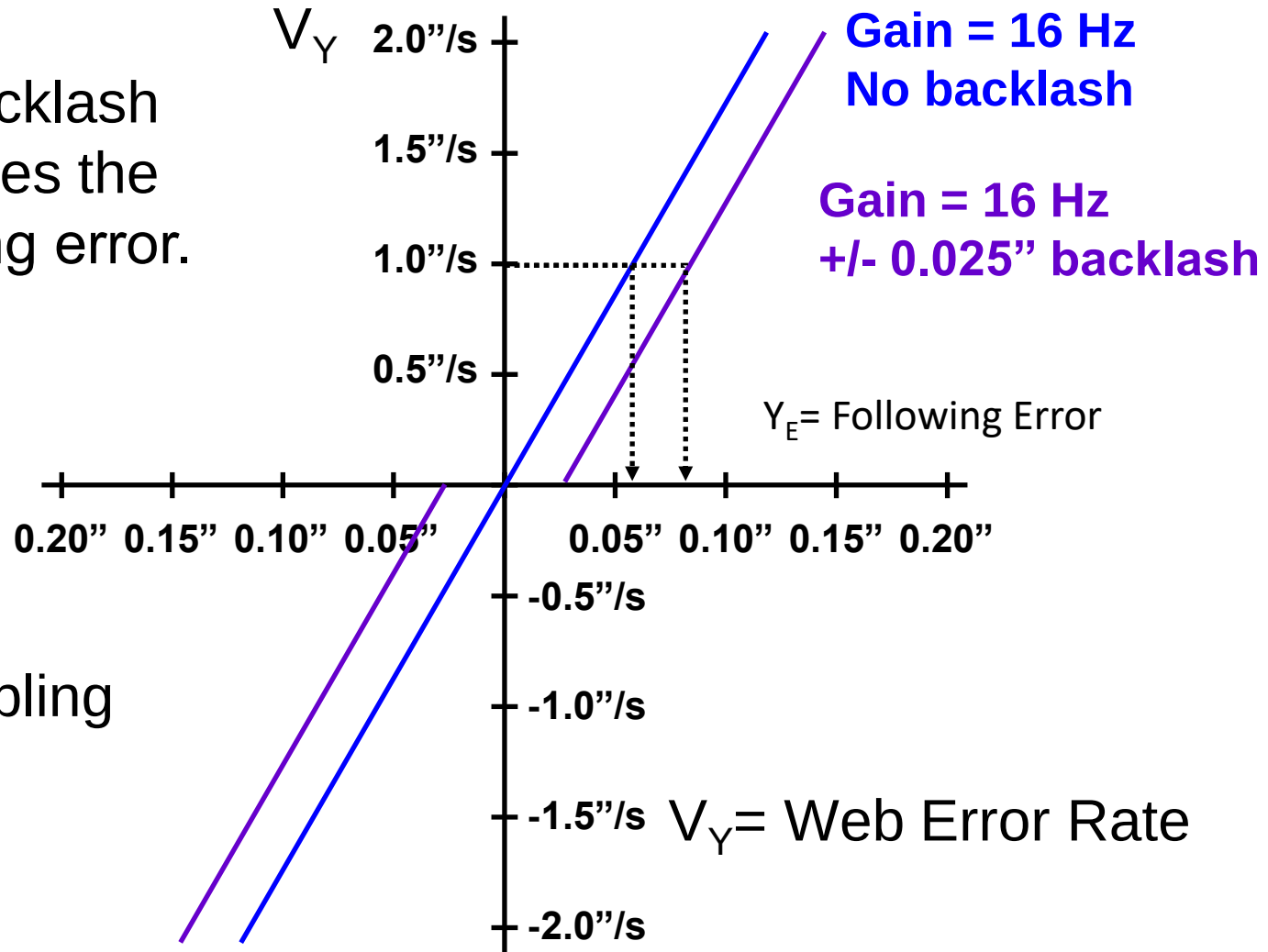
Gain is a function of actuator, sensor, and controller performance.



Backlash Creates Guiding Error

Any system backlash directly increases the guide's following error.

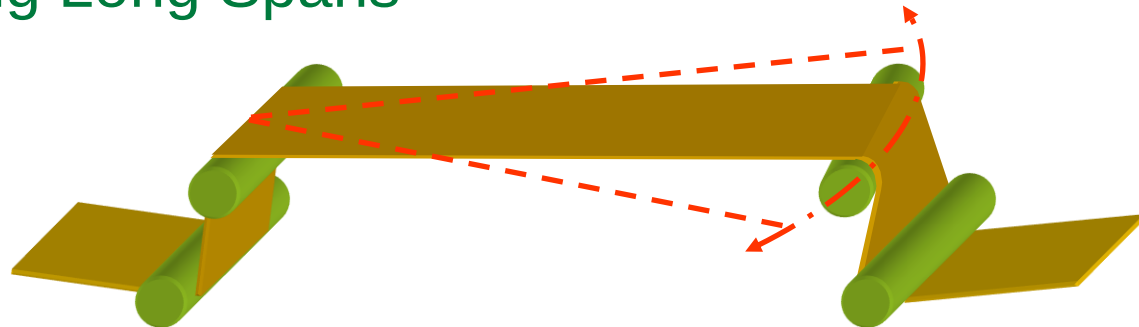
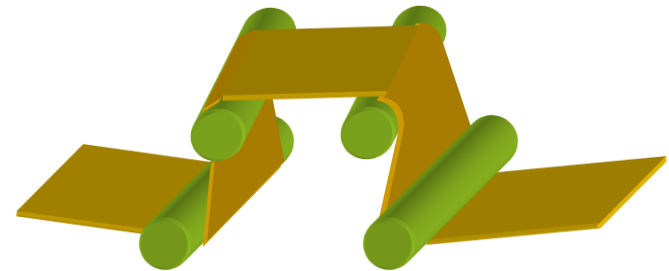
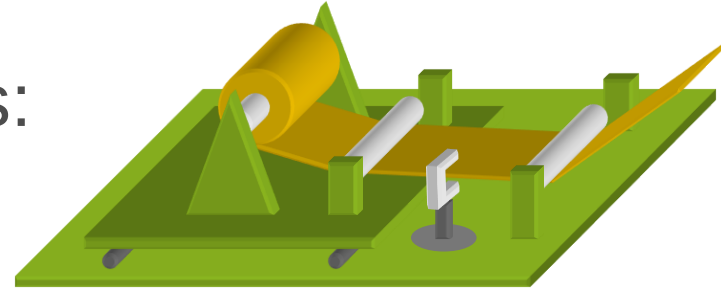
Backlash is a function of coupling and structural rigidity.



Three Active Guide Methods

Three Active Guide Options:

- Sidelay
 - Unwinds, Rewinds
- Displacement
 - Preferred intermediate guide
- Steering
 - Controlling Long Spans



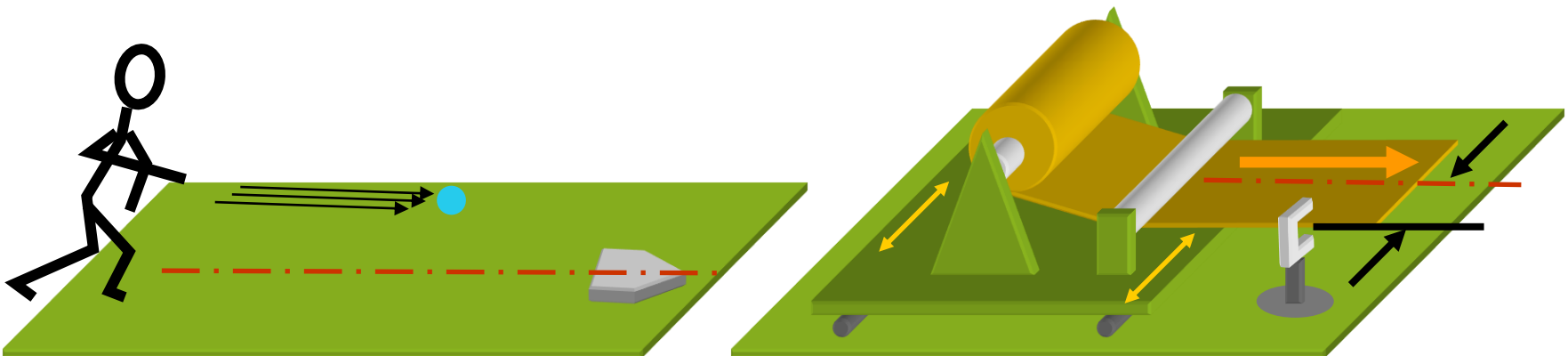
Unwind Sidelay Guidelines

1. Like a baseball pitcher:

“Throw the ball across home plate.”

“Dispense the web aligned to the sensor.”

2. Sensor is a fixed reference to machine centerline (or desired web lateral position).



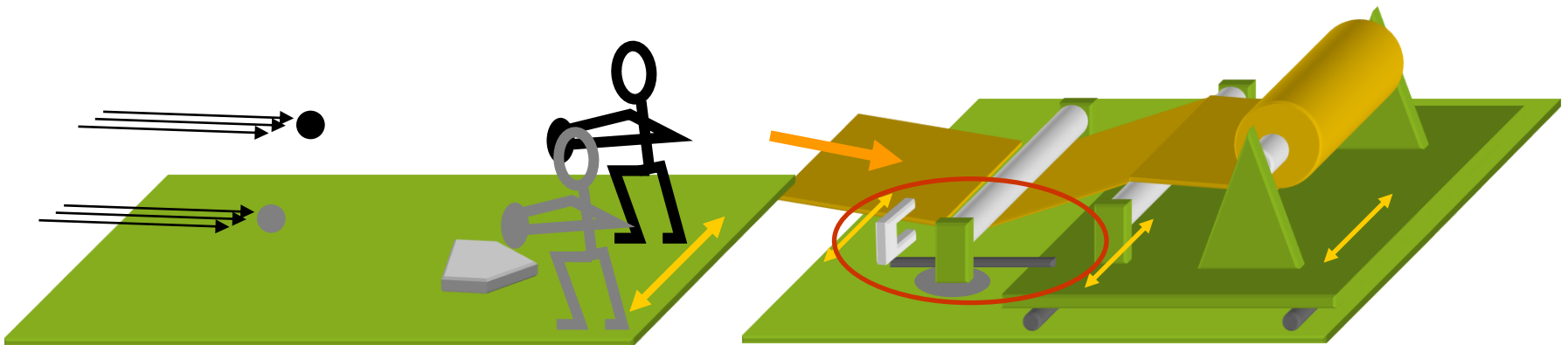
Rewind Sidelay Guidelines

1. Like a baseball catcher:

“Shift to catch the ball in your glove.”

“Shift to wind the web on the core.”

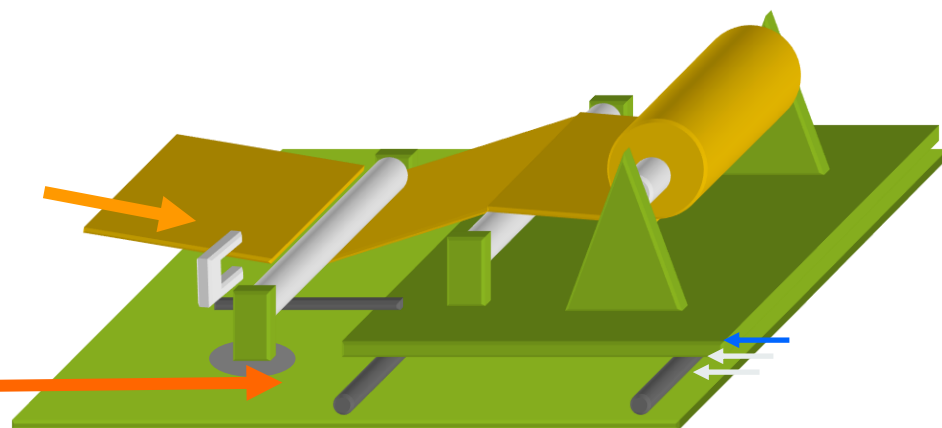
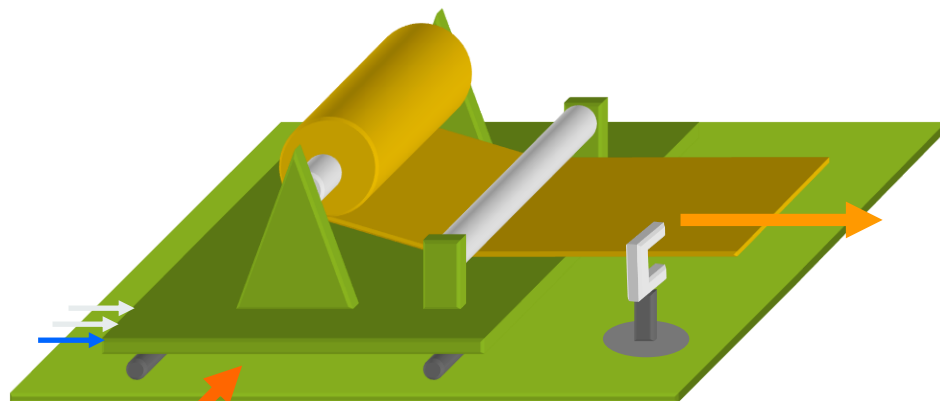
2. Sensor moves as reference to shifting core



General Sidelay Guidelines

Parallel, Lateral Shift

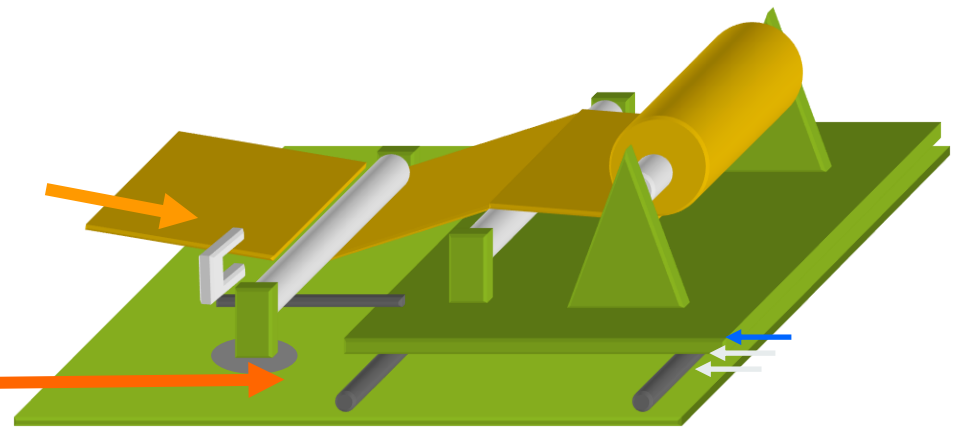
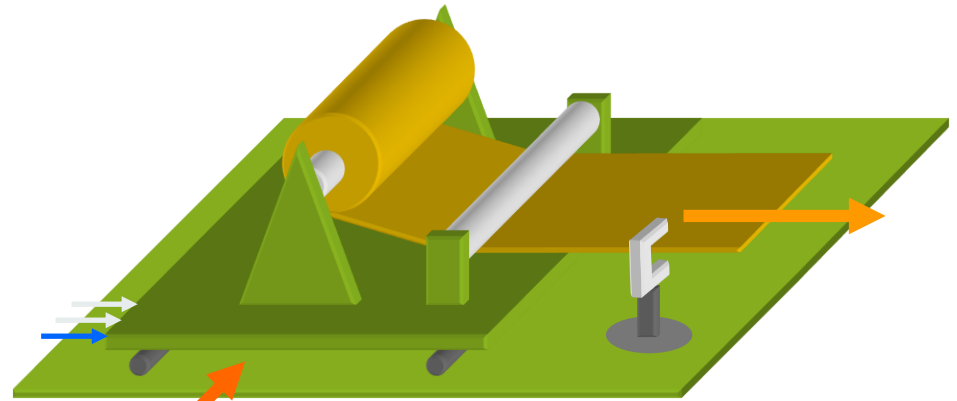
The sidelay actuator laterally shifts the wound roll and rollers, but holds parallel alignment.



General Sidelay Guidelines

Parallel, Lateral Shift

The sidelay actuator laterally shifts the wound roll and rollers, but holds parallel alignment.



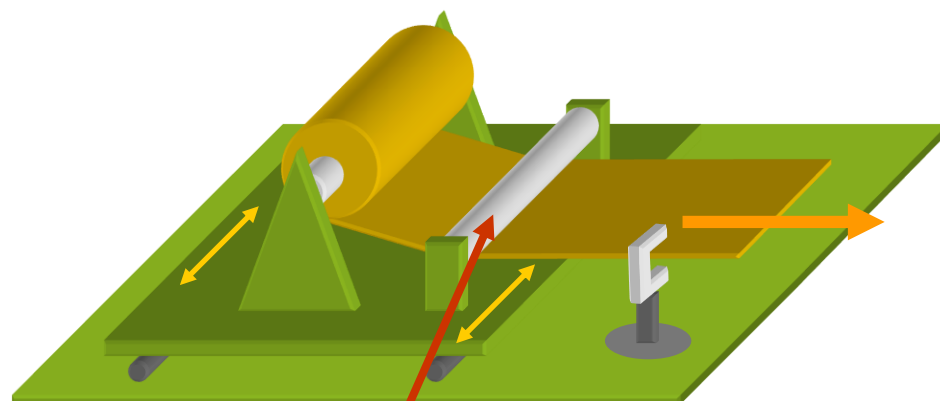
General Sidelay Guidelines

Deflector Roller

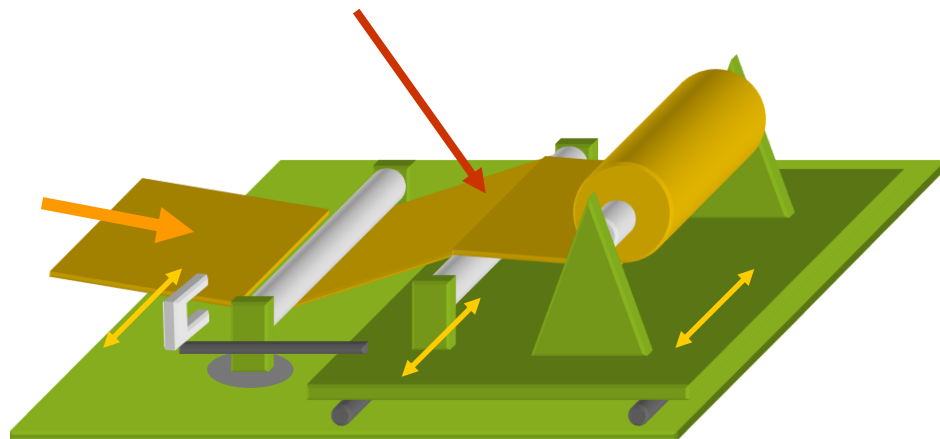
At least one roller shifts with the roll to isolate the shifting web span from the roll.

Sensor Span

Sense the web in a span that is not affected by roll diameter (anywhere other than web span leading to the deflector roller).

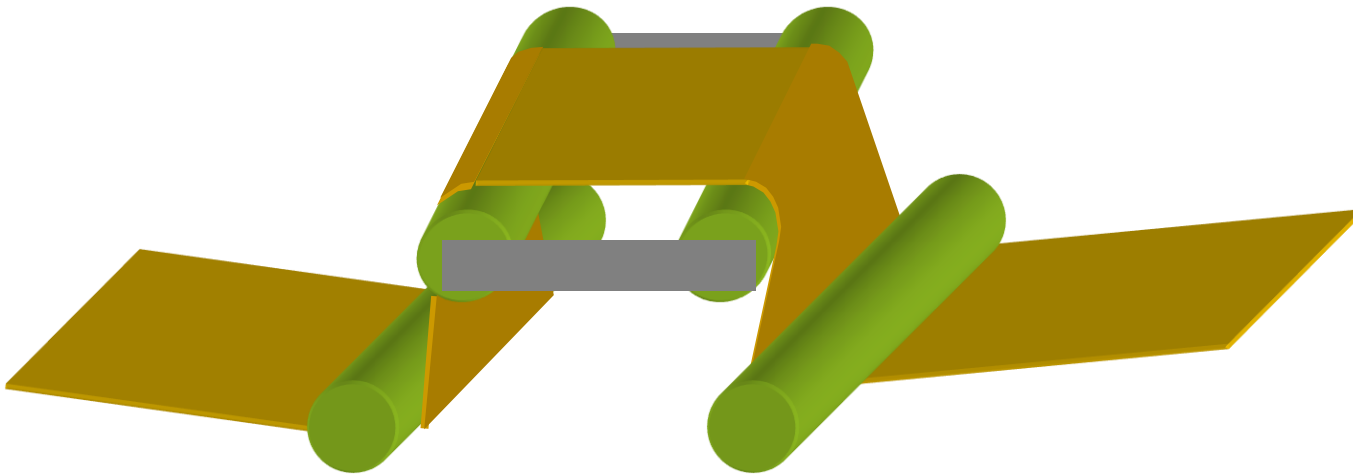


Deflector Roller



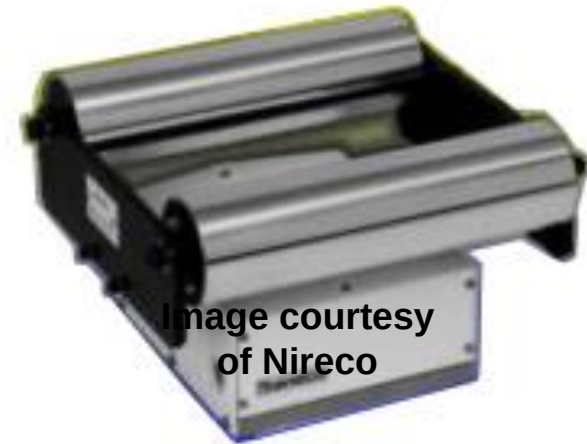
Displacement Guiding Summary

- Twist-Displace-Twist Design (no “bending”)
- Easy to Apply, Few Problems
- Easy to Fit into Existing Web Lines



Displacement Guides

Also known as
Offset Pivot Guides



Single Roller
Displacement Guide

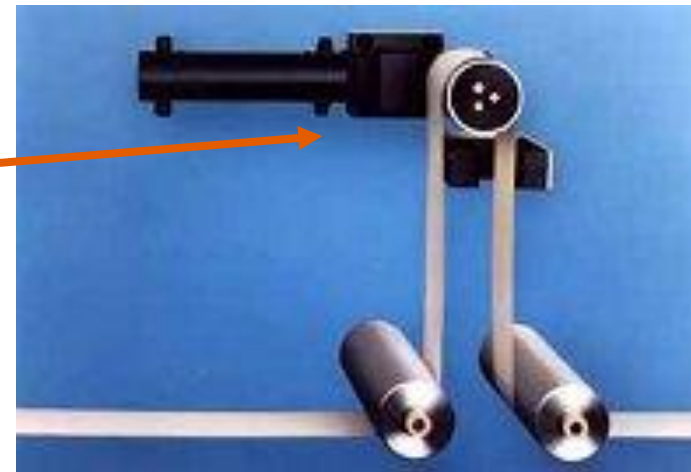
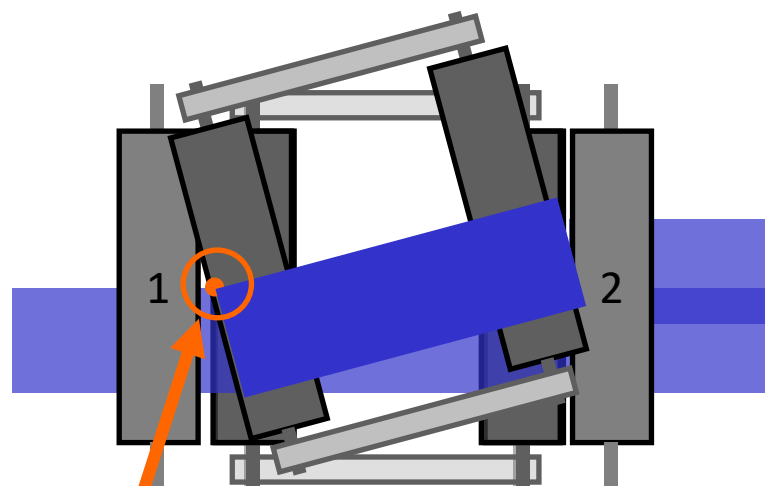
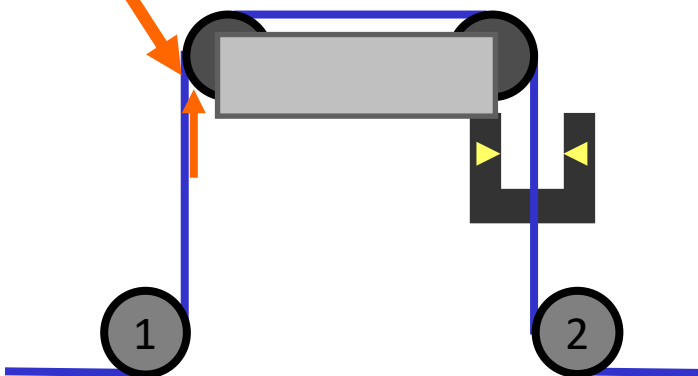


Image courtesy of Fife Corp.

Displacement Guide Geometry



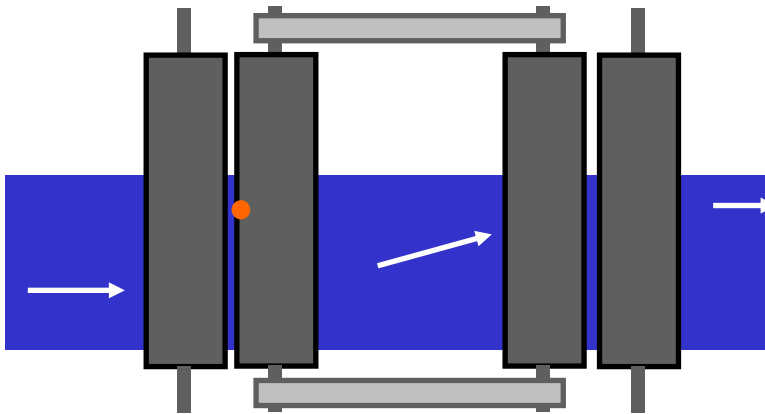
Pivot
Point



Though a displace guide includes only two rollers, two additional rollers are important to good performance.

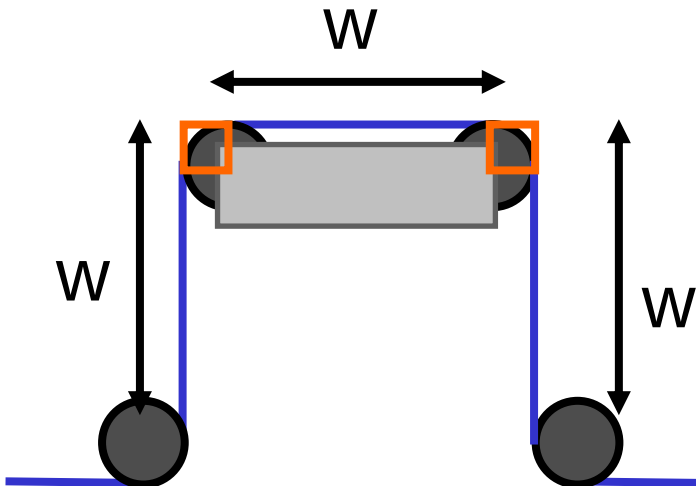
Displacement guide pivot point is centered on entry span.

Displacement Guide Geometry



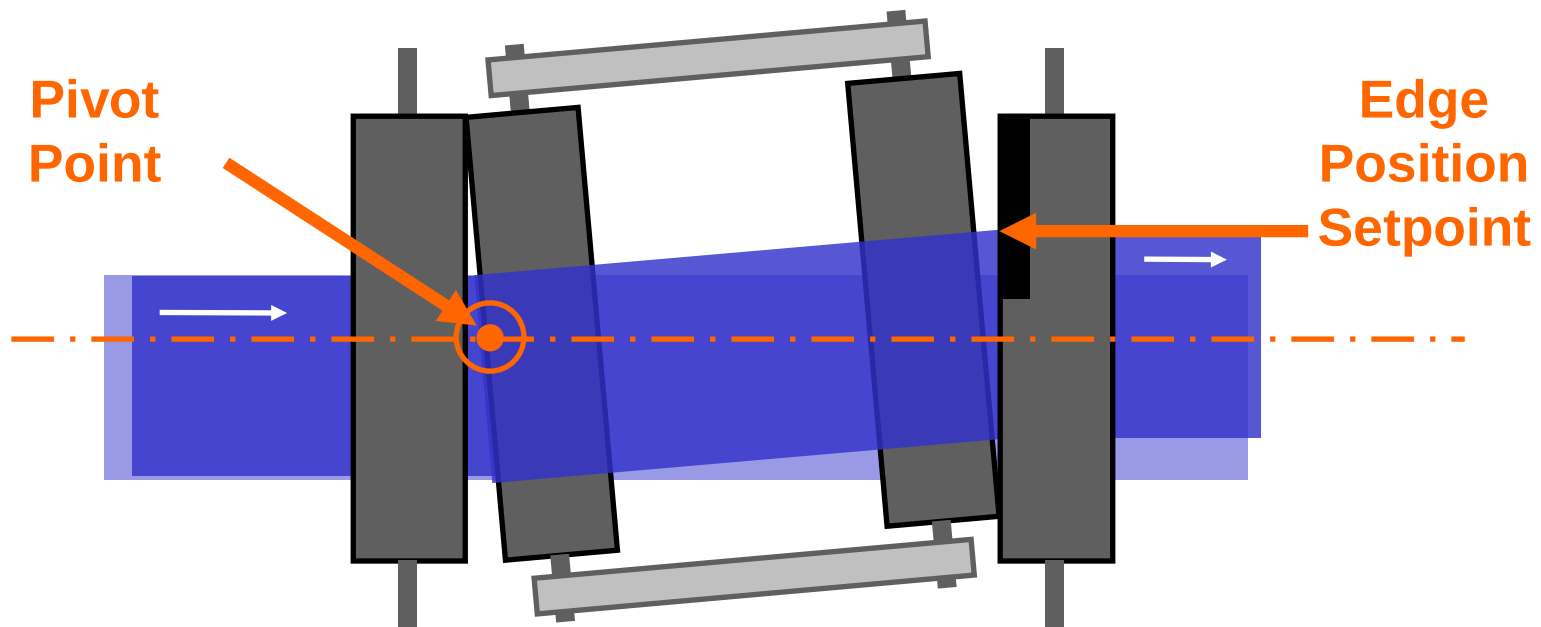
Entry and exit spans must be at right angle to displacement plane.

This creates the desired twist, displace, twist geometry.



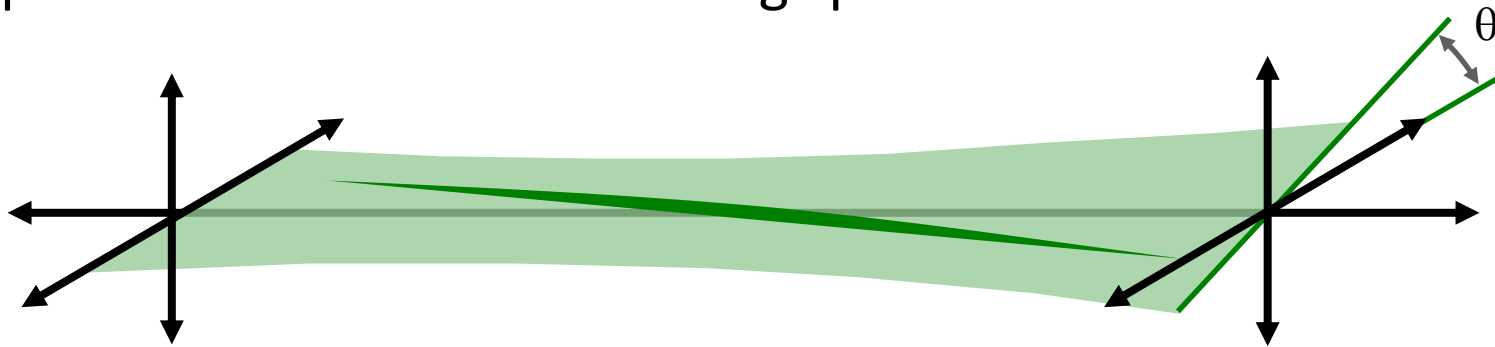
Typical setup has entry, displacement, and exit spans equal to web width (but this can be modified)

Displacement Guide Animation



Stresses from Twisting

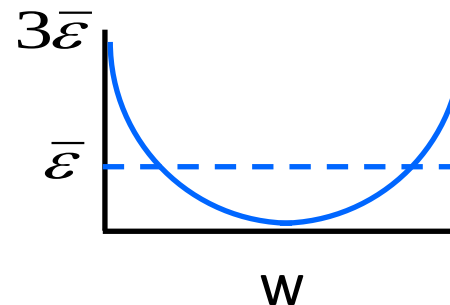
Web twisting is stable and non-deforming when applied to narrow webs and long spans.



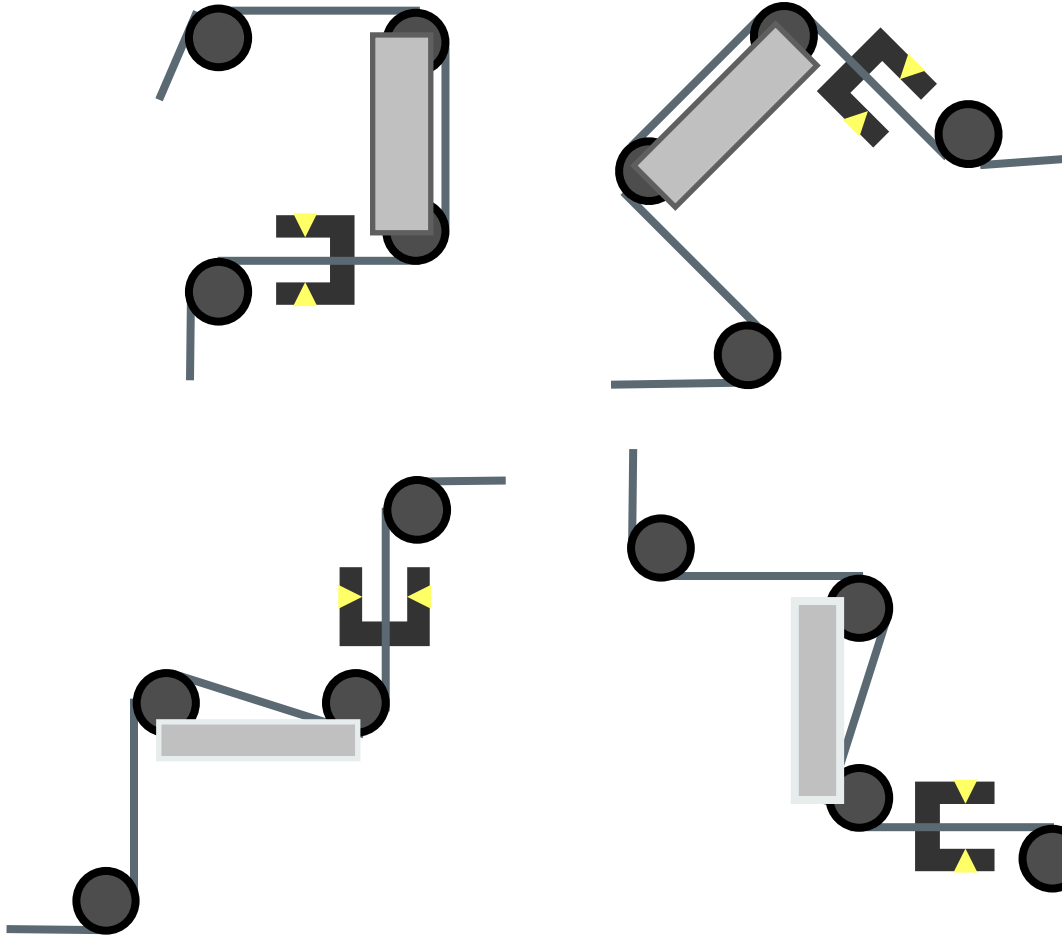
Assuming a parabolic tension distribution, the center web tension is zero at the minimum twisting span, L_{min} , or maximum twisting angle, θ_{crit} :

$$L_{MIN} = w\theta \sqrt{\frac{twE}{24F_T}} = \frac{0.2w\theta}{\varepsilon^{0.5}}$$

$$\theta_{crit} = \frac{L\varepsilon^{0.5}}{0.2w}$$



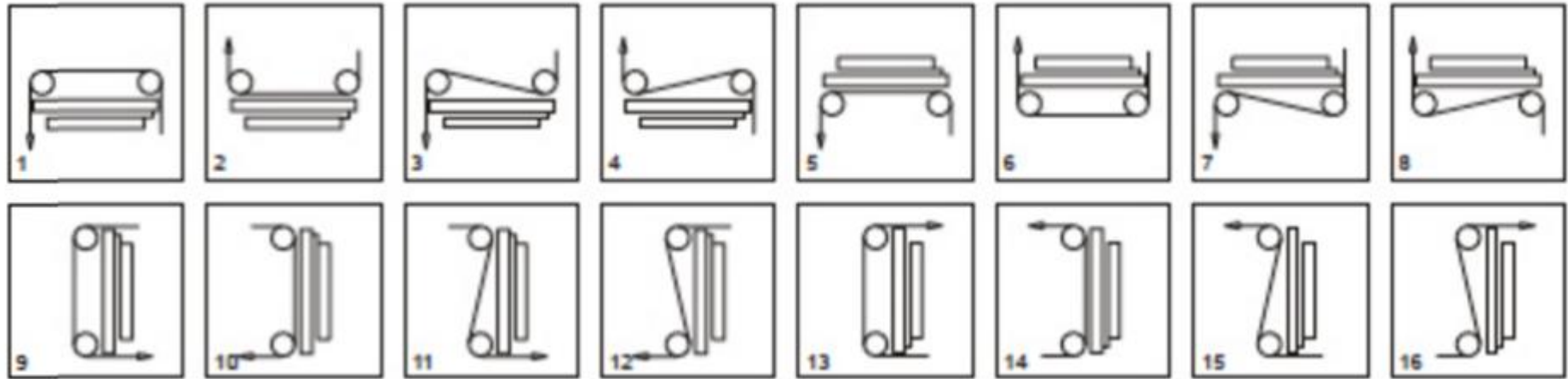
Displacement Guide Geometry



**Displacement plane
can be vertical or
angled.**

**Entry and exit spans
can be in opposite
direction.**

Maxcess Fife Offset Pivot Guide Web Paths



This diagram from Maxcess Fife shows the options for offset pivot guide geometry, including (<http://www.maxcessintl.com/offset-pivot-web-guides>):

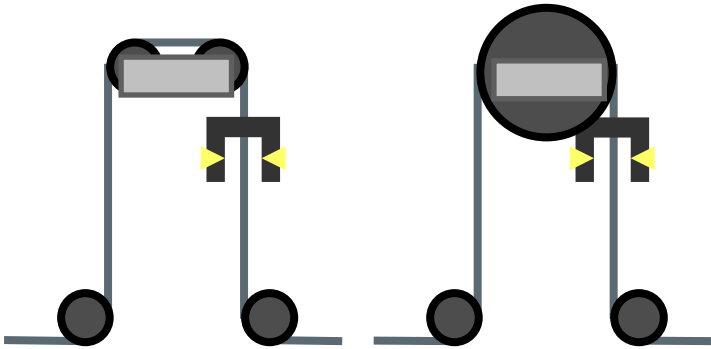
- Displacement plant (horizontal vs vertical)
- Reversing vs pass-through web path (a.k.a. tabletop vs 'S' paths)
- Location of guide pivot frame relative to web path

This does not show recommended lengths for entry, displacement, and exit spans

These spans are typically one web width each, but can be shorter for smaller corrections.

Also, longer displacement spans can allow for shorter entry and exit spans (and vice versa).

Displacement Guide Geometry



If displacement span is shortened, exit and entry spans should be lengthened.

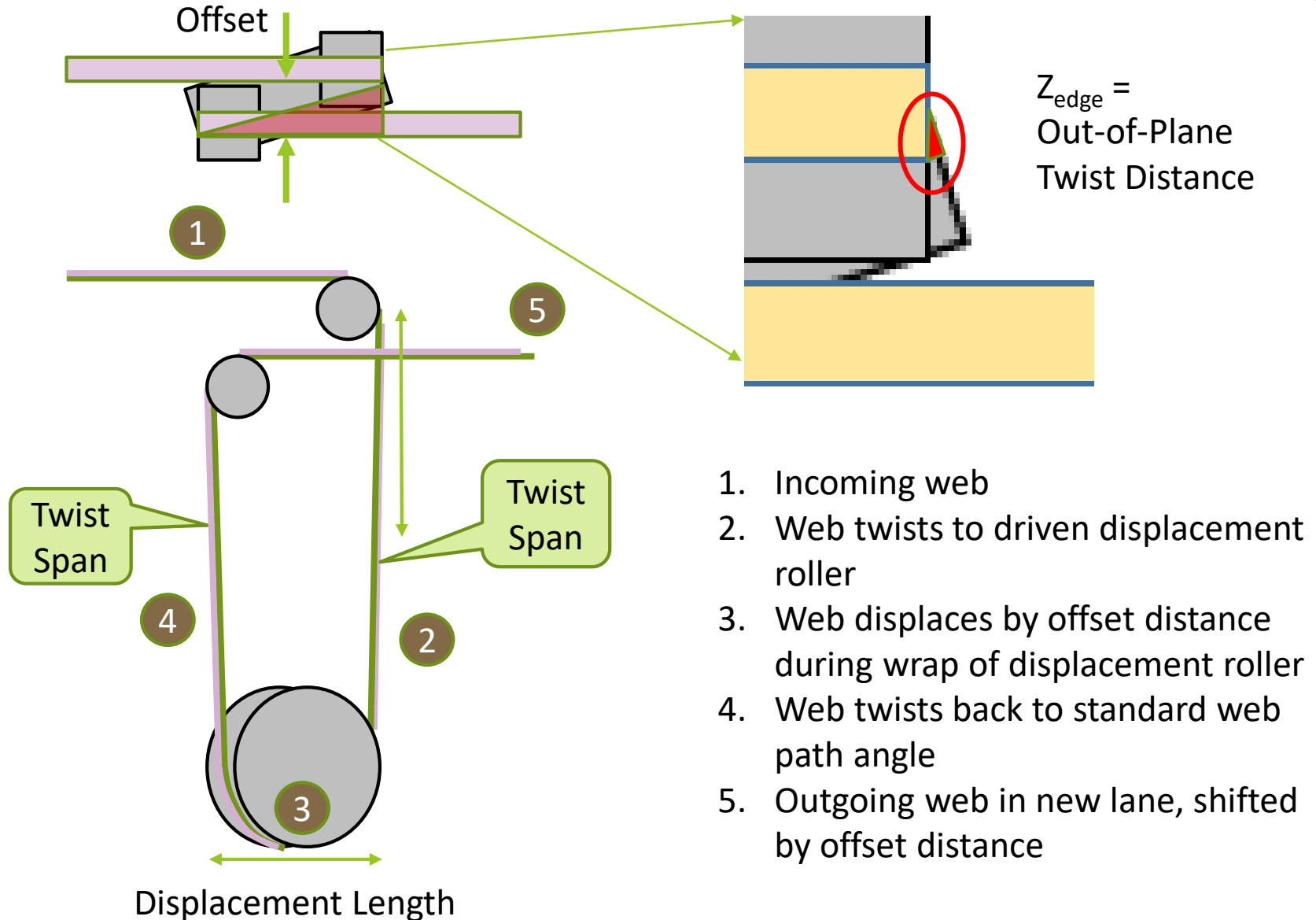
A single large roller can be used in place of two framed rollers.



For shorter exit and entry spans, displacement span should be lengthened.

Theory suggests this is the most precise guide geometry.

One-Side Contact Twist-Drive-Twist



Steering Guiding Summary

- Steering guides creates in-plane bending using the parallel entry rule
- Steering guides have more setup problems than displacement or sidelay guiding.
- TJW recommends steering guides **only** for long span corrections

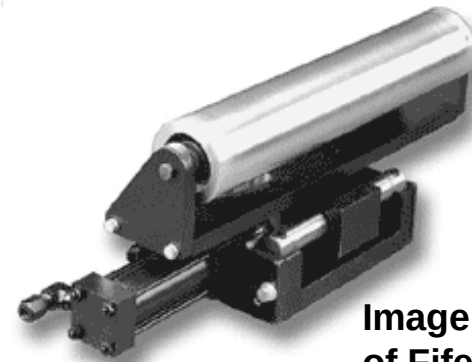
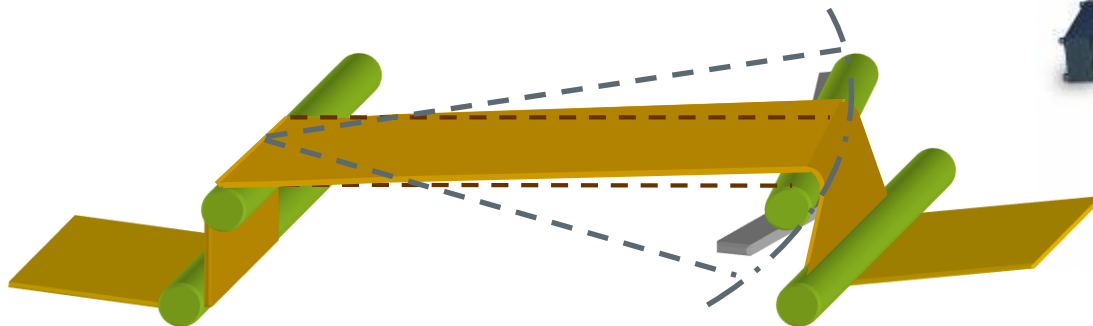


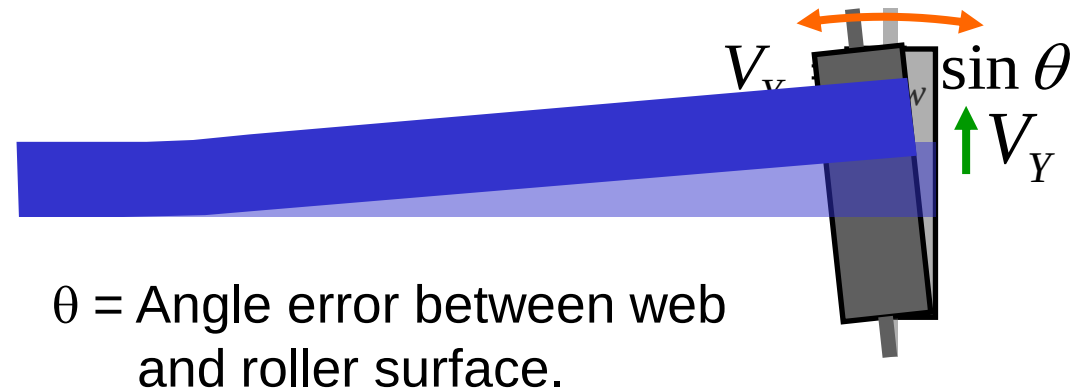
Image courtesy of Fife Corp.



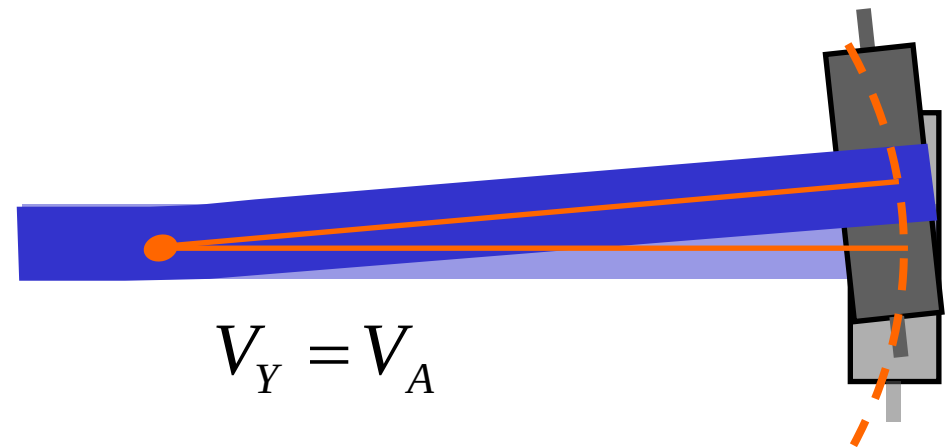
Images courtesy of Fife Corp.

Advantage to Radial Pivot

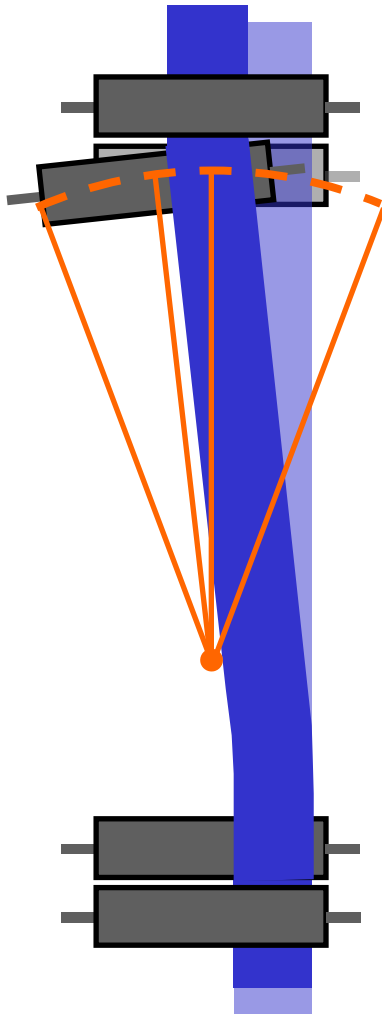
The correction rate of end- or center-pivoting rollers is proportional to web speed.



The correction rate of a radial pivot is proportional to actuation rate.



Steering Mechanism Components



Translation

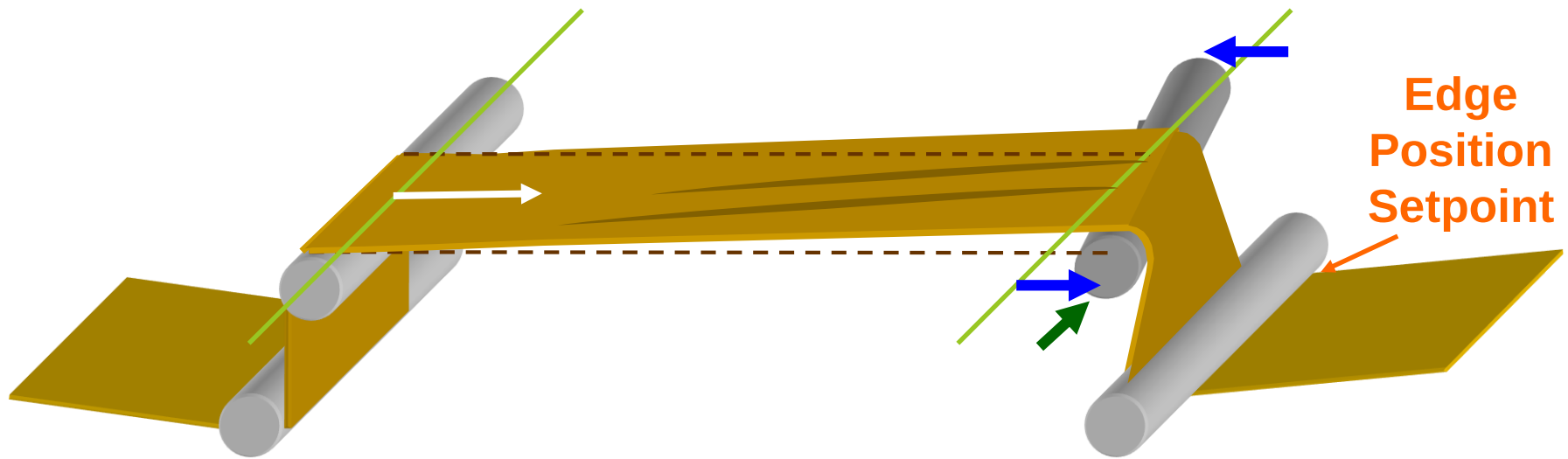
**Translation
transports web
to new location
at actuation
rate.**



Rotation

**Rotation
satisfies parallel
entry rule at new
web tracking
position.**

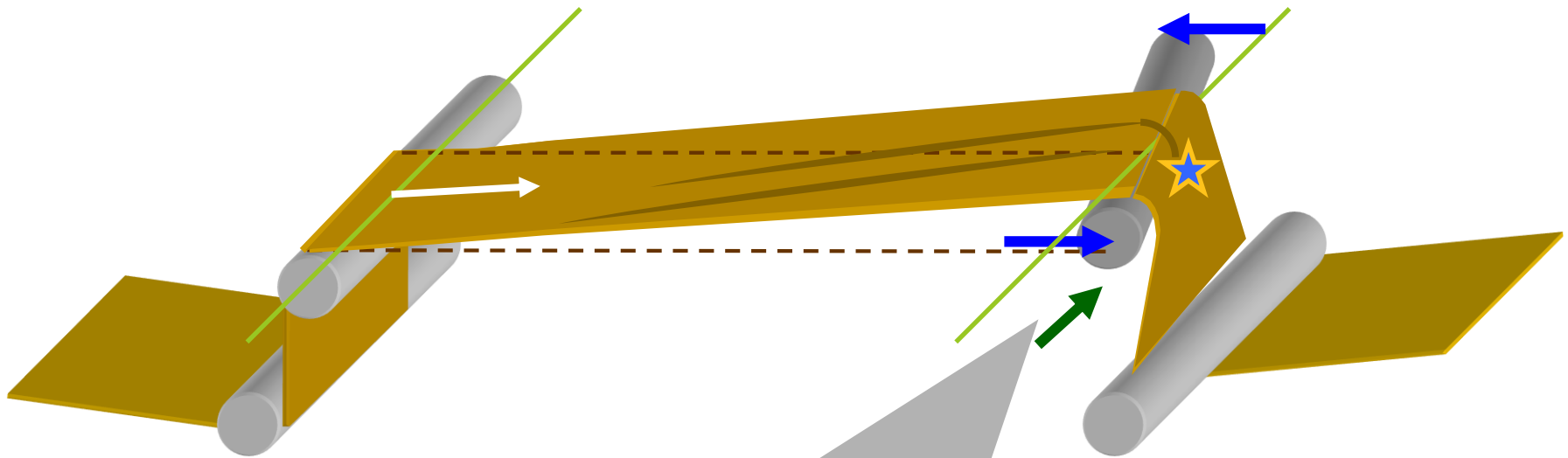
Steering Guide Animation



Rotation →

Translation ↗

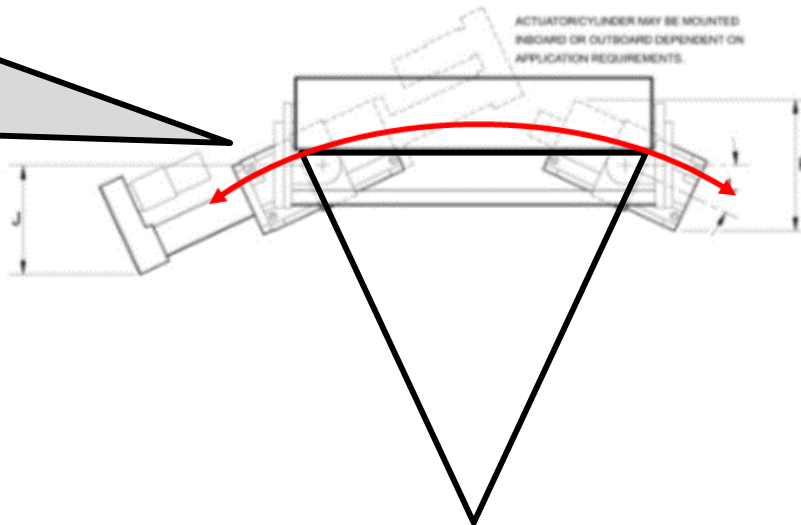
Steering Guide Animation



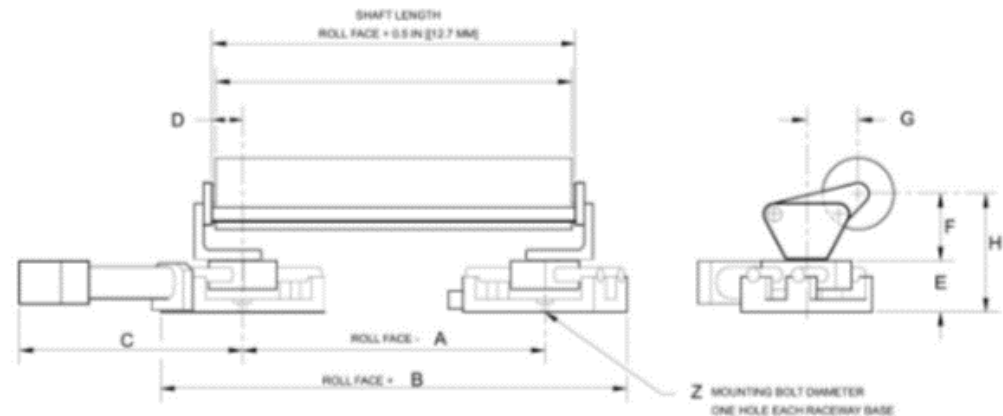
Too much steering correction and too short of entry span creates wrinkles!

Mechanics of Steering-Style Guide

The two linear slides are angled to create a motion as if the steering assembly was pivoting about a point in the upstream span.

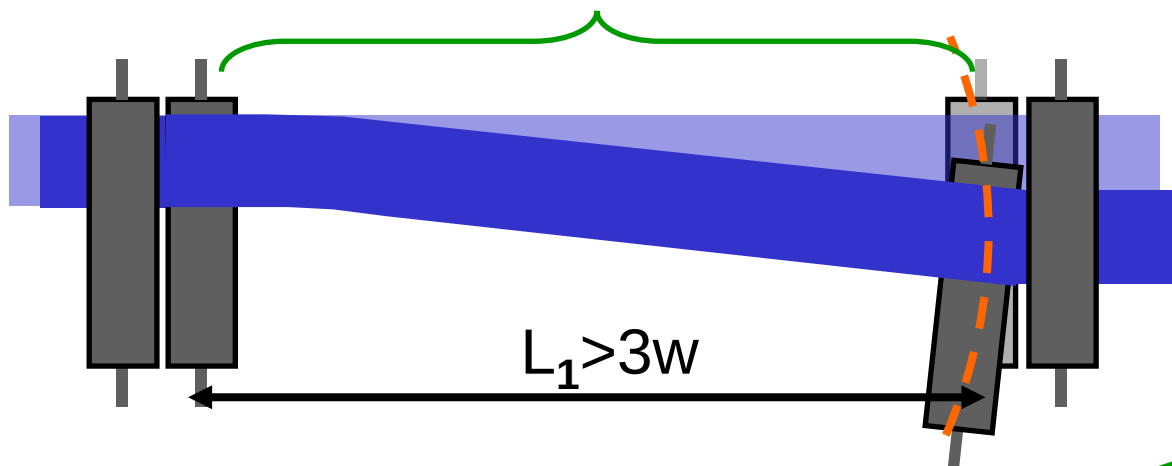


Most commonly, one or two rollers are mounted on the pivoting assembly, but in other cases, several rollers or near roller process equipment can also be on the shifting unit.

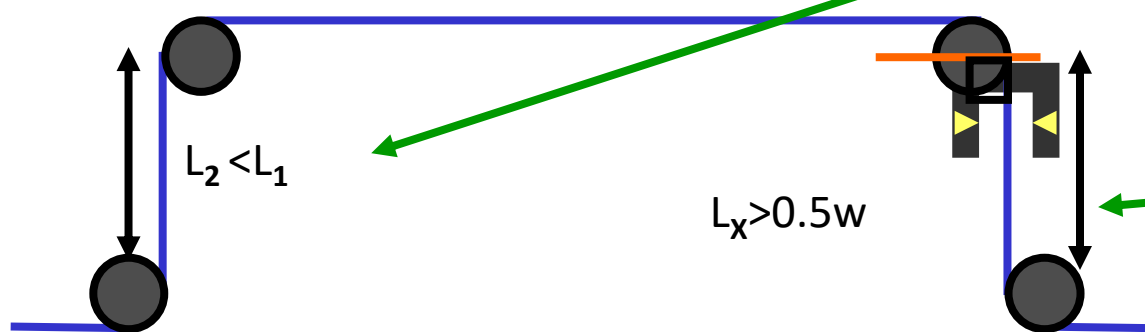


Steering Guide Geometry

Steering or entry span should be long, $L_1 > 3 \times \text{web width}$.

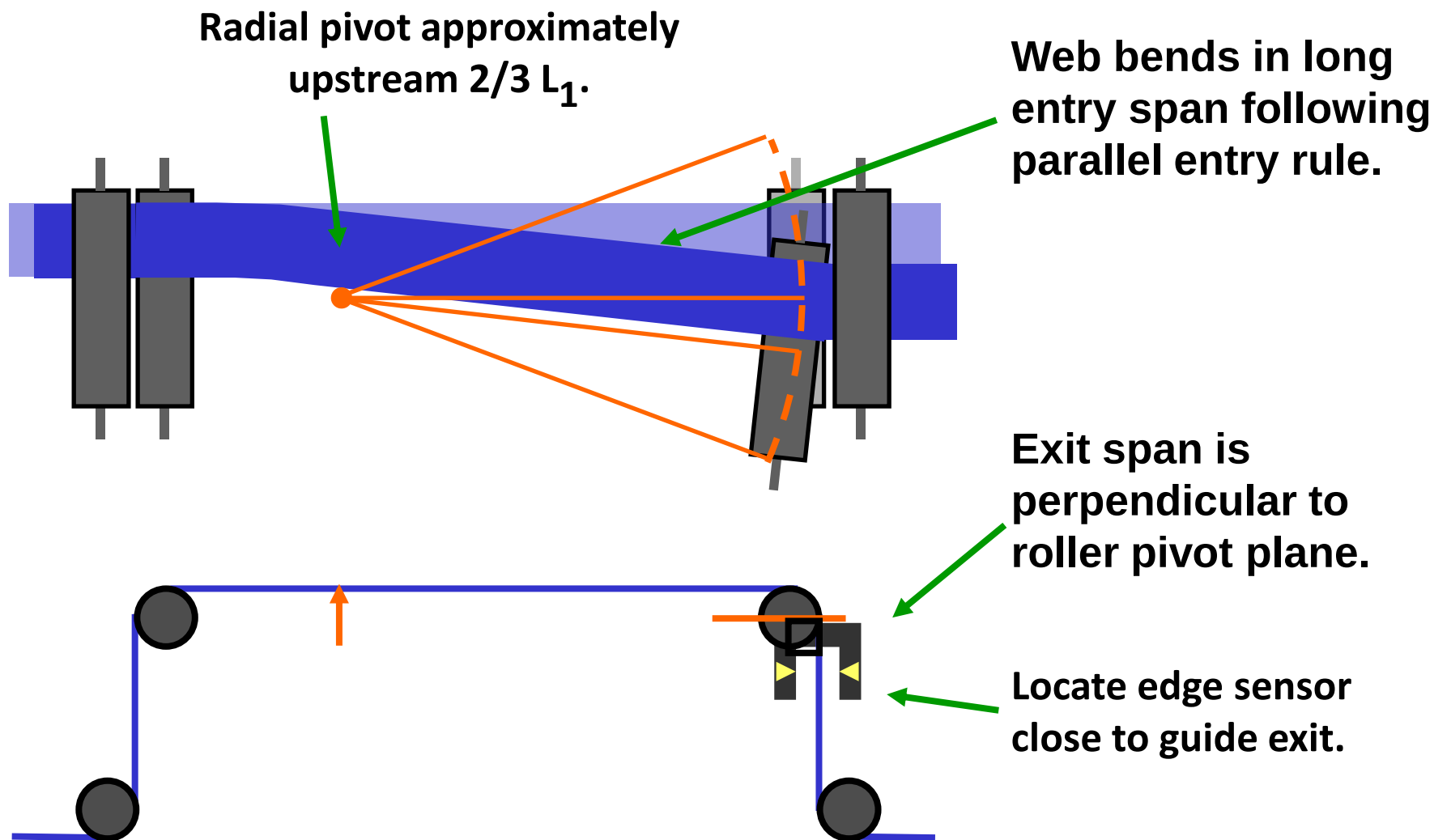


The pre-entry span should be shorter than the entry span, $L_2 < L_1$.



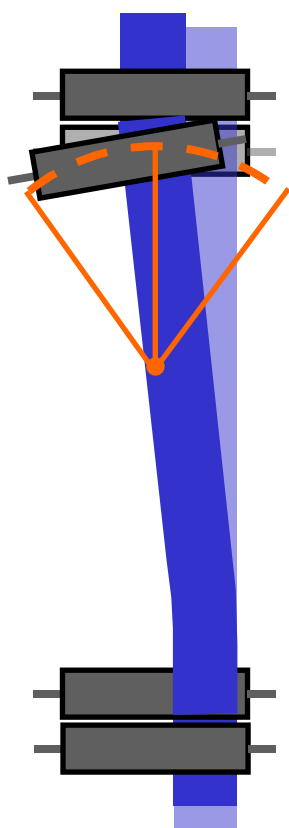
The exit span length should be at least half the web width.

Steering Guide Geometry

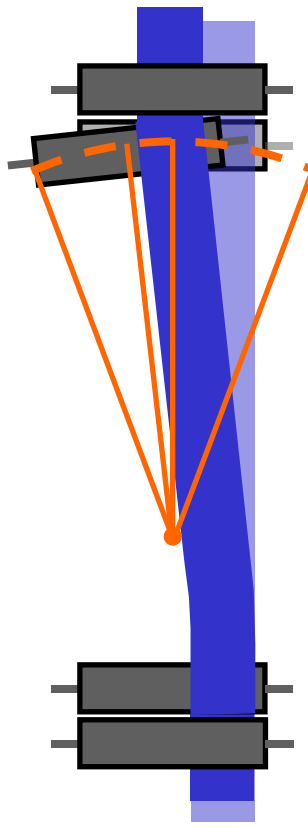


Avoid Over- & Under-Steering

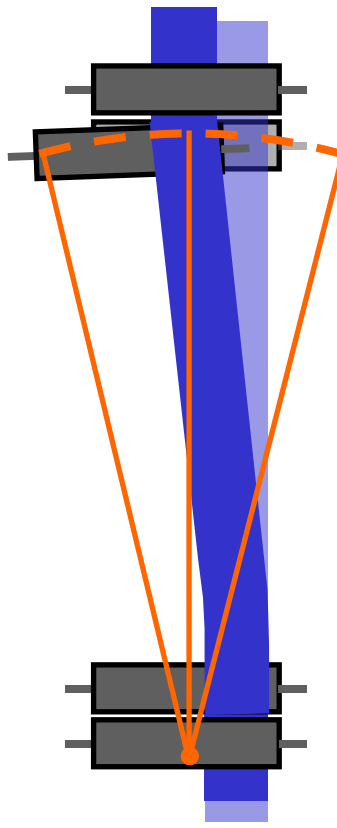
Over-Steering



Good Steering



Under-Steering



Steering guides must be tuned to the upstream web span.

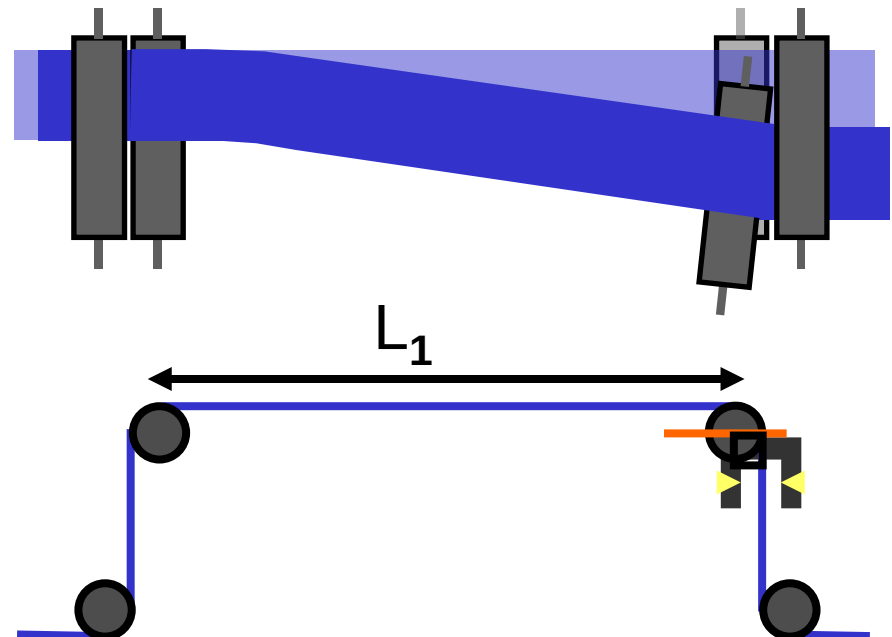
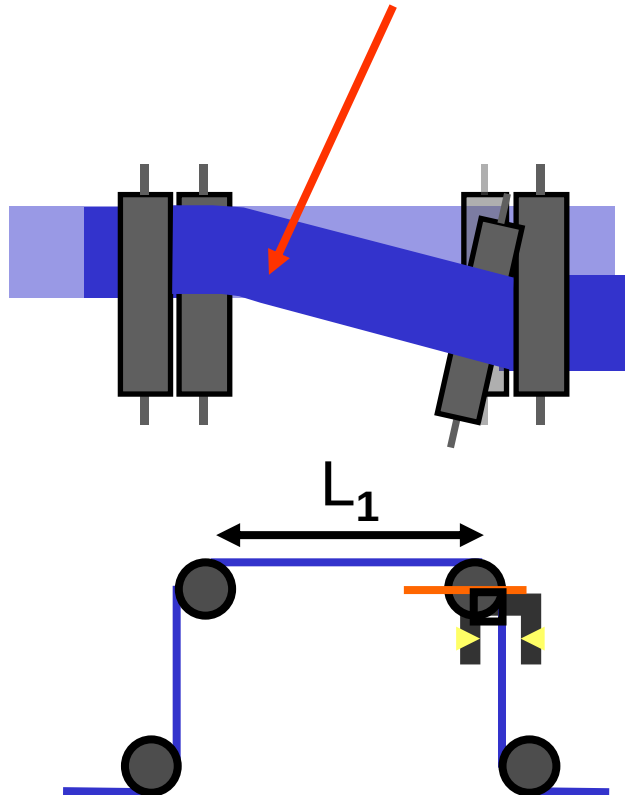
**When pivot radius is properly set,
roller and web move together laterally.**

Avoid Slack Edge

Some steering guidelines recommend an entry span to avoid bending to edge slackness, but for stiff webs, this will be an impossibly long length.
I don't use this criteria.

There will be a slack edge here if:

$$L_{1,\min} = \sqrt{\frac{3CtwE}{2T}} = \sqrt{\frac{3Cw}{2\varepsilon}}$$



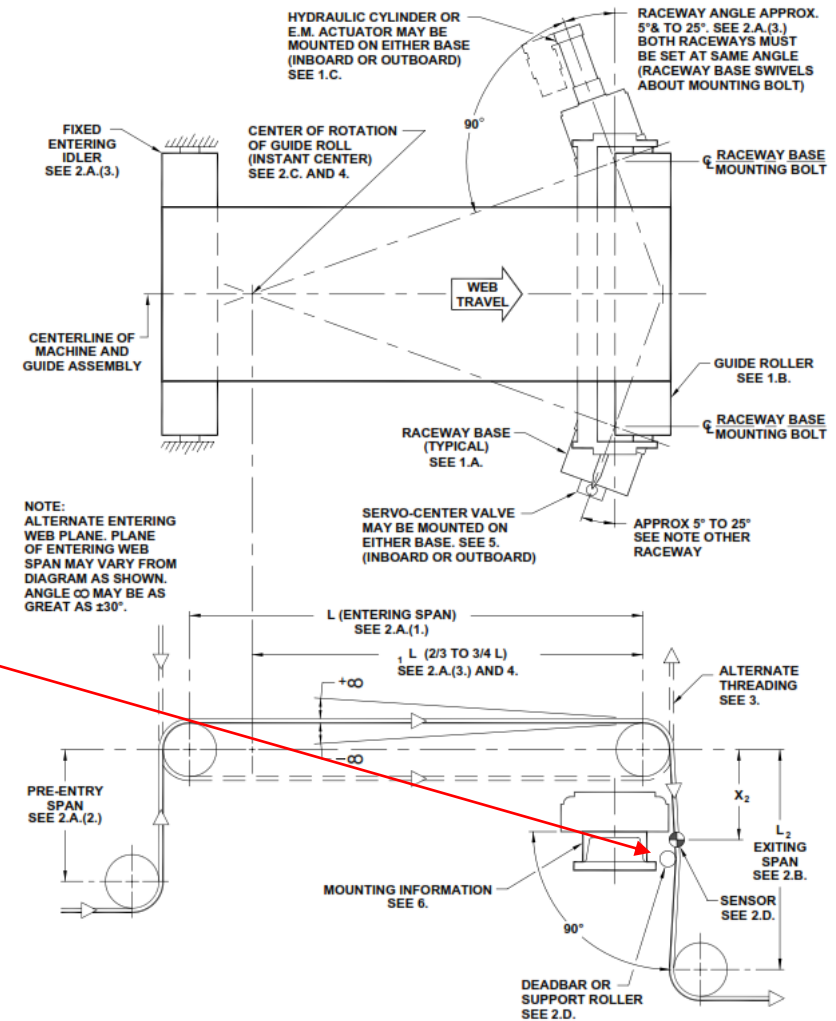
Variable Definitions

- C = correction lateral shift (in.)
- δ = deflection or misalignment (in.)
- E = modulus (lb/in²)
- L = span length (in.)
- L_1 = entry span length (in.)
- L_2 = pre-entry span length (in.)
- t = thickness (in.)
- T = tension (lbf/in)
- w = width (in.)

Fife Kamberoller Design Specifications

This diagram shows the preferred roller, slides, and web path geometry for steering-style web guiding.

The optional dead bar prevents loose edges or low tension webs from fluttering or jumping out of the edge sensor.



Courtesy of Maxcess Fife Corp.

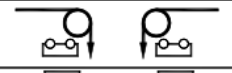
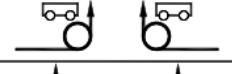
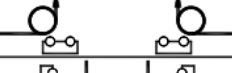
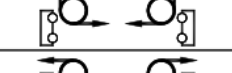
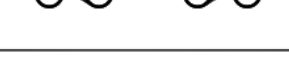
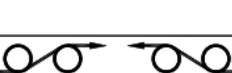
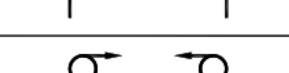
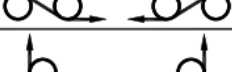
Figure 1.
KAMBEROLLER® INSTALL DESIGN SPECIFICATIONS

Maxcess Fife Kamberoller Layout Guide

Fife provides this diagram to confirm the web path for steering roller installations. Options include:

- ▶ Single or multi-rollers
- ▶ Web path in and out (both orientation and direction)
- ▶ Location or linear slides relative to roller (inside or outside web path)

I do not recommend options 6, 7, or 8 to avoid exit span steering and wrinkling.

1	A		9	
	B			
2	A		10	
	B			
3	A		11	
	B			
4	A		12	
	B			
5	NOT IN USE PRESENTLY		13	
6			14	
7			15	
8			16	

Courtesy of Maxcess Fife Corp.

Avoid “Straight-Through” Steering

The best geometry of a steering guides wraps the roller 90-degrees, where the pivoting motion bends the entering web and twists the exit web.

The ‘straight-through’ web paths show here will bend the web in the exit span, creating some post-guide correction. This can lead to instability or wrinkling.

Though these web paths may be acceptable for small corrections, including belt guiding, my preference is to avoid ‘straight-through’ steering web path geometry.

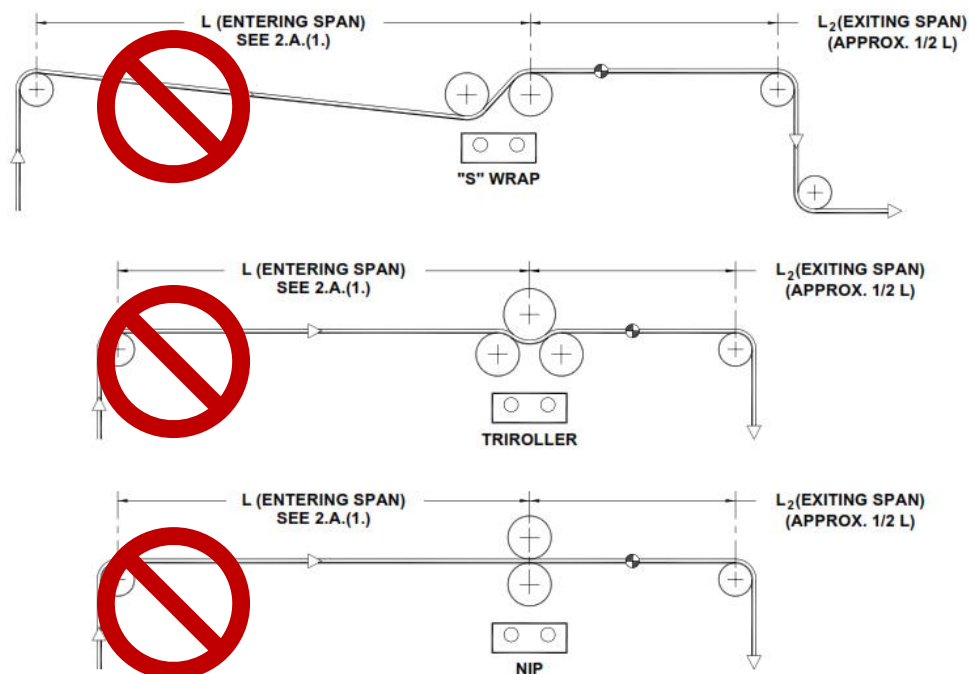
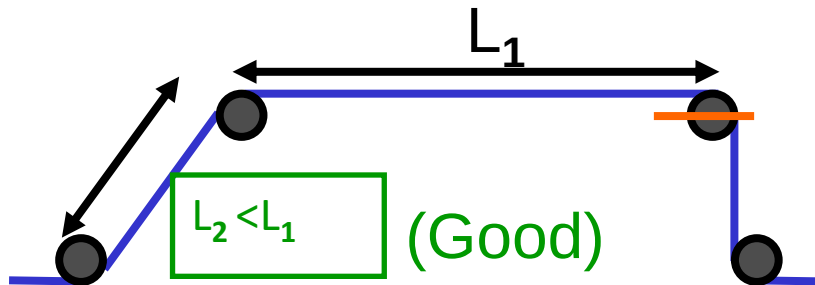
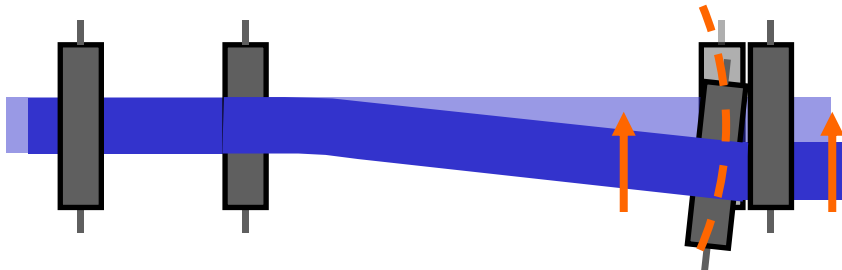


Figure 2.
KAMBEROLLER® WRAP STYLES

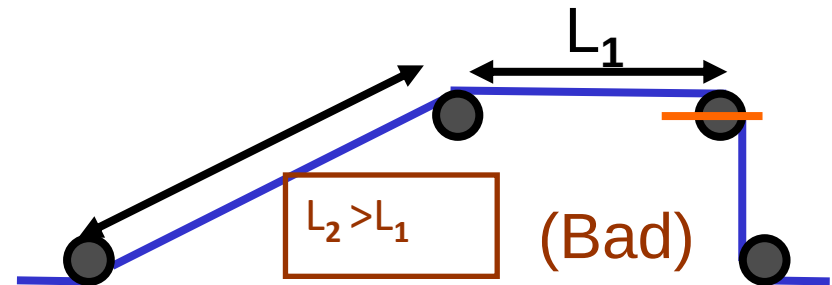
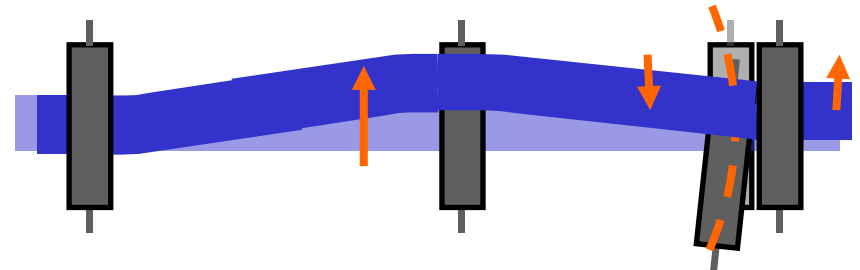
Avoid “Unstable” Span Geometry

Stable Geometry



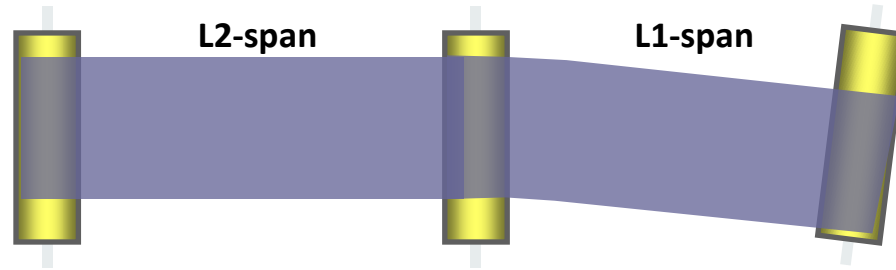
Steering roller entry span must be longer than pre-entry span.

Unstable Geometry

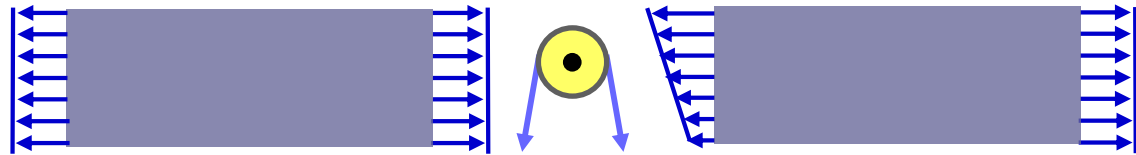


With unstable geometry the web may guide opposite of intentions.

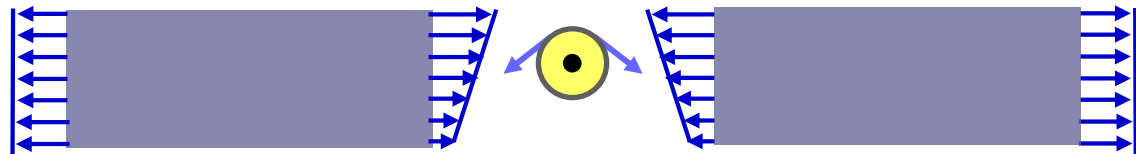
Moment Transfer



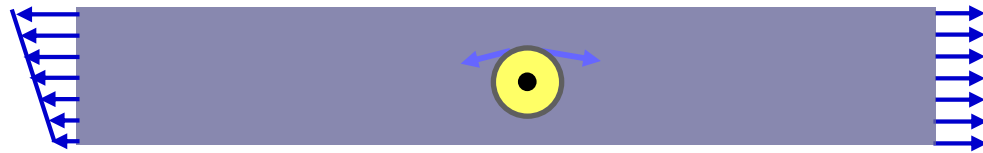
No moment transfer



Moment transfer + parallel entry

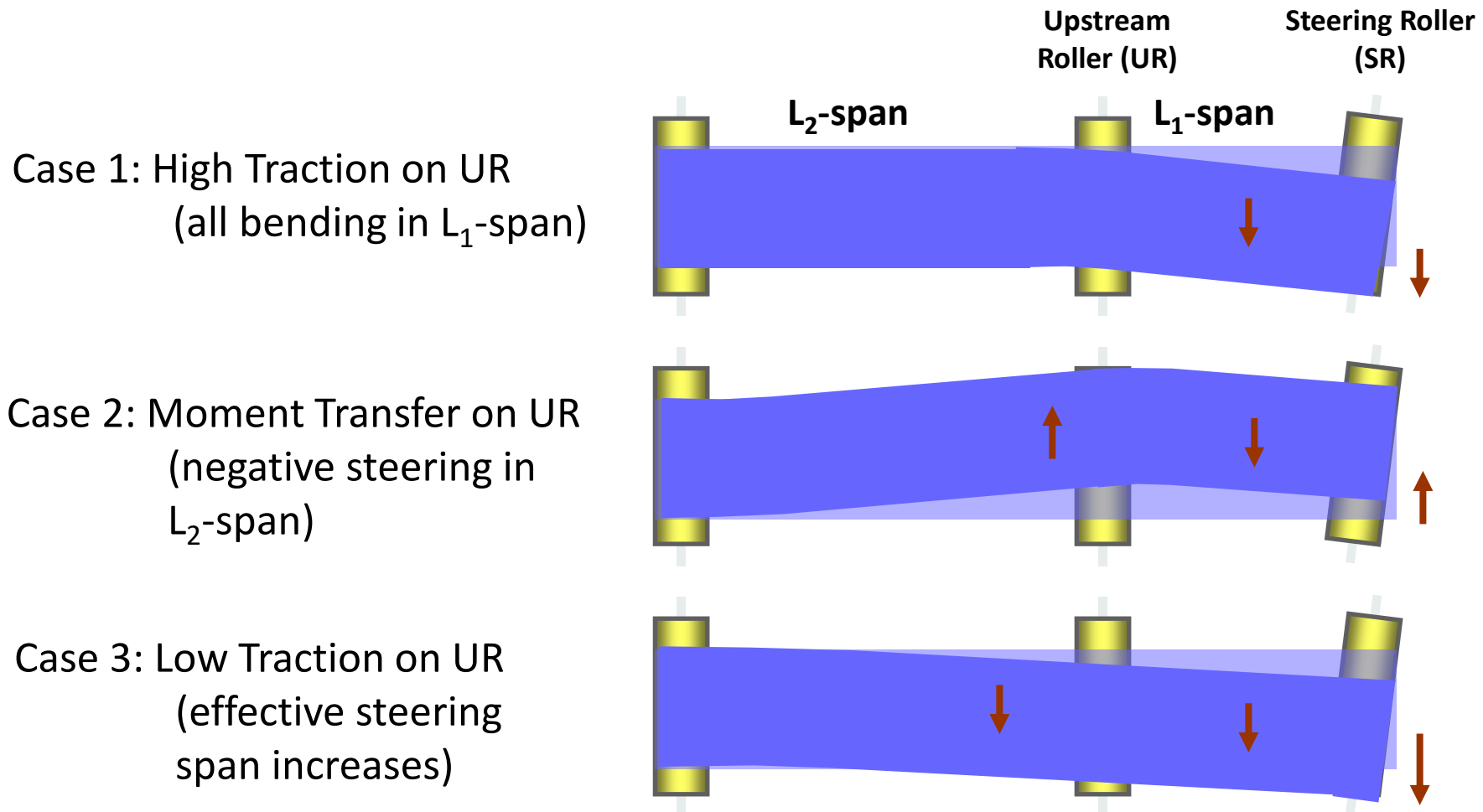


Moment transfer + No parallel entry

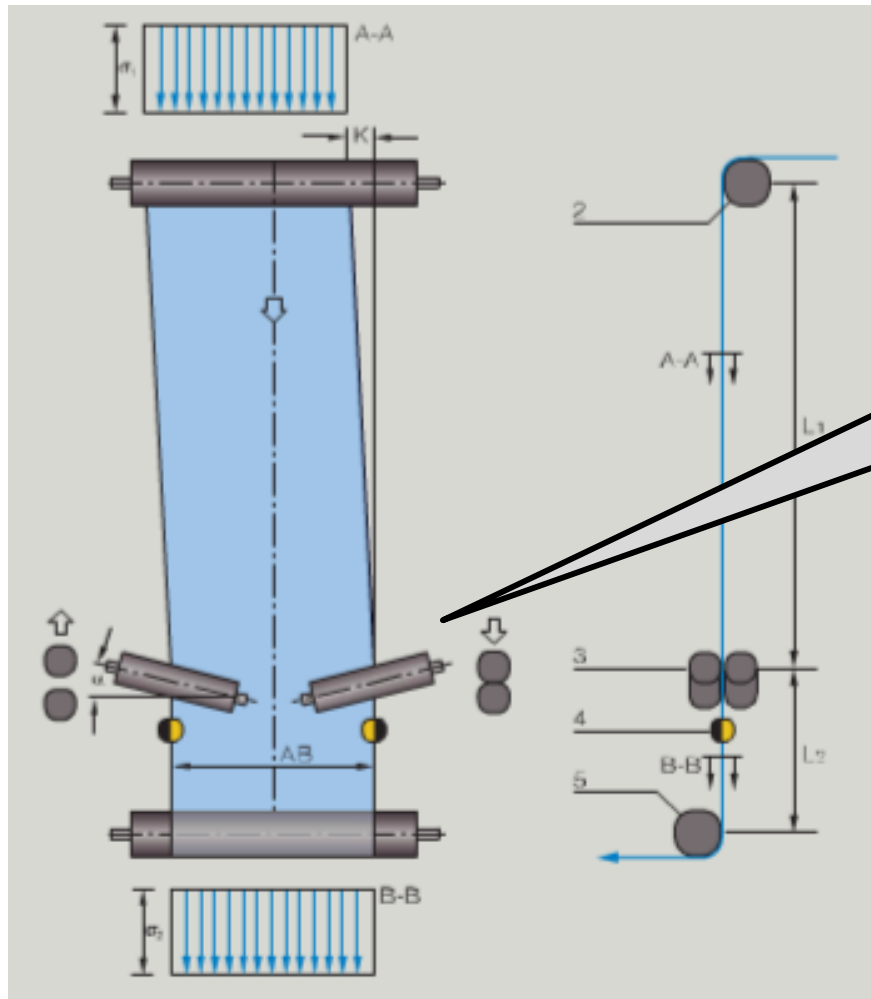


Unstable Web Path and Steering

Moment transfer and $L_2 > L_1$ can lead to negative steering.



Edge Nip Guiding (only for stiff webs or belts)



Nipping rollers are installed at the downstream edges of a long span. The nips are angled outward. When engaged, each nip pulls the web to its side. Differential nipping or loading can be used for closed-loop control.

ELTWIN KF 2020

Scanning: mechanical
Nip roller pressure: pneumatic



ELTWIN KF 2040

Scanning: mechanical
Nip roller pressure: electromagnetic



ELTWIN KF 2060

Scanning: optoelectronic
Nip roller pressure: electromagnetic
Auxiliary drive possible as an option
Solo control possible as an option



ELTWIN KF 2080

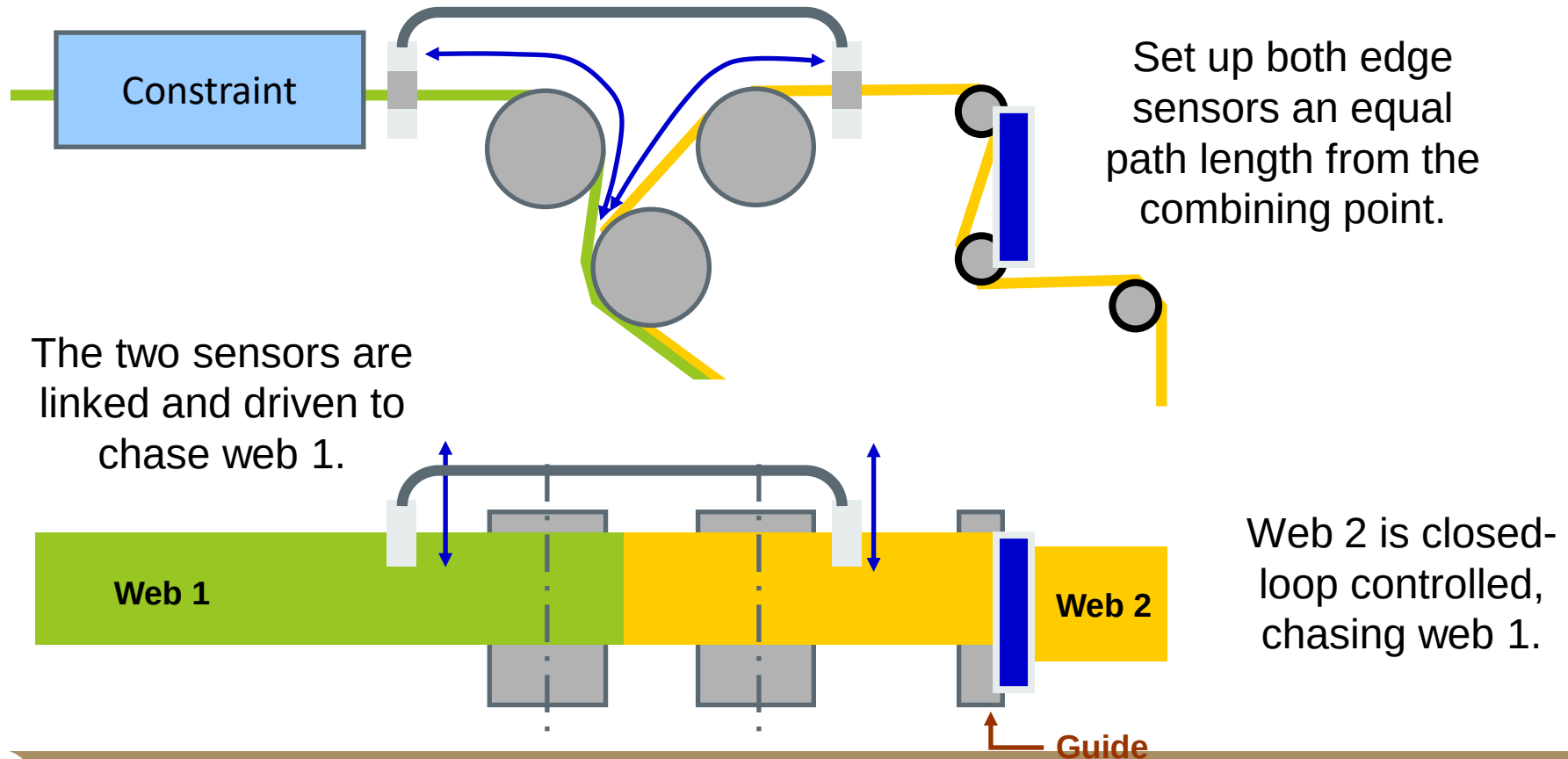
Scanning: broadband sensor
Nip roller pressure: electromagnetic
Auxiliary drive possible as an option
Oscillation possible as an option



Courtesy of Erhardt and Leimer

Chase Guiding

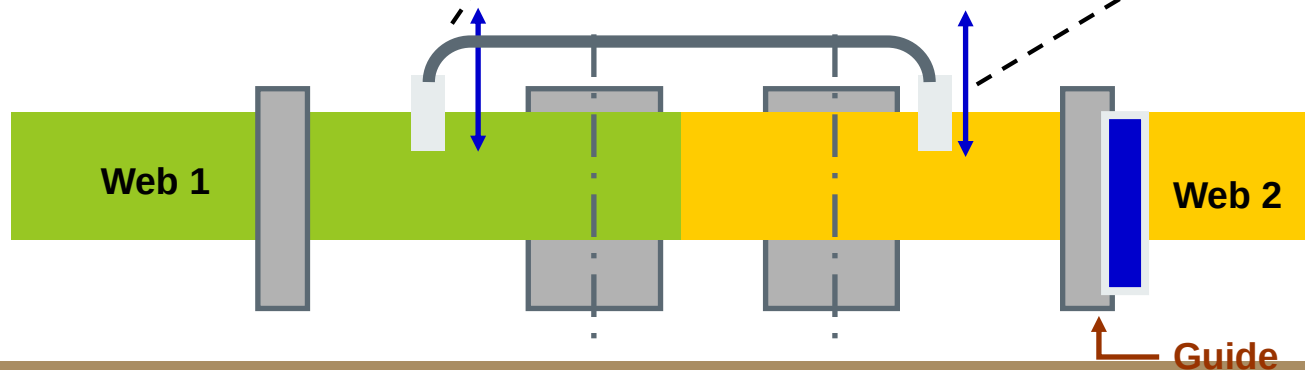
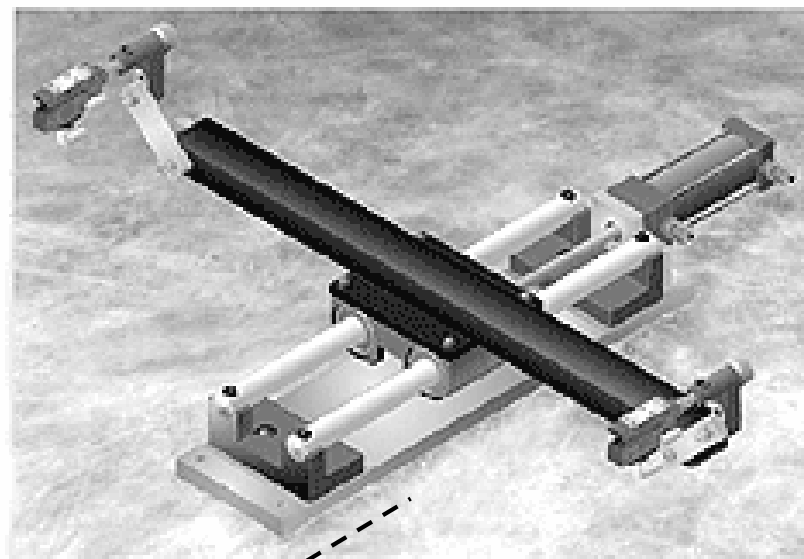
Chase guiding is used to align two webs, where system constraints make one web difficult to guide.



Chase Guiding Setpoint Control

Example of chase
guide dual sensor
actuator

One sensor
mechanically follows
the other.



Chase Guiding Setpoint Control

Alternately, using two wide array sensors...

...one sensor's setpoint electronically follows the other.

